Black Hole Dynamics and Improving Waveforms in Einstein–Maxwell–Dilaton–Axion Theory

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ABSTRACT

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This dissertation studies two complementary aspects of numerical relativity. First, we investigate extensions of general relativity by numerically simulating binary black hole mergers in alternative theories of gravity. In particular, we generalize our numerical relativity code to evolve black holes in Einstein Maxwell Dilaton Axion (EMDA) theory. Black holes in this theory can support additional scalar and pseudoscalar fields, or hair, that are absent in vacuum general relativity. We present a new method for generating approximate binary black hole initial data in EMDA theory and use head on collisions to demonstrate the stability of charged black holes in this setting. We then develop a modest catalog of inspiraling charged black hole waveforms and compare these waveforms with general relativistic reference waveforms. These comparisons show that the additional fields can modify the gravitational waveform, with differences that generally increase with charge. We also explore the broader EMDA theory space by varying the coupling parameters that control the interaction between the dilaton, axion, and electromagnetic fields.

Second, we study compact finite difference derivative operators for use in adaptive numerical simulations. These operators are of interest because they have the potential to improve the accuracy and efficiency of numerical relativity calculations. We construct several novel compact finite difference operators and test their performance on a variety of systems, including the Maxwell equations, the nonlinear sigma model, and the BSSN equations. These tests allow us to compare convergence, stability, and robustness across both linear and nonlinear problems. We show that compact finite difference operators can improve performance compared to their explicit finite difference counterparts.

Keywords: Numerical relativity, alternative theories of gravity, black holes, gravitational waves, compact finite difference methods, adaptive mesh refinement

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Chapter 1

Introduction

1.1 Introduction

Since Einstein introduced the theory of general relativity in 1915 [2], it has passed all observational tests [3, 4]. General relativity has fundamentally transformed our understanding of space and time, explaining phenomena ranging from the anomalous precession of Mercury to the operation of atomic clocks in GPS satellites [5], laid the foundations for cosmology, gravitational radiation as well as predicting the existence of compact objects such as neutron stars and black holes [6–8].

Gravitational waves were first directly detected by LIGO in 2015 [9]. These signals originate from some of the most energetic events in the universe, such as mergers of neutron stars and black holes. However, the signals detected by LIGO are extremely weak; the detectors are sensitive enough to measure relative length changes on the order of 10^{-21}, corresponding to displacements smaller than a proton diameter [10]. Despite this extraordinary sensitivity, gravitational wave signals remain weak relative to background noise. The highest signal to noise ratio (SNR) gravitational wave event reported to date is GW250114, with a network matched filter SNR of approximately 80 [11].

Interpreting gravitational wave detections requires accurate theoretical models of the expected signals. In practice, this is achieved by comparing observational data against waveform templates constructed from a combination of analytical approximations, phenomenological models, and numerical simulations. This template based comparison allows the physical properties of the source, including the component masses and spins, to be inferred statistically [12, 13]. Figure 1.1 shows the observed GW150914 signal compared with a numerical relativity waveform for a binary black hole merger.

Constructing theoretical models of binary black hole mergers is challenging because the relevant dynamics are strongly nonlinear and only partially accessible through analytic methods. In general relativity, exact analytic solutions are typically restricted to highly symmetric systems and cannot describe the full inspiral, merger, and ringdown of a generic binary black hole system. Numerical relativity therefore plays an essential role in modeling the late inspiral, merger, and ringdown, where

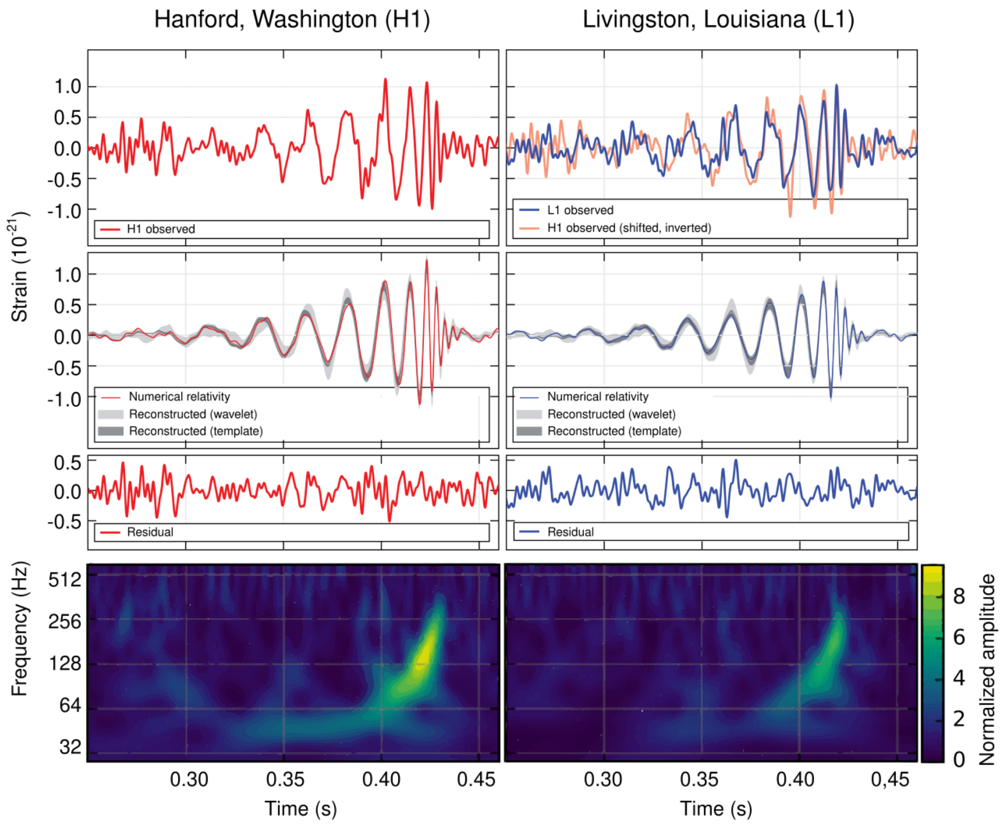


Figure 1.1 The first gravitational wave signal detected by LIGO, GW150914. The observed strain data from the two LIGO detectors are shown together with a numerical relativity waveform, identifying the source as a binary black hole merger.

fully nonlinear effects are important. A number of numerical relativity codes have been developed for this purpose, including the Einstein Toolkit, SpEC, GRChombo, BAM, GR Athena++, and Dendro-GR [14–19].

Simulating binary black hole mergers is computationally demanding. These simulations require solving a coupled system of nonlinear partial differential equations in three spatial and one time dimension, across multiple length scales, including the scale of the individual black holes, the orbital scale of the binary, and the radiation zone where gravitational waves are extracted [20, 21]. In this work, we make use of the Dendro numerical relativity framework, which combines adaptive mesh refinement with scalable parallel algorithms for binary black hole simulations [19, 22, 23].

For standard general relativity (GR), substantial progress has been made in modeling binary black hole mergers. Extensive waveform catalogs now cover a number of astrophysically relevant portions of the binary black hole parameter space. In GR, the parameter space for binary black holes covers each black hole's spin and the mass ratio of the system. Current catalogs have good coverage for near equal mass systems for a broad range of spins. Here we define the mass ratio as $q = M_{\text{larger}}/M_{\text{smaller}} \geq 1$. The LIGO Virgo KAGRA (LVK) collaboration has reported hundreds of compact binary candidates using waveform models informed by numerical relativity [15, 24, 25]. Future gravitational wave detectors, such as Einstein Telescope and Cosmic Explorer, are expected to achieve significantly greater sensitivity than current ground based detectors [26–28]. This increased sensitivity is expected to improve the signal to noise ratios of observed compact binary signals and to expand the range over which such systems can be detected. Meeting this challenge will require higher fidelity numerical simulations and more accurate waveform models [29, 30].

In general, higher fidelity simulations, particularly those with larger mass ratios, require finer grid resolution and therefore substantially greater computational cost. This trend is illustrated in Fig. 1.2. Current simulations with mass ratios $q < 8$ can require weeks of runtime on modern supercomputers, and further increases in resolution would significantly amplify this cost [15, 30].

To obtain more accurate waveforms, new techniques are needed. Running existing codes at higher resolution will be insufficent. The drawback of the brute force approach of simply increasing resolution is twofold. The first is that increasing the number of points inherently increases a simulation's cost. The second is that the size of the steps we take in time must also have a corresponding reduction. This, in turn, requires more timesteps, significantly increasing the total runtime and computational expense of a simulation.

For numerical relativity codes, spatial derivative operators are a major source of truncation error, since the evolution equations contain many derivative terms while algebraic operations are evaluated locally to roundoff accuracy. Therefore, one potential avenue for improving efficiency in numerical

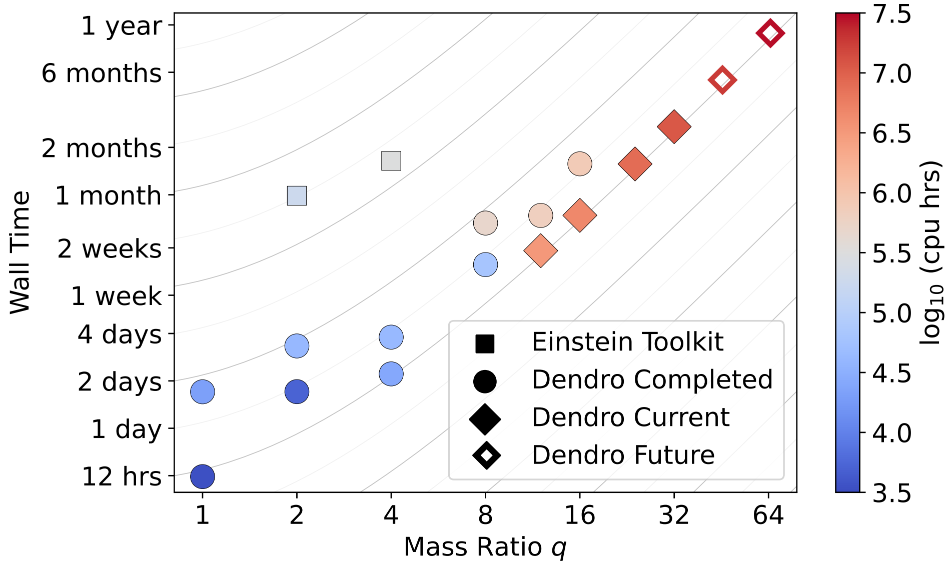


Figure 1.2 Wall time of numerical relativity simulations as a function of mass ratio. As the mass ratio increases, the computational cost increases dramatically because the smaller black hole must be resolved with sufficient grid points. With current codes, performing a large number of high mass ratio simulations would therefore be prohibitively expensive.

relativity codes lies in the computation of spatial derivatives. Most current implementations rely on explicit finite difference methods, while spectral methods, although highly accurate, are often difficult to implement. In this work, we explore compact finite difference (CFD) operators as an alternative. Compact schemes are implicit methods that can achieve higher spectral resolution than comparable explicit schemes [31]. Across a variety of test problems, including the nonlinear sigma model, Maxwell's equations, and Einstein's equations, the compact finite difference schemes generally outperform the corresponding explicit methods. In some cases, the compact schemes achieve comparable accuracy to explicit schemes at approximately half the resolution. Despite their success in computational fluid dynamics and related fields, their use in numerical relativity remains relatively unexplored [32].

In addition to the need for improved waveform modeling in GR, there is also a need for waveform models in alternative theories of gravity. Gravitational wave observations provide a powerful probe of gravity in the strongly nonlinear, dynamical regime, where deviations from general relativity (GR) may become apparent [4, 33, 34]. To search for such deviations, waveform templates from alternative theories of gravity are required [35–37]. While studies have been carried out in theories such as Chern-Simons gravity [38, 39], scalar-tensor theories [40–43], Einstein-dilaton-Gauss-Bonnet gravity [44–49], and Einstein-Maxwell-dilaton theory [50, 51], waveform catalogs in these frameworks remain limited [52, 53].

In this dissertation, we focus on Einstein-Maxwell-dilaton-axion (EMDA) theory. EMDA is of particular interest because it admits a well posed initial value formulation and arises as a low energy limit of string theory. In contrast to the no hair theorem in standard GR, which states that black holes can be completely defined by their mass spin and charge, EMDA black holes possess additional degrees of freedom associated with the dilaton and axion fields.

We present a numerical relativity framework for simulating binary black hole mergers in EMDA theory. We analyze the resulting waveforms and investigate whether deviations from GR may be observable in gravitational wave signals.

1.2 Background

We focus on the study of black hole dynamics. As such, we begin with the Einstein field equations:

$$G_{ab} = 8\pi T_{ab} + \Lambda g_{ab}. \tag{1.1}$$

The left hand side describes the geometry of spacetime, while the right hand side encodes the matter and energy content. [54, 55] We briefly examine each term in turn.

The tensor G_{ab} is the Einstein tensor, which is constructed from quantities that encode the curvature of spacetime. The stress energy tensor T_{ab} describes the distribution of matter and energy. The metric tensor g_{ab} defines distances and the causal structure of spacetime. The constant Λ is the cosmological constant, which is generally assumed to be very small. On the scales considered in this work, it is negligible, and we therefore set $\Lambda = 0$.

In standard general relativity simulations of binary black hole mergers, we consider vacuum spacetimes, for which

$$G_{ab} = 0. \tag{1.2}$$

Although this equation appears innocuous, it represents a complex system of partial differential equations.

The Einstein tensor G_{ab} is constructed from the Ricci tensor R_{ab}, the metric tensor g_{ab}, and the Ricci scalar R according to

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R. \tag{1.3}$$

The Ricci tensor and Ricci scalar are themselves contractions (contractions can be thought of as generalizations of dot products) of the Riemann curvature tensor $R^a{}_{bcd}$. The Ricci tensor is obtained by contracting the first and third indices,

$$R_{ab} = R^c{}_{acb}, \tag{1.4}$$

and the Ricci scalar is obtained by a further contraction,

$$R = R^a{}_a. \tag{1.5}$$

The Riemann tensor is built from the Christoffel symbols and their derivatives. With our sign convention, it may be written as

$$R^a{}_{bcd} = \partial_c \Gamma^a{}_{bd} - \partial_d \Gamma^a{}_{bc} + \Gamma^a{}_{ec} \Gamma^e{}_{bd} - \Gamma^a{}_{ed} \Gamma^e{}_{bc}, \tag{1.6}$$

where $\Gamma^a{}_{bc}$ are the Christoffel symbols. These are determined by the metric and its first derivatives:

$$\Gamma^a{}_{bc} = \frac{1}{2} g^{ad} \left(\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc} \right). \tag{1.7}$$

Together, these relations show how the compact equation $G_{ab} = 0$ encodes a system of nonlinear partial differential equations for the metric.

Although these geometric definitions expose the underlying structure of the Einstein equations, they still do not immediately resemble the second order system of partial differential equations used in numerical evolution. The equations can be reformulated into a strongly hyperbolic system of equations suitable for time evolution, together with a separate set of elliptic equations that constrain the solution. We defer a complete discussion and expanation of how these evolution equations are formulated into a set of coupled partial differential equations to Appendix A. A key point we make here is that these constraints, which for GR are the Hamiltonian and momentum constraints, must be satisfied at all times. The Hamiltonian constraint relates the intrinsic and extrinsic geometry of a spatial slice to the energy density on that slice, while the momentum constraints relate the extrinsic curvature to the momentum density. In this sense, they play a role analogous to enforcing the consistency of the energy and momentum content of the initial data, although they are not conservation laws in the ordinary Newtonian or Noetherian sense. This is important as the constraints can be understood as the equations that set the initial data which, subsequently, the time evolution equations evolve forward in time. In numerical simulations the constraints are not satisfied exactly because of finite resolution and truncation error. However, for a consistent numerical method, the constraint violations should decrease with increasing resolution and vanish in the continuum limit.

The preceding discussion about the constraints is somewhat dense and we have skipped several steps, so it may be useful to consider a simpler example in which we have a set of partial differential equations with constraints that set initial data and time evolution equations which evolve forward in time the dynamic degrees of freedom; namely Maxwell's equations. In familiar vector notation, Maxwell's equations are

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \tag{1.8}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{1.9}$$

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi \mathbf{J}, \tag{1.10}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}. \tag{1.11}$$

We present again the full $3+1$ decomposition of Maxwell's equations in tensor notation in Appendix A. However, an important feature is already visible in this form. There are eight scalar equations, but only six dynamical variables: E_x, E_y, E_z, B_x, B_y, and B_z. The remaining two equations constrain the allowed solutions.

Maxwell's equations can therefore be separated into evolution equations,

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi \mathbf{J}, \tag{1.12}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \tag{1.13}$$

and constraint equations,

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \tag{1.14}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{1.15}$$

The constraint equations may also be written as

$$C_E \equiv \nabla \cdot \mathbf{E} - 4\pi\rho = 0, \tag{1.16}$$

$$C_B \equiv \nabla \cdot \mathbf{B} = 0. \tag{1.17}$$

These equations illustrate an important feature of many systems of partial differential equations. The electric and magnetic fields cannot be chosen arbitrarily. Instead, the initial data must satisfy the constraint equations before the evolution equations can be used to advance the solution in time. If the constraints are satisfied initially, they are preserved by the continuum evolution.

General relativity has a similar mathematical structure, albeit nonlinear. As noted before, Einstein's equations can be separated into evolution equations and constraint equations. Therefore only initial data that satisfy the constraints represent physically valid solutions. Constructing such initial data is known as solving the Cauchy problem. More generally, the Cauchy problem is the initial value formulation of a system of partial differential equations, in which one specifies data on an initial spatial hypersurface that satisfy the constraint equations and then uses the evolution equations to determine the future development of the system.

Solving the Cauchy problem for Einstein's equations is nontrivial. In GR, the analogs of Gauss's law and the no monopole constraint are the Hamiltonian and momentum constraints. These constraints can be interpreted loosely as conditions related to energy and momentum conservation, and they must be satisfied by the initial data before the system can be evolved. Established techniques exist for constructing such data. In this work, we use the Bowen-York approach for solving the initial value problem. The Bowen-York approach is commonly used to ensure that, as part of the constraint equations, the vacuum momentum constraints reduce to a linear set of equations with a simple solution which can be superposed. The Hamiltonian constraint then reduces to an elliptic equation that must be solved numerically. Further details on this procedure are given in later chapters.

In EMDA theory, the Cauchy problem is further complicated by the presence of additional source terms. The stress energy tensor T_{ab} encodes contributions from matter and nongravitational fields. In vacuum GR, this tensor vanishes, but in EMDA it includes contributions from the electromagnetic field, the dilaton, and the axion together with their nonlinear couplings.

Once suitable initial data have been constructed and the system has been evolved, we extract the gravitational waveform. In this work, we compute the waveform using the Newman-Penrose formalism, in which outgoing gravitational radiation is represented by the complex Weyl scalar

Ψ_4 [56]. Ψ_4 is not what is measured directly by gravitational wave detectors but it is related. Details of how Ψ_4 is calculated are provided in Appendix D.

For our current purposes, the Weyl scalar, Ψ_4, is related to the gravitational wave strain, h, by:

$$\Psi_4 = \frac{\partial^2 h}{\partial t^2}, \tag{1.18}$$

To quantify the difference between two waveforms, we compute the overlap and the mismatch between them. The overlap is simply the inner product between two normalized waveforms:

$$O(h_1, h_2) = \frac{|\langle h_1, h_2 \rangle|}{\sqrt{\langle h_1, h_1 \rangle \langle h_2, h_2 \rangle}}, \tag{1.19}$$

The mismatch is then, simply

$$\mathcal{M}(h_1, h_2) = 1 - \max O(h_1, h_2). \tag{1.20}$$

where, again, O is the normalized waveform overlap and the maximum is taken over time shifts and polarization angles. The time shifts account for differences in the chosen oring in time for the simulations, or equivalently for small differences in the time at which the waveform reaches its maximum amplitude. Since the absolute time coordinate is arbitrary, two otherwise identical waveforms should not be considered different simply because one is shifted slightly forward or backward in time. We therefore allow a relative time shift when computing the overlap.

The polarization angle accounts for the arbitrary orientation of the waveform polarization basis. A polarization shift is equivalent to multiplying the complex waveform by an overall phase factor. Maximizing over this angle removes differences associated with the choice of polarization, rather than with the intrinsic dynamics of the binary.

The mismatch can be used to estimate the signal to noise ratio required to distinguish two waveforms. In the small mismatch limit, this distinguishability threshold is approximately

$$\rho \sim \frac{1}{\sqrt{2\mathcal{M}(h_1, h_2)}}. \tag{1.21}$$

This estimate provides a useful way to interpret waveform differences, although it should not be regarded as an absolute detection criterion. In practice, interpreting differences between waveforms remains challenging because the parameter space is often degenerate. For example, it may not be immediately clear whether observed differences arise from spin, charge, or other physical effects.

Improving the accuracy of waveform comparisons requires improving the numerical simulations that produce the waveforms. There are several approaches that can be taken to improve the accuracy of these simulations. As mentioned above, a significant source of (truncation) error comes from the numerical approximation of derivatives. We therefore also present our work incorporating compact finite-difference operators into our numerical relativity code. To begin, we review the finite-difference methods currently used in the code.

On a discrete grid, derivatives can be approximated using finite differences. For example, a second-order accurate centered approximation to the first derivative is

$$\left.\frac{df}{dx}\right|_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x} + O(\Delta x^2). \tag{1.22}$$

Higher order accurate formulas can be constructed by matching Taylor series expansions. For instance, a fourth order accurate centered stencil is

$$\left.\frac{df}{dx}\right|_i = \frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12\Delta x} + O(\Delta x^4). \tag{1.23}$$

These finite difference approximations are standard tools for numerical solutions of partial differential equations [20].

The above examples are explicit finite difference schemes, since the derivative at grid point i is computed directly from nearby function values. The derivatives are relatively straightforward to derive and easy to implement. Alternatively, one can approximate the first derivative using a compact, or implicit, finite difference scheme of the form

$$\beta f'_{i-2} + \alpha f'_{i-1} + f'_i + \alpha f'_{i+1} + \beta f'_{i+2} = c\,\frac{f_{i+3} - f_{i-3}}{6\Delta x} + b\,\frac{f_{i+2} - f_{i-2}}{4\Delta x} + a\,\frac{f_{i+1} - f_{i-1}}{2\Delta x}. \tag{1.24}$$

Here f_i' denotes the numerical approximation to df/dx at grid point i. Compact finite difference schemes of this type were developed to obtain better spectral resolution than comparable explicit schemes [31].

The coefficients are determined by matching Taylor series expansions to the desired order of accuracy. For the above stencil, the consistency conditions include

$$a+b+c = 1+2\alpha+2\beta, \qquad \text{second order,} \tag{1.25}$$

$$a+2^2b+3^2c = 2\frac{3!}{2!}\left(\alpha+2^2\beta\right), \qquad \text{fourth order,} \tag{1.26}$$

$$a+2^4b+3^4c = 2\frac{5!}{4!}\left(\alpha+2^4\beta\right), \qquad \text{sixth order.} \tag{1.27}$$

This approach is more complicated than an explicit finite difference stencil because the derivatives are obtained by solving an matrix system and doing a matrix inversion. The tradeoff is that compact schemes can provide improved accuracy and enhanced spectral resolution compared to explicit methods [31]. This improvement in spectral resolution is illustrated in Fig. 1.3. There are many possible compact finite difference operators that can be constructed. In this work, we describe our methodology for deriving accurate and stable compact derivative operators and evaluate their performance relative to their explicit counterparts.

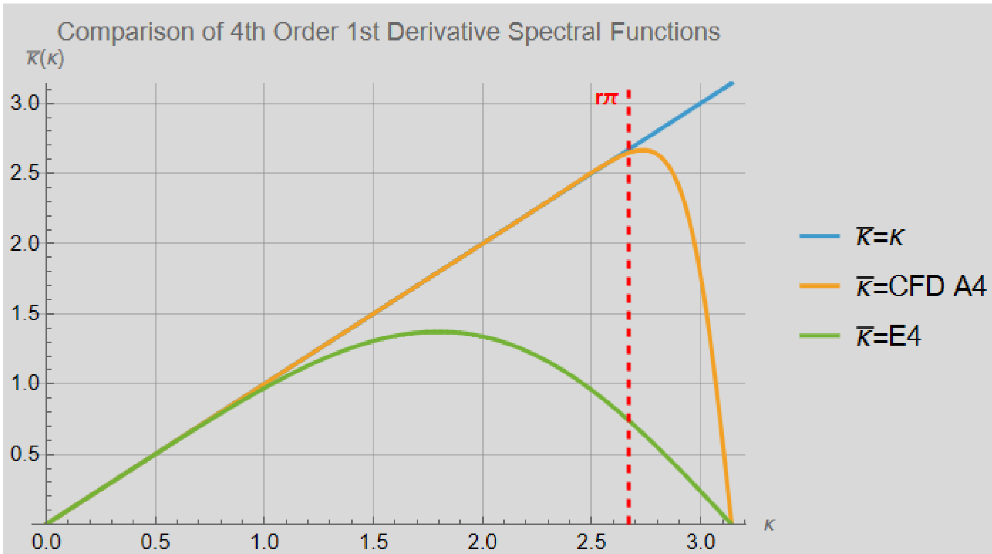


Figure 1.3 Spectral resolution of explicit and compact fourth order first derivative operators. The solid reference curve represents perfect resolution, and the dashed vertical line marks the cutoff frequency used in the optimization. The compact finite difference operator maintains better agreement with the ideal resolution curve over a wider range of wavenumbers.

1.3 Summary of Dissertation

This dissertation primarily presents results from binary black hole simulations in Einstein-Maxwell-dilaton-axion (EMDA) theory, along with a study of compact finite difference methods for numerical relativity. The dissertation is organized as follows. Chapter 2 discusses the formulation of the initial data and presents results from head on collision simulations. Chapter 3 presents inspiral and merger results in EMDA theory. Chapter 4 explores interpolated EMDA models in which the coupling strengths are varied. Chapter 5 investigates the implementation of compact finite difference operators in numerical relativity codes.

Several of the projects presented in this dissertation are being prepared for publication. The work described in Chapter 2, which concerns the stability and dynamics of Kerr–Sen black holes in Einstein–Maxwell–dilaton–axion gravity, has been submitted to a Journal Physical Review D. This manuscript was prepared in collaboration with Eric Hirschmann and David Neilsen at Brigham

Young University, Hyun Lim at Los Alamos National Laboratory, David Van Komen at the University of Utah, and Sebastian Vander Ploeg Fallon at Cornell University.

The binary black hole inspiral results presented in Chapter 3 are being prepared as a separate manuscript with the same group of collaborators. We expect to submit this work for publication within approximately one month. A third manuscript will present the interpolated black hole solutions that extend beyond the Kerr–Sen sector of the theory. This project is also being carried out with the EMDA collaboration, and we anticipate completing the manuscript by the end of 2026.

The compact finite-difference results presented in Chapter 4 are being prepared for publication in collaboration with Eric Hirschmann, David Van Komen, David Neilsen, Nathaniel Garey, and Luke Papenfuss. The manuscript is largely complete, and we expect to submit it for publication within approximately one month.

1.4 Conventions

In this work, we use geometrized units with $c = G = 1$. Unless otherwise stated, lengths and masses are expressed in units of a characteristic mass scale M. In physical units, a length scale $d = M$ corresponds to $d = GM/c^2$, while a time scale $t = M$ corresponds to $t = GM/c^3$.

We use the metric signature $(-,+,+,+)$ and assume the Einstein summation convention unless otherwise stated. Spacetime indices $a, b, c, \ldots$ run over $0, 1, 2, 3$, while spatial indices $i, j, k, \ldots$ run over $1, 2, 3$. The symmetric and antisymmetric parts of a tensor are defined as

$$T_{(ab)} \equiv \frac{1}{2}\left(T_{ab} + T_{ba}\right), \qquad T_{[ab]} \equiv \frac{1}{2}\left(T_{ab} T_{ba}\right). \tag{1.28}$$

Partial derivatives are denoted by $\partial_i g$ or equivalently $g_{,i}$. Covariant derivatives are written as $\nabla_a f_b$ or equivalently $f_{b;a}$. The Lie derivative of a tensor X^a along a vector field β^a is denoted by $\mathcal{L}_\beta X^a$.

1.4.1 Artificial intelligence assistance

The authors acknowledge the limited use of artificial intelligence tools, including ChatGPT and Gemini, during the preparation of this manuscript. These tools were used for spelling and grammar checks, LaTeX editing and debugging, and improving the clarity and readability of some text. They were also used to assist with code debugging and code development. The authors reviewed and verified all AI-assisted material and take full responsibility for the scientific content, calculations, code, interpretations, and conclusions presented in this work.

Chapter 2

Head On Collisions and Stability Tests in EMDA Theory

2.1 Introduction

While general relativity (GR) has passed all current experimental tests, there exist a large number of potentially viable extensions to it. Ongoing improvements in detector sensitivity, combined with numerical relativity will continue to enable increasingly precise tests of GR and proposed alternatives [4, 11, 34, 37, 57–63]. Indeed, advances in gravitational wave (GW) astronomy have led to a dramatic increase in the total number of detections of signals from compact binary mergers, thereby providing access to strongly gravitating, nonlinear domains. Since the first detection of gravitational waves, the Advanced LIGO and Virgo detectors have completed multiple observing runs, producing a growing catalog of compact binary mergers [24, 25, 64, 65]. All observed mergers to date are consistent with the predictions of GR within current uncertainties [9, 33, 64]. Nevertheless, compact binary coalescences continue to provide one of the most promising arenas for testing gravity beyond GR.

The increasing precision of gravitational wave observations has motivated effort to model compact object dynamics in theories beyond general relativity [52, 59, 66–68]. Numerical relativity is essential for this program because the late inspiral, merger, and ringdown probe regimes in which the dynamics are intrinsically nonlinear and analytic approximations are not uniformly reliable. Extending numerical relativity to alternative theories of gravity, however, requires careful control of the associated initial value problem. Reliable nonlinear simulations require evolution systems with suitable hyperbolic structure, controlled constraint propagation, and stable gauge dynamics. These considerations have become central in beyond GR numerical relativity, where the construction of robust evolution formulations is often a prerequisite for extracting physical predictions.

The challenge is particularly relevant for theories motivated by effective field theory or higher curvature corrections to GR. Such theories introduce higher derivatives, additional characteristic modes, or principal parts whose hyperbolicity depends on the background solution and coupling strength. In some cases, higher derivatives may also be associated with Ostrogradsky type instabilities, unless the theory is interpreted within a controlled effective description or reformulated appropriately. A significant number of previous studies have therefore been devoted to approaches that permit a more complete consideration of nonlinear aspects while maintaining control over the fundamental evolutionary problem. Examples of these include order reduction schemes in dynamical Chern Simons gravity [38, 39] and Einstein dilaton Gauss Bonnet theory [47, 69], modified gauge choices in Einstein scalar Gauss Bonnet [45,70], the so called “fixing the equations" approach [71], and auxiliary field formulations for solving higher curvature systems including quadratic gravity [53, 72–75] (see also [66] for a more in depth discussion and comparison of these and other approaches). These developments have expanded the range of beyond GR theories accessible to numerical relativity while emphasizing that dynamical consistency and observational phenomenology must be addressed together.

From this perspective, theories whose field equations can be written as second order evolution systems provide particularly useful laboratories for beyond GR numerical relativity. They allow one to study additional propagating degrees of freedom in fully nonlinear black hole (BH) spacetimes without relying entirely on perturbative or order reduced treatments of the gravitational field equations. Einstein–Maxwell–Dilaton–Axion (EMDA) [76, 77], arising as a low energy limit of heterotic string theory, is one such theory well suited to this purpose. Unlike many higher curvature modifications to GR, EMDA does not introduce higher derivatives into the metric field equations. Instead, it extends the Einstein Maxwell system by coupling the electromagnetic field to two additional propagating matter degrees of freedom: a dilaton and an axion. The resulting equations retain a structure closely related to the Einstein Maxwell system coupled to nonlinear matter fields. Consequently, EMDA can be formulated as a second order hyperbolic evolution problem using standard numerical relativity methods, together with appropriate evolution equations for the electromagnetic, dilaton, and axion fields. In the limit of vanishing axion coupling, EMDA reduces to Einstein–Maxwell–Dilaton (EMD) theory.

Having a well posed initial value problem for EMDA is a crucial ingredient for sensible simulations, but it is also advantageous as it has analytically known black hole solutions at specific points in the theory space. Certain of these are the stationary Kerr–Sen black holes. They generalize the Kerr–Newman family of general relativity in that they carry nontrivial dilaton and axion fields in addition to mass, angular momentum, and electromagnetic charge. As is well known, in standard GR coupled to Maxwell fields, stationary black holes are characterized by their mass, angular momentum, and electric charge [78, 79]. In contrast, extensions of GR such as EMDA can admit black holes with additional matter fields. Kerr–Sen solutions therefore provide a natural setting in which to explore whether these additional degrees of freedom behave as stable black hole hair in the nonlinear strong field regime. Related questions have been studied in scalar Gauss–Bonnet gravity, where scalarization and dynamical descalarization can occur during compact binary evolutions [48, 80]. The nonlinear behavior of scalar and axionic hair in EMDA is connected to several lines of previous work. Studies of EMD [50, 51] provide an important foundation for extending nonlinear simulations to the full EMDA system. Analytical investigations have examined aspects of the perturbative stability of EMDA black holes in certain parameter regimes [81]. In addition, astrophysical studies have explored possible observational signatures of Kerr–Sen BHs. For example, analyses of Event Horizon Telescope (EHT) shadow data suggest that Sgr A* may favor the Kerr–Sen solution, although similar data from M87* favor Kerr while remaining consistent with Kerr–Sen within current observational uncertainties [82, 83].

Fully nonlinear simulations of binary BHs in EMDA remain limited. In particular, it is not yet well understood whether dilaton and axion hair persist through merger, disperse via radiation, or settle into a stable remnant configuration. Addressing these questions is important for assessing the nonlinear stability of Kerr–Sen BHs and for determining whether EMDA can provide distinct strong field signatures for future GW observations. Although significant progress has been made in waveform modeling for selected beyond GR theories, including scalar Gauss Bonnet and Einstein dilaton Gauss Bonnet gravity, waveform coverage in alternative theories remains far less developed than in GR. Consequently, many current tests still rely on phenomenological parameterizations, post Newtonian approximations, or targeted numerical relativity simulations [48, 49, 84, 85].

In this study, we extend these investigations by simulating head on black hole mergers governed by the full EMDA equations. We analyze the resulting waveforms and examine the stability of the remnants, thereby gaining insight into the persistence of scalar hair within the theory. Specifically, we evolve several head on collisions of BHs in EMDA to determine the conditions under which the dilaton and axion fields survive merger.

The remainder of this chapter is organized as follows. In Sec. 3.2, we review the EMDA action, equations of motion, and the formulation used for numerical evolution. In Sec. 3.3, we describe our numerical methods, including the modified `Dendro-GR` infrastructure, the construction of approximate binary BH initial data, and the extraction of gravitational, electromagnetic, dilaton, and axion radiation. In Sec. 3.4, we present simulations of head on Kerr–Sen BH collisions, analyze the emitted radiation, and examine the persistence of scalar hair in the remnant. We also study initially unscalarized configurations to determine whether scalar hair can be generated dynamically. We conclude in Sec. 5.5 with a summary of our results and directions for future work. Several important details such as the equations of motion, the analytic form of the Kerr–Sen solution, initial data and wave extraction techniques and issues associated with code performance are relegated to a number of appendices in order not to disrupt the continuity of the discussion.

2.2 Einstein–Maxwell–Dilaton–Axion

The general theory we consider in this work is an extension of Einstein Maxwell on inclusion of two coupled scalar fields, the dilaton and the axion. The action for the EMDA system is

$$S = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{R}{8\pi} - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi}F^2 - \tfrac{1}{2}e^{4\alpha_1\phi}(\nabla\kappa)^2 - \kappa F_{ab}(*F)^{ab}\right], \tag{2.1}$$

where the first term is the standard Hilbert action and ϕ and κ are the dilaton and axion, respectively. F_{ab} is the U(1) field strength, and $(*F)_{ab}$ is its dual. The parameters α_0 and α_1 are real, positive constants that set the coupling strengths and allow us to consider a family of EMDA theories. Particular choices correspond to particular theories. For instance, $\alpha_0 = \alpha_1 = 0$ gives Einstein–Maxwell theory minimally coupled to scalar and axion fields, while $(\alpha_0, \alpha_1) = (\sqrt{3}, 0)$ corresponds to Kaluza–Klein. In this work we take $(\alpha_0, \alpha_1) = (1, 1)$, corresponding to a low energy limit of heterotic string theory. For these coupling values, the theory admits the Kerr–Sen family of rotating, charged black hole solutions. These solutions reduce to Kerr when the electromagnetic charge vanishes. In contrast to the Kerr–Newman solution of Einstein–Maxwell theory, the Kerr–Sen geometry is accompanied by nontrivial dilaton and axion fields whose structure is determined by the black hole charge and spin. We now summarize the Kerr–Sen black hole solution, which provides an exact rotating, charged black hole solution of EMDA theory. This solution serves as a useful analytic reference for the charged rotating black holes we consider.

2.2.1 Kerr–Sen black holes

We present the Kerr–Sen solution in the Einstein frame, written in Boyer–Lindquist type coordinates. The metric takes the form

$$ds^2 = -\left(1 - \frac{2mr\cosh^2\alpha}{\rho^2}\right) dt^2 + \rho^2 \left(\frac{dr^2}{\Delta} + d\theta^2\right) + \frac{\Sigma}{\rho^2}\sin^2\theta\, d\varphi^2 - \frac{4amr\cosh^2\alpha}{\rho^2}\sin^2\theta\, dt\, d\varphi \quad (2.2)$$

where the metric functions are defined by

$$\Delta = r^2 - 2mr + a^2, \quad (2.3)$$

$$\rho^2 = r^2 + a^2\cos^2\theta + 2mr\sinh^2\alpha, \quad (2.4)$$

$$\Sigma = \left(r^2 + a^2 + 2mr\sinh^2\alpha\right)^2 - \Delta a^2 \sin^2\theta. \quad (2.5)$$

The associated gauge, dilaton, and axion fields are given by

$$A_a dx^a = \frac{mr\sinh 2\alpha}{\sqrt{2}\rho^2}\left(dt - a\sin^2\theta\, d\varphi\right), \tag{2.6}$$

$$e^{-2\phi} = \frac{\rho^2}{r^2 + a^2\cos^2\theta}, \tag{2.7}$$

$$\kappa = -2ma\sinh^2\alpha\,\frac{\cos\theta}{r^2 + a^2\cos^2\theta}. \tag{2.8}$$

The parameters m, α, and a arise as integration constants. The physical mass M, electric charge Q, angular momentum J, and magnetic dipole moment μ are related to these parameters via

$$M = m\cosh^2\alpha, \tag{2.9}$$

$$Q = \frac{m}{\sqrt{2}}\sinh 2\alpha, \tag{2.10}$$

$$J = aM, \tag{2.11}$$

$$\mu = aQ. \tag{2.12}$$

With respect to the Boyer–Lindquist radial coordinate, r, the event horizon is located at the familiar

$$r_H = m + \sqrt{m^2 - a^2}, \tag{2.13}$$

and the condition for the existence of a nonextremal black hole is

$$m^2 > a^2 \quad \Leftrightarrow \quad M^2 > |J| + \frac{Q^2}{2}. \tag{2.14}$$

The inverse relations are

$$m = M - \frac{Q^2}{2M}, \tag{2.15}$$

$$\sinh 2\alpha = \frac{2\sqrt{2}QM}{2M^2 - Q^2}, \tag{2.16}$$

$$a = \frac{J}{M}. \tag{2.17}$$

The analytic Kerr–Sen solution provides a useful reference for understanding the isolated charged and rotating black holes that appear in EMDA theory. However, the binary systems considered in this work are dynamical, nonlinear, and do not admit closed form solutions. We therefore evolve the EMDA equations numerically. In the next section, we summarize the numerical formulation used in our simulations.

2.2.2 The equations of motion for EMDA

With the aim of studying the nonlinear dynamics of EMDA in black hole collisions, we express EMDA as a Cauchy problem and evolve the metric using the BSSN formulation [20, 86, 87], supplemented by evolution equations for the electromagnetic, dilaton, and axion fields. We also employ puncture gauge conditions [88, 89].

For completeness, we present here the full equations of motion and their $3+1$ decomposition. Starting from the action for EMDA as given in Eq. 2.1, we can derive the relevant equations of motion. These include the Einstein equations:

$$G_{ab} = 8\pi G\left[2\left(\nabla_a\phi\nabla_b\phi - \frac{1}{2}g_{ab}(\nabla\phi)^2\right) + 2e^{-2\alpha_0\phi}\left(F_{ac}F_b{}^c - \frac{1}{4}g_{ab}F^2\right) + \frac{1}{2}e^{4\alpha_1\phi}\left(\nabla_a\kappa\nabla_b\kappa - \frac{1}{2}g_{ab}(\nabla\kappa)^2\right)\right]. \tag{2.18}$$

with the matter equations of motion written as

$$0 = \nabla^a\nabla_a\phi + \frac{1}{2}\alpha_0 e^{-2\alpha_0\phi}F^2 - \frac{1}{2}\alpha_1 e^{4\alpha_1\phi}(\nabla\kappa)^2, \tag{2.19}$$

$$0 = \nabla^a\left(e^{4\alpha_1\phi}\nabla_a\kappa\right) - F_{ab}({}^*F)^{ab}, \tag{2.20}$$

$$0 = \nabla_a\left(e^{-2\alpha_0\phi}F^{ab}\right) + ({}^*F)^{ab}\nabla_a\kappa. \tag{2.21}$$

We include a form of hyperbolic constraint damping into the Maxwell equations so that violations propagate away at the speed of light. In particular, we introduce two scalar fields Φ and Ψ into Eq. (2.21) and the identity $\nabla_a(^*F)^{ab} = 0$ as follows:

$$\nabla_a \left(F^{ab} + g^{ab}\Psi \right) = \eta_1 n^a \Psi + 2\alpha_0 \nabla_a \phi \, F^{ab} - e^{2\alpha_0 \phi} (^*F)^{ab} \nabla_a \kappa, \tag{2.22}$$

$$\nabla_a \left((^*F)^{ab} + g^{ab}\Phi \right) = \eta_2 n^a \Phi. \tag{2.23}$$

where η_1 and η_2 are taken as positive constants and serve as tunable parameters to effect the damping of the electromagnetic constraints. Note that on the right hand sides, we have also included n^a, the usual unit timelike normal to the spatial hypersurfaces in anticipation of performing the usual $3+1$ split of spacetime. It is worth noting that these damped versions of Maxwell are now no longer strictly four covariant. We introduce the usual $3+1$ decomposition of spacetime by defining the line element as

$$\mathrm{d}s^2 = -\alpha^2 \mathrm{d}t^2 + \gamma_{ij} (\mathrm{d}x^i + \beta^i \mathrm{d}t)(\mathrm{d}x^j + \beta^j \mathrm{d}t) \tag{2.24}$$

with lapse, α, shift, β^i, 3-metric, γ_{ij}, unit timelike normal $n_a = (-\alpha, 0, 0, 0)$ and extrinsic curvature $K_{ab} = -\gamma_a{}^c \gamma_b{}^d \nabla_c n_d$.

Using the BSSN formulation, we solve the standard BSSN evolution equations for the usual $\chi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$ variables with EMDA matter terms.[1] These equations take the form

[1]Note that in this section only, χ represents the conformal factor and not the dimensionless spin parameter.

$$\partial_0 \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij}, \tag{2.25}$$

$$\partial_0 \chi = \tfrac{2}{3}\alpha \chi K, \tag{2.26}$$

$$\partial_0 K = [\ldots] + 4\pi\alpha (\rho + T), \tag{2.27}$$

$$\partial_0 \tilde{A}_{ij} = [\ldots] - 8\pi\chi\alpha \left(T_{ij} - \tfrac{1}{3}\gamma_{ij} T_k{}^k\right), \tag{2.28}$$

$$\partial_0 \tilde{\Gamma}^i = [\ldots] - 16\pi\alpha \chi^{-1} S^i, \tag{2.29}$$

where $\partial_0 \equiv \partial_t - \mathcal{L}_\beta$ with $\mathcal{L}_\beta$ the Lie derivative along the shift and the $[\ldots]$ are a stand in for the relevant right hand sides which can be found in standard references, e.g. [20]

The quantities S^i, T_{ij}, and ρ are defined as

$$\begin{aligned} S^i &\equiv -\gamma^{ia} T_{ab} n^b \\ &= 2\Pi D^i \phi + 2e^{-2\alpha_0 \phi} \epsilon^{ijk} E_j B_k + \tfrac{1}{2} e^{4\alpha_1 \phi} \Xi D^i \kappa, \end{aligned} \tag{2.30}$$

$$\begin{aligned} T_{ij} &\equiv T_{ab} \gamma^a{}_i \gamma^b{}_j \\ &= 2\left[D_i\phi D_j\phi - \tfrac{1}{2}\tilde{\gamma}_{ij}\left((D\phi)^2 - \Pi^2\right)\right] - 2e^{-2\alpha_0\phi}\left(E_i E_j + B_i B_j - \tfrac{1}{2}\tilde{\gamma}_{ij}\left(E_k E^k + B_k B^k\right)\right) \\ &\quad + \tfrac{1}{2} e^{4\alpha_1\phi}\left[D_i\kappa D_j\kappa - \tfrac{1}{2}\tilde{\gamma}_{ij}\left((D\kappa)^2 - \Xi^2\right)\right]. \end{aligned} \tag{2.31}$$

$$\rho \equiv \Pi^2 + (D\phi)^2 + e^{-2\alpha_0\phi}\left(E_a E^a + B_a B^a\right) + \tfrac{1}{4} e^{4\alpha_1\phi}\left(\Xi^2 + (D\kappa)^2\right). \tag{2.32}$$

We use standard puncture gauge conditions for the lapse and the shift:

$$\partial_t \alpha = \beta^i \partial_i - 2\alpha K, \tag{2.33}$$

$$\partial_t \beta^i = \beta^j \partial_j \beta^i + \tfrac{3}{4} B^i, \tag{2.34}$$

$$\partial_t B^i = \beta^j \partial_j B^i + \partial_t \tilde{\Gamma}^i - \beta^j \partial_j \tilde{\Gamma}^i - \eta B^i. \tag{2.35}$$

We also employ a slow start lapse condition [90] to reduce transients at early times. This amounts to adding the term

$$-\chi^{1/2}\left(\varpi\, e^{-t^2/2\sigma^2}\right)(\alpha-\chi^{1/2}) \tag{2.36}$$

to the equation for the lapse. Following [90], we take the constants $\varpi = 3/5$ and $\sigma = 20$.

For the matter equations we start with the dilaton and axion:

$$\partial_0\phi = -\alpha\Pi, \tag{2.37}$$

$$\partial_0\kappa = -\Xi \tag{2.38}$$

$$\begin{aligned}\partial_0\Pi =\,& \alpha K\Pi - \alpha\chi\,\tilde{\gamma}^{ij}\partial_i\partial_j\phi - \chi\,\tilde{\gamma}^{ij}\partial_i\alpha\,\partial_j\phi + \alpha\chi\,\tilde{\Gamma}^i\partial_i\phi + \frac{\alpha}{2}\tilde{\gamma}^{ij}\partial_i\chi\,\partial_j\phi \\ &+ \frac{\alpha\alpha_0}{\chi}e^{-2\alpha_0\phi}\tilde{\gamma}_{ij}\left(E^iE^j - B^iB^j\right) + \frac{\alpha\alpha_1}{2}e^{4\alpha_1\phi}\left(\chi\,\tilde{\gamma}^{ij}\partial_i\kappa\,\partial_j\kappa - \Xi^2\right).\end{aligned} \tag{2.39}$$

$$\begin{aligned}\partial_0\Xi =\,& \alpha K\Xi - \alpha\chi\,\tilde{\gamma}^{ij}\partial_i\partial_j\kappa - \chi\,\tilde{\gamma}^{ij}\,\partial_i\alpha\,\partial_j\kappa + \alpha\chi\,\tilde{\Gamma}^i\partial_i\kappa + \frac{\alpha}{2}\,\tilde{\gamma}^{ij}\,\partial_i\kappa\,\partial_j\chi \\ &- 4\alpha\alpha_1\chi\,\tilde{\gamma}^{ij}\,\partial_i\kappa\,\partial_j\phi + 4\alpha\alpha_1\Pi\Xi + \frac{4\alpha}{\chi}\,e^{-4\alpha_1\phi}\,\tilde{\gamma}_{ij}B^iE^j.\end{aligned} \tag{2.40}$$

The equations for the electromagnetic fields, including hyperbolic divergence cleaning become

$$\partial_0\Psi = -\alpha\,\partial_iE^i + \frac{3\alpha}{2\chi}\,E^i\partial_i\chi + 2\alpha_0\alpha\,E^i\partial_i\phi + \alpha e^{2\alpha_0\phi}\,B^i\partial_i\kappa - \alpha\eta_1\Psi. \tag{2.41}$$

$$\partial_0\Phi = \alpha\,\partial_iB^i - \frac{3\alpha}{2\chi}\,B^i\partial_i\chi - \alpha\eta_2\Phi. \tag{2.42}$$

$$\begin{aligned}\partial_0E^i =\,& \alpha KE^i + 2\alpha\alpha_0\chi^{1/2}\,\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\phi - 2\alpha\alpha_0E^i\Pi \\ &+ \alpha e^{2\alpha_0\phi}\chi^{1/2}\,\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\kappa - \alpha e^{2\alpha_0\phi}\Xi B^i - \alpha\chi\,\tilde{\gamma}^{ij}\partial_j\Psi - \chi^{1/2}\,\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\alpha \\ &- \alpha\chi^{1/2}\,\epsilon^{jik}B^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\,\epsilon^{jik}\tilde{\gamma}_{kl}\partial_jB^l + \alpha\chi^{-1/2}\,\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\chi.\end{aligned} \tag{2.43}$$

$$\begin{aligned}\partial_0B^i =\,& \alpha KB^i + \alpha\chi\,\tilde{\gamma}^{ij}\partial_j\Phi \\ &- \chi^{1/2}\,\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\alpha + \alpha\chi^{-1/2}\,\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\chi - \alpha\chi^{1/2}\,\epsilon^{ijk}E^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\,\epsilon^{ijk}\tilde{\gamma}_{kl}\partial_jE^l.\end{aligned} \tag{2.44}$$

Having written the EMDA system in first order in time form, we now describe how these equations are implemented and evolved numerically. In particular, we summarize the adaptive mesh infrastructure, code generation, and refinement strategy used in our simulations.

2.2.3 Code description

We perform our black hole merger simulations using `Dendro-GR`, a well developed numerical relativity framework. `Dendro-GR` combines an octree based adaptive mesh infrastructure with automatic code generation for the evolution equations [19, 22, 23]. The computational domain is refined dynamically using wavelet adaptive multiresolution (WAMR) in which the evolved fields are represented on an adaptive hierarchy of octree blocks. Wavelet coefficients computed from the evolved variables provide a local estimate of the truncation error or unresolved field content [19, 91, 92]. Regions where these wavelet coefficients exceed a prescribed tolerance are refined, while regions in which wavelet coefficients remain below a coarsening threshold are coarsened. This tends to concentrate resolution near the black holes, in the wave extraction region, and in regions where the electromagnetic, dilaton, or axion fields develop large spatial gradients, while smoother regions are evolved on coarser grids.

Spatial derivatives are computed using sixth order finite difference stencils, and time integration is performed using a fourth order Runge–Kutta scheme. We include Kreiss–Oliger dissipation to damp poorly resolved high frequency modes and improve numerical stability [93]. Additional details of the `Dendro-GR` infrastructure, including its WAMR implementation, scalability, and previous applications to binary black hole evolutions, can be found in [19, 22, 23, 30].

We now describe the diagnostic quantities used to monitor the quality of the evolutions. In particular, we track the Hamiltonian and momentum constraints together with the electromagnetic divergence constraints throughout the computational domain. These quantities provide a global check on the consistency of the evolution and are used to assess the efficacy of the WAMR refinement parameters.

2.2.4 Constraint quantities

We compute the L_2 norms of the Hamiltonian and momentum constraints, together with the electromagnetic constraints. When evaluating these norms, we excise small neighborhoods around the punctures to avoid unduly weighting the norms by these regions. Representative constraint violations for the full simulation are shown in Figs. 2.1 and 2.2.

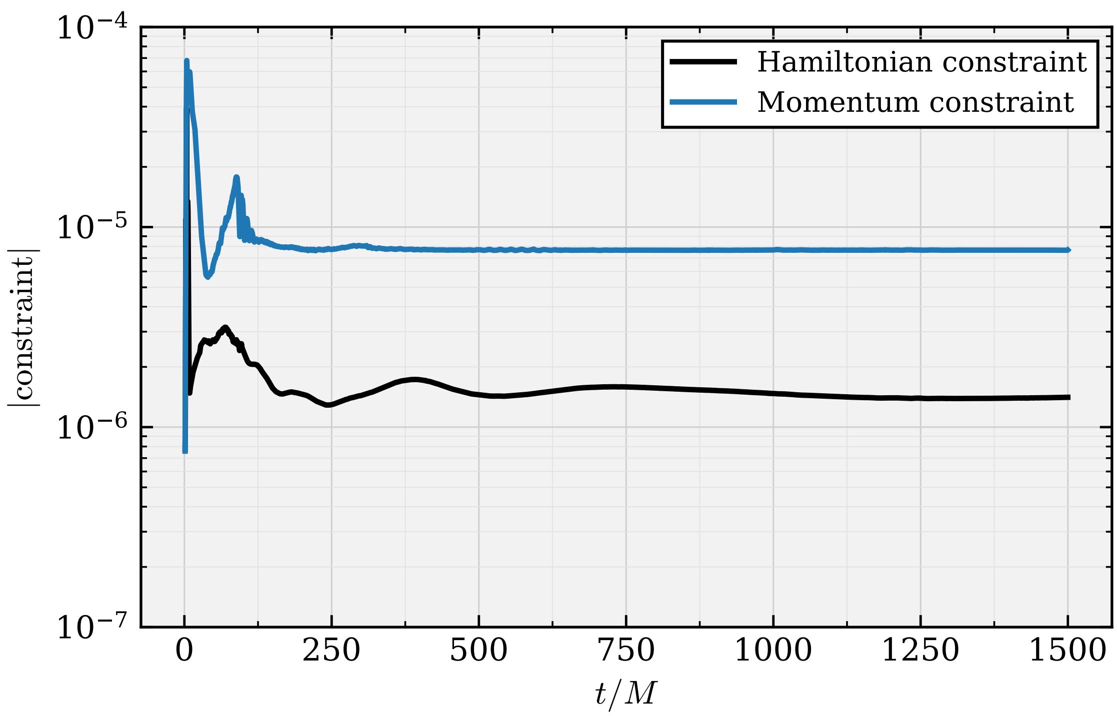


Figure 2.1 The Hamiltonian constraint $\mathcal{H}$ and summed momentum constraint violation measured over the full simulation. The L_2 norm is taken with respect to the number of points. The constraints remain controlled throughout the evolution, including the merger phase.

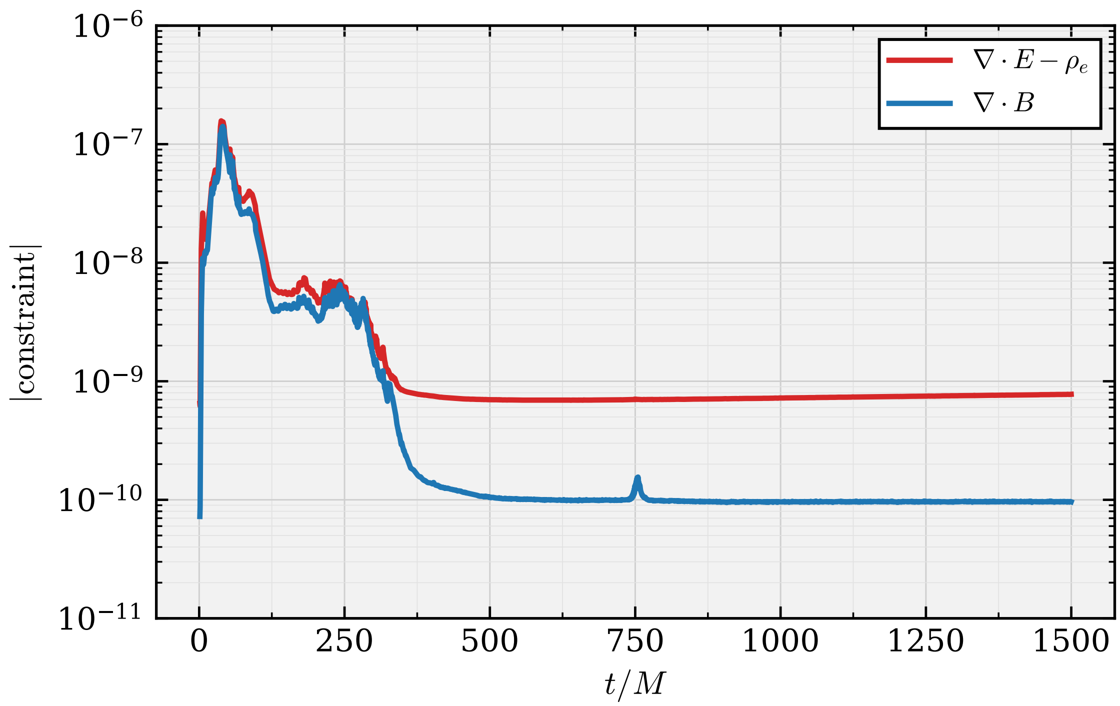


Figure 2.2 The volume-weighted L_2 norms of the electromagnetic constraints, namely the no-monopole constraint that involves the divergence of the magnetic field, $\nabla \cdot B$, and the Gauss constraint that involves the divergence of the electric field. Both are measured over the full simulation.

We also compute quasilocal quantities on apparent horizons using the `BHaHAHA` apparent horizon finder [94], which we have extended to evaluate the horizon mass, spin, and electric charge. These quantities provide independent diagnostics of remnant relaxation and of the conservation of the black hole charges during the evolution.

As a check on our overall numerical strategy, we perform a limited type of convergence study by varying both the wavelet tolerance and maximum refinement level for a short head on binary evolution. We confirm that with increased numbers of refinement levels and tighter refinement criteria constraint violations are reduced. The details of this study are presented below.

2.2.5 Code performance

A full convergence study for binary black hole evolutions is computationally expensive and is therefore not attempted here. Instead, we assess the behavior of the code by varying two parameters that directly affect the adaptive mesh refinement: the wavelet tolerance, ε, and the maximum refinement level, $\ell_{\max}$. The wavelet tolerance is the threshold value in our WAMR scheme on the coefficients of the wavelet expansion that determines whether refinement or coarsening occurs. The maximum refinement level sets the largest number of refinement levels that can be obtained in a simulation. We confirm that for both, making these refinement criteria more stringent leads to a corresponding reduction in the constraint violations.

The tests done here use the first $20M$ of a head-on evolution of two equal-mass, charged black holes. The black holes have an initial coordinate separation of $16M$, individual masses of $m_1 = m_2 = 0.50$, and charge-to-mass ratios $q_1/m_1 = q_2/m_2 = 0.1$. Both black holes have dimensionless spin $\chi = 0.4$ and are initially at rest. In computing the constraint diagnostics, we again exclude a small neighborhood of each puncture when calculating the norms. Throughout this section, the constraint measure shown is the sum of the volume weighted Hamiltonian constraint and the three volume weighted momentum constraints. This provides a single diagnostic for the overall constraint violation during the evolution.

Comparisons of simulations with respect to different wavelet tolerances are shown in Figure 2.3. Decreasing ε allows the grid to refine more aggressively in regions with larger truncation error. For the three different wavelet tolerances of $\varepsilon \in \{3\times10^{-5}, 1\times10^{-5}, 3\times10^{-6}\}$ we can see that the constraints are indeed reduced as ε decreases.

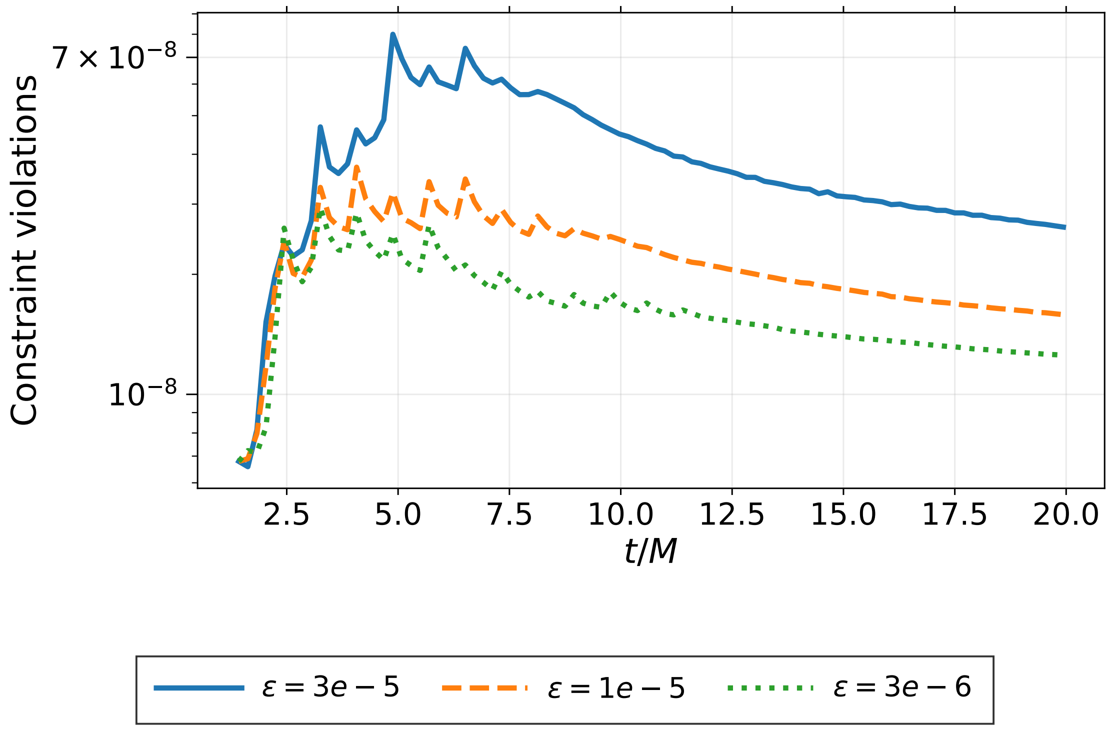


Figure 2.3 The sum of the volume-weighted Hamiltonian and momentum constraint violations over the first $(20M)$ of a head-on evolution of two equal-mass charged black holes with charge-to-mass ratio $(q/m = 0.1)$ and dimensionless spin $(\chi = 0.4)$. The simulations use wavelet tolerances of $\varepsilon \in 3\times10^{-5}, 1\times10^{-5}, 3\times10^{-6}$. These simulations were performed with a maximum refinement level of $\ell_{\max} = 15$. Lowering the wavelet tolerance improves the resolution of dynamically important regions and reduces the overall constraint violation.

Comparing simulations according to different maximum refinement levels is another means of changing the resolution in an AMR setting. Figure 2.4 compares simulations with maximum refinement levels of $\ell_{\max} \in \{14, 15, 16, 17\}$ corresponding to finest grid spacings of $h \in \{0.0325, 0.0162, 0.00813, 0.00407\}$, respectively. Increasing the allowed maximum depth provides additional resolution near the black holes and in regions where the fields develop sharp spatial structures. The resulting decrease in the constraints provides further evidence that the simulations improve with increased resolution.

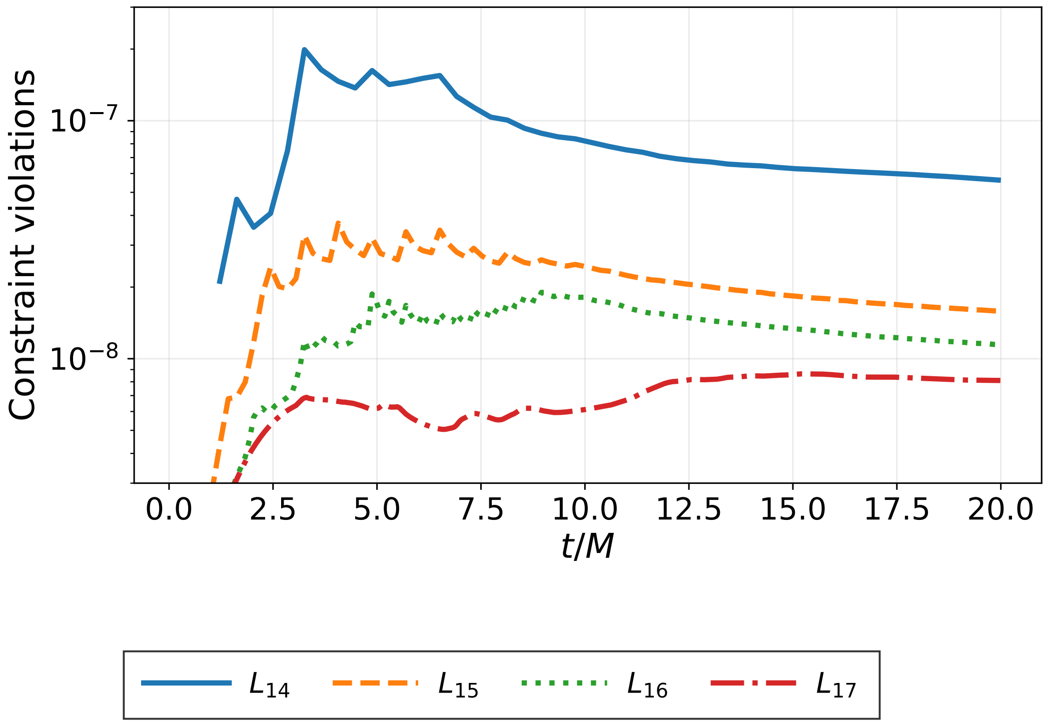


Figure 2.4 The sum of the volume-weighted Hamiltonian and momentum constraint violations over the first $(20M)$ of a head-on evolution of two equal-mass charged black holes with charge-to-mass ratio $(q/m = 0.1)$ and dimensionless spin $(\chi = 0.4)$ each. The simulations compare maximum refinement levels $\ell_{\max} \in 14, 15, 16, 17$, corresponding to finest grid spacings of $h \in 0.0325, 0.0162, 0.00813, 0.00407$, respectively. The wavelet tolerance is set to $\varepsilon = 1\times 10^{-5}$. Increasing the maximum refinement level reduces the overall constraint violation by allowing additional grid refinement in under-resolved regions.

Having checked that the constraint violations decrease as the adaptive refinement criteria are made more stringent, we now turn to the construction of the initial data used in the simulations. The initial data determine the starting geometry and matter fields and must satisfy the Einstein and Maxwell constraint equations.

2.3 Initial Data

It is necessary to have good initial data for our EMDA BH binaries. Of course, exact solutions for a binary system are not known. But we can generalize standard approaches for the construction of initial data in vacuum GR to these additional fields on making certain approximations. We follow closely the approach based on the conformal transverse-traceless decomposition used with punctures. Indeed, we extend this approach as it has been incorporated into the `TwoChargedPunctures` code of Bozzola and Paschalidis [95, 96]. We mostly follow their notation below.

As usual, we decompose the metric, but using notation slightly modified from the previous appendix. In particular we define

$$\gamma_{ij} = \psi^4 \bar{\gamma}_{ij}, \tag{2.45}$$

with ψ the conformal factor and the conformal metric assumed to be the flat metric, $\bar{\gamma}_{ij} = \eta_{ij}$. The latter assumption simplifies the following computations and is a reasonable choice for the systems we study. It is a limiting assumption preventing the construction of highly spinning black holes [97, 98]. Notwithstanding, Bowen–York black holes can reach spin values of order $\chi \sim 0.9$ [99]. As our simulations remain below this limit, the approximation should be reasonable. While we anticipate the appearance of "junk radiation" partly in consequence of this assumption [100], we find that it does decay as the fields relax to quasi-equilibrium which results in decreased constraint violation over the course of the evolutions. We adopt the maximal slicing condition, $K = 0$, and define a conformally rescaled tracefree extrinsic curvature, $\bar{A}_{ij} = \psi^2(K_{ij} - \frac{1}{3}\gamma_{ij}K)$. Ignoring the latter's radiative degrees of freedom, we can express it in terms of a vector, V^i, as

$$\bar{A}_{ij} = \partial_i V_j + \partial_j V_i - \frac{2}{3}\eta_{ij}\partial_k V^k. \tag{2.46}$$

The initial data problem then reduces to determining V^i and ψ for the gravitational pieces. In particular, the Hamiltonian and momentum constraint equations take the form

$$\partial_k \partial^k \psi + \tfrac{1}{8}\psi^{-7}\bar{A}_{ij}\bar{A}^{ij} = -2\pi\psi^5\rho, \tag{2.47}$$

$$\partial_k \partial^k V^i + \tfrac{1}{3}\eta^{ij}\partial_j(\partial_k V^k) = 8\pi S^i. \tag{2.48}$$

The momentum constraint is linear and independent of ψ. This permits dividing the solution between homogeneous (vacuum gravitational) and inhomogeneous (matter) contributions:

$$V^i = V^i_{\mathrm{GR}} + V^i_{\mathrm{M}}. \tag{2.49}$$

For V^i_{GR}, we use the standard Bowen–York solutions [101] while for the matter contribution, we must solve Eq. 2.48 with

$$S^i = 2\,\Pi\, D^i\phi + 2e^{-2\alpha_0\phi}\epsilon^{ijk}E_j B_k + \tfrac{1}{2}e^{4\alpha_1\phi}\,\Xi\, D^i\kappa. \tag{2.50}$$

If we assume a moment of time symmetry and neglect contributions from the magnetic field this source term vanishes. On imposing asymptotically flat boundary conditions, the solution to the matter part therefore becomes $V^i_{\mathrm{M}} = 0$.

We must also impose the electromagnetic constraints. At this point we depart somewhat from [95] because the EMDA version of these constraints are no longer linear, containing rather a nontrivial current that couples the dilaton and axion to the electromagnetic fields:

$$\nabla_i E^i = 2\alpha_0 E^i \partial_i\phi + e^{2\alpha_0\phi} B^i \partial_i \kappa \tag{2.51}$$

$$\nabla_i B^i = 0 \tag{2.52}$$

We follow [102, 103] and perform a conformal rescaling of the fields as

$$\bar{E}^i = \psi^6 E^i, \quad \bar{B}^i = \psi^6 B^i. \tag{2.53}$$

This results in the Maxwell constraints taking the form

$$\partial_i(e^{-2\alpha_0\phi}\bar{E}^i) = e^{2\alpha_0\phi}\psi^{-6}\bar{B}^i\partial_i\kappa, \quad \partial_i\bar{B}^i = 0. \tag{2.54}$$

Again neglecting the contribution from the magnetic field this results in homogeneous constraints

$$\partial_i(e^{-2\alpha_0\phi}\bar{E}^i) = 0, \quad \partial_i\bar{B}^i = 0. \tag{2.55}$$

To solve this we take a solution for the electric field in standard Einstein–Maxwell and rescale by $e^{-2\alpha_0\phi}$. This is independent of the conformal factor and therefore solvable separately from the spacetime variables. As these equations are linear, solutions may be superposed. Partly due to residual violations introduced by these approximate solutions, we have introduced hyperbolic divergence cleaning to drive these violations smaller over the course of the evolution. The electric and magnetic constraint behavior, including their damping, is demonstrated in Figure 2.2. The constraints remain well behaved and small throughout the evolution.

The Hamiltonian constraint above, not being linear, must be treated differently. We assume that we have two black holes charged under the U(1) gauge field as well as possessing scalar dilaton and axion fields. Following [95], we approximate this with a conformal factor of the form

$$\psi^2 = (1+u+\bar{\eta})^2 - \bar{\xi}^2 \tag{2.56}$$

for which we define

$$\bar{\eta} = \frac{m_1}{2R_1} + \frac{m_2}{2R_2}, \tag{2.57}$$

$$\bar{\xi} = \frac{q_1}{2R_1} + \frac{q_2}{2R_2}, \tag{2.58}$$

$$\bar{\kappa} = 1 + u + \bar{\eta}, \tag{2.59}$$

and where m_i, q_i and R_i here are the masses, electric charges and the coordinate distances from the origin of the BHs. This is intended to correspond to two electrically charged BHs with the singular part of the conformal factor separated out with u giving finite corrections, including the necessary corrections for the additional scalar fields.

On rearranging the Hamiltonian constraint in terms of u, we find

$$\partial_k\partial^k u = -\frac{1}{\bar{\kappa}}\Big(\partial_a\bar{\kappa}\partial^a\bar{\kappa} - \partial_a\bar{\xi}\partial^a\bar{\xi} - \partial_a\psi\partial^a\psi + \tfrac{1}{8}\psi^{-6}\bar{A}_{ij}\bar{A}^{ij} + 2\pi\psi^6\rho\Big), \tag{2.60}$$

with

$$\rho \equiv \left[\Pi^2 + (D\phi)^2\right] + e^{-2\alpha_0\phi}\left[E_a E^a + B_a B^a\right] + \tfrac{1}{4}e^{4\alpha_1\phi}\left[\Xi^2 + (D\kappa)^2\right]. \tag{2.61}$$

Using simplifying assumptions similar to above, namely a moment of time symmetry and neglecting the magnetic field, ρ reduces to

$$\rho = (D\phi)^2 + e^{-2\alpha_0\phi}\, E_a E^a + \tfrac{1}{4}e^{4\alpha_1\phi}\,(D\kappa)^2. \tag{2.62}$$

Again, this follows [95] quite closely, except that we have now included dilaton and axion contributions in ρ. Specifically, we employ a rescaled Reissner–Nordström electric field for the electric component and add a dilaton field and axion field appropriate for a single Kerr–Sen black hole superposed for each black hole.

The electric field is initially set to be:

$$E^i = \psi^{-6}\left[\frac{q_1}{R_1^2} + \frac{q_2}{R_2^2}\right]. \tag{2.63}$$

with a subsequent rescaling by $e^{2\alpha_0\phi}$ with the dilaton and axion fields initially set as:

$$\phi = \phi_1(r_1,\theta_1) + \phi_2(r_2,\theta_2) \tag{2.64}$$

$$\kappa = \kappa_1(r_1,\theta_1) + \kappa_2(r_2,\theta_2) \tag{2.65}$$

where Eqs. (2.7) and (2.8) give the forms of the dilaton and axion fields, respectively, for a single Kerr–Sen black hole (albeit in Boyer–Lindquist-type coordinates) for our current theory parameters $(\alpha_0,\alpha_1) = (1,1)$.

2.4 Radiative Quantities

To analyze the radiation emitted, we compute the Newman–Penrose scalar (Ψ_4), along with the electromagnetic radiation. We extract their multipolar components at ($r \approx 100M$). The extraction procedure is described below. A summary of the initial data parameters used in our simulations is provided in Appendix 2.5.2.

At an extraction radius R_{ex}, each radiative quantity is decomposed into spin weighted spherical harmonics with the appropriate spin weight:

$$\Psi_4(t,\theta,\varphi) = \sum_{\ell,m} \Psi_4^{\ell m}(t)\,{}_{-2}Y_{\ell m}(\theta,\varphi), \tag{2.66}$$

$$\Phi_1(t,\theta,\varphi) = \sum_{\ell,m} \Phi_1^{\ell m}(t)\,{}_{0}Y_{\ell m}(\theta,\varphi), \tag{2.67}$$

$$\Phi_2(t,\theta,\varphi) = \sum_{\ell,m} \Phi_2^{\ell m}(t)\,{}_{-1}Y_{\ell m}(\theta,\varphi), \tag{2.68}$$

$$\mathcal{D}(t,\theta,\varphi) = \sum_{\ell,m} \mathcal{D}^{\ell m}(t)\,{}_{0}Y_{\ell m}(\theta,\varphi), \tag{2.69}$$

$$\mathcal{A}(t,\theta,\varphi) = \sum_{\ell,m} \mathcal{A}^{\ell m}(t)\,{}_{0}Y_{\ell m}(\theta,\varphi). \tag{2.70}$$

Waveform comparisons in this work primarily use the dominant $(\ell,m) = (2,0)$ mode. To quantify the total radiation emitted in each field, we also compute the corresponding energy fluxes from the extracted multipolar amplitudes.

2.4.1 Radiated energy fluxes

The radiated energy fluxes in the gravitational, electromagnetic, dilaton, and axion channels are computed from the multipolar amplitudes. Following Ref. [103], we use

$$F_{\mathrm{GW}} = \frac{dE_{\mathrm{GW}}}{dt} = \lim_{r\to\infty} \frac{r^2}{16\pi} \sum_{\ell,m} \left| \int_{-\infty}^{t} \Psi_4^{\ell m}(t')\,dt' \right|^2, \tag{2.71}$$

$$F_{\mathrm{EM}} = \frac{dE_{\mathrm{EM}}}{dt} = \lim_{r\to\infty} \frac{r^2}{4\pi} \sum_{\ell,m} \left|\Phi_2^{\ell m}(t)\right|^2, \tag{2.72}$$

$$F_{\mathrm{D}} = \frac{dE_{\mathrm{D}}}{dt} = \lim_{r\to\infty} \frac{r^2}{4\pi} \sum_{\ell,m} \left|\mathcal{D}^{\ell m}(t)\right|^2, \tag{2.73}$$

$$F_{\mathrm{A}} = \frac{dE_{\mathrm{A}}}{dt} = \lim_{r\to\infty} \frac{r^2}{4\pi} \sum_{\ell,m} \left|\mathcal{A}^{\ell m}(t)\right|^2. \tag{2.74}$$

In numerical simulations, these quantities are evaluated at finite extraction radii. The expressions above should therefore be interpreted as asymptotic large radius definitions, with the finite radius values serving as numerical approximations.

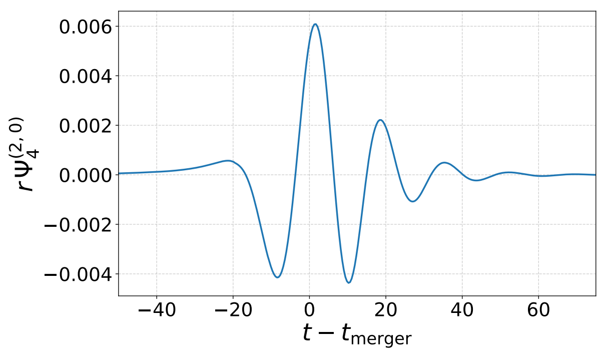
2.5 Results

We now describe the simulations of two broad classes of binary black holes in EMDA. The first consist of two Kerr–Sen black holes with at least one having nontrivial dilaton and axion fields attached. This is a nice, even stringent test of the persistence of these scalar fields through merger and the settling down of the final object to a new and different Kerr–Sen black hole. The second is the simulation of two Kerr–Newman black holes with perturbative dilaton and axion fields in the vicinity. This latter case is to test whether the merging black holes can acquire these scalar fields and whether the merger then results in a final Kerr–Sen remnant.

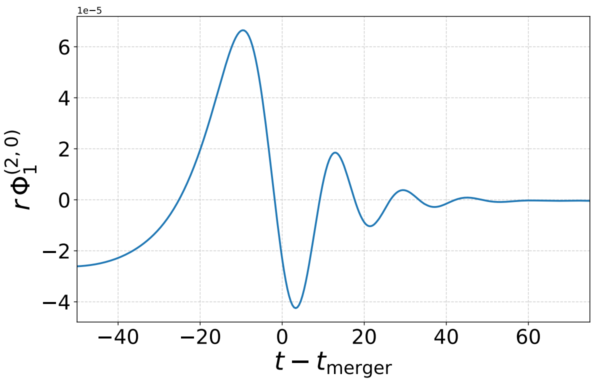
2.5.1 Initially scalarized black holes

We perform simulations across a range of binary Kerr–Sen black hole configurations in the initial data space. For the current work, we focus on equal mass binaries with low to moderate charge and low spin. In particular, the total mass of the system is normalized to $M = 1$. The charge to mass ratio on each black hole ranges between $0.0625 \leq q_i/m_i \leq 0.5$ where m_i and q_i are the initial masses and charges of the individual black holes. For spinning cases, the spins of both black holes are usually taken to be equal and aligned. The spin magnitude of each is taken to lie in the range $0.05 \leq \chi \leq 0.4$. A more complete summary of the parameter values used in our tests is provided in the next subsection.

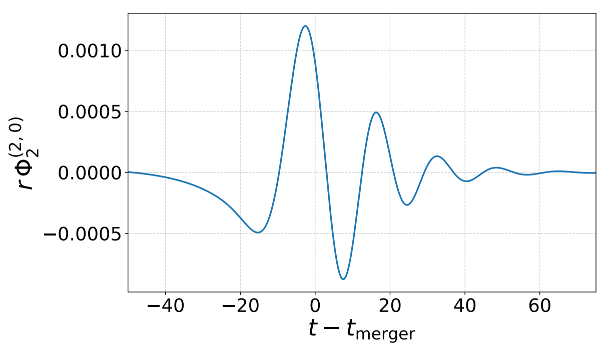
Having described the main region of parameter space explored here, we now summarize the specific tests performed and the qualitative late time behavior observed in each case.



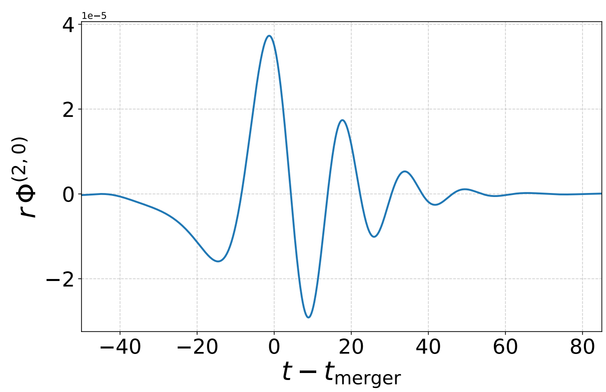
(a) Real part of the $(2,0)$ mode of Ψ_4 extracted at $R_{\text{ex}} = 100M$.



(b) Real part of the $(2,0)$ mode of Φ_1 extracted at $R_{\text{ex}} = 100M$.



(c) Real part of the $(2,0)$ mode of Φ_2 extracted at $R_{\text{ex}} = 100M$.



(d) Real part of the $(2,0)$ mode of ϕ extracted at $R_{\text{ex}} = 100M$.

Figure 2.5 The real parts of the $(2,0)$ modes of the gravitational radiation Ψ_4, electromagnetic radiation Φ_1 and Φ_2, and dilaton field ϕ, extracted at $R_{\text{ex}} = 100M$ for a binary with charge-to-mass ratio $q_i/m_i = 0.1$, dimensionless spin $\chi = 0.4$, and an initial coordinate separation of $16M$.

2.5.2 Summary of tests done

In addition to some of the tests mentioned earlier in the chapter, we have also performed a broader set of simulations to test how the late time dilaton and axion fields depend on the initial charges and spins of the binary components. In particular, we considered Kerr–Sen binaries with similar and opposite charges, spin aligned and spin anti-aligned configurations, as well as mergers involving neutral GR black holes. For binaries with opposite charges, the net charge vanishes and the remnant settles to a Kerr black hole, with neither a persistent dilaton nor axion field. For binaries with similar charges but oppositely aligned spins, the axion field decays, while the dilaton field remains nontrivial, corresponding to an EMD like remnant. We further find that when charged, spinless black holes are merged with Kerr black holes, both axion and dilaton fields persist. Finally, we find that mergers of initially nonscalarized Kerr–Newman black holes produce remnants with both dilaton and axion hair. The configurations considered are summarized in Table 2.1, where a green check mark indicates that the corresponding field remains persistent and nontrivial at late times.

Table 2.1 Binary black hole configurations.

(a) Kerr–Sen Same-Sign Charge and Spin-Aligned Black Hole Tests

(χ_1, q_1)	(χ_2, q_2)	Dilaton ϕ	Axion κ
$\chi_1 = 0.10$, $q_1 \in \{0.0624, 0.125, 0.25, 0.50\}$	$\chi_2 = 0.10$, $q_2 \in \{0.0624, 0.125, 0.25, 0.50\}$	✓	✓
$\chi_1 = 0.20$, $q_1 \in \{0.0624, 0.125, 0.25, 0.50\}$	$\chi_2 \in \{0.10, 0.20\}$, q_2 as above	✓	✓
$\chi_1 = 0.40$, $q_1 \in \{0.0624, 0.125, 0.25, 0.50\}$	$\chi_2 \in \{0.10, 0.20, 0.40\}$, $q_2 \in \{0.0624, 0.125, 0.25, 0.50\}$	✓	✓

(b) Collisions with Other Black Holes

Black Hole 1 (q, χ)	Black Hole 2 (q, χ)	ϕ	κ
Kerr–Sen (0.5, 0.4)	Kerr–Sen (0.5, 0.4)	✓	✓
Kerr–Sen (0.5, 0.4)	Kerr (0.0, 0.4)	✓	✓
Kerr–Sen (0.5, 0.4)	Kerr–Sen (−0.5, 0.4)	✗	✗
Kerr–Sen (0.5, −0.4)	Kerr–Sen (0.5, 0.4)	✓	✗
EMD (0.5, 0.0)	Kerr (0.0, 0.4)	✓	✓
KN (0.5, 0.4)†	KN (0.5, 0.4)†	✓	✓

†These configurations were initialized as nonscalarized Kerr–Newman black holes.

Abbreviations: EMD denotes Einstein–Maxwell–dilaton, corresponding here to a charged black hole with dilaton hair; KN denotes Kerr–Newman, corresponding to a black hole with charge and spin; and Kerr–Sen denotes a rotating charged black hole with dilaton and axion hair.

Having summarized the qualitative late time behavior across the full set of configurations, we now examine one representative case in more detail. This example illustrates the gravitational and electromagnetic radiation produced during a head on Kerr–Sen collision and provides a concrete example of the waveform and flux diagnostics used in the simulations.

Figure 2.5 shows the waveforms from one such representative head on collision of two identical Kerr–Sen black holes. For this system, both BHs have $\chi = 0.4$ and $q/m = 0.1$, are initially separated by $16M$ and fall together from rest. The waveforms are of the gravitational radiation, Ψ_4 and electromagnetic radiation, Φ_1 and Φ_2. The radiated energy as a function of time is shown in Figure 2.6. In this case, the GW is clearly the dominant form of energy released followed by the electromagnetic radiation. The scalar radiation is much smaller than either the EM or GW radiation. We conclude that there is very little energy radiated in the scalar field channels through the evolution. In all similar experiments, we find that the scalar fields on the black holes persist and do not radiate away through or following the merger. Nontrivial dilaton and axion fields are clearly present and remain attached to the black hole after the merger, as illustrated in Figure 2.7.

We monitor quasilocal quantities on the apparent horizon to verify that the remnant black hole is stable. In particular, we find that the horizon mass, dimensionless spin, and electric charge settle down to approximately constant values. Figure 2.8 shows the percent variation in these quantities after the final black hole has settled into a stationary configuration. In general, these quantities exhibit only small late time variations, indicating that the remnant has settled to a stable final configuration. Indeed, with these calculated black hole values, we are able to confirm that on comparison with the value of the dilaton on the horizon, the resulting black hole is another, larger mass Kerr–Sen black hole with dilaton and axion fields.

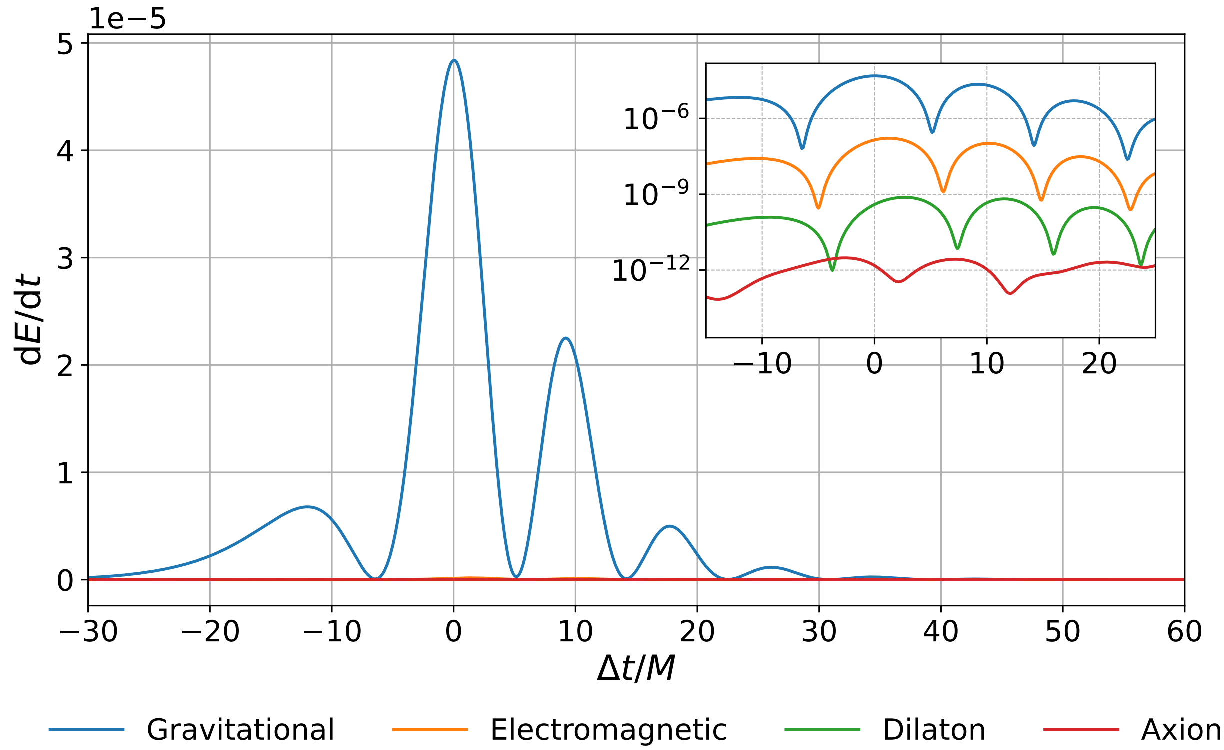
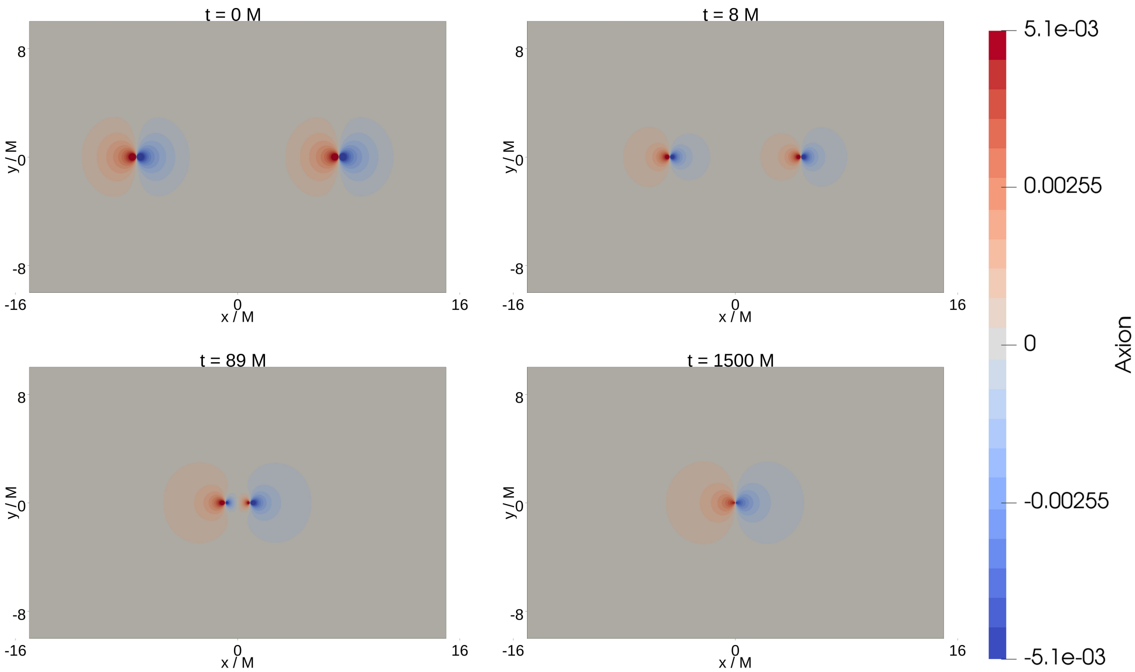


Figure 2.6 Plot of the radiated energy for the case $\chi = 0.4$ and $q_i/m_i = 0.1$. The curves have been aligned such that $t = 0$ corresponds to the global maximum. The axion and dilaton radiated energies remain at least three orders of magnitude smaller than the gravitational-wave or electromagnetic contributions.

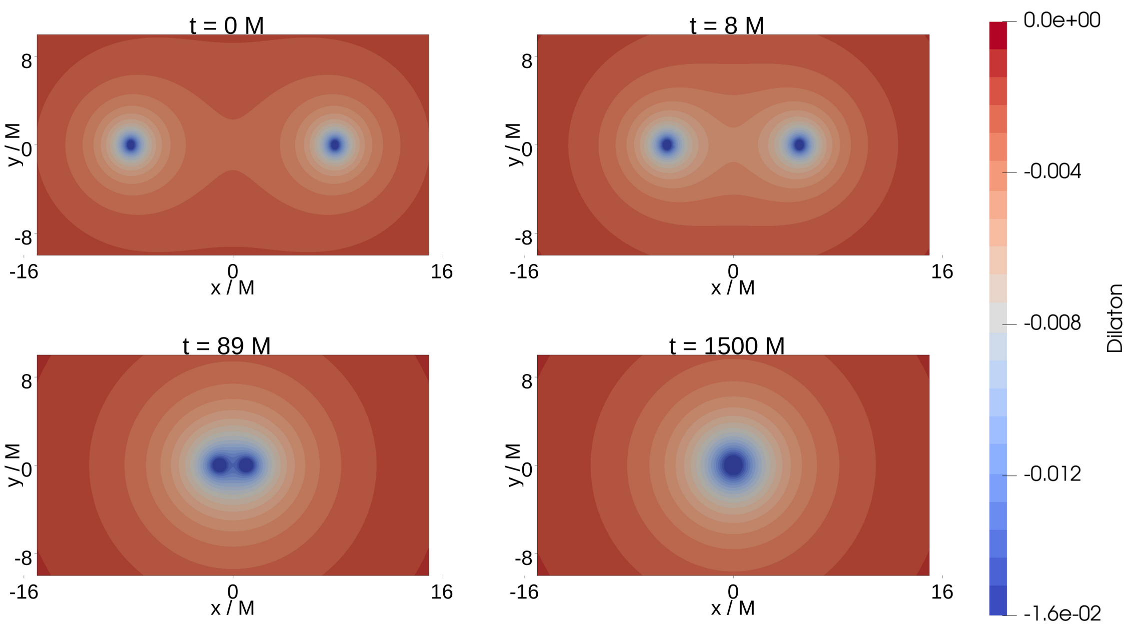
2.5.3 Initially unscalarized black holes

To further assess the stability of scalar hair, we conducted a series of tests initialized with a Gaussian distribution of axion and dilaton fields centered at the origin, with amplitudes ranging from 10^{-7} up to almost 10^{-2}. Despite not strictly satisfying the constraints, these superposed perturbative scalar fields propagate outward and enable us to examine whether initially unscalarized black holes can dynamically acquire and retain scalar fields. Our results confirm that they do: a Kerr–Newman black hole in EMDA rapidly scalarizes, developing both dilaton and axion hair. In addition, the change to the constraint violations both from the initial perturbations as well as through the subsequent evolution are small enough to leave us confident in our results.



[0.5em]

(a) Axion



(b) Dilaton

Figure 2.7 Scalar fields on the $z = 0$ plane for the case of $\chi = 0.4$ and $q_i/m_i = 0.1$ Kerr–Sen black holes. We show the amplitudes of the axion, κ (top four panels), and dilaton, ϕ (bottom four panels), fields at the beginning of the evolution (top left), shortly before merger (top right), just after merger (bottom left), and at a later time after merger (bottom right).

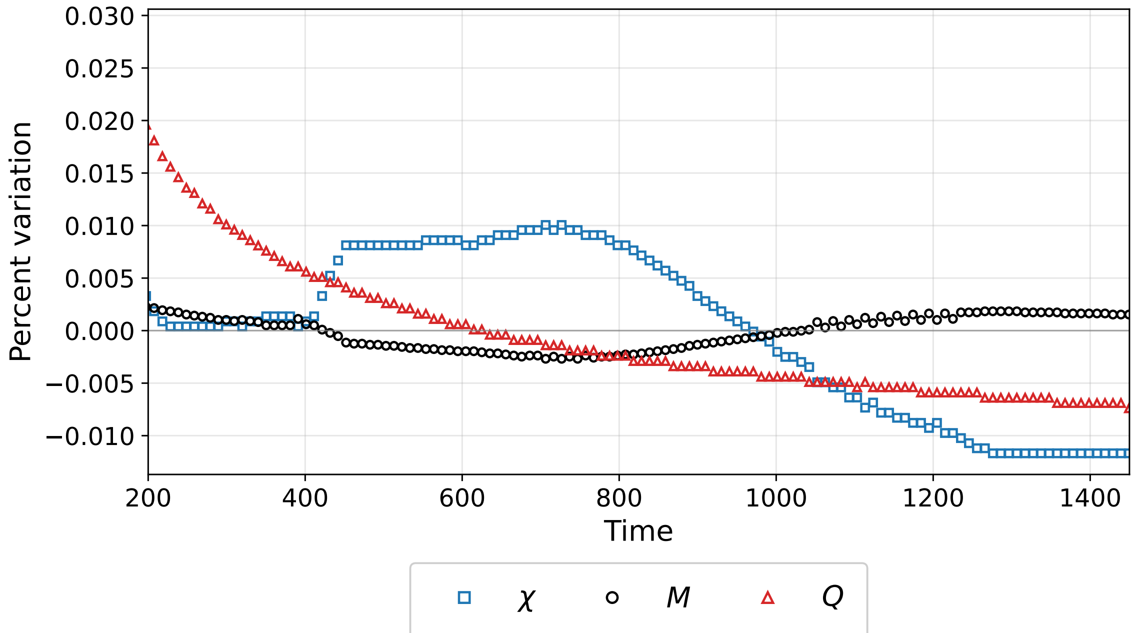
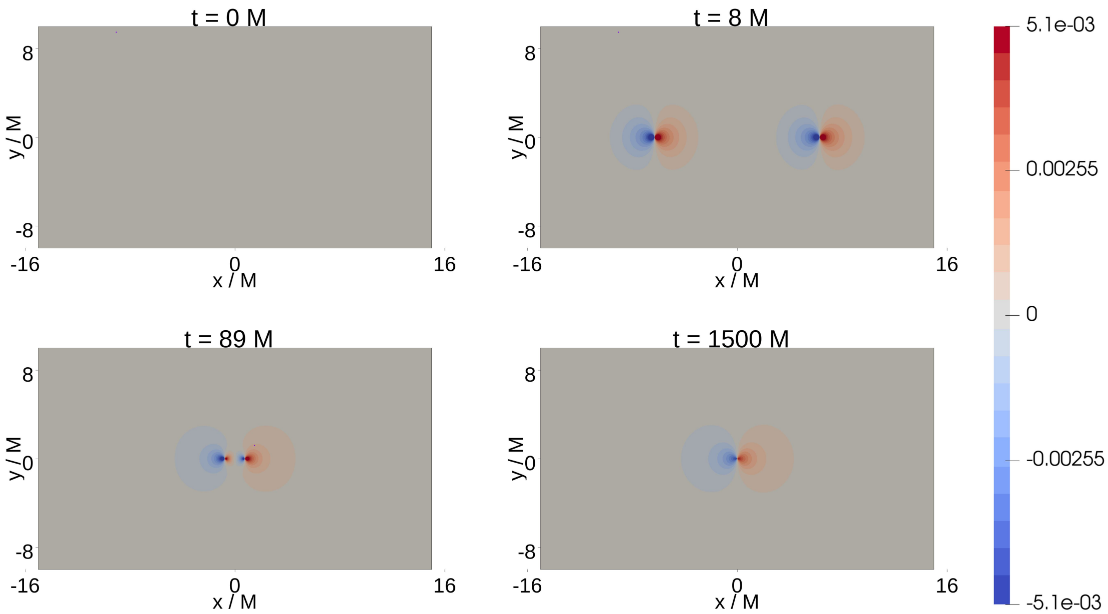


Figure 2.8 Percent deviation of the remnant black hole's horizon mass M, electric charge Q, and dimensionless spin χ from their calculated average values: $\langle M\rangle = 0.9682$, $\langle Q\rangle = 0.2001$, and $\langle\chi\rangle = 0.1961$. These quantities are measured on the apparent horizon of the remnant black hole for an initially spinning and charged binary black hole configuration with individual charge-to-mass ratios $q_i/m_i = 0.1$, initial dimensionless spins $\chi = 0.4$, and total mass $M = 1$.

Even in the absence of any initial scalar fields in the background, the system evolves to generate nontrivial scalar structure. This demonstrates that scalarization occurs dynamically and does not require pre-existing scalar hair. The scalarization occurs quickly with initially Kerr–Newman BHs acquiring Kerr–Sen aspects, i.e. scalar hair prior to merger. Mass and electromagnetic charge get radiated with a very small amount of scalar hair, but the final object is a Kerr–Sen BH. Figure 2.9 illustrates the evolution of the scalar fields for the case without initial scalar hair on the BHs.

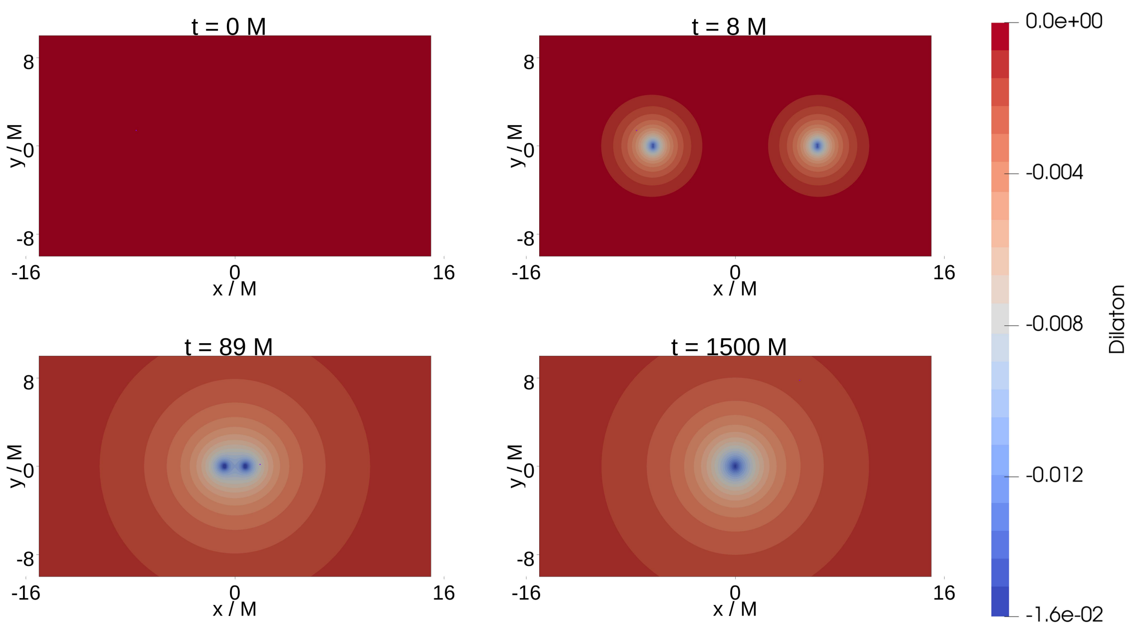
2.5.4 Constraint and refinement tests

As a consistency check on the evolutions, we monitor the constraint violations throughout the simulations. Representative full evolution constraint plots were presented in Sec. 2.2.4. These



[0.5em]

(a) Axion



(b) Dilaton

Figure 2.9 Scalar fields on the $z = 0$ plane for Kerr–Newman black holes with initial dimensionless spin $\chi = 0.4$ and charge-to-mass ratio $q_i/m_i = 0.1$. Initially, neither black hole carries dilaton or axion hair. Shown are the amplitudes of the scalar fields κ and ϕ at various stages of the evolution: the initial stage (top left), where both fields begin at zero; a short time later as the fields start to develop (top right); just before merger (bottom left); and some time after merger (bottom right).

diagnostics show that the Hamiltonian, momentum, and electromagnetic constraint violations remain controlled over the time scales considered here, including through merger and ringdown.

We also performed a limited refinement study for a short head on binary evolution, as described in Sec. 2.2.5. In this study, we varied both the wavelet tolerance and the maximum number of refinement levels. Both are WAMR related parameters that help determine the amount and extent of refinement. The wavelet tolerance was varied between 3×10^{-5} and 3×10^{-6}, and the maximum refinement level was varied between 14 and 17, corresponding to grid spacings between $h_{14} = 0.0325$ and $h_{17} = 0.00407$. Tightening the refinement criteria reduces the volume weighted Hamiltonian and momentum constraint violations, consistent with the expected behavior of the WAMR refinement scheme. Taken together, the full evolution constraint monitoring and refinement tests indicate that the simulations remain well behaved over the time scales considered here.

2.6 Conclusion

We have simulated a range of different parameters for the head on collision of Kerr–Sen black holes. We have primarily been concerned with the persistence of scalar hair through this highly nonlinear, dynamical process. In summary, our simulations show that the scalar fields do not radiate away so long as there is a nonzero charge (and spin for the axion). We have found that Kerr–Sen black holes are stable. In particular, we find that if the remnant black hole has a nonzero charge, the dilaton field persists. If it has both nonzero charge and spin, the axion field also persists after merger. These results suggest that Kerr–Sen black holes are stable under the conditions and time frames we considered.

A natural follow up study would be the analysis of waveforms produced by inspiraling black holes, an area where the available results remain relatively limited. Further, while we have assumed a particular member of the family of EMDA theories in this work, namely a low energy limit of heterotic string theory, there is a much larger theory space parametrized by the dilaton and axion couplings, α_0 and α_1, that remains largely unexplored. A more detailed investigation of this theory space could be intriguing; possibly giving rise to broader classes of stable black holes as well as detectable departures from the predicted general relativistic gravitational wave signal.

Chapter 3

Inspiraling Binary Black Hole Waveforms in EMDA Theory

3.1 Introduction

Gravitational wave observations now provide direct access to the strong field, dynamical regime of compact-binary coalescence. As detector sensitivities improve and high signal-to-noise ratio detections become more likely, the late inspiral, merger, and ringdown will offer increasingly stringent tests of the general relativistic description of black holes [4, 11, 34, 37, 61]. Recent detections have reached signal-to-noise ratios as high as 80, enabling tests of Hawking's area law and improved measurements of the remnant black hole [104]. These developments motivate numerical studies of how specific beyond-GR effects could appear in the late-inspiral, merger, and ringdown waveform.

For many alternative theories of gravity, analytic approximations are insufficient near merger, while numerical simulations require a well-posed initial-value formulation, stable gauge choices, and controlled constraint behavior as we mentioned in the prior chapter. Considerable effort has been devoted to developing formulations and approximation schemes that permit nonlinear simulations beyond GR [52, 59, 67, 68]. Examples include studies of dynamical Chern–Simons gravity [38, 39], Einstein-dilaton-Gauss-Bonnet theory [47, 69], modified gauge choices in Einstein-scalar-Gauss-Bonnet theory [45, 70], the "fixing the equations" approach [71], and auxiliary-field formulations for higher-curvature systems, including quadratic gravity [53, 72–75]. See also Ref. [66] for a more detailed comparison of these approaches.

In this chapter, we again focus on Einstein–Maxwell–dilaton–axion (EMDA) theory. EMDA extends Einstein–Maxwell theory by coupling the electromagnetic field to a dilaton and an axion, and it arises as a low-energy limit of heterotic string theory [76, 77]. The theory admits rotating charged black hole solutions, such as the Kerr–Sen spacetime, which differs from Kerr–Newman by the presence of nontrivial dilaton and axion fields. In the last chapter, we investigated the stability of black holes and black hole hair at one point in the space of EMDA theories [81, 105]. Beyond questions of stability and hair, EMDA black holes have also been studied through their potential observational signatures. In particular, black hole shadow studies using Event Horizon Telescope observations have not ruled out Kerr–Sen black holes [82, 83]. These results motivate further study of the dynamical signatures of EMDA and make the theory a useful setting in which to investigate how additional matter fields can influence compact-binary dynamics and gravitational radiation.

Charged binary black holes have also been studied in Einstein–Maxwell gravity [95, 100, 103, 106–108]. These simulations show that electromagnetic charge can modify the orbital dynamics, emitted radiation, and recoil of the remnant. In one analysis, the GW150914 signal was found to be compatible with charge-to-mass ratios as large as $Q/M \sim 0.3$ [100]. These studies demonstrate that charge alone can alter the inspiral, merger, radiation content, and remnant properties. In EMDA theory, charge also sources additional scalar and pseudoscalar fields, so charged binaries provide a natural setting in which to study how electromagnetic and beyond-GR matter and couplings affect the gravitational waveform.

The goal of this chapter is to take a step toward an exploratory waveform catalog for charged binary black holes within EMDA. We simulate inspiraling charged binary black holes and compare the resulting gravitational waveforms with neutral general relativistic reference waveforms having the same mass ratio and spin. The cases considered here include charge-to-mass ratios much larger than those expected for astrophysical black holes [109, 110]. Our aim is therefore not to construct immediately realistic astrophysical templates, but to identify regimes in which charge and the additional EMDA fields produce appreciable changes in the binary dynamics and gravitational wave signal.

These comparisons should be interpreted as fixed parameter waveform comparisons rather than as a complete parameter estimation study. Compact binary waveforms are highly degenerate, and differences between an EMDA waveform and a GR waveform may be partially mimicked by varying intrinsic binary parameters such as the mass ratio, spins, total mass scale, or eccentricity [111, 112]. Moreover, because the comparisons in this chapter are made against neutral GR rather than charged

Einstein–Maxwell binaries, the resulting differences should be understood as the combined effect of charge, modified orbital dynamics, and the additional EMDA fields, rather than as isolated dilaton or axion signatures.

The catalog presented here is therefore intended as an initial step toward a more complete study of EMDA binary black hole phenomenology. In particular, it identifies regions of parameter space where charged EMDA binaries produce appreciable waveform differences relative to neutral GR and motivates future comparisons among neutral GR, Einstein–Maxwell, and EMDA binaries.

3.2 Einstein–Maxwell–Dilaton–Axion Theory

We again consider Einstein–Maxwell–dilaton–axion (EMDA) theory, in which the Einstein–Maxwell system is supplemented by a scalar dilaton field ϕ and a pseudoscalar axion field κ. With the conventions used here, we restate the action as

$$S = \int \mathrm{d}^4 x, \sqrt{-g}\left[\frac{R}{8\pi} - 2\nabla_a \phi \nabla^a \phi - e^{-2\alpha_0 \phi} F_{ab} F^{ab} - \frac{1}{2} e^{4\alpha_1 \phi} \nabla_a \kappa \nabla^a \kappa - \kappa F_{ab} (*F)^{ab}\right], \tag{3.1}$$

where R is the Ricci scalar, F_{ab} is the $U(1)$ field strength, and $(*F)_{ab}$ is its dual. The constants α_0 and α_1 determine the coupling strengths between the electromagnetic field and the dilaton and axion fields.

As we mentioned in the prior chapter different choices of α_0 and α_1 correspond to different members of this class of theories. The parameter α_0 controls the exponential coupling between the dilaton and the electromagnetic field, while α_1 controls the exponential coupling of the dilaton to the axion kinetic term. For example, the choice $(\alpha_0, \alpha_1) = (\sqrt{3}, 0)$ gives the Kaluza–Klein value of the dilaton coupling [113, 114]. In the simulations presented here, we fix

$$(\alpha_0, \alpha_1) = (1, 1), \tag{3.2}$$

which corresponds to the coupling values associated with a low-energy limit of heterotic string theory [76, 77].

For this choice of couplings, the theory admits rotating charged black hole solutions of the Kerr–Sen type [76]. These solutions reduce to Kerr in the limit of vanishing electromagnetic charge, but for nonzero charge the black hole is accompanied by a nontrivial dilaton field and, in the rotating case, a nontrivial axion field. The structure of these fields is known analytically for a single isolated black hole; the solution we summarized in the previous chapter.

To simulate inspiraling binary black holes, we construct approximate charged binary black hole initial data and evolve the full EMDA system numerically. This allows us to compare the resulting gravitational waveforms against neutral general relativistic reference waveforms and to identify how the combined electromagnetic, dilaton, and axion fields influence the inspiral and merger.

3.3 Numerical Methods

To study the strong-field dynamics and gravitational radiation from inspiraling charged black hole binaries in Einstein–Maxwell–dilaton–axion (EMDA) theory, we evolve the full EMDA system as a Cauchy initial-value problem. The EMDA field equations and their $3+1$ decomposition are given in Eqs. 2.18–2.44 of the previous chapter. The spacetime metric is evolved using the BSSN formulation [20, 86, 87], together with puncture gauge conditions [88, 89]. The electromagnetic, dilaton, and axion fields are evolved using the corresponding matter evolution equations obtained from this $3+1$ decomposition.

The simulations are initialized with approximate charged binary black hole initial data and evolved through the late inspiral, merger, and ringdown. During the evolution, we extract gravitational radiation and monitor the matter fields, constraints, and quasilocal horizon quantities in order to assess the dynamics and numerical consistency of the solutions.

3.3.1 Numerical implementation

For the simulations presented here, we evolve the EMDA system using the same code described in Chapter 2, `Dendro-GR`. Further details on the `Dendro-GR` infrastructure, including its adaptive wavelet-refinement strategy, code-generation framework, scalability, and previous applications to binary black hole simulations, can be found in the previous chapter and in Refs. [19, 22, 23, 30]. In particular, `Dendro-GR` has been shown to perform well for intermediate mass ratio inspiral simulations, where adaptive refinement is especially useful for resolving disparate length scales [23, 30].

As before, throughout each evolution, we monitor the Hamiltonian and momentum constraints, as well as the electromagnetic divergence constraints. These quantities are used to track the consistency of the numerical solution and to assess whether the chosen refinement parameters adequately resolve the relevant dynamics.

We also extract quasilocal horizon quantities using a modified version of the `BHaHAHA` apparent-horizon finder [94]. In addition to locating the apparent horizons, this modified version computes the horizon mass, spin, and electric charge during the evolution.

3.3.2 Controlling the eccentricity

In this chapter, we focus on approximately quasicircular inspirals. For uncharged binary black holes, post-Newtonian methods can be used to choose initial momenta that reduce the orbital eccentricity. Post-Newtonian methods approximate general relativity by expanding the equations of motion in powers of v/c, where v is the characteristic orbital speed. They are most accurate when the objects are moving slowly compared with the speed of light and are separated by distances large compared with their gravitational radii. In compact binary systems, post-Newtonian theory provides analytic expressions for the orbital dynamics and gravitational waveform during the inspiral, before the late merger regime where full numerical relativity is needed. In the EMDA system, however, the presence of electromagnetic, dilaton, and axion fields complicates the construction of low-eccentricity initial data. Post-Newtonian treatments have been developed for Einstein–Maxwell theory and for several scalar-modified theories of gravity [115–118]. Here we adopt a simpler prescription based on the charged-binary argument used in [100].

At Newtonian order, both the gravitational and electromagnetic interactions are central forces. For two charged compact objects, the Coulomb interaction modifies the effective strength of the attractive force. Denoting the charge-to-mass ratio of each black hole by

$$\lambda_i = \frac{Q_i}{M_i}, \tag{3.3}$$

the leading-order effect of the electromagnetic interaction can be incorporated through the replacement of Newton's constant

$$G \to \tilde{G} = G\,(1 - \lambda_1 \lambda_2)\,. \tag{3.4}$$

In units where $G = 1$, this becomes

$$\tilde{G} = 1 - \lambda_1 \lambda_2. \tag{3.5}$$

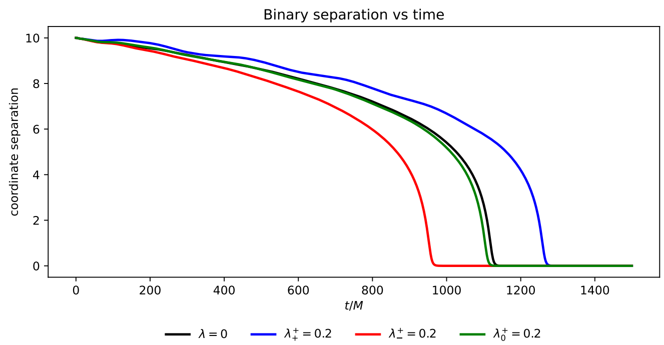
Thus, similar charges reduce the effective attraction, while opposite charges increase it. We first compute the momenta required for the corresponding uncharged binary to be approximately quasicircular using post-Newtonian estimates, and then rescale the orbital momenta according to the charge-dependent effective coupling. At leading Newtonian order, this corresponds to scaling the tangential momentum by

$$p_t \to p_t \sqrt{1 - \lambda_1 \lambda_2}. \tag{3.6}$$

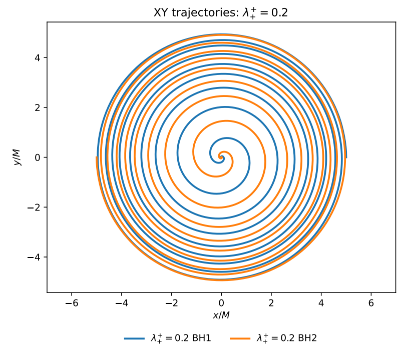
This prescription assumes that the dominant correction to the initial orbital parameters comes from the electromagnetic interaction. Possible direct corrections from the dilaton and axion fields are not included in the initial momentum estimate.

We focus on quasicircular orbits because gravitational wave emission tends to circularize isolated compact binaries during the inspiral [119, 120]. The initial momenta for the corresponding uncharged binaries are computed using the `NRPyPN` package [121]. This package provides post-Newtonian estimates for low-eccentricity binary black hole initial data compatible with Bowen–York/TwoPunctures data [122, 123]. We then apply the charge-dependent rescaling described above to obtain the momenta used in the charged evolutions.

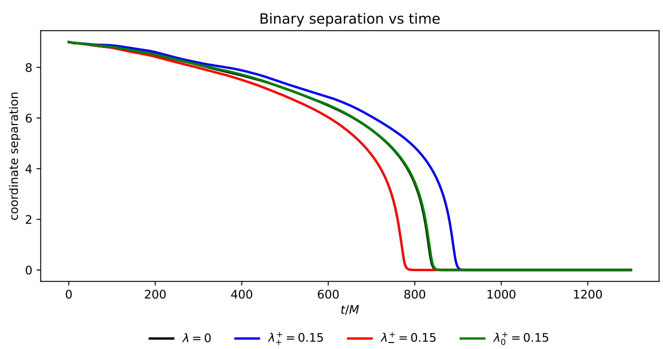
Although approximate, this prescription yields eccentricities that are adequate for the exploratory comparisons performed here, as illustrated in Figs. 3.1 and 3.2. This figure demonstrates that the black hole trajectories are approximately circular.



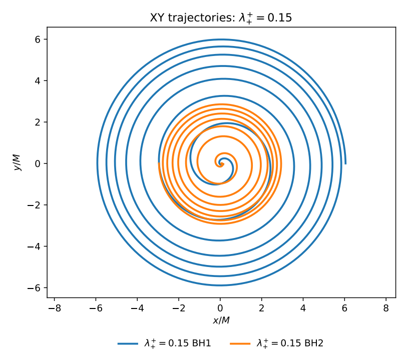
(a) $q = 1$: black hole separation.



(b) $q = 1$: orbital trajectories.

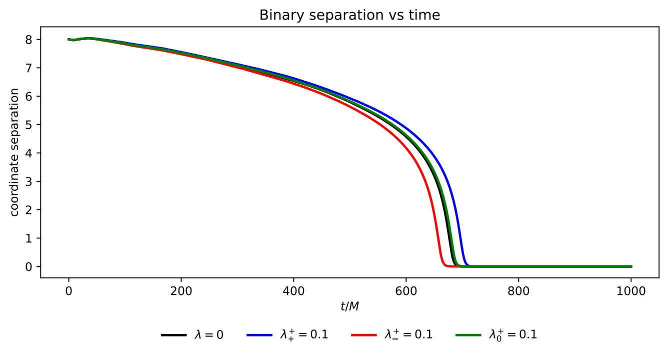


(c) $q = 2$: black hole separation.

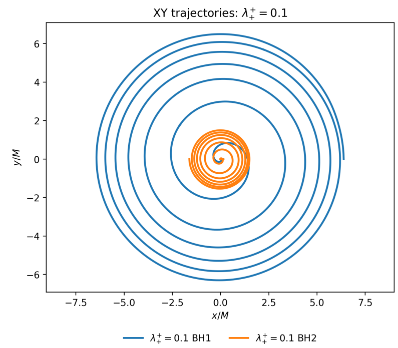


(d) $q = 2$: orbital trajectories.

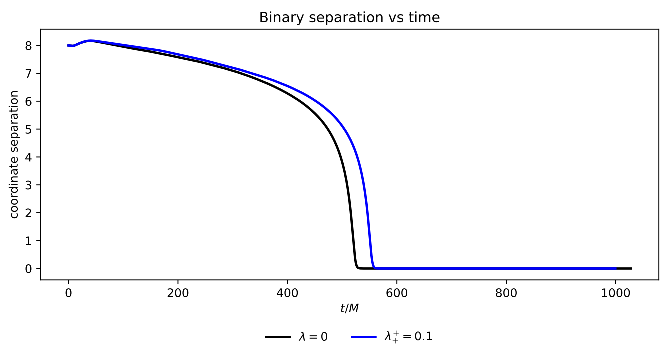
Figure 3.1 Orbital dynamics for the $q = 1$ and $q = 2$ configurations. For each mass ratio, the left panel shows the coordinate separation as a function of time, while the right panel shows representative coordinate trajectories in the orbital plane. The charge configuration affects both the inspiral time and the orbital phasing by modifying the effective interaction between the black holes. The trajectories are approximately quasicircular until the final plunge, which begins once the separation drops below roughly $6M$. Here λ denotes the magnitude of the charge-to-mass ratio, $|Q|/M$, assigned to each charged black hole; neutral black holes have $\lambda = 0$. The superscript and subscript indicate the charge configuration, with the larger black hole assigned the positive charge in unequal mass cases. Thus, λ_+^+ denotes two positively charged black holes, λ_-^+ denotes oppositely charged black holes, and λ_0^+ denotes one positively charged and one neutral black hole. The $\lambda = 0$ case is the GR reference waveform. In the trajectory plots, orange and blue denote the paths of the heavier and lighter black holes, respectively, with the initial data chosen so that the center of mass is at the origin.



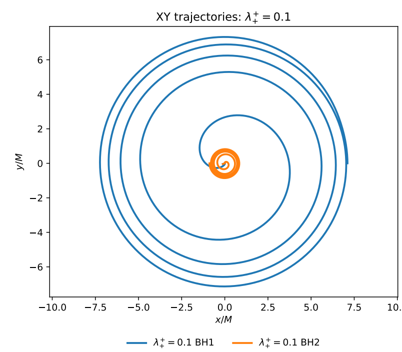
(a) $q = 4$: black hole separation.



(b) $q = 4$: orbital trajectories.



(c) $q = 8$: black hole separation.



(d) $q = 8$: orbital trajectories.

Figure 3.2 Orbital dynamics for the $q = 4$ and $q = 8$ configurations. For each mass ratio, the left panel shows the coordinate separation as a function of time, while the right panel shows representative coordinate trajectories in the orbital plane. The charge configuration affects both the inspiral time and the orbital phasing by modifying the effective interaction between the black holes. The trajectories are approximately quasicircular until the final plunge, which begins once the separation drops below roughly $6M$. Here λ denotes the magnitude of the charge-to-mass ratio, $|Q|/M$, assigned to each charged black hole; neutral black holes have $\lambda = 0$. The superscript and subscript indicate the charge configuration, with the larger black hole assigned the positive charge. Thus, $\lambda^+ * +$ denotes two positively charged black holes, $\lambda^+ * -$ denotes oppositely charged black holes, and λ^+_0 denotes one positively charged and one neutral black hole. The $\lambda = 0$ case is the GR reference waveform. In the trajectory plots, orange and blue denote the paths of the heavier and lighter black holes, respectively, with the initial data chosen so that the center of mass is near the origin.

We estimate the orbital eccentricity using an estimator based on the orbital frequency [123],

$$\varepsilon_\Omega(t) = \frac{1}{2}\left[\frac{\Omega(t) - \Omega_{e=0}(t)}{\Omega_{e=0}(t)}\right], \tag{3.7}$$

where $\Omega(t)$ is the measured orbital frequency and $\Omega_{e=0}(t)$ denotes the corresponding non-eccentric orbital frequency.

We use an orbital frequency based eccentricity estimator rather than one based directly on the coordinate separation. Coordinate separation measures can be strongly affected by gauge dynamics, whereas orbital frequency based estimators are commonly used in eccentricity reduction procedures and are generally less sensitive to these effects [124, 125]. The orbital frequency is nevertheless not a fully gauge invariant quantity. To construct the estimator, we track the coordinate positions of the two black holes throughout the inspiral, compute their coordinate separation, and use its evolution to determine the orbital frequency. The resulting orbital frequency data are then fit using `NonlinearModelFit` in `Mathematica` with a post-Newtonian inspired fitting model [123].

When fitting the orbital frequency data, it is useful to include as many orbits as possible in order to reduce the uncertainty in the eccentricity estimate. However, the early part of each simulation contains gauge settling effects that are not associated with the physical orbital eccentricity, while the late part of the simulation is affected by the onset of plunge. We therefore exclude the first $\sim 200M$ of the simulation and choose the end of the fitting window before the rapid late inspiral growth of the orbital frequency becomes dominant. In practice, this typically corresponds to ending the fit when the coordinate separation is of order $6M$. The need to have very small eccentricities is balanced by the cost of performing simulations with higher seperation that need more time to merge.

The measured orbital eccentricities for the catalog runs are shown in Table 3.3. The quoted eccentricities are extrapolated from the fit to $t = 0$. The initial coordinate separations are $r = 10M$, $r = 9M$, and $r = 8M$ for the $q = 1$, $q = 2$, and $q = 4$ or $q = 8$ configurations, respectively. Most of the lower spin and lower charge configurations have eccentricities of order 10^{-3}, while the largest eccentricities occur in the high spin and high charge equal mass cases.

Figure 3.3 shows an example of the orbital frequency eccentricity fit for one catalog run. This case has $q = 1$, $\chi = 0.4$, $\lambda = 0.3$, and charge configuration ++. The fit uses approximately four orbits, with $t_{\mathrm{min}} = 200M$ and $t_{\mathrm{max}} = 1300M$. The measured eccentricity extrapolated to $t = 0$ is $\varepsilon_\Omega = (14.44 \pm 0.33) \times 10^{-3}$. This comparatively large eccentricity is consistent with the fact that this is a high charge, similar charge configuration, for which the reduced effective attraction makes the initial momentum prescription less accurate.

In principle, lower eccentricities could be obtained through an iterative eccentricity-reduction procedure, in which the initial momenta are corrected after an initial eccentricity measurement and the simulation is repeated. We do not perform iterative eccentricity reduction for this exploratory catalog. Standard post-Newtonian eccentricity-reduction methods may also need modification in the EMDA case, since the electromagnetic, dilaton, and axion fields can affect the orbital dynamics. Developing eccentricity-reduction procedures that incorporate these effects is left for future work.

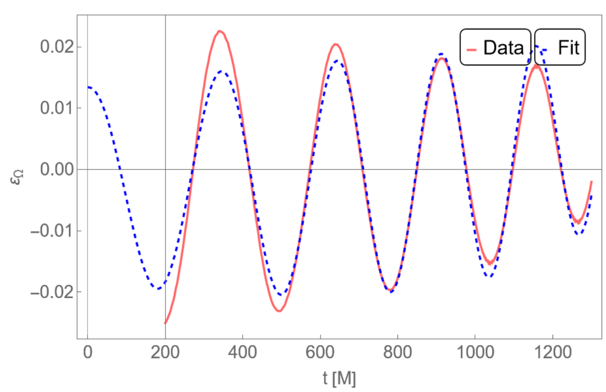


Figure 3.3 Measured orbital-frequency eccentricity and fitted eccentricity model for a binary black hole inspiral with $q = 1$, $\chi = 0.4$, $\lambda = 0.3$, and charge configuration ++. The fit is performed from $t_{\mathrm{min}} = 200M$ to $t_{\mathrm{max}} = 1300M$, excluding early gauge-settling effects and ending before the onset of plunge. The measured eccentricity extrapolated to $t = 0$ is $\varepsilon_\Omega = (14.44 \pm 0.33) \times 10^{-3}$.

3.3.3 Waveform mismatch and distinguishability

Following Refs. [100, 126–128], we use waveform overlaps and mismatches to quantify differences between the simulated gravitational wave signals.

A direct comparison between charged EMDA and neutral GR waveforms does not, by itself, isolate the contribution of the dilaton and axion fields. Part of the measured difference may arise from the electromagnetic charge and the associated change in the orbital dynamics, rather than from the additional EMDA fields alone. A more complete study would compare EMDA waveforms not only with neutral GR waveforms, but also with charged Einstein–Maxwell evolutions having the same charge configurations. Such a comparison is beyond the scope of the present work. Our goal here is instead to quantify the fixed parameter waveform differences between charged EMDA binaries and their neutral GR counterparts.

To quantify the difference between two gravitational wave signals, we compute the mismatch

$$\mathcal{M}(h_1, h_2) = 1 - [\max_{\Delta t, \Delta\varphi} \mathcal{O}(h_1, h_2)], \tag{3.8}$$

where the maximum is taken over relative time shifts Δt and polarization-angle shifts $\Delta\varphi$. The overlap $\mathcal{O}$ is defined by

$$\mathcal{O}(h_1, h_2) = \frac{(h_1, h_2)}{\sqrt{(h_1, h_1)(h_2, h_2)}}, \tag{3.9}$$

where (h_1, h_2) denotes the detector-noise-weighted inner product between the two waveforms. In practice, this inner product is evaluated over the finite frequency range used in the analysis, with the same preprocessing choices applied to each waveform pair.

The comparison performed here does not maximize over intrinsic binary parameters such as the masses, spins, eccentricity, or charge. These degeneracies could partially obscure the waveform differences found in a fixed parameter comparison and would likely make charge-induced deviations more difficult to identify in a full parameter estimation study. The mismatch values reported below should therefore be interpreted as a measure of waveform separation at fixed intrinsic parameters, rather than as evidence for a unique observational signature of EMDA theory.

As a rough estimate of distinguishability, we associate a mismatch $\mathcal{M}$ with the critical signal-to-noise ratio

$$\rho_{\text{crit}} \simeq \frac{1}{\sqrt{2\mathcal{M}}}. \tag{3.10}$$

This estimate gives the signal-to-noise ratio at which two waveforms with mismatch $\mathcal{M}$ would be expected to become distinguishable in an idealized fixed parameter comparison.

3.4 Results

We compare charged EMDA waveforms with neutral GR reference waveforms for a representative set of binary black hole configurations. The purpose of this comparison is to determine how the waveform differences depend on charge magnitude, charge configuration, mass ratio, and spin. In each case, the charged EMDA waveform is compared against a neutral reference waveform with the same mass ratio and spin.

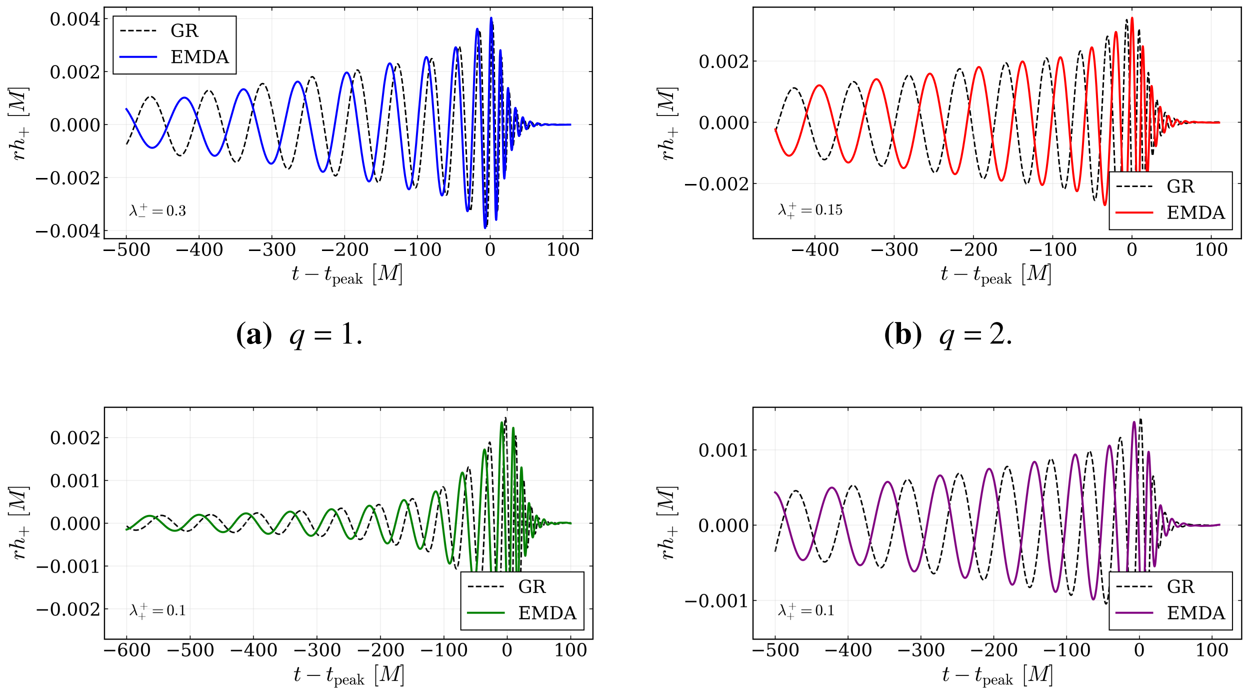
The simulation catalog considered in this chapter is summarized in Table 3.1. For each choice of mass ratio $q = m_1/m_2$, spin χ, and charge magnitude $\lambda = |Q|/M$, we consider one neutral configuration, $Q_1 = Q_2 = 0$, and several charged configurations. We label the charged cases by the signs of the two black hole charges. The cases $++$, $+-$, and $+0$ denote similar charges, opposite charges, and a configuration with one charged and one neutral black hole, respectively. For unequal mass binaries, the larger black hole is assigned the positive charge in the $+-$ and $+0$ cases.

Representative peak-aligned waveforms are shown in Fig. 3.4. The figure compares the strain component rh_+ for selected charged EMDA configurations against the corresponding neutral GR reference waveform. The waveforms are aligned at the peak amplitude in order to emphasize differences in the late inspiral, merger, and ringdown. Even after this alignment, the charged and neutral waveforms show visible phase and amplitude differences, indicating that charge and the additional EMDA fields can modify the binary dynamics prior to merger as well as the waveform near peak emission.

The full set of waveform mismatches is computed using the numerical relativity analysis toolkit `kuibit` [129]. kuibit is a numerical relativity library with functions to process gravitational waves. The results are given in Table 3.2. The mismatch values generally increase with charge magnitude, although the trend is not monotonic across all spin and charge configurations. For example, in the $q = 1$, $\chi = 0.4$, ++ sequence, the mismatch increases from 2.60×10^{-4} for $\lambda = 0.1$ to 5.28×10^{-3} for $\lambda = 0.3$. Similar growth with charge is seen in several other equal mass configurations.

The charge configuration also plays an important role. The ++, +−, and +0 cases do not produce equivalent waveform differences, even at the same charge magnitude. In many cases, the singly charged +0 configurations remain closer to the neutral reference waveform than the corresponding two-charged configurations. This is consistent with the expectation that the strength and symmetry of the charge distribution influence both the orbital dynamics and the matter field content of the spacetime. Opposite charges can also produce comparatively large mismatches, since the electromagnetic interaction increases the effective attraction and changes the inspiral rate.

The $\chi = 0.8$ equal mass configurations are somewhat anomalous compared with the lower-spin cases. While the mismatch generally grows with charge for the lower-spin sequences, the $\chi = 0.8$ cases show larger and less monotonic variations across the different charge configurations. These runs also have noticeably larger eccentricities, especially at the highest charge magnitudes, suggesting that the approximate charged initial-data prescription is being pushed beyond the regime where it gives uniformly low-eccentricity inspirals. We therefore include these cases as useful indicators of where large waveform differences can occur, but we do not interpret their detailed mismatch hierarchy as a clean physical trend.



(a) $q = 1$. **(b)** $q = 2$.

(c) $q = 4$. **(d)** $q = 8$.

Figure 3.4 Peak-aligned comparisons of the gravitational-wave strain component rh_+ for representative charged EMDA configurations against the corresponding neutral GR reference waveforms. In each panel, the time coordinate is shifted so that the waveform peak occurs at $t - t_{\text{peak}} = 0$. The remaining differences after peak alignment reflect changes in the accumulated phase, merger time, and waveform morphology.

The high-spin configurations nevertheless show the largest waveform separations in the catalog. For the $q = 1$, $\chi = 0.8$ cases, some mismatches reach the 10^{-2}–10^{-1} level. These large values indicate that the charged and neutral waveforms are substantially separated at fixed intrinsic parameters. However, because these cases are also more sensitive to eccentricity and initial-data systematics, they should not be interpreted as isolated signatures of the dilaton or axion fields.

The critical signal-to-noise ratio estimate associated with the mismatch values spans approximately $\rho_{\text{crit}} \sim 2.3$ to over 500 across the catalog. These values should be understood as fixed parameter distinguishability estimates: they indicate the signal-to-noise ratio at which the charged EMDA waveform and the corresponding neutral GR waveform would become distinguishable if the intrinsic binary parameters were held fixed. In a full parameter estimation study, changes in the masses, spins, eccentricity, or other parameters could partially obscure these differences. The mismatch values therefore quantify waveform separation within this catalog, rather than providing direct observational bounds on EMDA theory.

These results demonstrate that our EMDA evolution framework can produce a range of charged binary black hole simulations with relatively low eccentricity. For most of the catalog, the observed waveform trends are consistent with physical expectations: increasing the charge magnitude tends to increase the waveform mismatch, and different charge configurations produce distinct inspiral and merger behavior. Nevertheless, the highest-spin configurations require additional caution. These cases show larger residual eccentricities and noisier waveforms than the lower-spin runs, suggesting that the approximate charged initial-data prescription and the chosen grid resolution may be less reliable in this region of parameter space. Future work should revisit these cases with improved eccentricity reduction and higher-resolution simulations to determine whether the large waveform differences remain after these numerical effects are better controlled.

3.5 Conclusion

In this chapter, we have presented an exploratory waveform catalog for charged binary black hole simulations in Einstein–Maxwell–dilaton–axion (EMDA) theory. We compared the resulting charged EMDA waveforms with neutral general relativistic reference waveforms having the same mass ratio and spin. Across much of the catalog, the waveform differences increase with the charge magnitude, indicating that charge and the additional EMDA fields can produce appreciable changes in the gravitational wave signal. The trend is clearest in the lower-spin configurations, while the highest-spin cases show larger and less monotonic differences that are likely more sensitive to eccentricity and the approximate charged initial-data prescription.

Using the mismatch as a rough fixed parameter distinguishability estimate, the largest waveform separations in the catalog correspond to critical signal-to-noise ratios, with the smallest values approximately $\rho_{\rm crit} \sim 2.3$. These values indicate that the largest charged EMDA deviations from the corresponding neutral GR waveforms are substantial at fixed intrinsic parameters. However, they should not be interpreted as direct observational thresholds or constraints on EMDA theory. A full parameter estimation study would be required to determine whether these differences could be mimicked by changes in the masses, spins, eccentricity, total mass scale, or other binary parameters.

The waveform differences found here should also be interpreted with care because the comparison is made against neutral GR rather than against charged Einstein–Maxwell binaries. As a result, the reported mismatches include the combined effect of electric charge, modified orbital dynamics, and the additional dilaton and axion fields. They cannot be attributed uniquely to the dilaton or axion fields alone. A more complete comparison would include corresponding Einstein–Maxwell simulations with the same charge configurations, allowing the purely electromagnetic contribution to be separated from the additional EMDA-field effects.

These results demonstrate that EMDA theory provides a useful numerical relativity testbed for studying charged beyond-GR effects in binary black hole mergers. Even within this exploratory catalog, charged EMDA binaries can produce waveform differences that are large enough to motivate further study. Natural next steps include larger parameter surveys, improved charged initial data, iterative eccentricity reduction, longer inspirals, and convergence studies of the waveform mismatches. It would be especially useful to extend the high-mass-ratio portion of the catalog by performing additional $q = 8$ simulations with the full set of charge configurations, higher spins, and larger charge magnitudes. Direct comparisons among neutral GR, charged Einstein–Maxwell, and charged EMDA binaries would also help separate ordinary electromagnetic effects from waveform changes associated specifically with the dilaton and axion fields.

3.6 Simulation Catalog

The simulations used in this chapter are summarized in Table 3.1. The catalog samples several mass ratios, spin values, charge magnitudes, and charge configurations. For each mass ratio and spin, we include a neutral reference simulation and compare the charged EMDA evolutions against that corresponding neutral case.

The simulations used in this chapter are summarized in Table 3.1. The catalog samples several mass ratios, spin values, charge magnitudes, and charge configurations. For each mass ratio and spin, we include a neutral reference simulation and compare the charged EMDA evolutions against that corresponding neutral case. For the $q = 8$ simulations, we choose a negative spin value, $\chi = -0.2$, corresponding to spin anti-aligned with the orbital angular momentum. This choice reduces the inspiral time relative to an aligned-spin configuration, making the high-mass-ratio simulations less computationally expensive.

Table 3.1 EMDA binary black hole simulation catalog. Here $q = m_1/m_2$ denotes the mass ratio, χ denotes the dimensionless spin parameter, and $\lambda_i = Q_i/M_i$ denotes the charge-to-mass ratio of black hole i. The neutral case corresponds to $Q_1 = Q_2 = 0$. For charged configurations, the magnitude of the nonzero charge-to-mass ratios is denoted by λ, with the sign structure indicated by the charge configuration. The configurations ++, +−, and +0 denote similar charges, opposite charges, and cases in which only one black hole is charged, respectively. Because the neutral configuration is independent of λ, it is listed only once for each mass ratio and spin.

Mass Ratio q	Spin χ	Charge Magnitude λ	Neutral	++	+−	+0
1	0.0	0.1	✓	✓	✓	✓
1	0.0	0.2	–	✓	✓	✓
1	0.0	0.3	–	✓	✓	✓
1	0.4	0.1	✓	✓	✓	✓
1	0.4	0.2	–	✓	✓	✓
1	0.4	0.3	–	✓	✓	✓
1	0.8	0.1	✓	✓	✓	✓
1	0.8	0.2	–	✓	✓	✓
1	0.8	0.3	–	✓	✓	✓
2	0.0	0.05	✓	✓	✓	✓
2	0.0	0.15	–	✓	✓	✓
2	0.4	0.05	✓	✓	✓	✓
2	0.4	0.15	–	✓	✓	✓
4	0.4	0.05	✓	✓	✓	✓
4	0.4	0.10	–	✓	✓	✓
8	-0.2	0.10	✓	✓	–	–

Table 3.2 summarizes the fixed parameter waveform mismatches between the charged EMDA simulations and the corresponding neutral reference waveforms. Across much of the catalog, the mismatch increases with charge magnitude, although the trend is not strictly monotonic for every spin and charge configuration. The clearest growth occurs in the equal mass lower-spin cases. For example, for $q = 1$ and $\chi = 0.4$, the ++ mismatch grows from 2.60×10^{-4} at $\lambda = 0.1$ to 5.28×10^{-3} at $\lambda = 0.3$, while the $+-$ mismatch grows even more strongly, reaching 2.90×10^{-2}. The charge configuration also has a significant effect: the +0 cases are often closer to the neutral reference waveform than the two-charged configurations, while the $+-$ cases can produce relatively large mismatches because the opposite charges increase the effective attraction and accelerate the inspiral. The largest mismatches occur in the high-spin $q = 1$, $\chi = 0.8$ cases, where some values reach the 10^{-2}–10^{-1} level; however, these cases should be interpreted with caution because they also have larger residual eccentricities and noisier waveforms. For the unequal mass cases, especially $q = 2$ and $q = 4$, the mismatches are generally smaller, with many values below 10^{-3}. Overall, the table shows that charge can produce measurable waveform differences at fixed mass ratio and spin, but that the size of the effect depends strongly on charge magnitude, charge configuration, and spin.

Table 3.2 Mismatch of charged EMDA binary black hole waveforms relative to the corresponding neutral reference waveform. Here $q = m_1/m_2$, χ is the dimensionless spin parameter, and $\lambda_i = Q_i/M_i$. The column $|\lambda_i|$ gives the magnitude of the nonzero charge-to-mass ratio used in each charged configuration. The columns ++, +−, and +0 denote similar, opposite charges, and configurations in which only one black hole is charged, respectively. The mismatch is computed from the $(\ell, m) = (2,2)$ gravitational waveform using the numerical relativity analysis package `kuibit`, after optimizing over relative time and polarization shifts. The values in parentheses give the approximate fixed parameter critical signal-to-noise ratio, $\rho_{\rm crit} \simeq 1/\sqrt{2\mathcal{M}}$. These values should be interpreted as rough waveform-separation estimates rather than full observational detection thresholds.

Mass Ratio q	Spin χ	Charge Magnitude $\lvert\lambda_i\rvert$	++	+−	+0
1	0.0	0.10	3.53×10^{-4}; (37.6)	2.74×10^{-4}; (42.8)	1.55×10^{-4}; (56.8)
1	0.0	0.20	6.71×10^{-4}; (27.3)	1.45×10^{-3}; (18.5)	6.91×10^{-4}; (26.9)
1	0.0	0.30	1.94×10^{-3}; (16.0)	5.32×10^{-3}; (9.7)	7.54×10^{-4}; (25.8)
1	0.4	0.10	2.60×10^{-4}; (43.9)	2.87×10^{-4}; (41.8)	4.08×10^{-4}; (35.0)
1	0.4	0.20	7.10×10^{-4}; (26.5)	7.40×10^{-3}; (8.2)	7.42×10^{-4}; (26.0)
1	0.4	0.30	5.28×10^{-3}; (9.7)	2.90×10^{-2}; (4.1)	1.04×10^{-3}; (21.9)
1	0.8	0.10	2.77×10^{-4}; (42.5)	9.44×10^{-4}; (23.0)	1.03×10^{-3}; (22.0)
1	0.8	0.20	4.66×10^{-2}; (3.3)	9.30×10^{-2}; (2.3)	1.67×10^{-2}; (5.5)
1	0.8	0.30	6.19×10^{-2}; (2.8)	2.53×10^{-2}; (4.4)	7.99×10^{-3}; (7.9)
2	0.0	0.05	1.58×10^{-3}; (17.8)	1.34×10^{-3}; (19.3)	2.56×10^{-5}; (139.7)
2	0.0	0.15	1.83×10^{-3}; (16.5)	3.21×10^{-3}; (12.5)	7.47×10^{-4}; (25.9)
2	0.4	0.05	7.27×10^{-5}; (82.9)	1.33×10^{-4}; (61.3)	1.73×10^{-6}; (537.4)
2	0.4	0.15	6.17×10^{-4}; (28.5)	2.34×10^{-4}; (46.2)	7.58×10^{-5}; (81.2)
4	0.4	0.05	5.10×10^{-5}; (99.0)	7.34×10^{-5}; (82.5)	7.17×10^{-5}; (83.5)
4	0.4	0.10	3.17×10^{-4}; (39.7)	1.82×10^{-4}; (52.5)	8.80×10^{-5}; (75.4)
8	-0.2	0.10	1.14×10^{-3}; (21.0)	–	–

3.7 Measured Eccentricities

Table 3.3 summarizes the measured orbital eccentricities for the EMDA waveform catalog. The values are obtained from the orbital frequency eccentricity estimator described in Sec. 3.3.2 and are reported as $(\varepsilon_\Omega \pm \delta\varepsilon_\Omega) \times 10^3$.

The eccentricity estimates for the $q = 8$ configurations are poorly constrained compared with the rest of the catalog, as indicated by their large uncertainties. We therefore use these values only as rough indicators of the orbital behavior and do not place significant weight on them when interpreting the mismatch trends.

Table 3.3 Measured orbital eccentricities for the EMDA waveform catalog. The entries are reported as $(\varepsilon_\Omega \pm \delta\varepsilon_\Omega) \times 10^3$. Here $q = m_1/m_2$ denotes the mass ratio, χ denotes the dimensionless spin parameter, and $\lambda_i = Q_i/M_i$ denotes the charge-to-mass ratio of black hole i. The neutral case corresponds to $Q_1 = Q_2 = 0$. For charged configurations, the magnitude of the nonzero charge-to-mass ratios is denoted by λ, with the sign structure indicated by the charge configuration. The configurations ++, +−, and +0 denote similar charges, opposite charges, and cases in which only one black hole is charged, respectively.

Mass Ratio q	Spin χ	Charge λ	Neutral	++	+−	+0
1	0.0	0.10	1.59 ± 0.21	2.57 ± 0.17	1.97 ± 0.27	2.54 ± 0.23
1	0.0	0.20	–	6.26 ± 0.26	3.52 ± 0.35	4.41 ± 0.24
1	0.0	0.30	–	14.99 ± 0.40	2.08 ± 0.36	5.28 ± 0.45
1	0.4	0.10	1.39 ± 0.10	1.80 ± 0.11	1.30 ± 0.13	1.26 ± 0.11
1	0.4	0.20	–	5.17 ± 0.18	1.04 ± 0.22	2.68 ± 0.18
1	0.4	0.30	–	14.44 ± 0.33	4.89 ± 0.53	4.19 ± 0.19
1	0.8	0.10	3.47 ± 0.23	3.23 ± 0.18	2.31 ± 0.23	2.31 ± 0.21
1	0.8	0.20	–	4.95 ± 0.64	5.44 ± 0.71	4.57 ± 0.27
1	0.8	0.30	–	44.75 ± 0.70	22.70 ± 6.36	13.03 ± 2.02
2	0.0	0.05	2.16 ± 0.60	1.98 ± 0.22	1.22 ± 0.62	1.62 ± 0.31
2	0.0	0.15	–	4.65 ± 0.36	2.18 ± 0.33	3.05 ± 0.37
2	0.4	0.05	1.68 ± 0.11	1.46 ± 0.24	1.78 ± 0.14	1.48 ± 0.17
2	0.4	0.15	–	5.19 ± 0.15	1.35 ± 0.24	2.09 ± 0.18
4	0.4	0.05	1.66 ± 0.43	1.44 ± 0.32	1.93 ± 0.34	1.49 ± 0.50
4	0.4	0.10	–	1.01 ± 0.09	0.81 ± 0.21	0.79 ± 0.21
8	-0.2	0.10	4.88 ± 61.91	1.45 ± 5.10	–	–

3.8 Constraint Behavior

Throughout the evolutions, we monitor the Hamiltonian and momentum constraints. These quantities provide an important check of the numerical consistency of the simulations and help identify periods in which the adaptive mesh or finite-difference resolution is insufficient to resolve the evolving fields.

Figures 3.5 and 3.6 show representative constraint histories for the $q = 1$ ($\chi = 0.4$ $\lambda_+^+ = 0.1$) and $q = 8$($\chi = -0.2$ $\lambda_+^+ = 0.1$) configurations, respectively. In each case, the plotted quantities are the Hamiltonian constraint and the combined momentum-constraint violation. The constraint violations remain controlled throughout the inspiral.

The unequal mass $q = 8$ simulation provides a more demanding numerical test than the equal mass case because the smaller black hole introduces a shorter physical length scale. Nevertheless, the constraints remain bounded throughout the evolution.

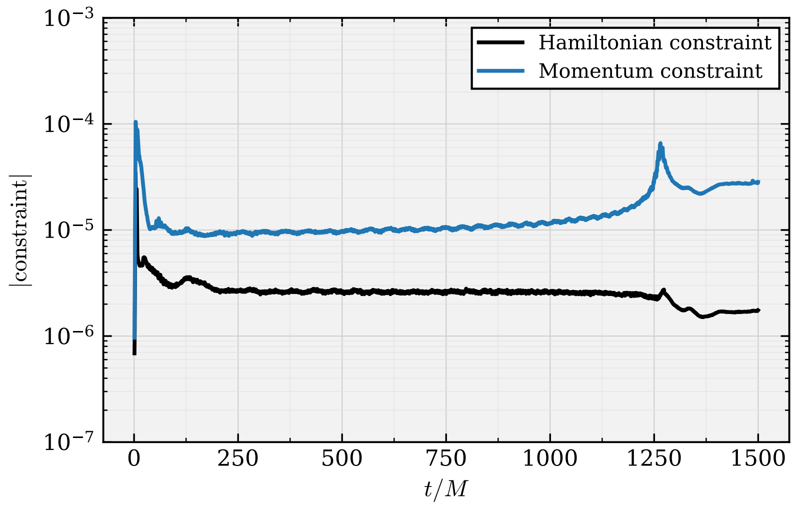


Figure 3.5 Hamiltonian and combined momentum-constraint violations for a representative $q = 1$ binary evolution. The constraints remain controlled throughout the inspiral.

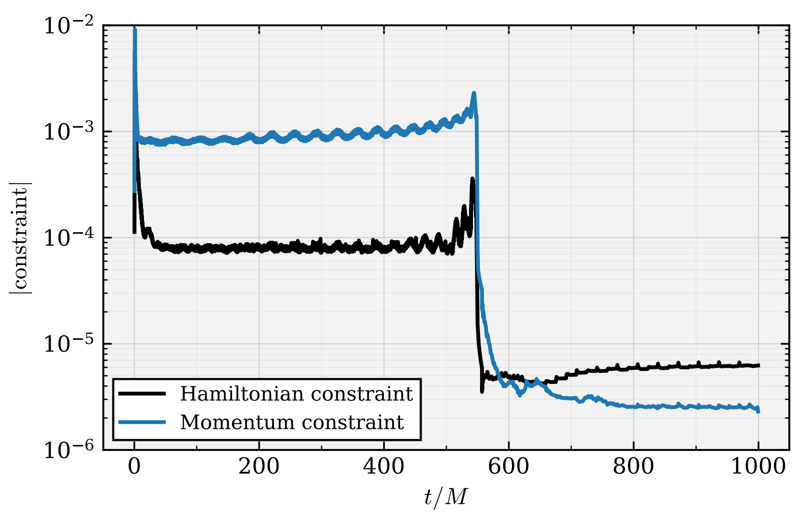


Figure 3.6 Hamiltonian and combined momentum-constraint violations for the $q = 8$ binary evolution. Although the unequal mass system places greater demands on the adaptive mesh, the constraints remain bounded throughout the simulation.

Chapter 4

Beyond Kerr–Sen Black Holes: Exploring the Theory Space of EMDA

4.1 Introduction

The growing number of detected binary black hole mergers provides an increasingly broad sample of gravitational wave signals with which to test general relativity in the strong field, dynamical regime. Continued improvements to the LIGO, Virgo, and KAGRA detectors are expected to further increase the rate of observed compact binary mergers, expanding the available catalog of gravitational wave events [24, 25]. As this catalog grows, gravitational wave observations provide an increasingly powerful way to search for small deviations from general relativity and to constrain alternative theories of gravity [4, 37, 61].

Numerical relativity plays an important role in this effort because the merger regime is highly nonlinear and theory dependent. However, numerical relativity models for alternative theories of gravity are less mature than their general relativistic counterparts. Simulating such theories presents several challenges. Some modified theories suffer from pathologies such as evolution systems that are not well posed or ghostlike degrees of freedom associated with higher derivative terms [130–132]. Numerical studies of these theories therefore require formulations in which the initial value problem can be posed consistently and the subsequent evolution remains well behaved. In particular, one must construct suitable initial data and evolve the system in a way that controls the constraints.

Many tests of gravity can be understood as constraints on additional features that are absent in vacuum general relativity. For example, previous work has used gravitational wave observations to bound higher curvature corrections, placing limits on the length scales over which such terms can become important [4, 36, 133]. Other studies have considered constraints on black hole charge [96, 100], as well as tests based on black hole shadows [82, 83]. In this chapter, we investigate another possible extension: Einstein–Maxwell–dilaton–axion (EMDA) theory. EMDA theory contains, in addition to the metric and electromagnetic field, a scalar dilaton field and a pseudoscalar axion field. These additional degrees of freedom can modify the structure of charged black holes and may leave observable imprints on binary mergers and their emitted radiation [50, 76, 77, 113].

Analytic black hole solutions in theories related to EMDA are known for special choices of the coupling parameters. These include the Kerr–Sen black hole for $\alpha_0 = \alpha_1 = 1$, Einstein–Maxwell–dilaton black holes for $\alpha_1 = 0$, and Kaluza–Klein black holes for $\alpha_0 = \sqrt{3}$ [76, 77, 113, 114]. To explore the broader theory space, we construct approximate initial data by scaling the scalar fields by their couplings and then solving the initial data problem. Although these data are not exact, they are expected to be close enough to the desired configuration that the black holes quickly relax toward a quasiequilibrium state early in the evolution. The stability of Kerr–Sen black holes has recently been studied in both perturbative and fully nonlinear settings [81, 105], and previous numerical work has explored portions of the EMD parameter space using similar techniques [50]. However, less is known about how varying the EMDA coupling parameters affects the nonlinear dynamics of binary black hole mergers.

Here, we focus primarily on the $\alpha_0 = \alpha_1$ part of the EMDA parameter space. By varying this common coupling, we study how the strength of the dilaton and axion interactions influences the merger dynamics and the resulting gravitational wave signals. This provides a first step toward understanding how binary black hole observations may constrain broader regions of EMDA theory beyond the analytically known special cases.

The goal of this chapter is not to produce a catalog level set of simulations. Instead, we use approximate charged binary black hole initial data to explore how the dynamics depend on the EMDA coupling parameters.

4.2 Theory space

As our starting point, we consider Einstein–Maxwell–dilaton–axion (EMDA) theory, in which the Einstein–Maxwell system is supplemented by a scalar dilaton field ϕ and a pseudoscalar axion field κ. The action is

$$S = \int \mathrm{d}^4x, \sqrt{-g}\Big[\frac{R}{8\pi} - 2\nabla_a\phi\nabla^a\phi - e^{-2\alpha_0\phi}F_{ab}F^{ab} - \frac{1}{2}e^{4\alpha_1\phi}\nabla_a\kappa\nabla^a\kappa - \kappa F_{ab}(*F)^{ab}\Big], \tag{4.1}$$

where R is the Ricci scalar, F_{ab} is the $U(1)$ field strength, and $(*F)_{ab}$ is its dual. The constants α_0 and α_1 determine the coupling strengths between the dilaton, axion, and electromagnetic field. Different choices of α_0 and α_1 correspond to different members of this class of theories. The choice $(\alpha_0, \alpha_1) = (1,1)$ corresponds to the coupling values associated with the low-energy limit of heterotic string theory and admits the Kerr–Sen black hole solution [76, 77]. Setting $\alpha_0 = 1, \alpha_1 = 0$ gives an Einstein–Maxwell–dilaton-type system, with the Kaluza–Klein value corresponding to $\alpha_0 = \sqrt{3}$ [113, 114]. The case $(\alpha_0, \alpha_1) = (0,0)$ removes the exponential dilaton dependence from the Maxwell and axion kinetic terms and provides a useful reference point for comparison.

In this chapter, we focus primarily on the one-parameter diagonal sector

$$\alpha_0 = \alpha_1. \tag{4.2}$$

For the comparisons presented here, we consider the diagonal EMDA values

$$(\alpha_0, \alpha_1) = (0,0), \quad (1,1), \quad (10,10), \quad (100,100). \tag{4.3}$$

We also include selected EMD benchmark cases,

$$(\alpha_0, \alpha_1) = (1,0), \quad (\sqrt{3},0), \tag{4.4}$$

to compare the diagonal EMDA evolutions with representative dilatonic cases. These choices allow us to compare a weakly coupled reference case, the standard EMDA/Kerr–Sen coupling, two larger-coupling EMDA cases, and known EMD limits.

Analytic stationary black hole solutions are not known for all of the coupling choices considered here. Moreover, even when stationary single black hole solutions are known, our binary configurations are constructed using approximate initial data rather than exact binary black hole solutions. We therefore construct approximate charged binary black hole initial data and evolve the full EMDA system numerically. During the early part of the evolution, the black holes and surrounding fields are allowed to relax dynamically before inspiral and merger.

This approach allows us to compare the resulting gravitational waveforms against a common reference waveform and to identify how the electromagnetic, dilaton, and axion fields influence the inspiral and merger dynamics. The known analytic black hole metrics and associated fields relevant to this theory space are summarized in Appendix B.1, with the Kerr–Sen black hole metric presented in the chapter "Head-On Collisions and Stability Tests in EMDA Theory."

4.3 Numerical Methods

To study the strong field dynamics and gravitational radiation from inspiraling charged black hole binaries in Einstein–Maxwell–dilaton–axion theory, we evolve the full EMDA system as a Cauchy initial value problem. The spacetime metric is evolved using the BSSN formulation [20, 86, 87], together with puncture gauge conditions [88, 89]. The electromagnetic, dilaton, and axion fields are evolved using the matter evolution equations obtained from the $3+1$ decomposition of the EMDA equations of motion. The full evolution system is summarized in earlier chapters. We use the same numerical code used in earlier chapters. In contrast to the head-on collisions considered there, the simulations presented here describe binary black holes with nonzero orbital angular momentum. We use the same assumptions and techniques to set the inititial data that we used in our chapter on inspiraling black holes. The black holes are assigned initial linear momenta chosen to approximate quasicircular inspiral. This allows us to follow the late inspiral, merger, and ringdown and to extract gravitational, electromagnetic, dilaton, and axion radiation from the resulting spacetime.

Because the initial data are approximate, the electromagnetic constraints do not vanish identically at the initial time. To control these constraint violations, we evolve the electromagnetic fields with hyperbolic divergence cleaning. In this approach, auxiliary damping fields propagate violations of the electric and magnetic constraints away from the strong field region and damp them during the evolution [134]. This prevents electromagnetic constraint violations from accumulating during the relaxation, inspiral, and merger. The expected damping of the constraint equations is shown in Figure 4.5.

4.3.1 Initial data

Constructing initial data for binary black holes in EMDA theory is challenging because exact binary black hole solutions are not known. In addition, for general choices of the coupling parameters, the corresponding stationary single black hole solutions are not available in closed analytic form. We therefore use an approximate initial data prescription guided by the known Kerr–Newman and Kerr–Sen solutions [76, 77].

As before, our starting point is a charged binary black hole configuration motivated by Kerr–Newman and constructed using approximate charged puncture data [95, 101, 102]. We supplement these data with approximate dilaton and axion fields motivated by the Kerr–Sen solution. Since we are interested in varying the coupling parameters, we rescale the scalar and pseudoscalar fields according to the chosen coupling. In particular, for the diagonal choice $\alpha_0 = \alpha_1$, we take the initial dilaton and fields to scale with α_0 and α_1, so that larger coupling values correspond to larger initial scalar field amplitudes. For the dilaton, we therefore set

$$\phi = \alpha_0 \left[\phi_1(r_1, \theta_1) + \phi_2(r_2, \theta_2) \right] . \tag{4.5}$$

$$\kappa = \alpha_1 \left[\kappa_1(r_1, \theta_1) + \kappa_2(r_2, \theta_2) \right]. \tag{4.6}$$

This prescription does not produce an exact solution of the EMDA constraint equations. Instead, it should be viewed as an approximate data construction designed to place the system near the expected charged black hole configuration for each coupling choice. After the evolution begins, the black holes and surrounding fields are allowed to relax dynamically before the late inspiral and merger.

For each coupling choice, we consider binaries with charge to mass ratio magnitude $\lambda = Q/M = 0.01$ and a small spin $\chi = 0.2$. We examine three representative charge configurations: ++, +−, and +0. In the ++ case, both black holes carry charge with the same sign; in the +− case, the black holes carry equal and opposite charges; and in the +0 case, only one black hole is charged. These configurations allow us to compare systems with different net charges and electromagnetic field structures while keeping the individual charge scale fixed.

We then use the subsequent evolution to study how changing the coupling strength affects the electromagnetic, dilaton, and axion fields, the binary dynamics, and the emitted gravitational radiation.

4.4 Results

4.4.1 Waveform comparison

Gravitational waves We first compare the gravitational waveforms produced by the different coupling choices. For this comparison, we use the Einstein–Maxwell case, $(\alpha_0, \alpha_1) = (0, 0)$, as the reference waveform and compare the dominant $(\ell, m) = (2, 2)$ mode against the corresponding modes from the EMD and EMDA evolutions.

For the charge considered here, the gravitational waveforms remain close to the Einstein–Maxwell reference waveform across all of the coupling choices. In most cases, the differences in the dominant strain are too small to interpret as clearly resolved coupling-dependent effects. Because the simulations use approximate initial data, the early relaxation of the black holes and surrounding fields can introduce waveform differences that are unrelated to the intended change in the coupling parameters. These effects, together with finite-resolution error, waveform-extraction error, and the short duration of the evolutions, likely dominate the very small differences observed among most of the runs.

The main robust conclusion is therefore qualitative. For the smaller coupling EMD and EMDA cases, the dominant gravitational waveform is effectively indistinguishable from the Einstein–Maxwell reference waveform at the accuracy of the present simulations. The strongest-coupling case, $(\alpha_0, \alpha_1) = (100, 100)$, produces the largest departure from the reference waveform. Even in this case, however, the dominant strain remains close to the Einstein–Maxwell waveform.

These results suggest that, for the relatively small charge considered here, coupling dependent effects in the dominant gravitational wave mode are subtle. Larger charges, longer inspirals, improved initial data, and explicit convergence tests may be required to separate physical coupling dependent effects from relaxation and numerical errors. The present comparisons should therefore be interpreted as an exploratory test of the EMDA coupling dependence rather than as precision estimates of waveform distinguishability.

Figure 4.1 shows a representative comparison between the Einstein–Maxwell reference waveform and the strongest coupling EMDA case. The two waveforms are peak-aligned, there is almost no difference in the wavefrom.

Electromagnetic radiation We also compare the electromagnetic radiative modes and Φ_2, as introduced in earlier chapters. Rather than computing a mismatch for these modes, we perform a direct comparison of the electromagnetic waveforms. The results show that the waveform differences

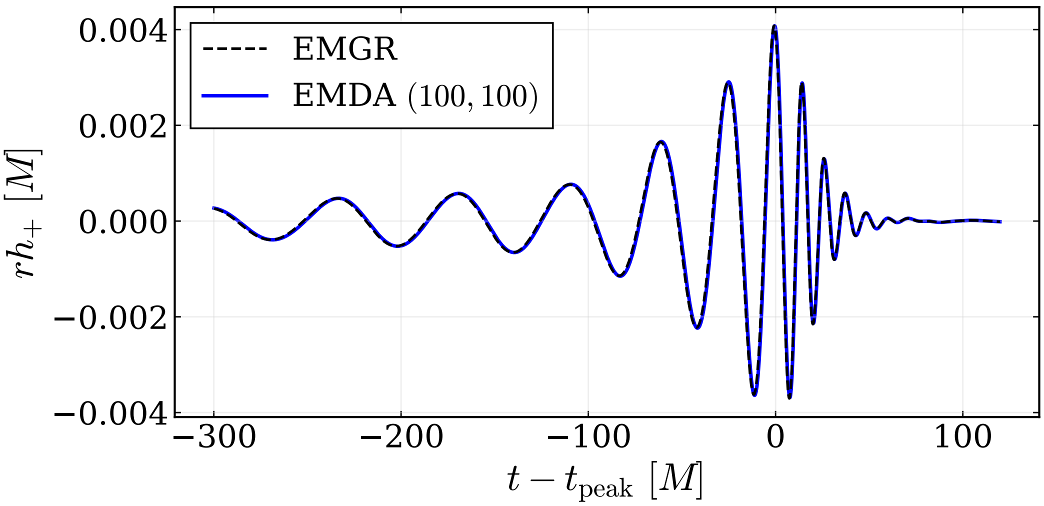
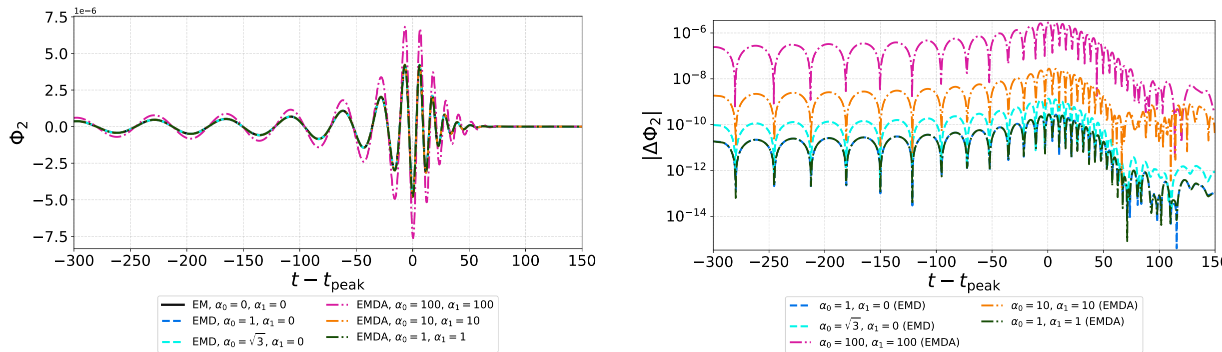


Figure 4.1 Peak-aligned comparison of the dominant gravitational-wave strain between the Einstein–Maxwell reference waveform, $(\alpha_0, \alpha_1) = (0, 0)$, and the strongest-coupling EMDA case, $(\alpha_0, \alpha_1) = (100, 100)$. The waveforms agree extremely well, with a mismatch on the order of 10^{-5}.

increase as the coupling strengths are increased. The electromagnetic signal retains the same basic structure across the coupling choices, but its amplitude changes. We display the ++ charge configuration here, although the other charge configurations show broadly similar behavior.

Dilaton radiation



(a) Real part of Φ_2^{22}.

(b) Absolute difference relative to the Einstein–Maxwell case.

Figure 4.2 Comparison of the $(\ell, m) = (2,2)$ mode of the electromagnetic scalar Φ_2, extracted at $R_{\rm ex} = 100M$. The left panel shows the peak-aligned real part of Φ_2^{22} across the coupling choices considered, while the right panel shows the absolute difference relative to the Einstein–Maxwell reference case, $(\alpha_0, \alpha_1) = (0,0)$, on a logarithmic scale. The waveform differences increase as the coupling is increased, although the change is primarily in the amplitude.

As the coupling strength increases, the amplitude of the emitted dilaton radiation also increases.

4.4.2 Constraints

Because the initial data are approximate, the constraint violations provide an important check on the quality of the evolutions. We monitor the Hamiltonian constraint, together with the electric and magnetic divergence constraints, throughout the simulations. These quantities are shown in Figs. 4.4–4.6. Across the set of coupling choices considered here, the constraints remain controlled during the initial relaxation, inspiral, and merger. The largest features occur during the early relaxation phase and near merger, where the fields vary most rapidly.

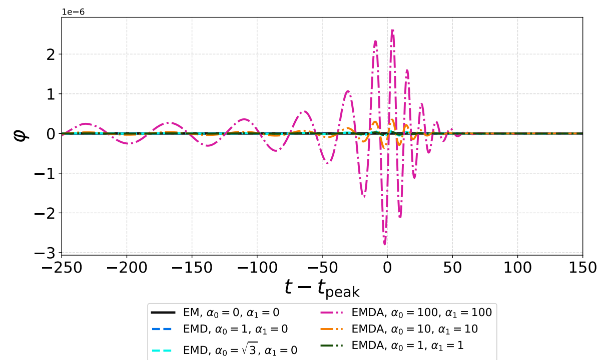


Figure 4.3 Peak-aligned comparison of the real part of the $(\ell, m) = (2,2)$ mode of the dilaton radiation, extracted at $R_{\text{ex}} = 100M$, for the coupling choices considered.

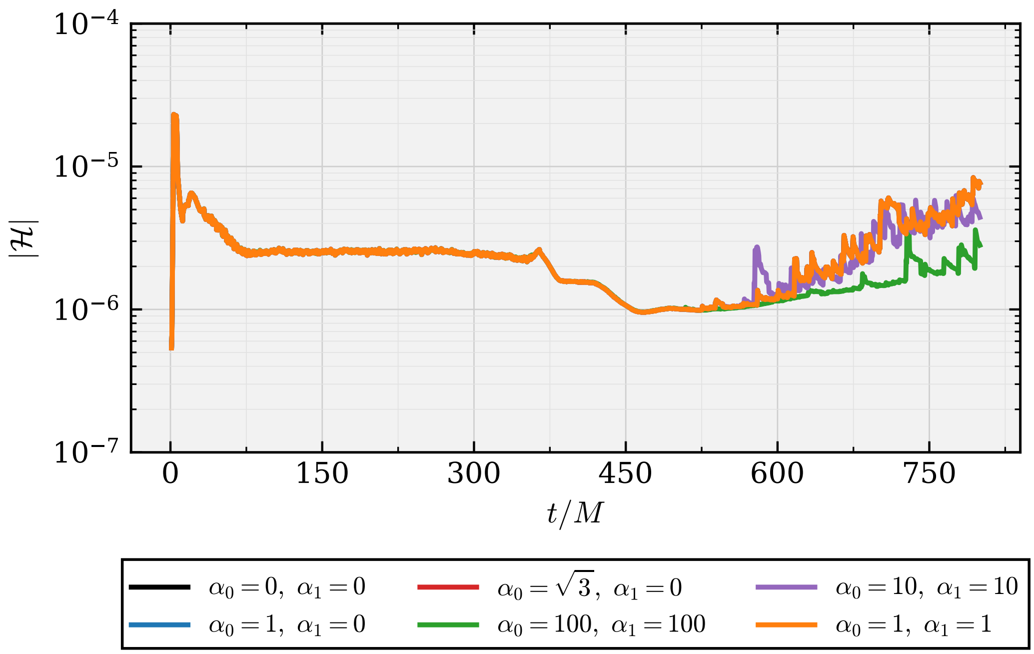


Figure 4.4 Hamiltonian constraint violation for the different coupling choices. The curves show the evolution of the Hamiltonian constraint norm for the Einstein–Maxwell reference case and the corresponding EMD and EMDA evolutions. There is an initial spike as the simulations begin, but the constraint quickly settles to a lower value. The Hamiltonian constraint does not show runaway growth and remains low throughout the evolution.

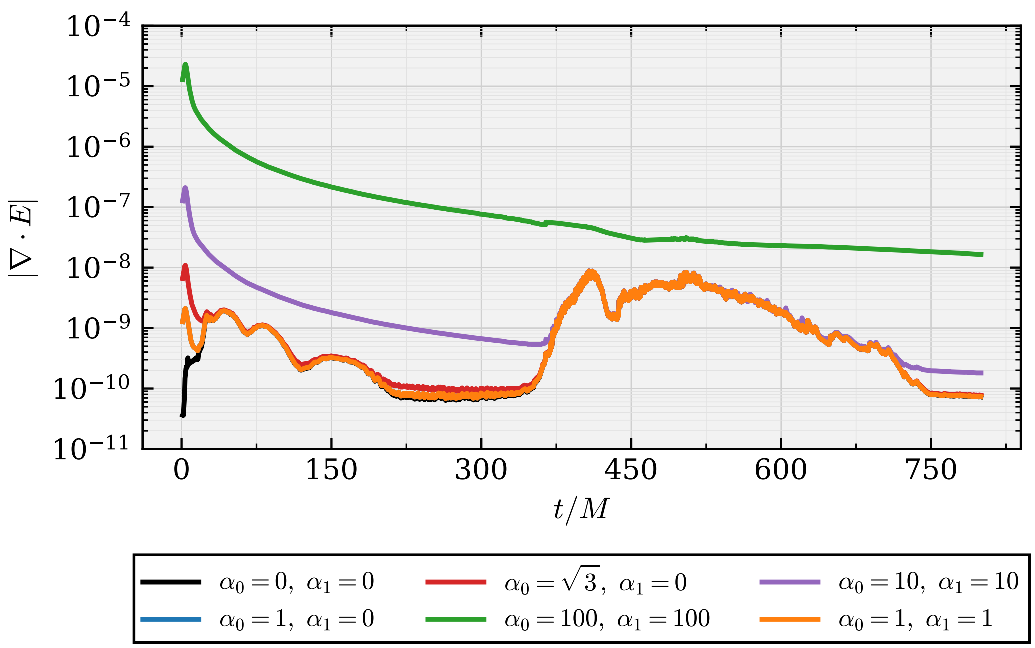


Figure 4.5 Electric divergence constraint violation for the different coupling choices. The curves show the evolution of the electric constraint norm across the Einstein–Maxwell, EMD, and EMDA evolutions. The constraint is initially largest for the strongest coupling, but hyperbolic cleaning reduces it over time. Overall, these constraints remain well behaved throughout the evolution.

The electromagnetic constraints remain bounded throughout the evolutions, indicating that the divergence-cleaning scheme prevents the electric and magnetic constraint violations from growing without bound. The Hamiltonian constraint shows the expected transient behavior during the initial relaxation and near merger, but it does not exhibit runaway growth. These results indicate that the simulations remain sufficiently well controlled for the qualitative waveform comparisons presented above.

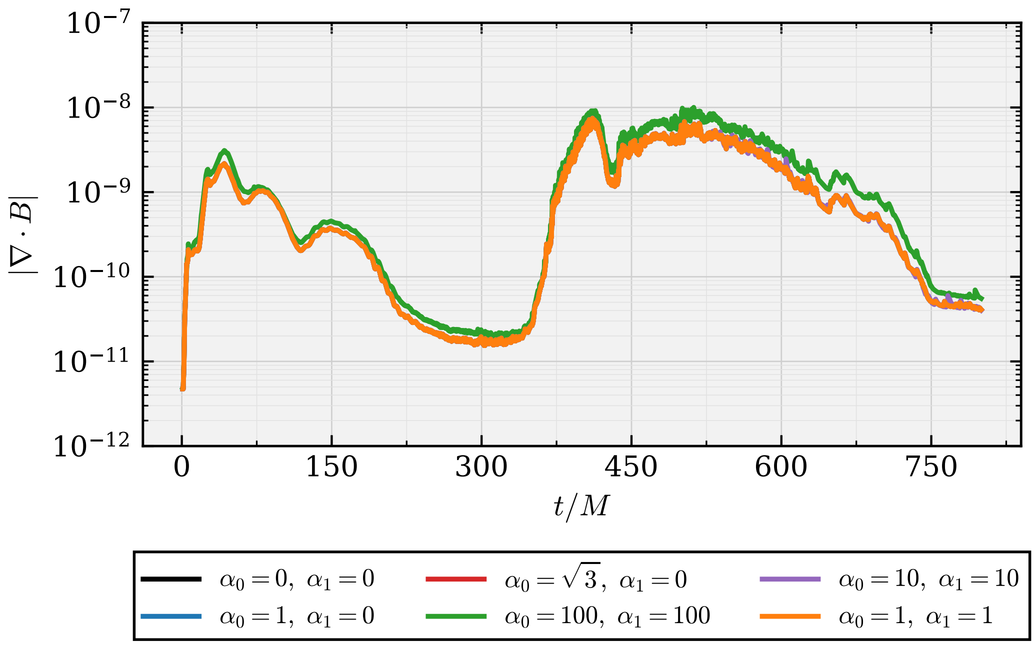


Figure 4.6 Magnetic divergence constraint violation for the different coupling choices. The curves show the evolution of the magnetic constraint norm across the Einstein–Maxwell, EMD, and EMDA evolutions. These values remain low throughout the simulation, providing confidence that the magnetic divergence constraint remains well behaved.

4.5 Conclusion

In this chapter, we studied binary black hole mergers in a broader region of Einstein–Maxwell–dilaton–axion theory by varying the coupling parameters that control the interactions between the electromagnetic, dilaton, and axion fields. We focused primarily on the diagonal sector, $\alpha_0 = \alpha_1$, and compared several representative coupling choices against the Einstein–Maxwell reference case. Because exact binary black hole initial data are not available for these theories, we used approximate charged binary black hole initial data and allowed the systems to relax dynamically before merger.

The evolutions are stable for the coupling choices considered here, and the electromagnetic and gravitational constraints remain controlled throughout the simulations. This provides evidence that the numerical framework can be used to explore binary black hole dynamics beyond the analytically known special cases of EMDA theory. In particular, these simulations provide a first step toward studying how nonlinear merger dynamics depend on the theory's coupling parameters.

For the small charge considered in this study, we found that the dominant gravitational wave signal is only weakly affected by the choice of coupling. The smaller coupling EMD and EMDA cases remained effectively indistinguishable from the Einstein–Maxwell reference waveform at the accuracy of the present simulations. The strongest coupling case, $(\alpha_0, \alpha_1) = (100, 100)$, produced the largest departure from the reference waveform, but even this difference remained small in the dominant strain.

The electromagnetic radiation shows a clearer dependence on the coupling parameters. Rather than computing electromagnetic mismatches, we directly compared the radiative modes Φ_2 against the Einstein–Maxwell reference case. These comparisons show that the electromagnetic waveform differences generally increase as the coupling strengths are increased. The overall structure of the electromagnetic signal remains similar across the coupling choices, but the amplitude changes. This behavior is shown for the ++ charge configuration, with the other charge configurations exhibiting broadly similar patterns.

Taken together, these results suggest that coupling dependent effects are subtle in the gravitational radiation at small charge, but are more visible in the electromagnetic sector. Because the simulations use approximate initial data, very small waveform differences may be affected by relaxation effects, finite resolution error, and waveform extraction uncertainty rather than only by physical coupling dependence. Larger charges, longer inspirals, improved initial data, and explicit convergence tests may be required to clearly separate physical effects from numerical and initial data uncertainties.

Future work could potentially look at improving the initial data by adding axion contributions and improving the initial electric field used in the TPID solver. We could also perform higher charge or higher spin runs to test a wider range of Kerr–Sen configurations. Another study could examine different parts of the EMDA parameter space, including cases where the two couplings are not equal, $\alpha_0 \neq \alpha_1$.

4.5.1 Mismatch summary

Table 4.1 Mismatch between the Einstein–Maxwell reference waveform, $(\alpha_0, \alpha_1) = (0, 0)$, and the waveforms obtained for different coupling choices. The comparisons use the dominant $\ell = m = 2$ gravitational strain mode. The mismatches are small, so most of these waveforms should be considered effectively indistinguishable within the accuracy of the present simulations. In this regime, finite resolution, approximate initial data, and other numerical effects may contribute more to the measured mismatch than the scalar and pseudoscalar fields themselves.

Theory	$q = 0.05$, $+0$	$q = 0.05$, $+-$	$q = 0.05$, $++$
EMD $(\alpha_0, \alpha_1) = (1, 0)$	3.85×10^{-8}	2.70×10^{-7}	4.43×10^{-14}
EMD $(\alpha_0, \alpha_1) = (\sqrt{3}, 0)$	3.53×10^{-12}	9.00×10^{-8}	2.89×10^{-12}
EMDA $(\alpha_0, \alpha_1) = (1, 1)$	2.59×10^{-8}	6.08×10^{-8}	4.92×10^{-14}
EMDA $(\alpha_0, \alpha_1) = (10, 10)$	7.06×10^{-8}	1.22×10^{-6}	6.14×10^{-10}
EMDA $(\alpha_0, \alpha_1) = (100, 100)$	9.62×10^{-6}	4.42×10^{-5}	3.91×10^{-6}

Chapter 5

Compact Finite Difference Operators for Adaptive Simulations

5.1 Introduction

The last decade has seen dramatic growth in the field of gravitational wave astronomy [9, 24, 25, 65, 135, 136]. LVK has discovered the gravitational wave signals from nearly four hundred compact mergers, the majority being binary black hole mergers [136]. Additional detections will continue to be made and new gravitational wavebands will open up with planned and proposed gravitational wave detectors [26–28]. With these, there will come an increasing need for more comprehensive and ever more accurate simulated gravitational waveforms [29, 137, 138].

While the exact level of precision required is not entirely clear, there is broad agreement that significant improvements in simulation fidelity will be required [29, 137, 138]. For example, future gravitational wave detectors such as LISA, Einstein Telescope, and Cosmic Explorer will have higher signal-to-noise ratios than current detectors [26–28]. This will increase accuracy requirements for simulated waveforms; some estimates suggest by as much as two orders of magnitude [29, 138–140]. While there are some ideas [19, 22, 30, 31, 90, 141–143] of how to improve the efficiency and accuracy of current codes, the size of the hoped for increase is daunting.

Current numerical relativity codes run on CPUs have probably wrung about as much efficiency and time savings as possible from standard finite difference and finite volume techniques. Adaptive methods have become fairly state-of-the-art and a number of codes have adopted octree partitioning to improve the distributed aspects of binary inspiral calculations [19, 22, 144–146]. It is natural to ask where accuracy improvements for waveform simulation might come from. Two possible areas where improvements in accuracy may be obtainable include moving to GPUs to speed up calculations and using improved, higher order accuracy algorithms [31, 142, 143, 147, 148]. A number of groups and codes have begun to develop for GPUs as have we [22]. But in this chapter we will focus on the latter of these, namely higher accuracy finite differencing schemes in the form of compact finite differences (CFDs) [31, 143, 149–153].

To the best of our knowledge, the first use of CFD operators in numerical relativity was by Daszuta [143]. His implementation focused on costs associated with communication across many processors and, partly in consequence of that, was not entirely satisfactory as it did not demonstrate the hoped for performance gains. Nonetheless, CFDs have been used for over thirty years in the engineering literature [31, 149, 153–155] with impressive accuracy improvements in the simulation of both linear and nonlinear systems of equations. So one might think that use of these in numerical relativity may be one route to improved accuracy in binary simulations and the gravitational radiation they source.

Our aim in the current chapter is to develop high order compact finite difference schemes together with boundary closures and test them on linear and nonlinear time evolution systems in three dimensions. Indeed, we are able to demonstrate the utility of these schemes and that, in the test cases considered here, an increase in the effective order of accuracy is possible.

The chapter is structured as follows. Section 5.2 provides background on compact finite difference schemes and discusses their advantages relative to explicit finite difference methods, together with the challenges associated with boundary closures and stability. Section 5.3 describes the construction and optimization of the compact finite difference operators, including the treatment of interior and boundary stencils for first and second derivatives. Section 5.4 tests these operators on a set of linear and nonlinear systems, including Burgers' equation, Maxwell's equations, the nonlinear sigma model, and the Einstein equations using the BSSN formalism. Finally, Sec. 5.5 summarizes our results and discusses future applications of compact finite differences in numerical relativity.

5.2 Background

Finite difference (FD) schemes are widely used in numerical relativity. This is partly due to their simiplicity and relative ease of implementation. Unfortunately, higher order FD schemes require large stencils, particularly when shifted stencils near boundaries may be required. In a given implementation, this can lead to a large spread in memory access. In general, this will complicate numerical treatments near boundaries and increase communication costs for ghost regions at both processor and refinement boundaries [22, 143, 151, 152]. As such, seeking higher accuracy gravitational waveforms using standard FDs would appear to incur increasing compute costs and thereby compromise those hoped for gains.

Compact finite difference (CFD) schemes are a generalization of Padé approximates [31] that couple derivatives at neighboring points at different spatial scales. In general, for a fixed stencil size, these will be higher-order than standard FD schemes [31, 143, 149]. When compared to explicit FDs, compact schemes of similar order require smaller stencils and can be significantly more accurate–often by a factor of 10–100× [31, 143, 149, 156]. A price to pay for this increase in accuracy is that CFD schemes are implicit and therefore require matrix inversion [31, 151, 152]. However, our use of DENDRO-GR permits us to partially sidestep this issue [19, 22]. In particular, DENDRO-GR's leaf nodes are of fixed size, and using vectorized libraries for matrix inversion gives an efficient calculation of CFDs that, we have found, are as fast as hand-coded C++ code for explicit FDs.

We here describe the development and evaluation of new families of CFD operators appropriate for use within distributed numerical relativity codes using adaptive meshes. This has required the careful consideration of two crucial issues, namely appropriate boundary closures for these CFD operators and the question of stability of the operators for both linear and nonlinear systems of equations.

For finite difference (or finite volume) based approaches, the stencil that defines derivative approximations will necessarily be different between the “interior" of the mesh and near the boundaries of the mesh. These boundaries can include physical boundaries as well as, in the case of adaptively refined meshes, refinement boundaries. Typically, shifted FD stencils together with some form of interpolation are used in the vicinity of either type of boundaries. However, this can introduce larger errors than those expected from uniform, centered approximations. Such remains true for CFDs. Following an ingenious approach of Wu and Kim [149, 153], we have developed CFD operators for both first and second derivatives with boundary closures which use a high order interpolating polynomial to permit the continued use of centered-like stencils at and near boundaries. We have generalized this method to include a set of free parameters in the specification of the CFD operators. Together with the use of spectral error quantities associated with both interior and boundary operators, we minimize the error with respect to the free parameters thereby allowing the construction of optimized families of CFD operators for a fixed order of accuracy.

The stability of the resulting families of CFD operators is then determined in two complementary ways. A version of linear stability is defined with respect to a simple advection (for first derivatives) or wave equation (for second derivatives) by considering the eigenvalue spectrum of the operator. Given our definitions, the linear stability of a CFD derivative is established if all the eigenvalues are restricted to a portion of the complex frequency plane (left half plane for first derivatives and lower half plane for second derivatives).

The operators are then further tested in simpler model systems of equations, such as Maxwell’s equations. Normed error quantities are monitored and checked that the CFD operators do, indeed, produce stable evolutions.

As a result, we have found families of stable, fourth and sixth order accurate, boundary closed operators. They demonstrate improved accuracy over their explicit counterparts, as shown in Fig. 5.5. This figure shows the errors in the x components of the magnetic and electric fields for the evolution of a magnetic dipole wave using a first order in time, first order in space formulation of the Maxwell equations. Several fourth order compact and explicit finite difference operators are compared at different resolutions. The key results are that the nominally fourth order CFD operators produce error profiles comparable to those of the sixth order explicit scheme and, more significantly, perform comparably to the fourth order explicit scheme when run at half the resolution.

Comparable improvements are obtained for the sixth order first derivative operators, as shown in Fig. 5.6. Although the sixth order CFD operators do not achieve the full factor of two improvement in resolution observed for the fourth order operators, they still reduce the error by approximately an order of magnitude relative to the sixth order explicit scheme. Similar behavior is also observed in the nonlinear sigma model and Teukolsky wave tests presented below.

Early indications of the accuracy improvements possible suggest that replacing an explicit FD with a CFD of the same nominal accuracy results in an evolution with comparable error at half the resolution (see Figure 5.5). For a time dependent problem across a 3D grid, this could result in a potential speedup or efficiency boost of up to 16×. Indeed, in the aerospace community, there is a rule of thumb that the use of CFDs can result in a factor of 10× improvement even for complex, nonlinear fluid flows. In the event that we don't realize exactly these same performance gains in the nonlinear Einstein equations, we still think that we can make a conservative estimate that the use of CFDs should give us at least a factor 2× improvement. Coupling this with speedups of > 2× that we have already seen from an earlier version of our GPU code [22], should allow a class of such BBH evolutions to be evolved that have heretofore been inaccessible (see Figure 1.2).

5.3 Constructing CFD Operators

5.3.1 First derivative operators

We begin by reviewing the construction of high order, compact finite difference (CFD) schemes. The basic idea behind CFDs is a straightfoward generalization of a standard, explicit, centered finite difference stencil, such as that for the fourth order approximation of the first derivative

$$f_i' = \frac{1}{12h}\left[f_{i-2} - 8f_{i-1} + 8f_{i+1} - f_{i+2}\right] + O(h^4), \tag{5.1}$$

to include values of the derivative at nearby grid points. For example, one such generalization might be

$$\alpha_2 f_{i-2}' + \alpha_1 f_{i-1}' + f_i' + \alpha_1 f_{i+1}' + \alpha_2 f_{i+2}' = \frac{1}{2h}\left[a_1(f_{i+1} - f_{i-1}) + a_2(f_{i+2} - f_{i-2})\right] + O(h^{2n}) \tag{5.2}$$

where the constant coefficients, α_1, α_2, a_1 and a_2 can be found be demanding a particular order of accuracy for the approximation ($2n$ in this example). In this case, by matching Taylor coefficients, up to an eighth order scheme ($n = 4$) may be defined with four linear conditions on the four coefficients.

Similar such interior schemes are readily obtained. Of course, one one need not solve for all of the coefficients thereby allowing some of them to be left as free parameters which are then allowed to be subject to conditions other than the order of accuracy of the scheme.

One might consider potential complications to include the treatment of boundaries and how this might be extended beyond one dimension. We will consider boundaries below. The extension to higher dimensions is straightforward with the operators in higher dimensions being the tensor product of the one dimensional versions in each direction.

As our intent is to approximate a first derivative, $f'(x)$. we assume a uniform grid in 1D with $N+1$ points and spacing h. The earlier example can now be generalized to the following matrix system:

$$P_{ij} f_j' = \frac{1}{h} Q_{ij} f_j \tag{5.3}$$

where f_j is the vector of function values, f'_j the vector of derivative values, and the matrices P_{ij} and Q_{ij} define the nature of the coupling between neighboring derivative and function values, respectively. Of course, for an explicit finite difference scheme P would be the identity. Anything else will constitute some form of what we will refer to as a compact finite difference scheme. As in the example above, the interior points are straightforwardly obtained by matching the coefficients of a Taylor expansion of both sides in powers of the grid spacing h. For the interior points, we always use centered stencils such as that in Eq.(5.2) Note that this particular example will result in the interior portions of P and Q both being pentadiagonal. As mentioned, if all four coefficients are solved for, the accuracy of the interior scheme would be eighth order. But if we were to extend this to having septadiagonal P and Q matrices for the interior, we could go to twelfth order accuracy, in principle. We will find it useful, however, to choose a lower order of accuracy, solve for a subset of the coefficients by Taylor matching and to allow constraints other than accuracy to fix the remaining or “free" parameter values in our CFD schemes.

Of course, in our P and Q matrices are boundary and near boundary rows or stencils that we must incorporate. Centered stencils, of course, will no longer be usable near non-periodic boundaries. For explicit schemes, shifted stencils are often used. Applied here and akin to our above interior example, we might have

$$\beta_{-1} f'_{i-1} + f'_i + \beta_1 f'_{i+1} + \beta_2 f'_{i+2} + \cdots = \frac{1}{h}\left(b_{-1} f_{i-1} + b_0 f_i + b_1 f_{i+1} + \cdots\right) + O(h^n) \tag{5.4}$$

for a right shifted stencil. Note that a choice of an overall normalization is made with the coefficient of f'_i being 1. Were we to choose to maintain the pentadiagonality of P, we would have three unknown coefficients. For an accuracy of order n, we would then need $n-3$ coefficients on the right hand side. Depending on other constraints, it is unlikely that Q could likewise remain a pentadiagonal matrix while also having the near boundary derivative CFD scheme retain an order of accuracy comparable to that of the interior. In general, we will aim to have boundary schemes that have an order of accuracy similar to that of the interior while preserving the tridiagonality or pentadiagonality of P. As a result, the Q matrices will in general not be tri- or pentadiagonal. This shouldn't matter too much as computationally it is P that we will need to invert and having a simple structure will make the calculation of P^{-1} faster. The price to pay is that the Q matrix will not share as simple a structure.

5.3.2 Optimization

Interior Scheme

One advantage of compact finite difference schemes are that they have better spectral characteristics than standard finite differences. While not as good as spectral methods themselves, we can optimize CFDs for improved behahvior at large wavenumbers. We will choose to sacrifice some accuracy in an attempt to improve and optimize for these sorts of properties. To this end, we will construct the spectral or transfer function associated with the CFD scheme, and optimize on an error measure built from the spectral function. [31, 149]

Consider a general interior (central) CFD scheme such as

$$\bar{f}_i' + \sum_{j=1}^{m_L} \alpha_j \bar{f}_{i+j}' = \frac{1}{h} \sum_{j=1}^{m_R} a_j (f_{i+j} - f_{i-j}) \tag{5.5}$$

where m_L and m_R indicate the number of CFD parameters to be solved for on the left and right hand sides, or that are present in the P and Q matrices, respectively. Note that $\bar{f}$ is assumed to be the original f minus a truncated remainder of order $O(h^n)$ that allows us to use the equality above. On Fourier transforming this, rearranging and setting $\kappa = kh$ to be the rescaled wavenumber, we get the expression

$$\bar{\kappa}(\kappa) \equiv \kappa \frac{\tilde{\bar{f}}(k)}{\tilde{f}(k)} = \frac{\sum_{j=1}^{m_R} a_j \sin(j\kappa)}{1 + 2\sum_{j=1}^{m_L} \alpha_j \cos(j\kappa)} \tag{5.6}$$

where tildes designate the Fourier transform. This is the spectral or transfer function and measures the dispersion present in this (interior) CFD scheme. For small κ, there is little dispersion. As

$\kappa \to \pi$, clearly $\bar{\kappa} \to 0$ indicating maximum dispersion for the highest frequency modes. Figure 5.1 plots this for several possible interior schemes.

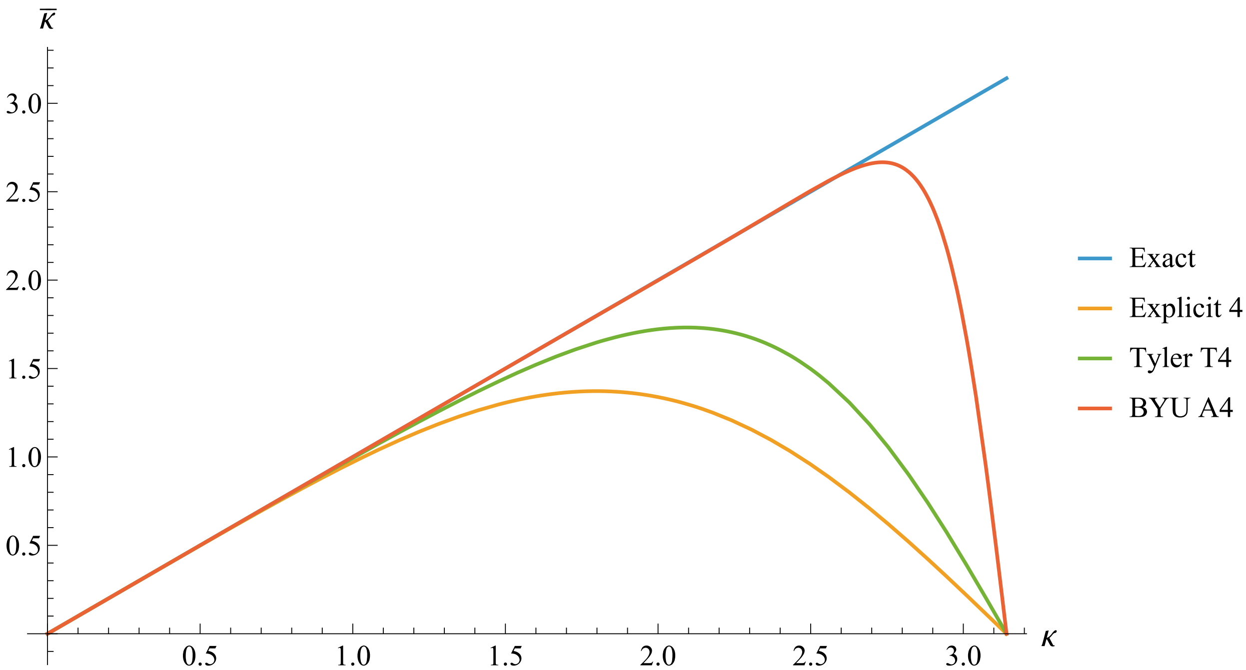


Figure 5.1 Comparison of spectral functions for several 4th order schemes. In particular, we note the explicit 4th order finite difference, a tridiagonal 4th order compact finite difference [1], and a spectrally-optimized pentadiagonal 4th order compact finite difference derived in this work.

We construct the following error quantity from the spectral function, namely

$$E(\alpha_i, a_i) = \int_0^{r\pi} \left[\bar{\kappa} - \kappa\right]^2 D(\kappa)^2 \, d\kappa \tag{5.7}$$

where $D(\kappa)$ is simply the denominator in the spectral function, Eq. (5.6). In general, an arbitrary weighting function can be used, but this form is chosen so that the integral can be computed analytically [157, 158]. The parameter $r \in (0, 1]$ in the upper bound of the integral is referred to as the "fractional cutoff wavenumber," and represents the maximum modified wavenumber κ over which the spectral error is optimized. It has been suggested [159] to have a lower cutoff wavenumber around 0.5 or 0.75 to avoid overly optimizing on higher wavenumbers, while others [149] have used higher cutoff wavenumbers such as 0.839. In this chapter, we develop several families of CFD operators for several values of the cutoff wavenumber from 0.6 to 0.85.

We minimize the error E by substituting the solutions of the Taylor constraint equations up to a given order of accuracy for our "fixed parameters" and by imposing constraints of the form

$$\frac{\partial E}{\partial a_i} = 0, \tag{5.8}$$

for our "free parameters." As the error quantity is quadratic with respect to its arguments, the resulting constraint equations form a linear system with respect to these free parameters, and hence yield a unique solution for a spectrally-optimized CFD scheme.

Boundary Scheme

If we wish to maintain the banded-diagonal form of these CFD operators on the boundary, we must develop a boundary closure using shifted stencils. However, it has been shown that centered stencils provide greater benefits in terms of accuracy and stability. Following the approach of Wu et al. [153], we use a polynomial to extrapolate our function to grid points which lie beyond the boundary of our numerical domain.

If we have interior coefficients α_1 through α_m on the LHS and a_1 through a_k on the RHS, then the order of the extrapolation polynomial must be at least $\ell = 3k+m+1$. We increase the order of this polynomial by one in order to introduce an extra free parameter in order to optimize the spectral behavior at and near the boundary. Following [153], we construct the extrapolation polynomial,

$$P(\hat{x}) = \sum_{s=0}^{\ell} p_s \hat{x}^s, \tag{5.9}$$

where $\hat{x} \equiv (x_j - x_0)/h$ and we use the information from the function and its derivatives:

$$\begin{cases} P(j) = f_j, \quad j = 0, 1, \ldots, 2k \\ \dfrac{dP}{dx}(j) = f_j', \quad j = 0, 1, \ldots, k+m \\ \dfrac{d^n P}{dx^n}(\hat{y}) = 0. \end{cases}$$

The parameter $\hat{y}$ is a value which will be used for optimizing the boundary closures. The quantity n is an arbitrary integer which should be at least greater than or equal to the order of accuracy of the scheme and at most less than or equal to the order of the extrapolation polynomial. We have found that a lower value of n leads to a larger family of minima in our boundary error quantity and thus a larger family of error-optimized boundary closures.

Once we have solved the system above to find the coefficients of the extrapolation polynomial, we construct a centered stencil for each of the boundary nodes, substituting in the extrapolation polynomial at each of the grid points which lie beyond the domain. Finally, we can rearrange the resulting equations into the shifted stencils which can be implemented, and the shifted stencil boundary coefficients can be read off in terms of the free parameter $\hat{y}$ only.

As with the approach to optimizing the interior scheme, we Fourier transform the shifted boundary stencils to define a spectral function. For the mth boundary node, we find:

$$\bar{\kappa}_m(\kappa) = \frac{1}{A_m(\kappa)^2 + B_m(\kappa)^2}\left[A_m(\kappa)D_m(\kappa) - B_m(\kappa)C_m(\kappa) - i(A_m(\kappa)C_m(\kappa) + B_m(\kappa)D_m(\kappa))\right], \tag{5.10}$$

where:

$$A_m(\kappa) = \sum_{j=0}^{3} \gamma_{mj} \cos(j\kappa), \quad B_m(\kappa) = \sum_{j=0}^{3} \gamma_{mj} \sin(j\kappa), \tag{5.11}$$

$$C_m(\kappa) = \sum_{j=0}^{6} a_{mj} \cos(j\kappa), \quad D_m(\kappa) = \sum_{j=0}^{6} a_{mj} \sin(j\kappa). \tag{5.12}$$

Note that because the boundary stencils are shifted, their spectral functions have real and imaginary components, corresponding to both dispersive and dissipative behavior in the numerical scheme.

At this point, many different error quantities could be constructed in order to optimize specific features of the boundary closure. For example, an error quantity could be chosen to optimize only on the imaginary part of the spectral function in order to minimize unwanted dissipative modes. In the present chapter, we use the error quantity

$$E_m = \int_0^{r\pi} \left[\left| \Re(N_m(\kappa)) + i\Im(N_m(\kappa)) \right| - \kappa D_m(\kappa) \right]^2 d\kappa \tag{5.13}$$

where $N_m(\kappa)$ and $D_m(\kappa)$ represent the numerator and denominator of the spectral function $\bar{\kappa}_m(\kappa)$, respectively; i.e., we have set:

$$\Re(N_m(\kappa)) = A(\kappa)D(\kappa) - B(\kappa)C(\kappa), \tag{5.14}$$

$$\Im(N_m(\kappa)) = -A(\kappa)C(\kappa) - B(\kappa)D(\kappa), \tag{5.15}$$

$$D_m(\kappa) = A(\kappa)^2 + B(\kappa)^2. \tag{5.16}$$

Note that each boundary node has a spectral function and thus error quantity which is independent from each other. As each of the boundary coefficients a_{mj} and γ_{mj} only depend on the free parameter $\hat{y}$, then each error quantity also only depends on $\hat{y}$. Since the error functions are decoupled, we opt to use a separate free parameter $\hat{y}_m$ for each boundary node. Clearly, even with a similar weighting function to the interior scheme, this error integral is not analytically integrable, so we must numerically integrate the error quantity over a grid of possible values for $\hat{y}_m$. Wu and Kim [153] suggests testing $\hat{y}_m \in (-6, 6)$, but we have found that the minima in the error are typically found for $\hat{y}_m \in (-1, 5)$.

Since each boundary error function $E_m(\hat{y}_m)$ has several minima, we obtain a family of spectrally-optimized boundary closures by taking all possible combinations of optimal $\hat{y}_m$ values.

Symbolic Matrices

This is specifically the formulation of the 1A family of matrices.

$$\mathbf{P}=\begin{pmatrix} 1 & \gamma_{01} & \gamma_{02} & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ \gamma_{10} & 1 & \gamma_{12} & \gamma_{13} & 0 & 0 & 0 & 0 & \cdots & 0 \\ \gamma_{20} & \gamma_{21} & 1 & \gamma_{23} & \gamma_{24} & 0 & 0 & 0 & \cdots & 0 \\ 0 & \beta & \alpha & 1 & \alpha & \beta & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \beta & \alpha & 1 & \alpha & \beta & 0 \\ 0 & \cdots & 0 & 0 & 0 & \gamma_{24} & \gamma_{23} & 1 & \gamma_{12} & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \gamma_{13} & \gamma_{12} & 1 & \gamma_{10} \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \gamma_{02} & \gamma_{01} & 1 \end{pmatrix}$$

$$\mathbf{Q}=\begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} & a_{04} & a_{05} & a_{06} & 0 & \cdots & 0 \\ a_{10} & a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & 0 & \cdots & 0 \\ a_{20} & a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} & 0 & \cdots & 0 \\ -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 \\ 0 & \cdots & 0 & -a_{26} & -a_{25} & -a_{24} & -a_{23} & -a_{22} & -a_{21} & -a_{20} \\ 0 & \cdots & 0 & -a_{16} & -a_{15} & -a_{14} & -a_{13} & -a_{12} & -a_{11} & -a_{10} \\ 0 & \cdots & 0 & -a_{06} & -a_{05} & -a_{04} & -a_{03} & -a_{02} & -a_{01} & -a_{00} \end{pmatrix}$$

5.4 Test cases

Having sketched the process of computing boundary-closed CFD operators for a given order of accuracy, we here present several tests of these operators as applied in specific systems of PDEs.

We begin with a 1D nonlinear system, namely Burgers' equation. We also consider Maxwell's equations in full 3D together with hyperbolic divergence cleaning. We return to nonlinear problems with a 3D nonlinear sigma model and conclude with the Einstein equations with the evolution of linearized Teukolsky gravitational waves.

In each case, the basic results are similar. Namely, the use of CFDs at some accuracy order perform better than a comparable explicit FD scheme, sometimes by more than an order of magnitude. Effectively, the CFDs are providing comparable error profiles at half the resolution of the FDs.

5.4.1 One-dimensional nonlinear Burgers' equation

Our first test uses the nonlinear one-dimensional viscous Burgers' Equation,

$$\frac{\partial f}{\partial t} = -f\frac{\partial f}{\partial x} + \frac{\partial^2 f}{\partial x^2}, \quad -\infty < x < \infty, \quad t \geq 0, \tag{5.17}$$

where the solution $f(x,t)$ depends only on x and t, and the viscosity coefficient is fixed at 1 for simplicity.

This admits a closed form solution against which we will compare the numerical solution. It is

$$f(x,t) = \frac{2}{1+e^{x-t}}. \tag{5.18}$$

The initial condition, of course, is thus

$$f(x,t=0) = \frac{2}{1+e^{x}}. \tag{5.19}$$

In solving this problem numerically, we approximate both first and second spatial derivatives using CFDs and explicit FDs separately. The time integration is done using a fourth order Runge-Kutta scheme as will be true for all subsequent tests as well.

To focus in this test problem on the interior parts of the CFDs, we take a domain of $-100 \leq x \leq 100$ and limit the simulation time to $0 \leq t < 50$. Because the solution approaches constant states far from the transition region near $x = t$, fixed Dirichlet boundary conditions provide an accurate approximation on the truncated domain. Appropriate boundary-closed operators continue to be used, but because we are limiting interactions with the boundaries, the results are more reflective of the behavior of the interior CFDs.

The continuous domain $[-100, 100]$ is divided into N uniform cells of width h. Using a grid defined at the cell boundaries, we will have $N+1$ grid points to represent the domain. Defining $f_i(t) = f(-100+ih, t)$ to be the ith point of the solution along the domain, with i ranging from 0 to N, we can write Eq. 5.17 in the semidiscrete form:

$$\frac{d\phi_i}{dt} = -\phi_i \mathbf{D}_1 \phi_i + \mathbf{D}_2 \phi_i, \tag{5.20}$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ denote the discrete first- and second-derivative operators, respectively.

A standard fourth-order Runge–Kutta (RK4) method is implemented to advance our solution in time. At each time step, including the intermediary stages in the integration, the Dirichlet boundary conditions are imposed on the first and last entries of the solution. This process is repeated for select resolutions of N, $2N$, and $4N$, and the L_2-norm of the error when compared with the analytic solution is calculated. Results are plotted in Figures 5.2 and 5.3.

Both figures show the absolute error of these evolutions using CFD and FD (first and second derivative) operators at three different resolutions. The low, medium and high resolutions have $N = 200, 400, 800$, respectively. In Figure 5.2, “E4" stands for an explicit fourth order operator. The “A4" and “C4" designations are members of two families of fourth order operators that were derived in previous sections. From the Figure, it is clear that for a given type of difference operator, there is convergence from low to high resolution. Indeed, the convergence is fourth order for each as it should be. But, in addition, the plot shows that CFD operators result in an overall error that is one to two orders of magnitude better than the corresponding explicit FDs. Even more significantly, it can be seen that the medium resolution CFDs have an improved error profile over the high resolution FD scheme. Effectively, the CFDs are providing improved accuracy at lower resolution.

A similar story holds from Figure 5.3 on comparing a standard sixth order explicit (E6) finite difference scheme for both first and second derivatives with an earlier derived sixth order CFD

(C6). Note that the C6 operator is an optimized 6th order compact scheme for the first and second derivatives. Likewise, the A4 and C4 operators are two of the optimized compacts schemes at 4th order for the first derivative and second derivatives. In both cases, the compact schemes at low resolutions perform almost as well or better than the explicit schemes at a resolution with twice as many points. This shows a definite improvement in accuracy with the compact schemes.

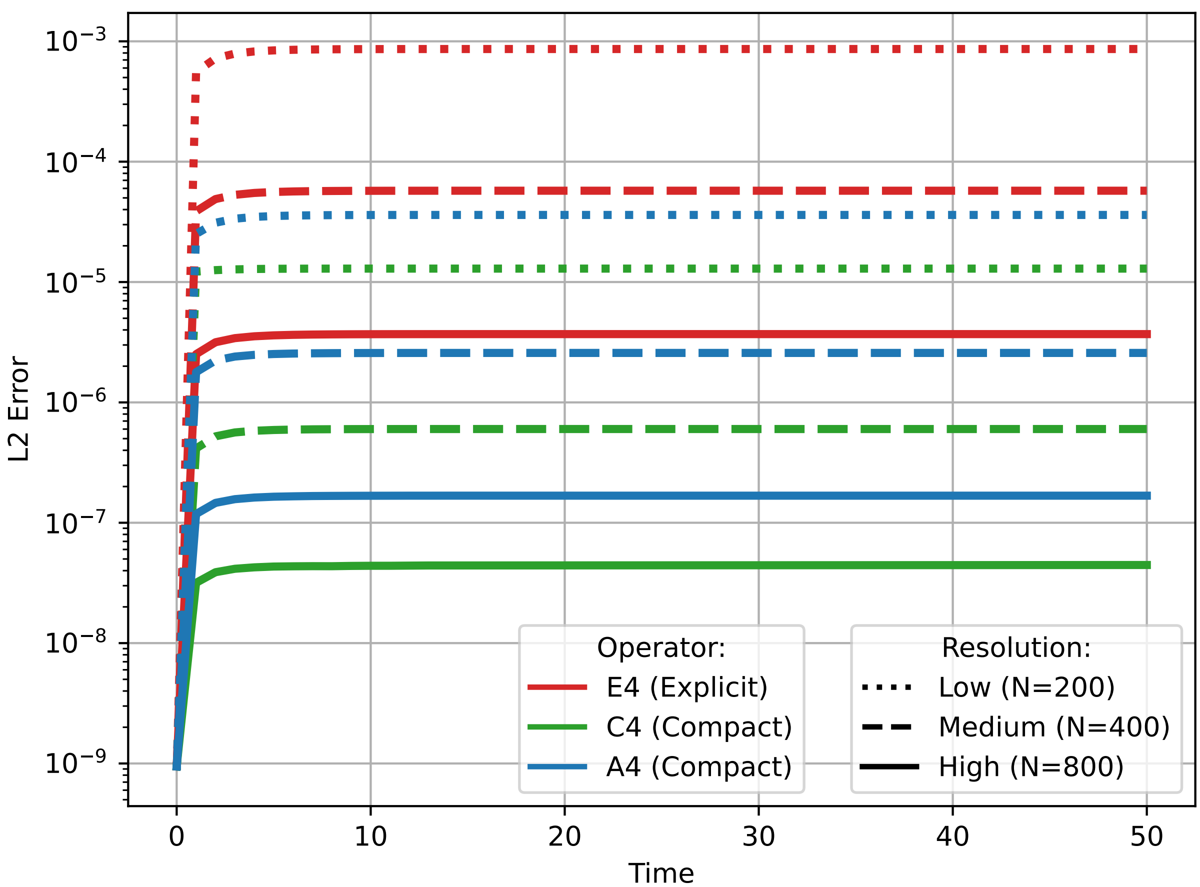


Figure 5.2 Analytic error for Burgers' equation at different resolutions using fourth-order operators. The dotted lines correspond to the coarsest resolution, the dashed lines correspond to the medium resolution, and the solid lines correspond to the highest resolution. Notice that the compact operators perform comparably to the sixth-order explicit operator while using only half the resolution. In the case of the C4 operator, the error is even smaller than that of the comparable explicit scheme at twice the resolution.

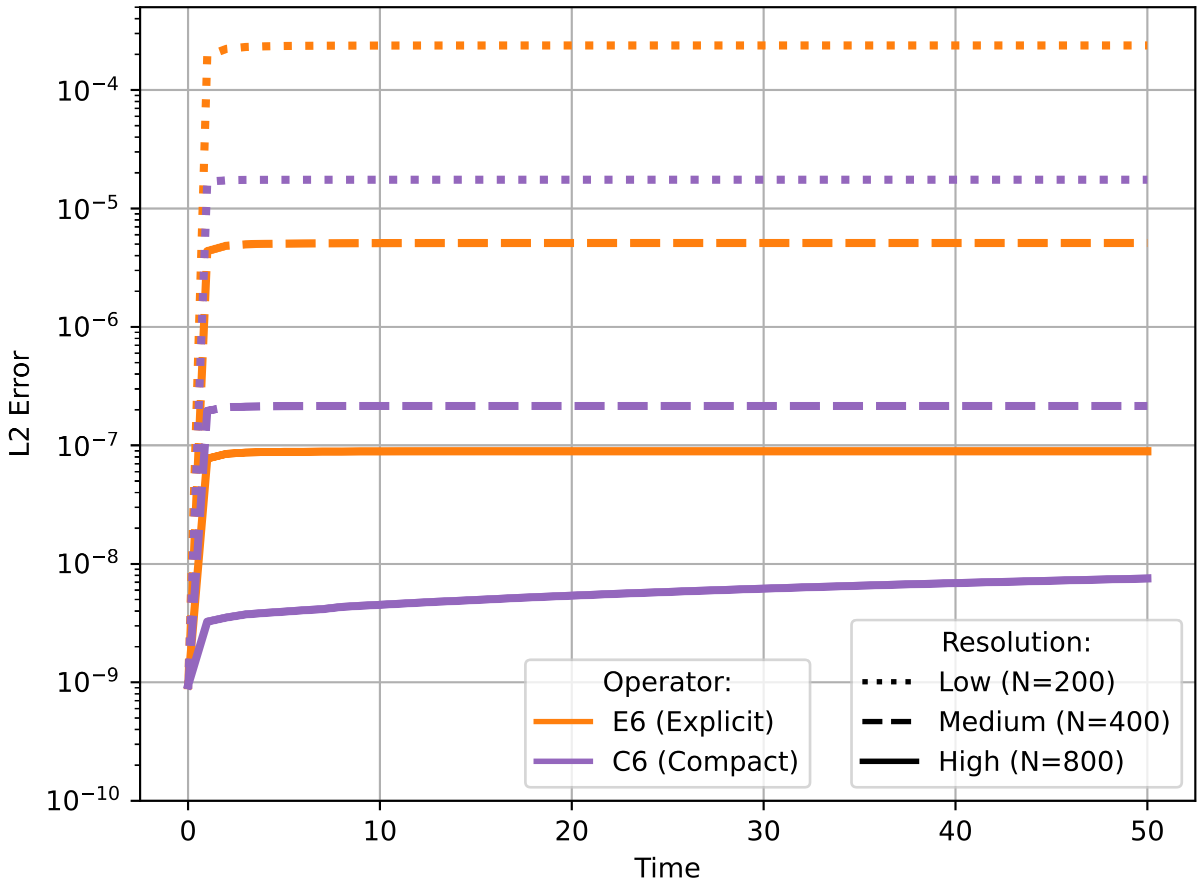


Figure 5.3 Analytic error for Burgers' equation at different resolutions using sixth-order operators. The dotted lines correspond to the coarsest resolution, the dashed lines correspond to the medium resolution, and the solid lines correspond to the highest resolution. The sixth-order CFD operators perform almost as well as the comparable explicit operators while again using a grid with half the resolution.

5.4.2 Maxwell equations in 3D

Our next test involves evolving the Maxwell equations in full 3D. These provide a useful physical system for testing and comparing different computational methods. They form a linear system with known solutions, while retaining sufficient structure to make the numerical tests nontrivial. The

Maxwell equations contain hyperbolic evolution equations for the electric and magnetic fields, **E** and **B**, respectively,

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B}, \qquad \frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \tag{5.21}$$

as well as constraint equations that must hold at the initial time and all subsequent times in order to have self-consistent fields and sources:

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \cdot \mathbf{B} = 0. \tag{5.22}$$

While the above are most familiar, these equations can be written in a variety of different formulations. For example, Knapp et al. [160] considered a formulation intended to reproduce features encountered when solving the Einstein field equations in numerical relativity.

In our current test, we consider the standard Maxwell system with additional constraint damping fields, Ψ and Φ, which are used to damp violations of the electric and magnetic constraints:

$$\partial_t E_i = \epsilon_{ijk} \nabla^j B^k - 4\pi J_i - \partial_i \Psi, \tag{5.23}$$

$$\partial_t B_i = -\epsilon_{ijk} \nabla^j E^k + \partial_i \Phi, \tag{5.24}$$

$$\partial_t \Psi = 4\pi \rho_e - \partial_i E^i - \kappa_1 \Psi, \tag{5.25}$$

$$\partial_t \Phi = \partial_i B^i - \kappa_2 \Phi, \tag{5.26}$$

$$0 = \nabla_i E^i - 4\pi \rho_e, \tag{5.27}$$

$$0 = \nabla_i B^i. \tag{5.28}$$

For the current test, we use an exact solution of Maxwell's equations, namely a propagating dipole wave. In particular, we choose initial data from the exact solution given by Knapp et al. [160]:

$$A^{\hat{\phi}} = \mathcal{A} \sin\theta \left[\frac{e^{-\lambda u^2} - e^{-\lambda v^2}}{r^2} - 2\lambda \frac{u e^{-\lambda u^2} + v e^{-\lambda v^2}}{r} \right]. \tag{5.29}$$

The corresponding electric field is

$$E^{\hat{\phi}} = 2\lambda \mathcal{A} \sin\theta \Big[\frac{u e^{-\lambda u^2} - v e^{-\lambda v^2}}{r^2} + \frac{1}{r} \big[(1 - 2\lambda u^2) e^{-\lambda u^2} + (1 - 2\lambda v^2) e^{-\lambda v^2} \big] \Big]. \tag{5.30}$$

Here

$$u = t - r \quad \text{and} \quad v = t + r \tag{5.31}$$

are the retarded and advanced time coordinates, respectively. The parameter $\mathcal{A}$ sets the initial amplitude, while λ controls the width of the pulse. The corresponding Cartesian components are obtained using the standard coordinate transformation from spherical to Cartesian coordinates.

In our experiments, we set $\mathcal{A} = 1.0$ and $\lambda = 0.5$. A representative evolution is shown in Fig. 5.4. Again, we calculate the normed difference of the analytic and numerical solutions at three resolutions. Fig. 5.5 shows these for fourth order CFDs compared against both fourth and sixth order explicit first derivative FD operators. While we show the absolute errors associated with the x component of the electric and magnetic fields, the remaining Maxwell fields demonstrate similar behavior. It is clear that, as with the Burgers' equation test, the CFDs have a better error profile than the standard explicit FD schemes. They all demonstrate the expected convergence but, in addition, the medium resolution CFDs perform as well or better than the high resolution explicit FDs of the same nominal order of accuracy. Further, when compared with the explicit operators of higher accuracy (E6), the nominally fourth order CFDs perform just as well. This can be reinterpreted as saying that the CFDs effectively achieve an extra level of resolution with the same order of accuracy. Fig. 5.6 shows a similar result for sixth order first derivative CFDs.

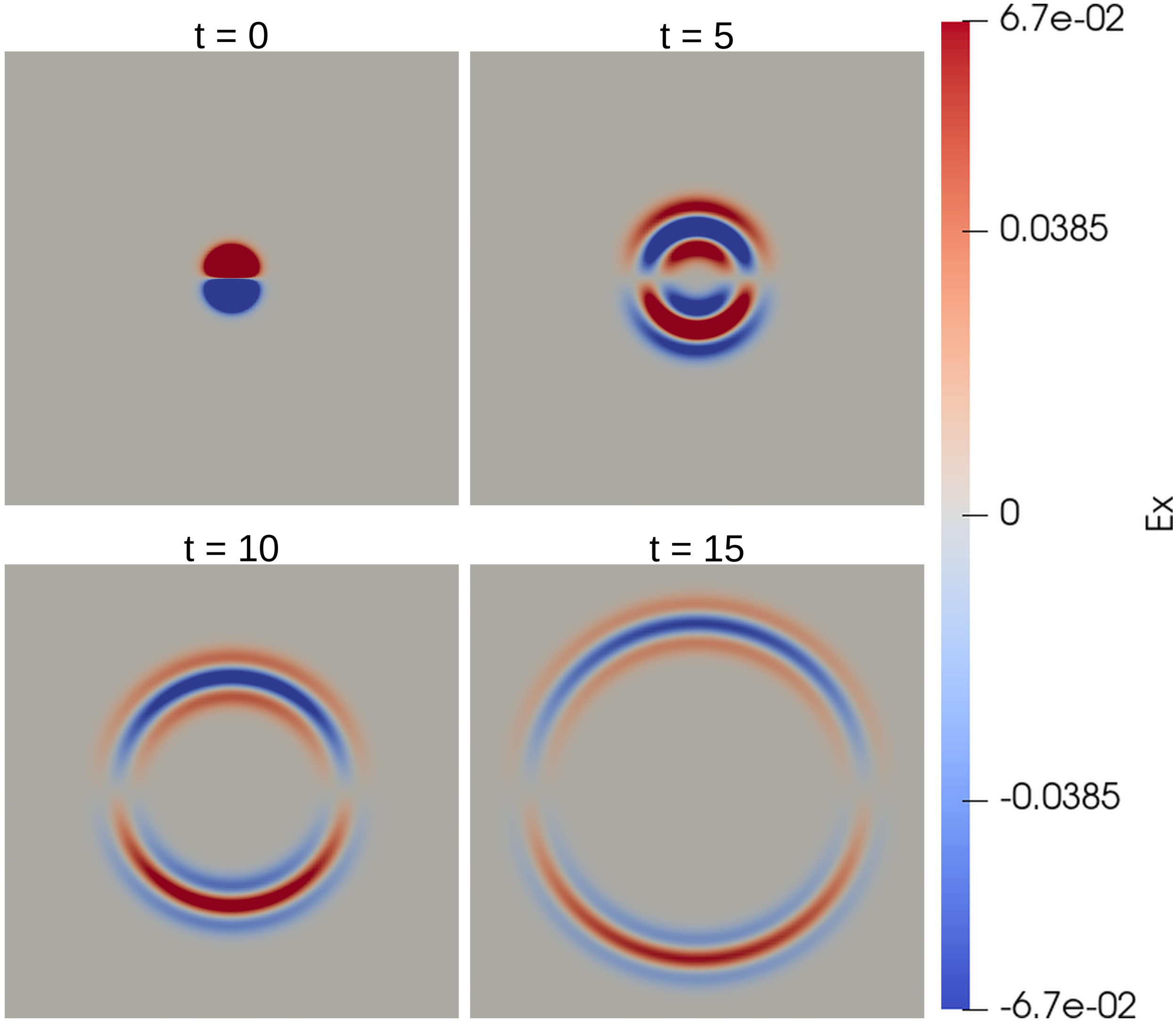


Figure 5.4 Evolution of the electric field component E_x for the dipolar Maxwell test problem. The panels show representative snapshots during the evolution, with a shared color scale shown on the right. The initial dipolar pulse gradually propagates outward.

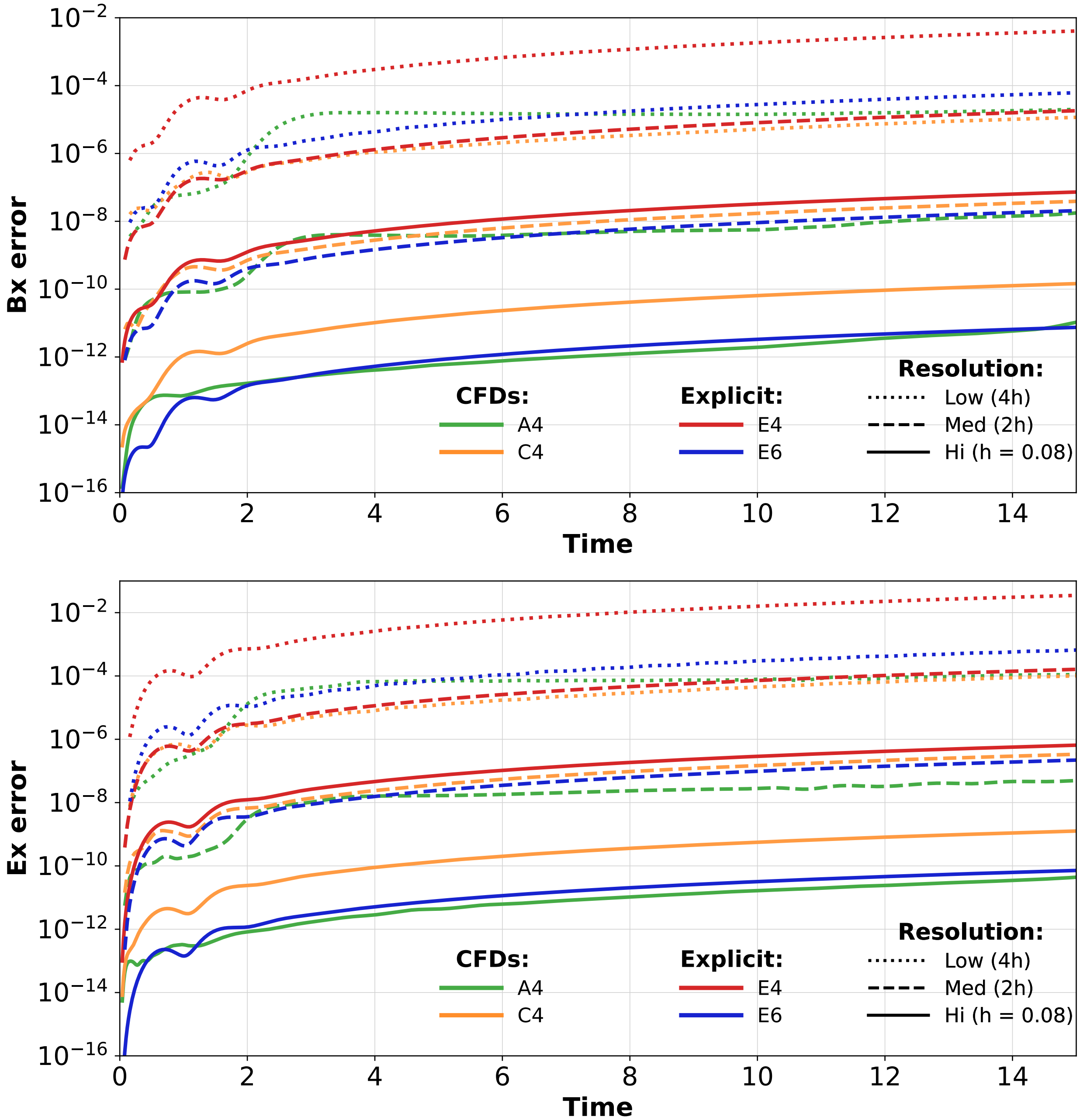


Figure 5.5 Comparison of the errors in the magnetic and electric field components, B_x and E_x, for different fourth-order compact and explicit finite difference schemes. We plot the L_2 error of the numerical solution across the grid relative to the analytic solution. Dotted lines denote the low-resolution runs, dashed lines denote the medium-resolution runs, and solid lines denote the high-resolution runs. The grid spacing is halved between each successive resolution. The fourth-order compact operators perform comparably to their explicit counterparts at half the resolution and achieve error levels comparable to those of the sixth-order explicit operators. This behavior is observed for both the A4 and C4 families of operators tested here.

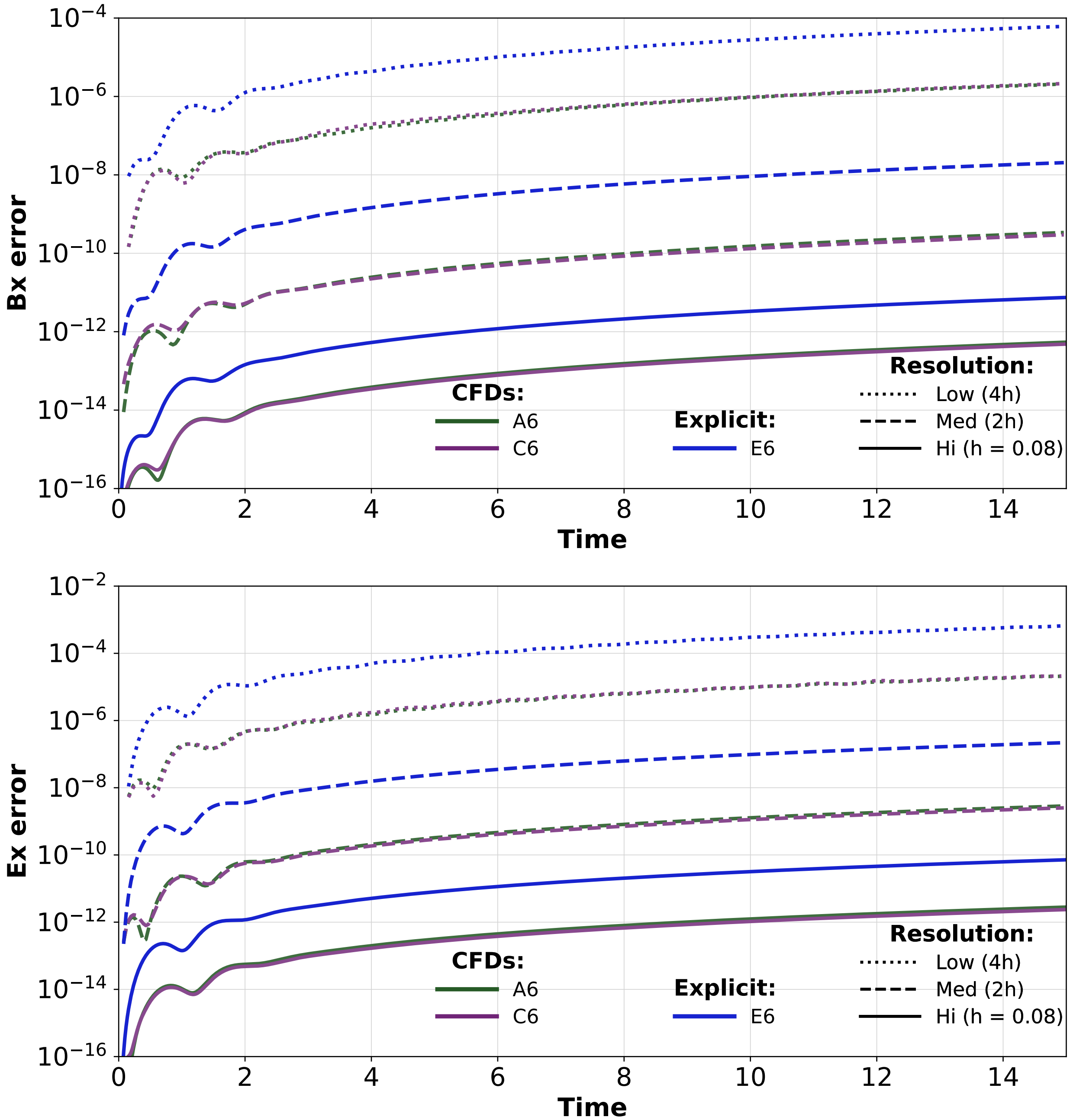


Figure 5.6 Comparison of the errors in the magnetic and electric field components, B_x and E_x, for different sixth-order compact and explicit finite difference schemes. We plot the L_2 error of the numerical solution across the grid relative to the analytic solution. Dotted lines denote the low-resolution runs, dashed lines denote the medium-resolution runs, and solid lines denote the high-resolution runs. The grid spacing is halved between each successive resolution. Although the sixth-order CFD operators do not provide a full factor-of-two improvement in resolution over their explicit counterparts of the same order, they still produce more than an order-of-magnitude improvement in accuracy. This behavior is observed for both the A6 and C6 families of operators tested here.

5.4.3 Nonlinear sigma model

The next problem for which we test our CFDs is a nonlinear sigma model. In this case, we test our second derivative CFDs, again in full 3D, but now with nonlinear source terms. We write the system as first order in time and second order in space. We evolve the scalar field χ together with its conjugate momentum ϕ. The evolution equations are

$$\partial_t \phi = \nabla^2 \chi - \frac{\sin(2\chi)}{r^2}, \tag{5.32}$$

$$\partial_t \chi = \phi. \tag{5.33}$$

where ∇^2 denotes the flat space Laplacian. The term $\sin(2\chi)/r^2$ provides a useful test of the stability and accuracy of the numerical method in nonlinear systems.

The initial scalar profile is taken to be a spherically symmetric Gaussian centered at the origin. The wave speeds in all three spatial directions are set to unity. In particular, the scalar field is initialized as

$$\chi(x, y, z) = A \exp\left[-\frac{(\tilde{r} - R)^2}{\delta^2}\right], \tag{5.34}$$

where

$$\tilde{r} = \sqrt{\epsilon_x (x - x_c)^2 + \epsilon_y (y - y_c)^2 + \epsilon_z (z - z_c)^2}. \tag{5.35}$$

The conjugate momentum is initially set to vanish, i.e. $\phi(x, y, z) = 0$.

For the simulations considered here, we choose

$$A = 1, \qquad \delta = 3, \qquad x_c = y_c = z_c = 0, \tag{5.36}$$

together with

$$\epsilon_x = \epsilon_y = \epsilon_z = 1, \qquad R = 0. \tag{5.37}$$

In this case we do not have an analytic solution and do not have a measure of the absolute error. We can, however, consider the relative difference in the numerical solutions at different resolutions.

These convergence like tests are shown in Figs. 5.7 and 5.8. Again, we see a similar story play out, albeit with respect to the relative errors in the solution between resolution levels. The explicit fourth order FD scheme sees a decrease in the error on going to higher resolution. The two nominally fourth order CFD examples also show evidence of convergence. However, they have at least an order of magnitude better relative error profile as compared to the explicit case. From Fig. 5.8 we see similar behavior, if not quite as dramatic.

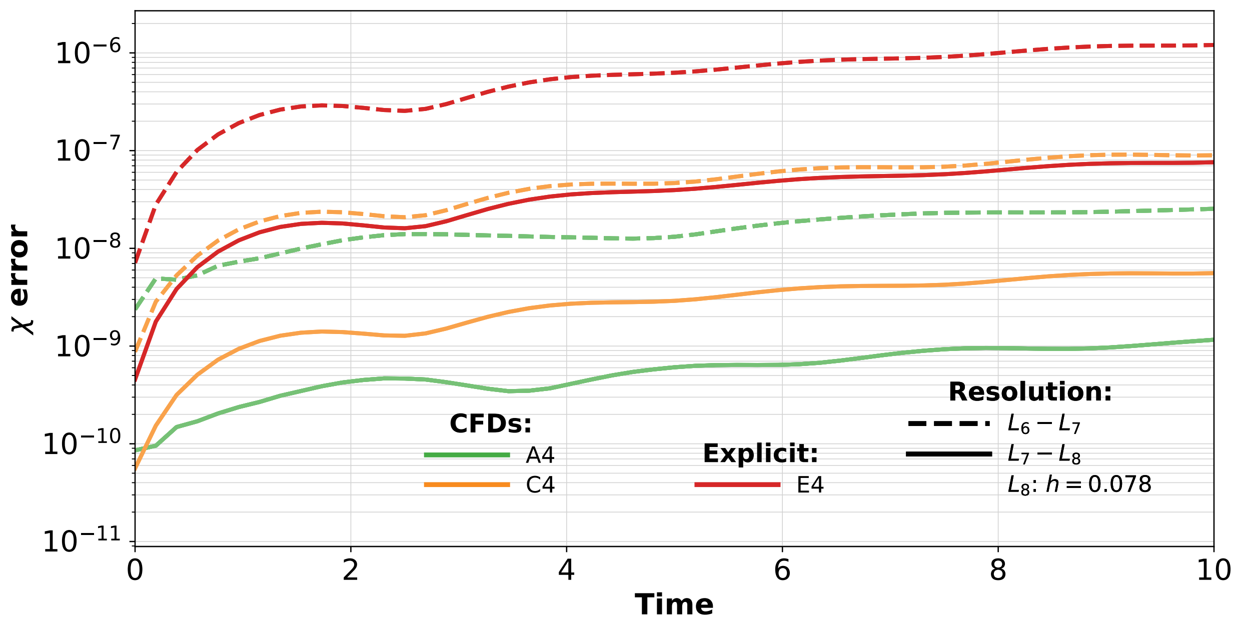


Figure 5.7 Convergence of the nonlinear sigma model field ϕ for the fourth-order compact and explicit finite difference schemes. The plotted quantity is the difference between successive resolutions. Dotted lines denote the low- and medium-resolution comparison, dashed lines denote the medium- and high-resolution comparison, and solid lines denote the highest-resolution run. The grid spacing is halved between each successive resolution. The CFD operators remain stable in this nonlinear system. The fourth-order A4 and C4 operators perform comparably to the explicit fourth-order operator while using approximately half the resolution.

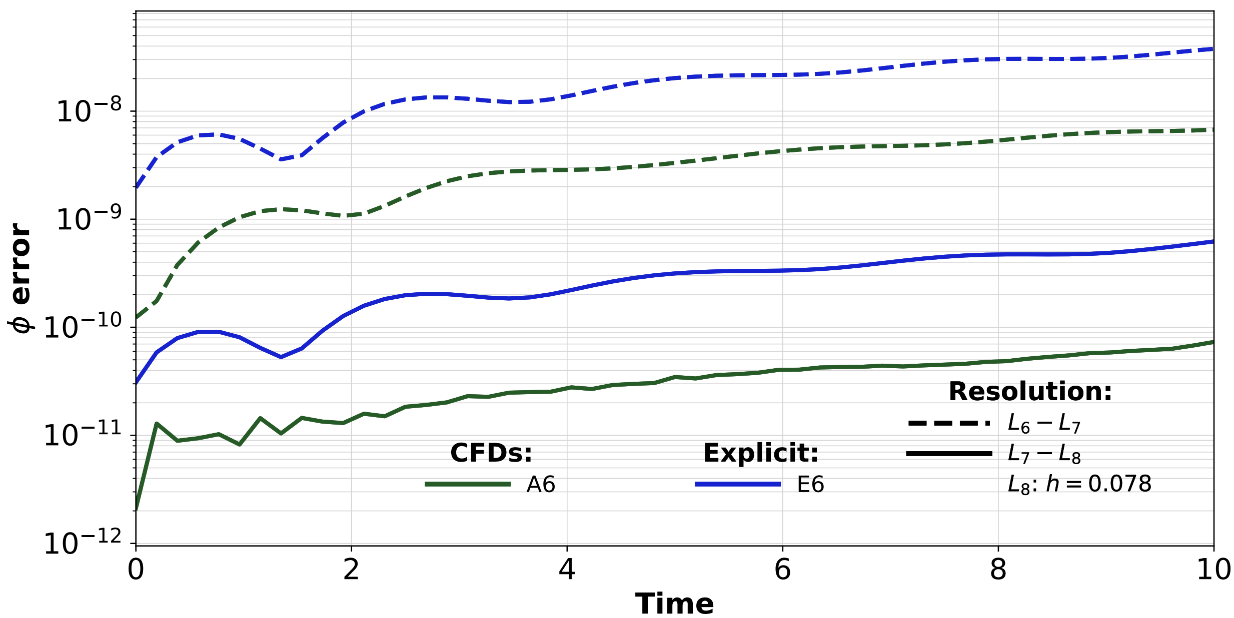


Figure 5.8 Convergence of the nonlinear sigma model field ϕ for the sixth-order compact and explicit finite difference schemes. The plotted quantity is the difference between successive resolutions. Dotted lines denote the low- and medium-resolution comparison, dashed lines denote the medium- and high-resolution comparison, and solid lines denote the highest-resolution run. The grid spacing is halved between each successive resolution. The sixth-order compact operators do not provide the same factor-of-two improvement in effective resolution observed for the fourth-order operators, but they still achieve close to an order-of-magnitude improvement in error.

5.4.4 Gravitational waves

Our final test for our CFD operators is to evolve linear Teukolsky wave initial data within full general relativity using the BSSN formalism. Details on Teukolsky wave initial data and evolutions can be found in [161–163]. The BSSN formulation is described in detail elsewhere, for example in Ref. [20]. These data describe a wave solution to the linearized Einstein equations. It represents a linearized gravitational wave. For sufficiently small amplitude, it is expressible as a small perturbation of the metric. One can increase the size of the perturbation and enter the nonlinear evolutionary regime. For our purposes, it provides a useful test problem for wave propagation in a numerical relativity setting. We emphasize that for the evolution system, we do not solve the linearized Einstein equations but rather the full Einstein equations with these linearized Teukolsky waves as initial data. We use this to test our CFDs.

For the results shown here, we perform direct convergence tests rather than comparing against an analytic solution. The computational domain extends over $x, y, z \in [-40, 40]$, and the initial perturbation is centered at radius $r_0 = 20.0$. We use outgoing, even-parity Teukolsky wave initial data in the quadrupolar mode $(\ell, m) = (2, 0)$ with amplitude $\mathcal{A} = 10^{-2}$. The radial profile is a compactly supported C^k polynomial with $k = 6$ and half-width $w = 2.0$. A representative visualization of the resulting evolution is shown in Fig. 5.9.

The relative errors in the evolutions are shown in Figs. 5.10 and 5.11. Again, these CFDs are for both first and second derivatives and are the optimized operators with boundary closures. In the plots, one of the metric coefficients and one of the extrinsic curvature coefficients are shown. Other metric and extrinsic curvature components behave similarly. Again, the nominally fourth order CFDs perform comparably to the explicit FDs at half the resolution. The sixth order CFDs do not perform quite so well, but continue to have almost an order of magnitude improvement in their relative error profiles. This provides compelling evidence that CFDs are promising candidates for inclusion in algorithms to improve accuracy in numerical relativity.

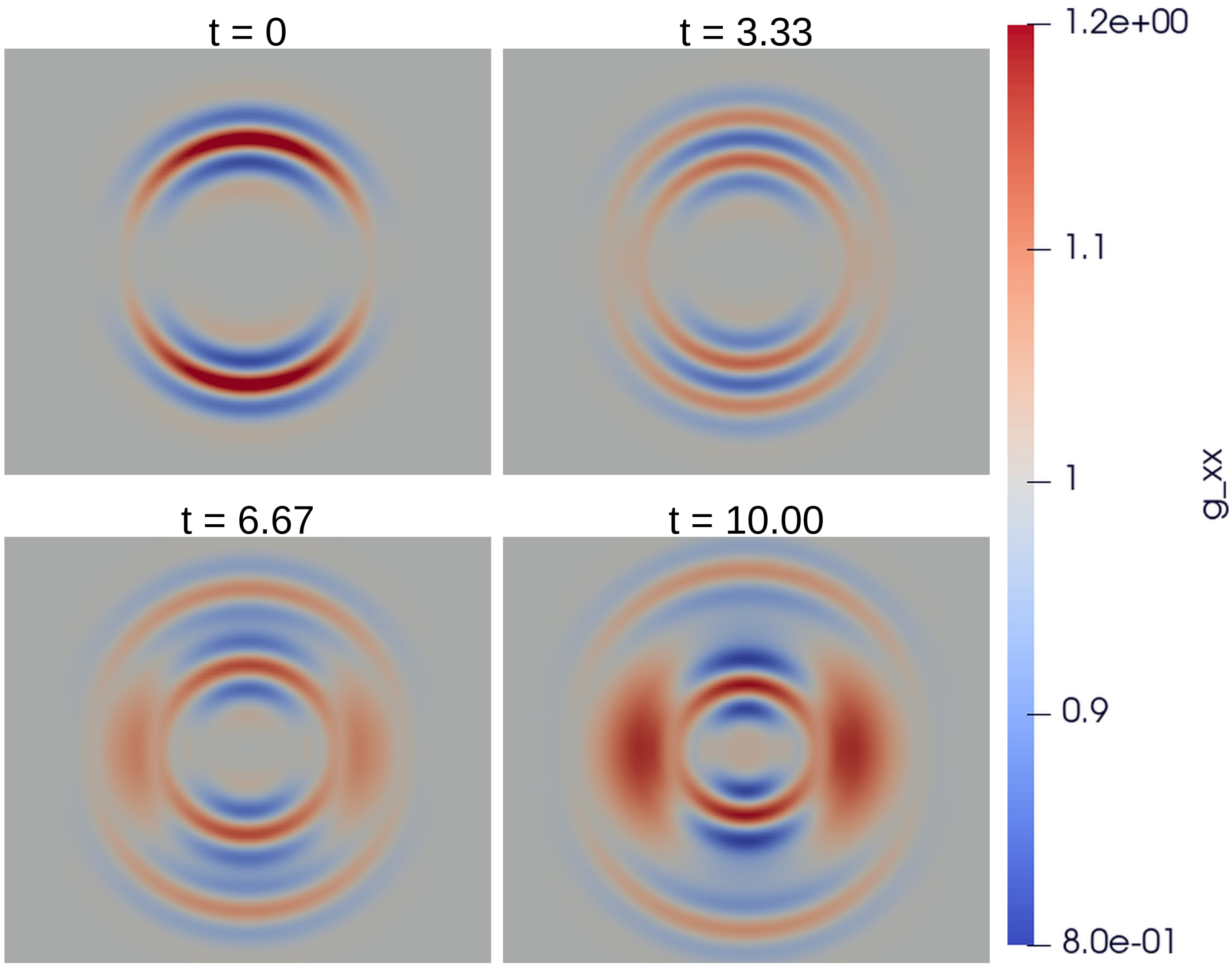


Figure 5.9 Evolution of the metric component g_{xx} for the Teukolsky wave test problem. The panels show snapshots at $t = 0$, $t = 3.33$, $t = 6.67$, and $t = 10.00$. The shared color scale on the right applies to all four panels.

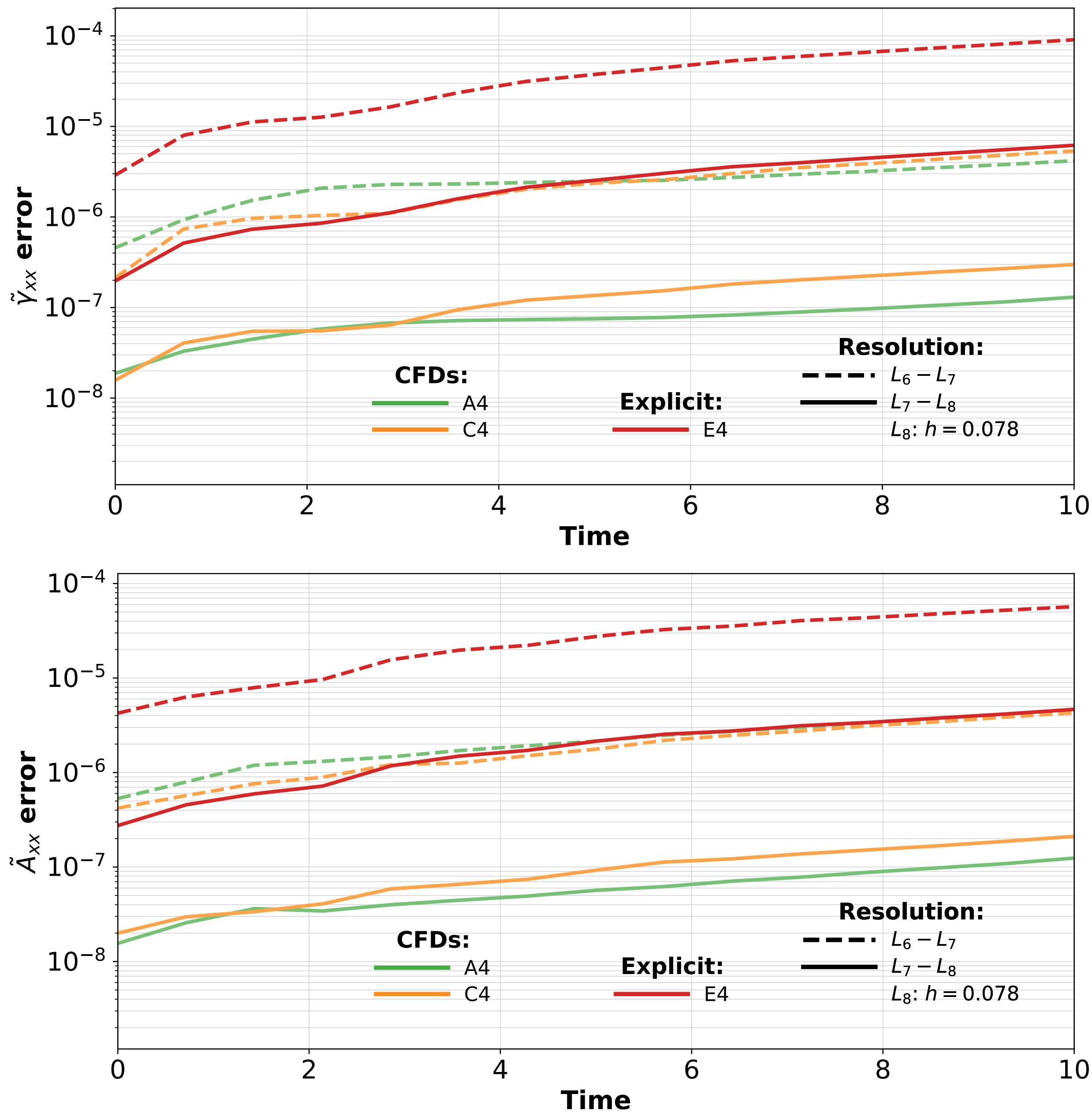


Figure 5.10 Convergence of the metric component g_{xx} and the conformal extrinsic curvature component $\tilde{A}_{xx}$ for the fourth-order compact and explicit finite difference schemes. The plotted quantities are the differences between successive resolutions. Dotted lines denote the low- and medium-resolution comparison, dashed lines denote the medium- and high-resolution comparison, and solid lines denote the highest-resolution run. The grid spacing is halved between each successive resolution. The compact operators remain stable during the Teukolsky wave evolution. As in the nonlinear sigma model, the fourth-order A4 and C4 operators perform comparably to the explicit fourth-order operator while using approximately half the resolution.

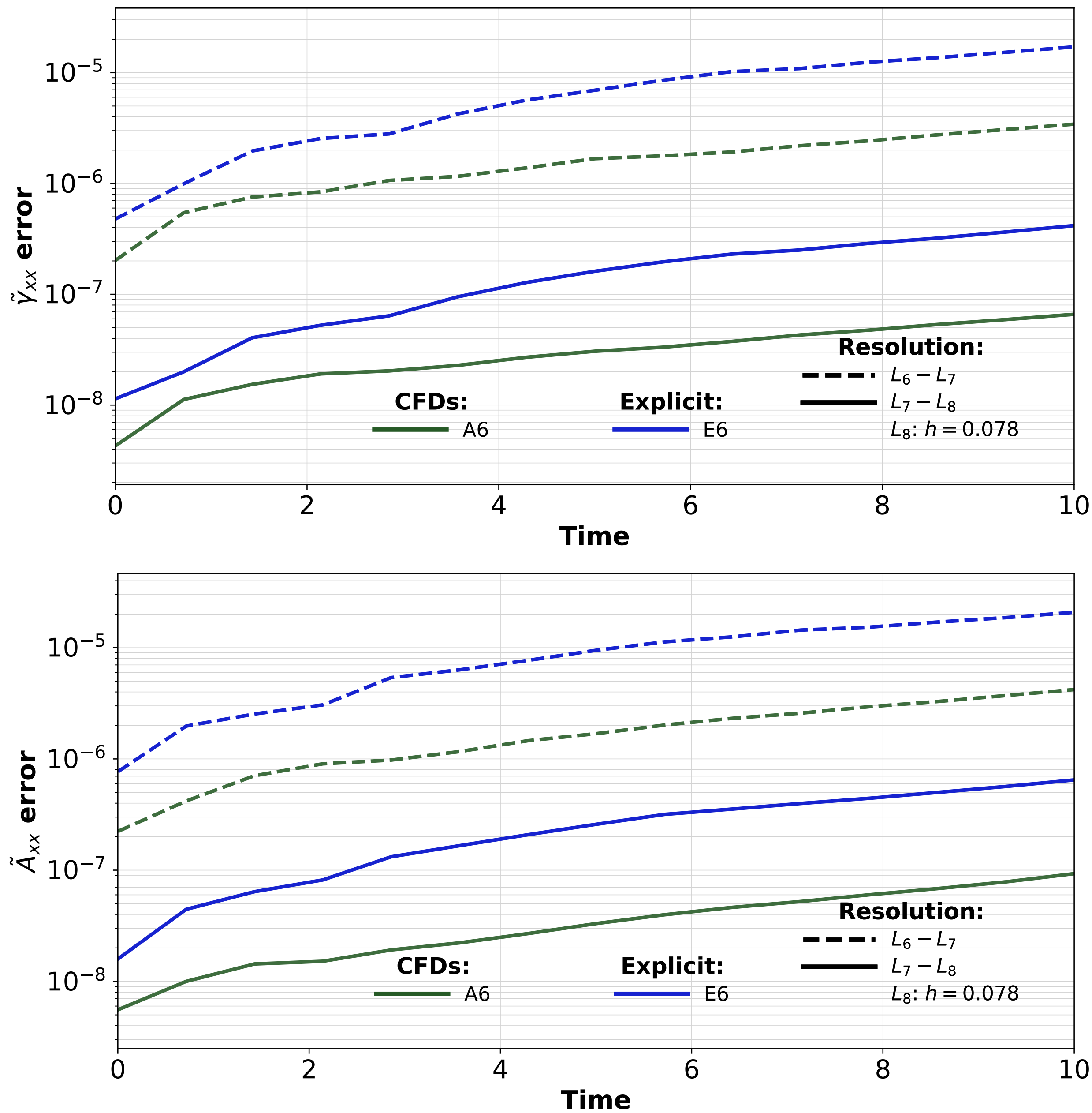


Figure 5.11 Convergence of the metric component g_{xx} and the conformal extrinsic curvature component $\tilde{A}_{xx}$ for the sixth-order compact and explicit finite difference schemes. The plotted quantities are the differences between successive resolutions. Dotted lines denote the low- and medium-resolution comparison, dashed lines denote the medium- and high-resolution comparison, and solid lines denote the highest-resolution run. The grid spacing is halved between each successive resolution. As in the nonlinear sigma model, the sixth-order CFD operators do not achieve the full resolution improvement seen for the fourth-order operators, but they still provide close to an order-of-magnitude improvement in accuracy.

5.5 Conclusion

The CFD operators consistently produced smaller errors than explicit finite difference operators of the same nominal order. In the fourth order tests, the CFD operators frequently achieved errors comparable to those of the explicit operators evaluated at twice the resolution. They also produced error levels comparable to sixth order explicit operators in several of the Maxwell tests. The sixth order CFD operators did not always provide a full factor of two improvement in effective resolution, but they commonly reduced the error by approximately an order of magnitude relative to their explicit counterparts.

These improvements were observed in each of our test system, including the 1D Burger's equation, the 3D Maxwell system, the 3D nonlinear sigma model, and 3D evolutions of Teukolsky gravitational wave initial data using the BSSN equations. The results therefore indicate that the benefits of the CFD operators are not limited to simple linear systems. In particular, the operators remained stable and retained their accuracy improvements in nonlinear and numerical relativity test problems.

Compact finite differences require the solution of an implicit matrix system, but the fixed size leaf nodes used by Dendro-GR allow these systems to be solved efficiently using vectorized linear algebra routines. The additional cost of the compact derivative calculation may therefore be offset by the ability to obtain a given error at a lower grid resolution. For three dimensional time dependent simulations, even a modest reduction in the required resolution could lead to substantial savings in computational cost.

The next step is to apply these operators to fully nonlinear binary black hole simulations. These tests will determine how the operators behave near punctures, across refinement boundaries, and during long inspiral evolutions. They will also allow direct comparisons of waveform accuracy and computational cost between compact and explicit finite difference methods. The results presented here suggest that compact finite differences provide a promising route toward more accurate gravitational waveforms without requiring a corresponding increase in computational expense. We already have some tentative results showing that CFDs perform well in full binary black hole systems, which we present in Appendix F.2. In those cases, we show that the fourth order CFD operators perform better than their explicit counterparts.

Chapter 6

Conclusion

6.1 Conclusion

Throughout this dissertation, we have pursued two complementary themes in numerical relativity. The first is the numerical modeling of black hole spacetimes in Einstein–Maxwell–dilaton–axion theory, an extension of general relativity that includes electromagnetic, dilaton, and axion fields. Within this setting, we studied the nonlinear stability of charged black holes and investigated how these additional fields can affect gravitational waveforms from binary black hole mergers. This work provides an example of how numerical relativity can be used to explore strong field dynamics in theories beyond general relativity.

The second theme is the improvement of numerical relativity methods through the use of compact finite difference operators. These methods are designed to improve the accuracy and efficiency of simulations of strong field gravitational systems. In this final chapter, we summarize the main conclusions from each part of the dissertation and discuss directions for future work.

In the first EMDA portion of this dissertation, we developed a method for constructing approximate binary black hole initial data in Einstein–Maxwell–dilaton–axion theory. This allowed us to evolve head-on collisions of charged black holes across a range of initial spins, charges, and field configurations. These simulations were used to study whether the additional fields present in EMDA theory remain associated with black holes in the fully nonlinear merger regime.

We found that scalar hair persists through merger for initially scalarized Kerr–Sen black holes. When the black holes carry nontrivial electric charge, the dilaton field remains present after merger. When both charge and spin are present, the remnant also retains an axion field. The horizon quantities of the final black hole settle to approximately constant values, supporting the interpretation that the remnant approaches another Kerr–Sen black hole. We also showed that initially unscalarized black holes can dynamically develop scalar hair. In particular, initially Kerr–Newman-like black holes in EMDA evolve toward configurations with nontrivial scalar structure. These results suggest that scalar and axionic hair can be robust features of charged black holes in EMDA theory.

For inspiraling black hole binaries, we found that increasing the charge produces small changes in the gravitational waveform relative to the corresponding neutral GR case. These waveform differences provide a first indication of how EMDA fields may affect binary black hole signals in the strong field regime. We also constructed a modest waveform catalog from the simulations performed in this dissertation, including several charge configurations, spins, and mass ratios. In particular, this catalog includes EMDA binary black hole mergers with mass ratios as large as $q = 8$, demonstrating that these simulations can be extended beyond the equal mass case.

In the broader EMDA theory space study, we investigated how binary black hole evolutions change as the coupling parameters are varied. We explored a range of EMDA coupling choices and compared these results with simulations in Einstein–Maxwell theory, Kaluza–Klein-inspired theory, and related Einstein–Maxwell–dilaton models. This allowed us to search for significant differences in the dynamics and to assess how theory-dependent effects appear in numerical evolutions. For the small charges considered in that study, the dominant gravitational wave signal was only weakly affected by the coupling choice, with the largest differences appearing in the strongest-coupling cases.

In the compact finite difference work, we developed novel compact derivative operators and evaluated their performance against comparable explicit finite difference operators. In the tests considered here, the fourth-order compact operators achieved accuracy comparable to fourth-order explicit operators while using roughly half the resolution. We also demonstrated that sixth-order compact derivatives can outperform their explicit counterparts. These improvements were shown for both first- and second-derivative operators and were tested across three systems: Maxwell theory, a nonlinear sigma model, and the BSSN equations. The improved accuracy of these operators suggests that compact finite differences may provide a useful tool for improving the fidelity of numerical relativity simulations.

Several directions remain for future work. First, compact finite difference operators should be implemented and tested in full binary black hole evolutions. Preliminary work in this direction has already shown some advantages, but additional simulations are needed to determine whether these operators provide a robust improvement over standard explicit finite differences in realistic merger calculations. If these advantages persist, compact finite differences could provide a path toward faster and higher fidelity gravitational wave simulations, which will be increasingly important as next-generation detectors require more accurate waveform models.

Second, future EMDA simulations could be used to make closer contact with gravitational wave observations. One promising approach would be to simulate systems similar to high-significance observed binary black hole mergers whose masses and spins are relatively well constrained. By comparing EMDA waveforms against observed signals, one could place constraints on the possible dilaton and axion charges of the merging black holes. Such a study would help connect the theoretical and numerical results of this dissertation to observational tests of deviations from general relativity.

Taken together, the results of this dissertation demonstrate how numerical relativity can both expand the study of black hole dynamics beyond general relativity and improve the computational tools needed to model gravitational wave sources with increasing accuracy.

EWH: I've gone back and forth on this, but I'm thinking that for the dissertation, the best place for this section should actually be at the end of the introduction rather than the conclusions. Do you mind making the shift?

Appendix A

Mathematical Overview

A.1 Scalar Wave Equation In 3 + 1 Form

As a simple example of a 3 + 1 decomposition, we first consider the scalar wave equation. This example illustrates the basic idea of rewriting a covariant equation as an initial value problem suitable for numerical time evolution.

We begin with the scalar wave equation with a source term,

$$\Box\psi = \nabla_a \nabla^a \psi = 4\pi\rho. \tag{A.1}$$

This equation is written in covariant form. In flat spacetime, using the Minkowski metric with signature $(-,+,+,+)$, it becomes

$$\Box\psi = \eta^{ab}\partial_a\partial_b\psi = \left(-\partial_t^2 + D^2\right)\psi, \tag{A.2}$$

where D^2 denotes the spatial Laplacian. Thus,

$$\partial_t^2\psi = D^2\psi - 4\pi\rho. \tag{A.3}$$

To write this as a first order system in time, we introduce

$$\Pi = \partial_t\psi. \tag{A.4}$$

The wave equation can then be written as

$$\partial_t\psi = \Pi, \tag{A.5}$$

$$\partial_t\Pi = D^2\psi - 4\pi\rho. \tag{A.6}$$

This form makes the time evolution structure explicit: given initial data for ψ and Π, the system determines their future evolution.

This scalar field system provides a useful contrast with the systems considered later. The scalar wave equation has a well posed initial value problem and contains no constraint equations. In particular, the initial data for ψ and Π can be freely specified, subject only to suitable smoothness and boundary conditions. This is not the case for Maxwell's equations or general relativity, where the initial data must also satisfy constraint equations.

A.2 Electricity and Magnetism

The scalar wave equation provides a simple example of an evolution system with no constraints on the initial data. We now consider Maxwell's equations, which provide the next step in complexity: the system naturally separates into evolution equations and constraint equations. This example illustrates, in a familiar setting, the same basic structure that appears later in general relativity.

We present the decomposition in flat spacetime, following an approach similar to that of *Numerical Relativity: Starting from Scratch* [21]. Working in flat spacetime allows us to emphasize the basic evolution and constraint structure before introducing the additional geometric terms that arise in curved spacetime.

We begin by introducing the Faraday tensor in terms of the electromagnetic vector potential A_a,

$$F_{ab} = \nabla_a A_b - \nabla_b A_a. \tag{A.7}$$

This definition makes the antisymmetry of the Faraday tensor explicit,

$$F_{ab} = -F_{ba}. \tag{A.8}$$

It also ensures that the homogeneous Maxwell equation,

$$\nabla_{[a} F_{bc]} = 0, \tag{A.9}$$

is satisfied automatically.

The remaining Maxwell equation is

$$\nabla_b F^{ab} = 4\pi j^a, \tag{A.10}$$

where j^a is the four current. In flat spacetime, we write its components schematically as

$$j^a = (\rho, J^i), \tag{A.11}$$

where ρ is the charge density and J^i is the spatial current. Charge conservation is expressed by the continuity equation

$$\nabla_a j^a = 0. \tag{A.12}$$

We now decompose the vector potential into an electromagnetic scalar potential Φ and a spatial vector potential A_i. With metric signature $(-,+,+,+)$, we write this schematically as

$$A_a = (-\Phi, A_i). \tag{A.13}$$

Using $E_i = F_{i0}$, we have

$$E_i = F_{i0} = \nabla_i A_0 - \nabla_0 A_i. \tag{A.14}$$

In flat spacetime, this becomes

$$E_i = -\partial_t A_i - D_i \Phi, \tag{A.15}$$

where D_i denotes the spatial derivative. Taking the divergence gives

$$D_i E^i = -\partial_t D_i A^i - D_i D^i \Phi. \tag{A.16}$$

The time component of Eq. (A.10) gives

$$D_i E^i = 4\pi\rho, \tag{A.17}$$

which is Gauss's law. This equation is a constraint equation: it restricts the allowed initial data for the electric field and charge density, rather than providing an independent time evolution equation.

Equation (A.15) can also be rearranged to give

$$\partial_t A_i = -E_i - D_i \Phi. \tag{A.18}$$

This equation gives the time evolution of the spatial vector potential once the scalar potential, or equivalently a gauge condition, has been chosen.

We now consider the spatial components of Maxwell's equations. From the definition

$$B^i = \frac{1}{2}\epsilon^{ijk}F_{jk}, \tag{A.19}$$

and the relation $F_{ij} = D_i A_j - D_j A_i$, the magnetic field can be written in terms of the vector potential as

$$B^i = \epsilon^{ijk}D_j A_k. \tag{A.20}$$

This form automatically satisfies the magnetic constraint in flat spacetime,

$$D_i B^i = 0. \tag{A.21}$$

The spatial components of Eq. (A.10) give the evolution equation for the electric field. In tensor component form, this is Ampere's law,

$$\partial_t E_i = \epsilon_i{}^{jk}D_j B_k - 4\pi J_i, \tag{A.22}$$

where $\epsilon_i{}^{jk}$ is the spatial Levi Civita tensor. Substituting $B^i = \epsilon^{ijk}D_j A_k$ gives the equivalent potential form

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi J_i. \tag{A.23}$$

Thus, in this flat spacetime example, Maxwell's equations separate into evolution equations,

$$\partial_t A_i = -E_i - D_i\Phi, \tag{A.24}$$

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi J_i, \tag{A.25}$$

and constraint equations,

$$D_i E^i = 4\pi\rho, \tag{A.26}$$

$$D_i B^i = 0. \tag{A.27}$$

The constraints restrict the allowed initial data, while the evolution equations determine how that data changes in time. If the constraints are satisfied initially and the continuity equation holds, the Maxwell evolution equations preserve the constraints.

This illustrates the same basic structure that appears in the $3+1$ decomposition of general relativity. Part of the system provides evolution equations, while the remaining equations constrain the allowed initial data. In general relativity, the analogous decomposition involves the spacetime metric, lapse, shift, spatial metric, extrinsic curvature, and spatial covariant derivative. The conservation laws are also tied to the geometric structure of the theory through the Bianchi identities.

A.3 Spacetime and Kinematics

Before proceeding further, we briefly discuss the metric in more detail. The metric encodes the notion of distance and separation in spacetime. As a simple example, the Minkowski metric describes flat spacetime. In Cartesian coordinates, and using the signature $(-,+,+,+)$, the line element is

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2. \tag{A.28}$$

Equivalently, the metric components can be written in matrix form as

$$\eta_{ab} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{A.29}$$

where the rows and columns are ordered as (t,x,y,z). The negative sign in the time component reflects our choice of signature.

The same flat spacetime can be expressed in different coordinate systems. For example, in spherical coordinates the Minkowski line element becomes

$$ds^2 = -dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2. \tag{A.30}$$

In these coordinates, the metric components are

$$\eta_{ab} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix}, \tag{A.31}$$

where the rows and columns are ordered as (t, r, θ, ϕ). Although these representations look different, they describe the same flat spacetime. This illustrates that the components of the metric depend on the choice of coordinates.

Before discussing curvature, we recall the definition of the covariant derivative and the Christoffel symbols. For scalars, the ordinary partial derivative is sufficient. For tensors, however, the derivative must also account for changes in the basis vectors. This is done by introducing the connection coefficients, or Christoffel symbols.

For a vector A^b, the covariant derivative is

$$\nabla_a A^b = \partial_a A^b + \Gamma^b{}_{ac} A^c. \tag{A.32}$$

For a covector A_b, the corresponding expression is

$$\nabla_a A_b = \partial_a A_b - \Gamma^c{}_{ab} A_c. \tag{A.33}$$

For the Levi-Civita connection, which is metric compatible and torsion free, the Christoffel symbols are

$$\Gamma^a{}_{bc} = \frac{1}{2} g^{ad} \left(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}\right). \tag{A.34}$$

A natural question is how to determine whether a given metric describes flat spacetime. A necessary and sufficient local condition is that the Riemann curvature tensor vanishes everywhere. Thus, curvature provides a way, independent of coordinates, to distinguish flat spacetime from genuinely curved spacetime.

We will now build up the mathematical foundation needed to construct the so-called ADM decomposition and the resulting ADM equations.[1] The freedom to choose a foliation of spacetime is needed to formulate the Cauchy problem in general relativity and in alternative theories of gravity. We foliate the spacetime into a family of spacelike hypersurfaces (Σ_t), which may be interpreted as slices of constant coordinate time (t).

The gradient of the time function is

$$\nabla_a t = \partial_a t. \tag{A.35}$$

This covector is normal to the constant time hypersurfaces. We define the future directed unit normal n_a to the slices by

$$n_a = -\alpha \nabla_a t, \tag{A.36}$$

where α is the lapse function. The normalization condition is

$$n_a n^a = g^{ab} n_a n_b = -1. \tag{A.37}$$

Because the hypersurfaces are spacelike, their normal is timelike, and we therefore normalize it to negative unity. The lapse α measures the rate at which proper time advances relative to coordinate time for observers moving normal to the slices.

In coordinates adapted to the foliation,

$$\nabla_a t = \partial_a t = \frac{\partial t}{\partial x^a}, \tag{A.38}$$

so the only nonzero component is the time component. Therefore,

$$n_a = (-\alpha, 0, 0, 0). \tag{A.39}$$

For an observer moving normal to the slice, the proper time interval satisfies

$$d\tau = \alpha\, dt. \tag{A.40}$$

[1]ADM stands for Arnowitt–Deser–Misner; after Richard Arnowitt, Stanley Deser, and Charles Misner. The ADM equations arise from a $(3+1)$ decomposition of the Einstein equations.

The coordinate time evolution vector t^a can be decomposed into a part normal to the slice and a part tangent to the slice:

$$t^a = \alpha n^a + \beta^a. \tag{A.41}$$

Here β^a is the shift vector. It is purely spatial, so

$$n_a \beta^a = 0. \tag{A.42}$$

Equivalently,

$$\alpha n^a = t^a - \beta^a. \tag{A.43}$$

In coordinates adapted to the foliation, $\beta^a = (0, \beta^i)$ and $t^a = (1,0,0,0)$. Therefore,

$$n^a = \frac{1}{\alpha}(1, -\beta^i). \tag{A.44}$$

The lapse controls the separation between neighboring slices in proper time, while the shift describes how the spatial coordinates move from one slice to the next.

With these definitions, the spacetime line element can be written in ADM form as

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}\left(dx^i + \beta^i dt\right)\left(dx^j + \beta^j dt\right). \tag{A.45}$$

This form makes explicit the split between the time direction, described by α and β^i, and the spatial geometry, described by the spatial metric γ_{ij}.

The spacetime metric can be decomposed into spatial and temporal parts using the normal vector. The spatial metric is

$$\gamma_{ab} = g_{ab} + n_a n_b, \tag{A.46}$$

and the corresponding projection operator is

$$\gamma^a{}_b = \delta^a{}_b + n^a n_b. \tag{A.47}$$

This operator projects spacetime tensors onto the spatial hypersurfaces.

The way the spatial hypersurfaces are embedded in spacetime is described by the extrinsic curvature. With the sign convention used in this work, we define

$$K_{ab} = -\gamma_a{}^c \gamma_b{}^d \nabla_c n_d. \tag{A.48}$$

Equivalently, K_{ab} measures the spatially projected change of the normal vector from point to point on the hypersurface.

The Lie derivative measures how a tensor field changes as it is dragged along the flow generated by a vector field. For a covector A_d, the Lie derivative along n^a is

$$\mathcal{L}_n A_d = n^c \nabla_c A_d + A_c \nabla_d n^c. \tag{A.49}$$

In the $3+1$ decomposition, the relation

$$t^a = \alpha n^a + \beta^a \tag{A.50}$$

allows coordinate time evolution to be separated into evolution normal to the slice and spatial transport along the shift.

A.4 General Relativity

We now apply the $3+1$ decomposition introduced above to Einstein's equations. The fundamental dynamical field is the spacetime metric g_{ab}, which satisfies

$$G_{ab} + \Lambda g_{ab} = 8\pi T_{ab}. \tag{A.51}$$

For the remainder of this discussion we set $\Lambda = 0$, as is appropriate for the asymptotically flat spacetimes considered in this work.

At first, Einstein's equation appears to provide ten equations for the ten independent components of the spacetime metric. However, these equations are not all independent evolution equations. The Einstein tensor satisfies the contracted Bianchi identity,

$$\nabla_a G^{ab} = 0. \tag{A.52}$$

Together with Einstein's equation, this implies the local conservation law

$$\nabla_a T^{ab} = 0. \tag{A.53}$$

In the $3+1$ decomposition, this structure appears as four constraint equations together with evolution equations for the spatial metric and extrinsic curvature. This is analogous to the way Maxwell's equations contain both evolution equations and constraints, but in general relativity the constraints restrict the geometry of the initial spatial slice.

Using the foliation defined above, the spacetime metric is decomposed into spatial and temporal parts according to

$$\gamma_{ab} = g_{ab} + n_a n_b, \tag{A.54}$$

where n^a is the future directed unit normal to the spatial hypersurfaces. Equivalently,

$$g_{ab} = \gamma_{ab} - n_a n_b. \tag{A.55}$$

The tensor γ_{ab} acts as the intrinsic metric on each spatial slice. Since it is purely spatial, its contraction with the normal vanishes:

$$\gamma_{ab} n^b = 0. \tag{A.56}$$

The inverse spacetime metric can similarly be written as

$$g^{ab} = \gamma^{ab} - n^a n^b. \tag{A.57}$$

The extrinsic curvature relates the embedding of the spatial slice to the time evolution of the spatial metric. With the sign convention adopted in this work,

$$K_{ab} = -\frac{1}{2}\mathcal{L}_n \gamma_{ab}. \tag{A.58}$$

Using the decomposition $t^a = \alpha n^a + \beta^a$, this gives

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}. \tag{A.59}$$

Thus, the extrinsic curvature measures how the spatial metric changes in time, up to the lapse and shift. The spatial metric γ_{ij} and the extrinsic curvature K_{ij} form the basic dynamical variables in the ADM decomposition, while α and β^i encode coordinate freedom.

We also introduce the spatial covariant derivative D_i, which is compatible with the spatial metric γ_{ij}. For a spatial vector V^j,

$$D_i V^j = \partial_i V^j + {}^{(3)}\Gamma^j{}_{ik} V^k, \tag{A.60}$$

where

$${}^{(3)}\Gamma^i{}_{jk} = \frac{1}{2}\gamma^{il}\left(\partial_j \gamma_{lk} + \partial_k \gamma_{lj} - \partial_l \gamma_{jk}\right) \tag{A.61}$$

is the Christoffel symbol associated with γ_{ij}.

The projections of the spacetime curvature are related to the intrinsic and extrinsic geometry of the spatial slices through the Gauss–Codazzi relations. These relations provide the geometric bridge between the four dimensional spacetime curvature and the three dimensional geometry of each spatial hypersurface.

The Gauss equation is

$$\gamma^a{}_i \gamma^b{}_j \gamma^c{}_k \gamma^d{}_l {}^{(4)}R_{abcd} = {}^{(3)}R_{ijkl} + K_{ik}K_{jl} - K_{il}K_{jk}. \tag{A.62}$$

Here ${}^{(4)}R_{abcd}$ is the four dimensional spacetime Riemann tensor, while ${}^{(3)}R_{ijkl}$ is the three dimensional Riemann tensor associated with γ_{ij}.

The Codazzi equation is

$$\gamma^a{}_i \gamma^b{}_j \gamma^c{}_k n^d {}^{(4)}R_{abcd} = D_j K_{ik} - D_i K_{jk}. \tag{A.63}$$

This relates the mixed spatial normal projection of the spacetime curvature to spatial derivatives of the extrinsic curvature.

The Ricci equation, which involves two projections along the normal direction, is

$$\gamma^a{}_i \gamma^b{}_j n^c n^d {}^{(4)}R_{acbd} = \mathcal{L}_n K_{ij} + \frac{1}{\alpha} D_i D_j \alpha + K_{ik} K^k{}_j. \tag{A.64}$$

To obtain the ADM equations, we insert these decompositions into Einstein's equation and equate the projections of the Einstein tensor with the corresponding projections of the stress energy tensor in Eq. (A.51).

Projecting Einstein's equation twice along the normal direction gives the Hamiltonian constraint:

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho. \tag{A.65}$$

Here R is the scalar curvature associated with γ_{ij},

$$K = \gamma^{ij}K_{ij} \tag{A.66}$$

is the trace of the extrinsic curvature, and

$$\rho = n^a n^b T_{ab} \tag{A.67}$$

is the energy density measured by an observer moving normal to the spatial slice.

A mixed projection, with one index projected along the normal direction and one projected onto the spatial slice, gives the momentum constraint:

$$D_j \left(K^{ij} - \gamma^{ij}K\right) = 8\pi j^i. \tag{A.68}$$

Here

$$j^i = -\gamma^{ia}n^b T_{ab} \tag{A.69}$$

is the momentum density measured by the normal observer.

A complete spatial projection gives the evolution equation for the extrinsic curvature:

$$\partial_t K_{ij} = \alpha\left(R_{ij} - 2K_{ik}K^k{}_j + KK_{ij}\right) - D_i D_j \alpha - 4\pi\alpha\left[2S_{ij} - \gamma_{ij}\left(S - \rho\right)\right] + \mathcal{L}_\beta K_{ij}. \tag{A.70}$$

Here R_{ij} is the Ricci tensor associated with γ_{ij}, $S_{ij} = \gamma^a{}_i\gamma^b{}_j T_{ab}$ is the spatial projection of the stress energy tensor, and $S = \gamma^{ij}S_{ij}$ is its trace.

Together with Eq. (A.59), these equations form the ADM system. The variables γ_{ij} and K_{ij} are the dynamical variables, while α and β^i encode the gauge freedom associated with the choice of spacetime coordinates.

$$\boxed{\begin{aligned}
&R+K^2-K_{ij}K^{ij}=16\pi\rho, && \text{Hamiltonian constraint,}\\
&D_j\left(K^{ij}-\gamma^{ij}K\right)=8\pi j^i, && \text{Momentum constraint,}\\
&\partial_t\gamma_{ij}=-2\alpha K_{ij}+\mathcal{L}_\beta\gamma_{ij}, && \text{Metric evolution,}\\
&\partial_t K_{ij}=\alpha\left(R_{ij}-2K_{ik}K^k{}_j+KK_{ij}\right)-D_iD_j\alpha && \\
&\qquad -4\pi\alpha\left[2S_{ij}-\gamma_{ij}(S-\rho)\right]+\mathcal{L}_\beta K_{ij}, && \text{Extrinsic curvature evolution.}
\end{aligned}} \tag{A.71}$$

A.4.1 Reformulating the Einstein equations

The ADM equations provide a natural $3+1$ decomposition of Einstein's equations, but the standard ADM system is only weakly hyperbolic and is not generally well suited for stable numerical evolution. It is therefore useful to reformulate the equations into a system with better hyperbolic properties. The Baumgarte–Shapiro–Shibata–Nakamura (BSSN) formulation achieves this by introducing conformal variables and promoting certain combinations of spatial derivatives of the metric to independent evolved quantities [20, 86, 87].

A Maxwell analogy

We first illustrate the basic idea with an analogy from electromagnetism. In a flat spacetime $3+1$ decomposition, Maxwell's equations can be written in terms of the electric field E_i and vector potential A_i as

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi J_i, \tag{A.72}$$

together with

$$\partial_t A_i = -E_i - D_i\Phi. \tag{A.73}$$

Here Φ is the electromagnetic scalar potential, and J_i is the spatial current. These evolution equations are supplemented by Gauss's law,

$$D_i E^i = 4\pi\rho. \tag{A.74}$$

We define the electric constraint violation by

$$\mathcal{C} \equiv D_i E^i - 4\pi\rho. \tag{A.75}$$

For constraint satisfying data, $\mathcal{C} = 0$. In exact arithmetic, if the constraint is satisfied initially, it remains satisfied during the evolution. In a numerical evolution, however, truncation error and roundoff error generally produce small but nonzero values in $\mathcal{C}$. To that end we reformulate the evolution system so that constraint violations propagate in a controlled way.

We now introduce the auxiliary variable

$$\Gamma \equiv D_i A^i. \tag{A.76}$$

This definition introduces an additional auxiliary constraint,

$$\mathcal{G} \equiv \Gamma - D_i A^i. \tag{A.77}$$

For the original Maxwell system, $\mathcal{G} = 0$. We can nevertheless promote Γ to an independent evolved variable. Taking a time derivative of its defining relation gives

$$\partial_t\Gamma = D_i\partial_t A^i. \tag{A.78}$$

Using the evolution equation for A_i, we find

$$\partial_t\Gamma = -D_i E^i - D_i D^i\Phi. \tag{A.79}$$

Since $C = 0$ for constraint satisfying data, we may add an arbitrary multiple of C to the right hand side without changing the continuum solutions. Thus, we write

$$\partial_t \Gamma = -D_i E^i - D_i D^i \Phi + \eta C, \tag{A.80}$$

where η is a constant. Using

$$C = D_i E^i - 4\pi\rho, \tag{A.81}$$

this can also be written as

$$\partial_t \Gamma = (\eta - 1) D_i E^i - D_i D^i \Phi - 4\pi\eta\rho. \tag{A.82}$$

With this new variable, the Maxwell evolution system becomes

$$\partial_t A_i = -E_i - D_i \Phi, \tag{A.83}$$

$$\partial_t E_i = -D^2 A_i + D_i \Gamma - 4\pi J_i, \tag{A.84}$$

$$\partial_t \Gamma = -D_i E^i - D_i D^i \Phi + \eta C. \tag{A.85}$$

Here $D^2 \equiv D_i D^i$. This system is equivalent to the original Maxwell system when the constraints are satisfied, but it changes how constraint violations behave.

To see this, start from

$$C = D_i E^i - 4\pi\rho. \tag{A.86}$$

Taking a time derivative gives

$$\partial_t C = D_i \partial_t E^i - 4\pi \partial_t \rho. \tag{A.87}$$

Using charge conservation,

$$\partial_t \rho = -D_i J^i, \tag{A.88}$$

we obtain

$$\partial_t C = D_i \partial_t E^i + 4\pi D_i J^i. \tag{A.89}$$

Substituting the evolution equation for E_i gives

$$\partial_t C = D^i \left(-D^2 A_i + D_i \Gamma - 4\pi J_i\right) + 4\pi D_i J^i \tag{A.90}$$

$$= -D^i D^2 A_i + D^i D_i \Gamma. \tag{A.91}$$

Assuming that the spatial derivatives commute, this becomes

$$\partial_t C = -D^2 D^i A_i + D^2 \Gamma. \tag{A.92}$$

Therefore,

$$\partial_t C = D^2 \left(\Gamma - D_i A^i\right) = D^2 \mathcal{G}. \tag{A.93}$$

We next compute the evolution of $\mathcal{G}$. Taking a time derivative gives

$$\partial_t \mathcal{G} = \partial_t \Gamma - D_i \partial_t A^i. \tag{A.94}$$

Using the evolution equations for Γ and A_i, we find

$$\partial_t \mathcal{G} = \left(-D_i E^i - D_i D^i \Phi + \eta C\right) - \left(-D_i E^i - D_i D^i \Phi\right) \tag{A.95}$$

$$= \eta C. \tag{A.96}$$

Combining this equation with

$$\partial_t C = D^2 \mathcal{G} \tag{A.97}$$

gives

$$\partial_t^2 C = D^2 \partial_t \mathcal{G} = \eta D^2 C. \tag{A.98}$$

Equivalently,

$$\boxed{\left(-\partial_t^2 + \eta D_i D^i\right) C = 0}. \tag{A.99}$$

Thus, for $\eta > 0$, electric constraint violations propagate with speed $\sqrt{\eta}$. Analytically, this reformulation does not change the physical solutions of Maxwell's equations, because the added terms vanish for constraint satisfying data. Numerically, however, it changes the behavior of constraint violations by allowing them to propagate rather than remain fixed in place.

The BSSN formulation uses a related idea. It introduces auxiliary variables, most notably the conformal connection functions, and adds suitable multiples of the constraints to improve the hyperbolic structure and numerical behavior of the Einstein evolution system.

BSSN Equations

We begin with the ADM equations:

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho, \qquad \text{Hamiltonian constraint,} \tag{A.100}$$

$$D_j\left(K^{ij} - \gamma^{ij}K\right) = 8\pi j^i, \qquad \text{Momentum constraint,} \tag{A.101}$$

$$\partial_t\gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta\gamma_{ij}, \qquad \text{Metric evolution,} \tag{A.102}$$

$$\begin{aligned}\partial_t K_{ij} = {}& \alpha\left(R_{ij} - 2K_{ik}K^k{}_j + KK_{ij}\right) - D_iD_j\alpha \\ & - 4\pi\alpha\left[2S_{ij} - \gamma_{ij}(S-\rho)\right] + \mathcal{L}_\beta K_{ij}, \qquad \text{Extrinsic curvature evolution.}\end{aligned} \tag{A.103}$$

The Ricci tensor can be written in several equivalent forms. In terms of the spatial metric γ_{ij}, it contains second derivatives of the metric: **EWH: Can you put the next equation on a single line?**

$$R_{ij} = -\frac{1}{2}\gamma^{kl}\left(\partial_k\partial_l\gamma_{ij} + \partial_i\partial_j\gamma_{kl} - \partial_i\partial_l\gamma_{kj} - \partial_k\partial_j\gamma_{il}\right) + \gamma^{kl}\left(\Gamma^m{}_{il}\Gamma_{mkj} - \Gamma^m{}_{ij}\Gamma_{mkl}\right). \tag{A.104}$$

The term $\gamma^{kl}\partial_k\partial_l\gamma_{ij}$ acts like a Laplace operator on the metric. However, the remaining second derivative terms prevent the system from immediately taking a simple wave equation form. In analogy with the Maxwell example above, we introduce an auxiliary variable related to spatial derivatives of the metric.

For the physical spatial metric, define the contracted Christoffel symbol

$$\Gamma^i \equiv \gamma^{jk}\Gamma^i{}_{jk}. \tag{A.105}$$

In terms of this quantity, the Ricci tensor may be written as **EWH: Can you put the next equation on a single line?**

$$R_{ij} = -\frac{1}{2}\gamma^{kl}\partial_k\partial_l\gamma_{ij} + \gamma_{k(i}\partial_{j)}\Gamma^k + \Gamma^k\Gamma_{(ij)k} + 2\gamma^{lm}\Gamma^k{}_{m(i}\Gamma_{j)kl} + \gamma^{kl}\Gamma^m{}_{il}\Gamma_{mkj}. \tag{A.106}$$

The contracted Christoffel symbol is related to the divergence of the inverse metric by

$$g^{bc}\Gamma^a{}_{bc} = -\frac{1}{\sqrt{|g|}}\partial_b\left(\sqrt{|g|}g^{ab}\right). \tag{A.107}$$

Applying this identity to the spatial metric gives

$$\Gamma^i = -\frac{1}{\sqrt{\gamma}}\partial_j\left(\sqrt{\gamma}\gamma^{ij}\right). \tag{A.108}$$

The BSSN formulation applies this idea to a conformally related metric. We introduce the conformal decomposition

$$\gamma_{ij} = e^{4\varphi}\tilde{\gamma}_{ij}. \tag{A.109}$$

The conformal factor is chosen so that **EWH: Can you put the next equation on a single line?**

$$\tilde{\gamma} = 1, \tag{A.110}$$

where $\tilde{\gamma}$ is the determinant of $\tilde{\gamma}_{ij}$. Taking the determinant of the conformal decomposition gives

$$\varphi = \frac{1}{12}\ln\gamma, \tag{A.111}$$

where γ is the determinant of the physical spatial metric γ_{ij}.

The conformal connection functions are then defined by

$$\tilde{\Gamma}^i \equiv \tilde{\gamma}^{jk}\tilde{\Gamma}^i{}_{jk}. \tag{A.112}$$

Since $\tilde{\gamma} = 1$, Eq. (A.108) gives the simpler expression

$$\tilde{\Gamma}^i = -\partial_j\tilde{\gamma}^{ij}. \tag{A.113}$$

In the BSSN formulation, these $\tilde{\Gamma}^i$ are promoted to independent evolved variables. This is directly analogous to introducing $\Gamma = D_i A^i$ in the Maxwell example: the new variable agrees with a spatial derivative of the fundamental field when the auxiliary constraint is satisfied, but treating it independently allows the evolution system to be rewritten with improved hyperbolic properties.

The physical Ricci tensor is split into two pieces,

$$R_{ij} = \tilde{R}_{ij} + R^{\varphi}_{ij}, \tag{A.114}$$

where $\tilde{R}_{ij}$ is computed from the conformal metric $\tilde{\gamma}_{ij}$, and R^{φ}_{ij} contains terms involving the conformal factor φ.

EWH: Can you put the next equation on a single line? The conformal Ricci tensor is

$$\tilde{R}_{ij} = -\frac{1}{2}\tilde{\gamma}^{kl}\partial_k\partial_l\tilde{\gamma}_{ij} + \tilde{\gamma}_{k(i}\partial_{j)}\tilde{\Gamma}^k + \tilde{\Gamma}^k\tilde{\Gamma}_{(ij)k} + 2\tilde{\gamma}^{lm}\tilde{\Gamma}^k{}_{m(i}\tilde{\Gamma}_{j)kl} + \tilde{\gamma}^{kl}\tilde{\Gamma}^m{}_{il}\tilde{\Gamma}_{mkj}. \tag{A.115}$$

Here $\tilde{\Gamma}^i{}_{jk}$ is the Christoffel symbol associated with the conformal metric $\tilde{\gamma}_{ij}$, and indices on conformal quantities are raised and lowered with $\tilde{\gamma}_{ij}$.

The conformal factor contribution is

$$R^{\varphi}_{ij} = -2\left(\tilde{D}_i\tilde{D}_j\varphi + \tilde{\gamma}_{ij}\tilde{D}^k\tilde{D}_k\varphi\right) + 4\left[\left(\tilde{D}_i\varphi\right)\left(\tilde{D}_j\varphi\right) - \tilde{\gamma}_{ij}\left(\tilde{D}^k\varphi\right)\left(\tilde{D}_k\varphi\right)\right], \tag{A.116}$$

where $\tilde{D}_i$ is the covariant derivative compatible with $\tilde{\gamma}_{ij}$.

The contracted conformal Christoffel symbols are defined by

$$\tilde{\Gamma}^i \equiv \tilde{\gamma}^{jk}\tilde{\Gamma}^i{}_{jk} = -\frac{1}{\sqrt{\tilde{\gamma}}}\partial_j\left(\sqrt{\tilde{\gamma}}\,\tilde{\gamma}^{ij}\right). \tag{A.117}$$

Since the conformal metric is chosen to satisfy $\tilde{\gamma} = 1$, this reduces to

$$\tilde{\Gamma}^i = -\partial_j\tilde{\gamma}^{ij}. \tag{A.118}$$

In the BSSN formulation, the quantities $\tilde{\Gamma}^i$ are promoted to independent evolved variables.

The extrinsic curvature is decomposed into its trace and trace free parts:

$$K_{ij} = A_{ij} + \frac{1}{3}\gamma_{ij}K, \tag{A.119}$$

where

$$K = \gamma^{ij}K_{ij}. \tag{A.120}$$

The trace free part is then conformally rescaled according to

$$A_{ij} = e^{4\varphi}\tilde{A}_{ij}, \tag{A.121}$$

or equivalently,

$$\tilde{A}_{ij} = e^{-4\varphi}\left(K_{ij} - \frac{1}{3}\gamma_{ij}K\right). \tag{A.122}$$

The evolved BSSN variables are therefore

$$\varphi, \qquad \tilde{\gamma}_{ij}, \qquad \tilde{A}_{ij}, \qquad K, \qquad \tilde{\Gamma}^i. \tag{A.123}$$

These variables are supplemented by the algebraic constraints

$$\tilde{\gamma} = 1, \qquad \tilde{\gamma}^{ij}\tilde{A}_{ij} = 0, \tag{A.124}$$

and by the auxiliary connection constraint

$$\mathcal{G}^i \equiv \tilde{\Gamma}^i + \partial_j\tilde{\gamma}^{ij} = 0. \tag{A.125}$$

Together with appropriate gauge conditions for the lapse and shift, these variables form the standard BSSN formulation.

The BSSN equations are

$$
\boxed{
\begin{aligned}
&\partial_t \varphi = -\frac{1}{6}\alpha K + \beta^i \partial_i \varphi + \frac{1}{6}\partial_i \beta^i, && \text{Conformal exponent evolution,}\\
&\partial_t K = -\gamma^{ij} D_i D_j \alpha + \alpha\left(\tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2\right) + 4\pi\alpha(\rho + S) + \beta^i \partial_i K, && \text{Trace evolution,}\\
&\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + \beta^k \partial_k \tilde{\gamma}_{ij} + \tilde{\gamma}_{ik}\partial_j \beta^k + \tilde{\gamma}_{kj}\partial_i \beta^k - \frac{2}{3}\tilde{\gamma}_{ij}\partial_k \beta^k, && \text{Conformal metric evolution,}\\
&\partial_t \tilde{A}_{ij} = e^{-4\varphi}\left[-D_i D_j \alpha + \alpha\left(R_{ij} - 8\pi S_{ij}\right)\right]^{\mathrm{TF}} && \\
&\qquad\qquad + \alpha\left(K\tilde{A}_{ij} - 2\tilde{A}_{il}\tilde{A}^l{}_j\right) && \\
&\qquad\qquad + \beta^k \partial_k \tilde{A}_{ij} + \tilde{A}_{ik}\partial_j \beta^k + \tilde{A}_{kj}\partial_i \beta^k - \frac{2}{3}\tilde{A}_{ij}\partial_k \beta^k, && \text{Trace free curvature evolution,}\\
&\partial_t \tilde{\Gamma}^i = -2\tilde{A}^{ij}\partial_j \alpha + 2\alpha\left(\tilde{\Gamma}^i{}_{jk}\tilde{A}^{kj} - \frac{2}{3}\tilde{\gamma}^{ij}\partial_j K - 8\pi\tilde{\gamma}^{ij} j_j + 6\tilde{A}^{ij}\partial_j \varphi\right) && \\
&\qquad\qquad + \beta^j \partial_j \tilde{\Gamma}^i - \tilde{\Gamma}^j \partial_j \beta^i + \frac{2}{3}\tilde{\Gamma}^i \partial_j \beta^j + \frac{1}{3}\tilde{\gamma}^{li}\partial_l \partial_j \beta^j + \tilde{\gamma}^{lj}\partial_j \partial_l \beta^i, && \text{Conformal connection evolution.}
\end{aligned}
}
\tag{A.126}
$$

Gauge and Slicing Conditions

The BSSN equations do not fully determine the evolution until the coordinate gauge is specified. In particular, one must choose evolution conditions for the lapse α and the shift β^i. These choices determine how the spatial slices are placed in spacetime and how the spatial coordinates move from one slice to the next. We summarize several common choices below.

Geodesic slicing. A simple choice is

$$\alpha = 1, \qquad \beta^i = 0. \tag{A.127}$$

This is called geodesic slicing. Although this gauge is simple, it is usually a poor choice for numerical relativity simulations of black holes. The acceleration of the normal observers is

$$a_a = n^b \nabla_b n_a = D_a \ln \alpha. \tag{A.128}$$

For geodesic slicing, $\alpha = 1$, so $a_a = 0$. Thus, the normal observers follow geodesics. In a strong gravitational field, these freely falling observers can focus and form coordinate singularities, making geodesic slicing unsuitable for most black hole evolutions.

Maximal slicing. Another possibility is to control the convergence of the normal observers by controlling the mean curvature of the spatial slices. In maximal slicing, one imposes

$$K = 0. \tag{A.129}$$

If this condition is maintained throughout the evolution, then the slicing condition requires

$$\partial_t K = 0, \tag{A.130}$$

up to shift terms. Using the trace evolution equation, this condition gives an elliptic equation for the lapse. Schematically,

$$D^i D_i \alpha = \alpha \left(K_{ij} K^{ij} + 4\pi(\rho + S) \right), \tag{A.131}$$

for $K = 0$. Maximal slicing has several desirable properties, including singularity avoidance. In particular, the lapse tends to collapse toward zero near strong field regions, causing those parts of the slice to advance more slowly in coordinate time. This behavior is known as collapse of the lapse. However, maximal slicing also has disadvantages, such as grid stretching, which can produce sharp coordinate gradients that are difficult to resolve numerically.

Harmonic coordinates. Harmonic coordinates are defined by requiring the spacetime coordinates to satisfy wave equations,

$$\Box x^a = 0. \tag{A.132}$$

Equivalently, one may impose

$$\Gamma^a = g^{bc} \Gamma^a{}_{bc} = 0. \tag{A.133}$$

A related condition is harmonic slicing, in which only the time component of this condition is imposed. If we also set $\beta^i = 0$, then

$$\Gamma^0 = -\frac{1}{\alpha\sqrt{\gamma}}\partial_t\left(\frac{\sqrt{\gamma}}{\alpha}\right) = 0. \tag{A.134}$$

This gives the lapse evolution equation

$$\partial_t \alpha = -\alpha^2 K. \tag{A.135}$$

Harmonic slicing is closely related to wave coordinates and often leads to well behaved hyperbolic evolution systems. However, modified harmonic slicing conditions are typically more useful for black hole evolutions.

$1+\log$ slicing and the Gamma driver condition. A common generalization of harmonic slicing is the Bona–Masso family of slicing conditions,

$$\partial_t \alpha = -\alpha^2 f(\alpha) K. \tag{A.136}$$

For $f(\alpha) = 1$, this reduces to harmonic slicing. For $f(\alpha) = 0$, the lapse is constant in time, giving geodesic slicing when $\alpha = 1$. Larger values of $f(\alpha)$ make the lapse respond more strongly to the local value of K.

A particularly useful choice is

$$f(\alpha) = \frac{2}{\alpha}. \tag{A.137}$$

This gives

$$\partial_t \alpha = -2\alpha K, \tag{A.138}$$

which is the standard $1+\log$ slicing condition. This choice has singularity avoidance properties similar to maximal slicing while retaining good behavior in nearly flat regions.

When the shift is nonzero, it is common to include an advective term and write

$$\left(\partial_t - \beta^i \partial_i\right)\alpha = -\alpha^2 f(\alpha) K. \tag{A.139}$$

For $f(\alpha) = 2/\alpha$, this becomes

$$\left(\partial_t - \beta^i \partial_i\right) \alpha = -2\alpha K. \tag{A.140}$$

We also need an evolution condition for the shift. A common choice is the Gamma driver condition,

$$\left(\partial_t - \beta^j \partial_j\right) \beta^i = \mu B^i, \tag{A.141}$$

together with

$$\left(\partial_t - \beta^j \partial_j\right) B^i = \left(\partial_t - \beta^j \partial_j\right) \tilde{\Gamma}^i - \eta B^i. \tag{A.142}$$

Here B^i is an auxiliary variable, while μ and η are parameters that control the characteristic speed and damping of the shift condition. The combination of $1 + \log$ slicing and the Gamma driver shift condition is called the moving puncture gauge. A major advantage of this gauge is that it allows black hole spacetimes to be evolved without excising the singularity from the computational domain.

The parameter μ controls how quickly the shift responds to changes in the conformal connection functions, while η damps the auxiliary variable B^i and suppresses oscillatory behavior in the shift.

Appendix B

EMDA Black Holes

B.1 Schwarzschild Metric

We present the Schwarzschild solution in standard Schwarzschild coordinates. This solution describes a static, spherically symmetric, uncharged black hole in general relativity. The metric takes the form

$$ds^2 = -\left(1-\frac{2M}{r}\right)dt^2 + \left(1-\frac{2M}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\varphi^2. \tag{B.1}$$

Equivalently, this can be written as

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \tag{B.2}$$

where

$$f(r) = 1 - \frac{2M}{r}, \tag{B.3}$$

$$d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2. \tag{B.4}$$

The Schwarzschild solution is characterized by a single physical parameter, the ADM mass M. Since the spacetime is nonrotating and uncharged, the electric charge, angular momentum, and magnetic dipole moment vanish:

$$Q = 0, \tag{B.5}$$

$$J = 0, \tag{B.6}$$

$$\mu = 0. \tag{B.7}$$

In the context of EMDA, the absence of additional matter fields is trivially expressed as the the vanishing of the associated gauge, dilaton, and axion fields:

$$A_a dx^a = 0, \tag{B.8}$$

$$\phi = 0, \tag{B.9}$$

$$\kappa = 0. \tag{B.10}$$

With respect to the Schwarzschild radial coordinate r, the event horizon is located at

$$r_H = 2M. \tag{B.11}$$

The curvature singularity occurs at

$$r = 0, \tag{B.12}$$

while the apparent coordinate singularity at $r = 2M$ corresponds to the event horizon rather than a physical singularity.

The Schwarzschild solution may be obtained as the nonrotating, uncharged limit of the Kerr–Sen solution by taking

$$a \to 0, \qquad \alpha \to 0, \qquad m \to M. \tag{B.13}$$

In this limit, the Kerr–Sen metric functions reduce to

$$\Delta = r^2 - 2Mr, \tag{B.14}$$

$$\rho^2 = r^2, \tag{B.15}$$

$$\Sigma = r^4, \tag{B.16}$$

and the Kerr–Sen line element becomes the Schwarzschild line element.

B.2 Reissner–Nordström Metric

We present the Reissner–Nordström solution in standard Schwarzschild like coordinates. This solution describes a static, spherically symmetric, electrically charged black hole in general relativity coupled to electromagnetism. The metric takes the form

$$ds^2 = -\left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\varphi^2. \tag{B.17}$$

Equivalently, this can be written as

$$ds^2 = -f(r)dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2, \tag{B.18}$$

where

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}, \tag{B.19}$$

$$d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2. \tag{B.20}$$

The associated electromagnetic potential may be written as

$$A_a dx^a = -\frac{Q}{r}\, dt, \tag{B.21}$$

up to a gauge transformation. Equivalently, one may choose the opposite sign convention,

$$A_a dx^a = \frac{Q}{r}\, dt, \tag{B.22}$$

depending on the convention used for the Maxwell tensor. In either case, the corresponding electric field has magnitude proportional to Q/r^2.

The Reissner–Nordström solution is characterized by two physical parameters: the ADM mass M and the electric charge Q. Since the spacetime is nonrotating, the angular momentum and magnetic dipole moment vanish:

$$J = 0, \tag{B.23}$$

$$\mu = 0. \tag{B.24}$$

In the context of EMDA, the absence of additional scalar fields is trivially expressed as the the vanishing of the associated dilaton and axion fields:

$$\phi = 0, \tag{B.25}$$

$$\kappa = 0. \tag{B.26}$$

With respect to the Schwarzschild like radial coordinate r, the horizons are located at the roots of $f(r) = 0$, namely:

$$r_\pm = M \pm \sqrt{M^2 - Q^2}. \tag{B.27}$$

The outer event horizon is therefore

$$r_H = r_+ = M + \sqrt{M^2 - Q^2}. \tag{B.28}$$

The condition for the existence of a nonextremal Reissner–Nordström black hole is

$$M^2 > Q^2. \tag{B.29}$$

Equivalently, in terms of the dimensionless charge to mass ratio,

$$\lambda = \frac{Q}{M}, \tag{B.30}$$

the nonextremality condition is

$$|\lambda| < 1. \tag{B.31}$$

The curvature singularity occurs at

$$r = 0. \tag{B.32}$$

The apparent coordinate singularities at $r = r_\pm$ correspond to the inner and outer horizons.

The Reissner–Nordström solution reduces to the Schwarzschild solution in the uncharged limit,

$$Q \to 0. \tag{B.33}$$

In this limit,

$$f(r) \to 1 - \frac{2M}{r}, \tag{B.34}$$

$$r_+ \to 2M, \tag{B.35}$$

$$r_- \to 0, \tag{B.36}$$

and the line element becomes the Schwarzschild line element.

B.3 Kerr Metric

We present the Kerr solution in Boyer–Lindquist coordinates. This solution describes a stationary, axisymmetric, uncharged rotating black hole in general relativity. The metric takes the form **EWH: Can you put the next equation on a single line?**

$$ds^2 = -\left(1 - \frac{2Mr}{\rho^2}\right) dt^2 + \rho^2 \left(\frac{dr^2}{\Delta} + d\theta^2\right) + \frac{\Sigma}{\rho^2} \sin^2\theta\, d\varphi^2 - \frac{4Mar}{\rho^2} \sin^2\theta\, dt\, d\varphi, \tag{B.37}$$

where the metric functions are defined by

$$\Delta = r^2 - 2Mr + a^2, \tag{B.38}$$

$$\rho^2 = r^2 + a^2 \cos^2\theta, \tag{B.39}$$

$$\Sigma = \left(r^2 + a^2\right)^2 - \Delta a^2 \sin^2\theta. \tag{B.40}$$

The Kerr solution is characterized by two physical parameters: the ADM mass M and the angular momentum J. The spin parameter a is related to these quantities by

$$a = \frac{J}{M}. \tag{B.41}$$

Since the Kerr black hole is uncharged, the electric charge and magnetic dipole moment vanish:

$$Q = 0, \tag{B.42}$$

$$\mu = 0. \tag{B.43}$$

In the absence of additional matter fields, the associated gauge, dilaton, and axion fields are trivial:

$$A_a dx^a = 0, \tag{B.44}$$

$$\phi = 0, \tag{B.45}$$

$$\kappa = 0. \tag{B.46}$$

With respect to the Boyer–Lindquist radial coordinate r, the event horizons are located at the roots of $\Delta = 0$:

$$r_\pm = M \pm \sqrt{M^2 - a^2}. \tag{B.47}$$

The outer event horizon is therefore

$$r_H = r_+ = M + \sqrt{M^2 - a^2}. \tag{B.48}$$

The condition for the existence of a nonextremal Kerr black hole is

$$M^2 > a^2 \quad \Leftrightarrow \quad M^4 > J^2. \tag{B.49}$$

Equivalently, in terms of the dimensionless spin parameter

$$\chi = \frac{J}{M^2} = \frac{a}{M}, \tag{B.50}$$

the nonextremality condition is

$$|\chi| < 1. \tag{B.51}$$

The curvature singularity occurs where

$$\rho^2 = r^2 + a^2 \cos^2\theta = 0, \tag{B.52}$$

which corresponds to the Kerr ring singularity at

$$r = 0, \qquad \theta = \frac{\pi}{2}. \tag{B.53}$$

The Kerr solution may be obtained as the uncharged limit of the Kerr–Sen solution by taking

$$\alpha \to 0, \qquad m \to M. \tag{B.54}$$

In this limit, the Kerr–Sen fields become trivial,

$$A_a dx^a = 0, \tag{B.55}$$

$$\phi = 0, \tag{B.56}$$

$$\kappa = 0, \tag{B.57}$$

and the Kerr–Sen metric functions reduce to the Kerr metric functions

$$\Delta = r^2 - 2Mr + a^2, \tag{B.58}$$

$$\rho^2 = r^2 + a^2 \cos^2\theta, \tag{B.59}$$

$$\Sigma = \left(r^2 + a^2\right)^2 - \Delta a^2 \sin^2\theta. \tag{B.60}$$

B.4 Kerr–Newman Metric

We present the Kerr–Newman solution in Boyer–Lindquist coordinates. This solution describes a stationary, axisymmetric, electrically charged rotating black hole in Einstein–Maxwell theory. The metric takes the form **EWH: Can you put the next equation on a single line?**

$$ds^2 = -\left(1 - \frac{2MrQ^2}{\rho^2}\right)dt^2 + \rho^2\left(\frac{dr^2}{\Delta} + d\theta^2\right) + \frac{\Sigma}{\rho^2}\sin^2\theta\, d\varphi^2 - \frac{2a(2MrQ^2)}{\rho^2}\sin^2\theta\, dt\, d\varphi, \tag{B.61}$$

where the metric functions are defined by

$$\Delta = r^2 - 2Mr + a^2 + Q^2, \tag{B.62}$$

$$\rho^2 = r^2 + a^2 \cos^2\theta, \tag{B.63}$$

$$\Sigma = \left(r^2 + a^2\right)^2 - \Delta a^2 \sin^2\theta. \tag{B.64}$$

The associated electromagnetic potential may be written as

$$A_a dx^a = -\frac{Qr}{\rho^2}\left(dt - a\sin^2\theta\, d\varphi\right), \tag{B.65}$$

up to a gauge transformation and sign convention.

The Kerr–Newman solution is characterized by three physical parameters: the ADM mass M, electric charge Q, and angular momentum J. The spin parameter a is related to the angular momentum by

$$a = \frac{J}{M}. \tag{B.66}$$

The magnetic dipole moment is

$$\mu = aQ. \tag{B.67}$$

In the absence of additional scalar fields, the dilaton and axion are trivial:

$$\phi = 0, \tag{B.68}$$

$$\kappa = 0. \tag{B.69}$$

With respect to the Boyer–Lindquist radial coordinate r, the horizons are located at the roots of $\Delta = 0$:

$$r_\pm = M \pm \sqrt{M^2 - a^2 - Q^2}. \tag{B.70}$$

The outer event horizon is therefore

$$r_H = r_+ = M + \sqrt{M^2 - a^2 - Q^2}. \tag{B.71}$$

The condition for the existence of a nonextremal Kerr–Newman black hole is

$$M^2 > a^2 + Q^2. \tag{B.72}$$

Equivalently, using $a = J/M$, this condition may be written as

$$M^4 > J^2 + M^2 Q^2. \tag{B.73}$$

In terms of the dimensionless spin and charge parameters

$$\chi = \frac{J}{M^2} = \frac{a}{M}, \tag{B.74}$$

$$\lambda = \frac{Q}{M}, \tag{B.75}$$

the nonextremality condition becomes

$$\chi^2 + \lambda^2 < 1. \tag{B.76}$$

The curvature singularity occurs where

$$\rho^2 = r^2 + a^2 \cos^2\theta = 0, \tag{B.77}$$

which corresponds to the ring singularity

$$r = 0, \qquad \theta = \frac{\pi}{2}. \tag{B.78}$$

The Kerr–Newman solution reduces to the Kerr solution in the uncharged limit,

$$Q \to 0, \tag{B.79}$$

for which

$$\Delta \to r^2 - 2Mr + a^2, \tag{B.80}$$

$$r_+ \to M + \sqrt{M^2 - a^2}. \tag{B.81}$$

It reduces to the Reissner–Nordström solution in the nonrotating limit,

$$a \to 0, \tag{B.82}$$

for which

$$\Delta \to r^2 - 2Mr + Q^2, \tag{B.83}$$

$$r_+ \to M + \sqrt{M^2 - Q^2}. \tag{B.84}$$

Finally, in the uncharged and nonrotating limit,

$$Q \to 0, \qquad a \to 0, \tag{B.85}$$

the Kerr–Newman solution reduces to the Schwarzschild solution.

B.5 Einstein–Maxwell–Dilaton Black Holes

We present a static, spherically symmetric black hole solution in Einstein–Maxwell–Dilaton theory. In Schwarzschild like coordinates, the magnetically charged solution takes the form **EWH: Can you put the next equation on a single line?**

$$ds^2 = -\left(1-\frac{r_+}{r}\right)\left(1-\frac{r_-}{r}\right)^{1-\alpha_1} dt^2 + \left(1-\frac{r_+}{r}\right)^{-1}\left(1-\frac{r_-}{r}\right)^{\alpha_1-1} dr^2 + r^2\left(1-\frac{r_-}{r}\right)^{\alpha_1} d\Omega^2, \tag{B.86}$$

where

$$d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2. \tag{B.87}$$

The coupling dependent parameter (in this appendix only) α_1 is defined by

$$\alpha_1 = \frac{2\alpha_0^2}{1+\alpha_0^2}, \tag{B.88}$$

where α_0 is the dilaton coupling constant.

For the magnetic solution, the Maxwell field and dilaton are

$$F_{\theta\varphi} = Q_m \sin\theta, \tag{B.89}$$

$$e^{-2\alpha_0\phi} = e^{-2\alpha_0\phi_0}\left(1-\frac{r_-}{r}\right)^{\alpha_1}, \tag{B.90}$$

where Q_m is the magnetic charge and ϕ_0 is the asymptotic value of the dilaton at spatial infinity.

The constants r_+ and r_- are related to the ADM mass M, magnetic charge Q_m, and asymptotic dilaton value ϕ_0 by

$$2M = r_+ + (1-\alpha_1)\, r_-, \tag{B.91}$$

$$2Q_m^2 = e^{2\alpha_0\phi_0} r_+ r_- \,(2-\alpha_1)\,. \tag{B.92}$$

The surface $r = r_+$ corresponds to the event horizon, while $r = r_-$ corresponds to a curvature singularity. For $r > r_-$, the dilaton is larger than its asymptotic value and monotonically decreases toward ϕ_0 at spatial infinity.

The theory also admits an electrically charged solution related to the magnetic solution by a discrete electromagnetic duality. Under this duality, the metric is unchanged, while the Maxwell field and dilaton transform according to

$$F_{ab} \to e^{-2\alpha_0\phi}(*F)_{ab}, \qquad \phi \to -\phi. \tag{B.93}$$

For the electric solution, the Maxwell field and dilaton are

$$F_{tr} = \frac{Q_e}{r^2}, \tag{B.94}$$

$$e^{2\alpha_0\phi} = e^{2\alpha_0\phi_0}\left(1 - \frac{r_-}{r}\right)^{\alpha_1}, \tag{B.95}$$

where Q_e is the electric charge. The constants r_+ and r_- then satisfy

$$2M = r_+ + (1-\alpha_1)\, r_-, \tag{B.96}$$

$$2Q_e^2 = e^{-2\alpha_0\phi_0} r_+ r_- \,(2-\alpha_1)\,. \tag{B.97}$$

In this case, the solution has ADM mass M, electric charge Q_e, and asymptotic dilaton value ϕ_0. The event horizon is located at $r = r_+$, and the curvature singularity is located at $r = r_-$. For $r > r_-$, the dilaton monotonically increases toward ϕ_0 at spatial infinity.

Since this solution is static and spherically symmetric, the angular momentum and magnetic dipole moment vanish:

$$J = 0, \tag{B.98}$$

$$\mu = 0. \tag{B.99}$$

Several familiar black hole solutions are recovered as special limits. In the limit of vanishing dilaton coupling,

$$\alpha_0 \to 0, \qquad \alpha_1 \to 0, \tag{B.100}$$

the solution reduces to the Reissner–Nordström black hole. In the uncharged limit,

$$Q_m \to 0 \qquad \text{or} \qquad Q_e \to 0, \tag{B.101}$$

one has $r_- \to 0$, and the solution reduces to the Schwarzschild black hole with

$$r_+ = 2M. \tag{B.102}$$

Appendix C

Two Puncture Initial Data Solver

C.1 TPID Solver

Our charged-puncture initial-data solver is based on existing TwoPunctures-style methods. The discussion here follows closely the presentation of Ref. [95], as well as the original TwoPunctures paper [164]. The solver is used to construct initial data for two charged black holes separated along the x axis, with punctures located at $x = \pm b$. The user specifies the bare parameters of the punctures, including their masses, charges, linear momenta, and spins. These parameters determine the source terms that enter the constraint equations.

The TPID solver uses a pseudo-spectral collocation method. In contrast to a finite-difference method, where variables are stored on a grid and derivatives are approximated using local stencils, a pseudo-spectral method represents the unknown correction functions using global basis functions. The equations are then enforced at a discrete set of collocation points. Once the solution is obtained at these points, it is interpolated onto the numerical grid used for the time evolution.

The coordinate system used by TwoPunctures is designed to cover all of $\mathbb{R}^3$ with a single compactified computational domain. The physical coordinates (x, y, z) are parameterized by computational coordinates (A, B, ϕ), where $A, B \in [-1, 1]$ and $\phi \in [0, 2\pi]$. The transformation is given by

$$x = b\frac{(1+A)^2+4}{(1+A)^2-4}\frac{2B}{1+B^2}, \tag{C.1}$$

$$y = b\frac{4(1+A)}{4-(1+A)^2}\frac{1-B^2}{1+B^2}\cos\phi, \tag{C.2}$$

$$z = b\frac{4(1+A)}{4-(1+A)^2}\frac{1-B^2}{1+B^2}\sin\phi. \tag{C.3}$$

These coordinates form a cylindrical-like coordinate system around the x axis, with the mapping depending on both A and B. The punctures are located on the x axis, while the outer boundary of the compactified domain corresponds to spatial infinity. In particular, the limit $A \to 1$ maps to spatial infinity. This compactification makes it straightforward to impose the boundary condition that the correction to the conformal factor vanishes at infinity.

The computational coordinates are discretized using n_A, n_B, and n_ϕ collocation points. An example of this grid is shown in Fig. C.1. In the A and B directions, these points are chosen from the roots of Chebyshev polynomials, which concentrate resolution near the punctures and in regions where the solution varies rapidly. The unknown correction functions are expanded spectrally, and the expansion coefficients are determined by requiring the constraint equations to be satisfied at the collocation points.

Because the constraint equations are nonlinear, the resulting algebraic system is solved iteratively using a Newton–Raphson method. At each iteration, the residuals of the constraint equations are evaluated at the collocation points. These residuals measure the extent to which the current approximate solution fails to satisfy the constraint equations. For an exact solution, the residuals would vanish at every collocation point. In practice, the Newton–Raphson iteration is continued until a chosen norm of the residuals falls below a prescribed tolerance, indicating that the constraints are satisfied to the desired numerical accuracy. The resulting spectral solution is then interpolated from the TwoPunctures collocation points onto the evolution grid used in the numerical relativity simulation.

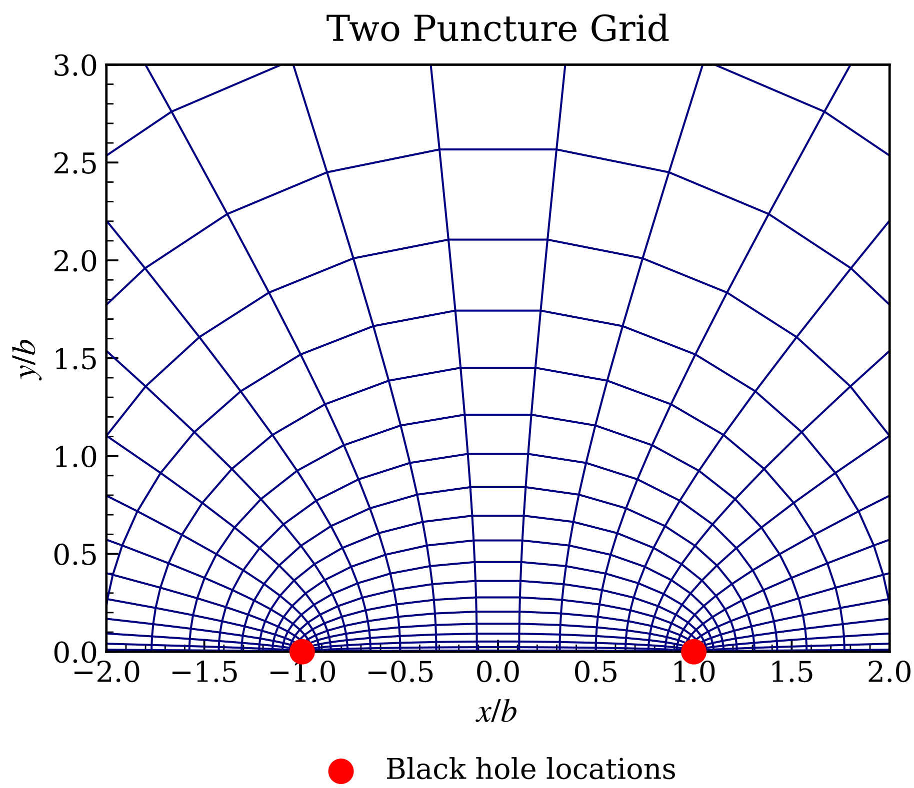


Figure C.1 Example of the compactified TwoPunctures grid in a slice through the two punctures. The physical coordinates are shown in units of the half-separation b, with the black-hole punctures located at $x = \pm b$. The coordinate lines illustrate how the pseudo-spectral collocation points are distributed in the compactified domain before the resulting initial data are interpolated onto the evolution grid. The outer compactified boundary corresponds to spatial infinity and is outside the finite plotting window.

Appendix D

Computation of Ψ_4

D.1 Wave Extraction

This appendix summarizes the wave extraction procedure used for the gravitational, electromagnetic, dilaton, and axion radiation. Gravitational radiation is extracted from the Newman–Penrose scalar Ψ_4, while electromagnetic radiation is extracted from the Maxwell Newman–Penrose scalars Φ_1 and Φ_2. For the dilaton and axion fields, we define analogous outgoing scalar quantities by projecting their gradients along an outgoing null direction.

D.1.1 Null tetrad and spatial triad

The Newman–Penrose scalar Ψ_4 is defined by projecting the Weyl tensor onto a null tetrad:

$$\Psi_4 = C_{abcd} k^a \bar{m}^b k^c \bar{m}^d. \tag{D.1}$$

where C_{abcd} is the Weyl tensor. The null tetrad is constructed from the future directed unit normal n^a to the spatial hypersurface and an orthonormal spatial triad r^a, θ^a, and φ^a. In the wave zone, these spatial vectors approach the usual radial, polar, and azimuthal directions.

With metric signature $(-,+,+,+)$, we define

$$\begin{aligned} l^a &= \frac{1}{\sqrt{2}}\left(n^a + r^a\right), \\ k^a &= \frac{1}{\sqrt{2}}\left(n^a - r^a\right), \\ m^a &= \frac{1}{\sqrt{2}}\left(\theta^a + i\varphi^a\right), \\ \bar{m}^a &= \frac{1}{\sqrt{2}}\left(\theta^a - i\varphi^a\right). \end{aligned} \tag{D.2}$$

These vectors satisfy

$$l^a k_a = -1, \qquad m^a \bar{m}_a = 1, \tag{D.3}$$

with all other tetrad inner products vanishing. In this appendix, l^a denotes the ingoing radial null tetrad vector, while ℓ is reserved for the spherical harmonic mode index. The outgoing radial null tetrad is k^a.

This construction ensures that r^a, θ^a, and φ^a are orthonormal with respect to the spatial metric γ_{ij} on each extraction sphere.

D.1.2 Gravitational radiation

Gravitational radiation is extracted from the Newman–Penrose scalar Ψ_4, which is constructed from the Weyl tensor. The Weyl tensor is the trace free part of the spacetime curvature. In four dimensions, the Riemann tensor has twenty independent components. Ten are determined by the Ricci tensor, while the remaining ten are contained in the Weyl tensor. Thus, the Weyl tensor captures the part of the curvature associated with tidal fields and gravitational radiation.

With the conventions used here, the four dimensional Weyl tensor is

$$C_{abcd} = R_{abcd} + \frac{1}{2}\left(g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}\right) + \frac{1}{6}\left(g_{ac}g_{db} - g_{ad}g_{cb}\right)R. \tag{D.4}$$

The Weyl tensor satisfies the same index symmetries as the Riemann tensor,

$$C_{abcd} = -C_{bacd} = -C_{abdc} = C_{cdab}, \tag{D.5}$$

and obeys the first Bianchi identity,

$$C_{abcd} + C_{acdb} + C_{adbc} = 0. \tag{D.6}$$

Unlike the Riemann tensor, however, the Weyl tensor is trace free:

$$g^{ac}C_{abcd} = 0. \tag{D.7}$$

This trace free property removes the Ricci curvature components, leaving the part of the spacetime curvature not determined locally by the matter content through the Einstein equations.

Projecting the Weyl tensor onto the outgoing null tetrad gives

$$\Psi_4 = C_{abcd}k^a\bar{m}^bk^c\bar{m}^d. \tag{D.8}$$

At future null infinity, and for an appropriate outgoing tetrad, Ψ_4 encodes the outgoing gravitational radiation. With the convention used here, the complex strain is

$$h = h_+ - i h_\times, \tag{D.9}$$

where h_+ and $h_\times$ are the two gravitational wave polarizations. It is related to Ψ_4 by

$$\Psi_4 = \ddot{h}_+ - i\ddot{h}_\times = \ddot{h}, \tag{D.10}$$

up to the overall sign convention chosen for the null tetrad. Here an overdot denotes a derivative with respect to coordinate time, so that $\ddot{h}$ is the second time derivative of the strain. The strain can therefore be obtained by integrating Ψ_4 twice in time:

$$h(t) = \int^t dt' \int^{t'} dt'', \Psi_4(t''). \tag{D.11}$$

In practice, this integration must be performed carefully because direct time domain integration can introduce secular drifts. We therefore use frequency domain integration methods, such as fixed frequency integration, to control low frequency numerical artifacts.

D.1.3 Electromagnetic, dilaton, and axion radiation

The outgoing electromagnetic radiation is extracted using Newman–Penrose scalars associated with the Maxwell tensor [56, 106]. With the null tetrad defined above, we use

$$\Phi_1 = \frac{1}{2} F_{ab} \left(l^a k^b + \bar{m}^a m^b \right), \tag{D.12}$$

$$\Phi_2 = F_{ab} \bar{m}^a k^b. \tag{D.13}$$

At large radius, Φ_2 contains the outgoing radiative electromagnetic degree of freedom, while Φ_1 is useful for monitoring the Coulomb part of the electromagnetic field.

We define analogous extracted quantities for the dilaton and axion fields by projecting their gradients along the outgoing null direction:

$$\mathcal{D} = k^a \nabla_a \phi, \qquad \mathcal{A} = k^a \nabla_a \kappa. \tag{D.14}$$

Here $\mathcal{D}$ and $\mathcal{A}$ denote the extracted dilaton and axion radiation variables, respectively. These quantities are not Newman–Penrose scalars in the electromagnetic or gravitational sense, but they provide a useful measure of the outgoing scalar radiation carried by the dilaton and axion fields.

D.1.4 Multipolar decomposition

The extracted radiation fields are decomposed into angular modes on coordinate spheres at fixed extraction radius. For a spin-weighted field $f(t,r,\theta,\varphi)$, we write

$$f(t,r,\theta,\varphi) = \sum_{\ell,m} f_{\ell m}(t,r)\, {}_sY_{\ell m}(\theta,\varphi), \tag{D.15}$$

where ${}_sY_{\ell m}$ are spin-weighted spherical harmonics and s is the spin weight of the extracted quantity. The mode coefficients are obtained from

$$f_{\ell m}(t,r) = \int f(t,r,\theta,\varphi)\, {}_s\bar{Y}_{\ell m}(\theta,\varphi)\, d\Omega. \tag{D.16}$$

For the gravitational waveform, Ψ_4 has spin weight $s = -2$, so its modes are computed using ${}_{-2}Y_{\ell m}$. The electromagnetic scalar Φ_2 has spin weight $s = -1$, while Φ_1 has spin weight $s = 0$. The dilaton and axion radiation variables $\mathcal{D}$ and $\mathcal{A}$ are scalar quantities and are therefore decomposed using ordinary spherical harmonics, corresponding to $s = 0$.

In practice, the angular integrals are evaluated numerically on each extraction sphere. The resulting mode coefficients provide the time series used to analyze the gravitational, electromagnetic, dilaton, and axion radiation.

D.1.5 Waveform mismatch

To quantify the similarity between two gravitational waveforms, we compute a mismatch. A purely numerical comparison measures the waveform difference directly, while a detector based comparison weights the difference by the noise curve of a particular detector.

For two complex waveforms $h_1(t)$ and $h_2(t)$, a simple normalized overlap is

$$\mathcal{O} = \frac{|\langle h_1, h_2 \rangle|}{\sqrt{\langle h_1, h_1 \rangle \langle h_2, h_2 \rangle}}, \tag{D.17}$$

where a simple time domain inner product is

$$\langle h_1, h_2 \rangle = \int_{t_1}^{t_2} h_1(t) h_2^*(t)\, dt. \tag{D.18}$$

The mismatch is then

$$\mathcal{M} = 1 - \mathcal{O} \tag{D.19}$$

In practice, the overlap is maximized over relative time and phase shifts:

$$\mathcal{M} = 1 - \max_{\Delta t, \Delta\phi} \frac{\left|\left\langle h_1, h_2^{\Delta t, \Delta\phi} \right\rangle\right|}{\sqrt{\langle h_1, h_1 \rangle \langle h_2, h_2 \rangle}}. \tag{D.20}$$

Here $h_2^{\Delta t, \Delta\phi}$ denotes the waveform h_2 shifted in time by Δt and rotated by an overall phase $\Delta\phi$. This maximization removes trivial offsets in arrival time and phase, leaving a measure of the physical disagreement between the waveform shapes. A smaller mismatch indicates better agreement.

For detector weighted comparisons, one instead uses the frequency domain inner product

$$(h_1, h_2) = 4\,\mathrm{Re} \int_{f_{\min}}^{f_{\max}} \frac{\tilde{h}_1(f) \tilde{h}_2^*(f)}{S_n(f)}\, df, \tag{D.21}$$

where $\tilde{h}(f)$ is the Fourier transform of $h(t)$ and $S_n(f)$ is the one sided detector noise power spectral density. The detector weighted overlap is

$$\mathcal{O} = \frac{(h_1, h_2)}{\sqrt{(h_1, h_1)\,(h_2, h_2)}}, \tag{D.22}$$

again maximized over relative time and phase shifts. The corresponding mismatch is

$$\mathcal{M} = 1 - [\max_{\Delta t, \Delta\phi} \mathcal{O}]. \tag{D.23}$$

This detector weighted mismatch is the more relevant quantity when assessing whether two waveforms could be distinguished by a particular gravitational wave detector.

Appendix E

Dendro Notes

E.1 Wavelet Adaptive Mesh Refinement

Dendro employs wavelet adaptive mesh refinement (WAMR) to construct a sparse representation of the evolved fields while maintaining resolution in regions where the solution contains significant spatial structure. The discussion here closely follows [91]. The basic idea is to represent the computational domain using a hierarchy of nested dyadic grids. In one spatial dimension, these grids may be written as

$$V_j = \left\{x^{j,k} = 2^{-j}\, k\, \Delta x\right\}, \tag{E.1}$$

where j denotes the refinement level, k indexes the points on that level, and Δx is the spacing on the coarsest grid. If the physical domain has length L and the base grid V_0 contains $N+1$ points, then

$$\Delta x = \frac{L}{N}. \tag{E.2}$$

The grids are nested, so points present on a coarse level also appear on all finer levels. In particular, points with even index on level $j+1$ coincide with points on level j,

$$x^{j+1,2k} = x^{j,k}. \tag{E.3}$$

The multidimensional AMR hierarchy used in the simulations is built from the same dyadic refinement idea.

Figure E.1 illustrates this nested structure in one dimension. The black points denote grid points inherited from coarser levels, while the red points denote the additional points introduced at each refinement level. These additional basis functions span the spaces (W_j), which represent the new information introduced when passing from resolution level $(j-1)$ to level (j).

Let $u^{j,k}$ denote the value of a field u at the grid point $x^{j,k}$. Since the grids are nested, values at points inherited from the previous level satisfy

$$u^{j+1,2k} = u^{j,k}. \tag{E.4}$$

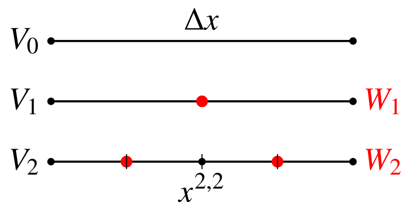


Figure E.1 Schematic illustration of the nested dyadic grids used in wavelet adaptive mesh refinement. Points present on coarser grids are retained on finer grids, while the red points indicate the additional degrees of freedom introduced at each refinement level. These additional points correspond to the space W_j.

The values at newly introduced points may be estimated by interpolation from the previous level,

$$u^{j+1,2k+1} = \sum_m h^{j+1,2k+1}_{j,m} u^{j,m}, \tag{E.5}$$

where $h^{j+1,2k+1}_{j,m}$ are the interpolation weights. In practice only a small number of these coefficients are nonzero. For the fourth order interpolation used here, the interior interpolation stencil is

$$u^{j+1,2k+1} = -\frac{1}{16}u^{j,k-1} + \frac{9}{16}u^{j,k} + \frac{9}{16}u^{j,k+1} - \frac{1}{16}u^{j,k+2}. \tag{E.6}$$

Near physical boundaries the same interpolation idea is used, but the stencil is shifted so that it remains inside the domain, leading to modified one sided interpolation weights.

Equivalently, this interpolation can be described in terms of scaling functions. A basis function on level j may be represented using basis functions on the finer level $j+1$ as

$$\phi_{j,k}(x) = \sum_l c_l\, \phi_{j+1,l}(x), \tag{E.7}$$

where the coefficients c_l encode the interpolation stencil. For the fourth order stencil above, the corresponding nonzero two scale coefficients are

$$c_l = \left\{\dots, 0, -\frac{1}{16}, 0, \frac{9}{16}, 1, \frac{9}{16}, 0, -\frac{1}{16}, 0, \dots\right\}. \tag{E.8}$$

Each basis function may be viewed as a scaled and translated copy of a single fundamental interpolating function,

$$\phi_{j,k}(x) = \phi\left(\frac{2^j x}{\Delta x} - k\right). \tag{E.9}$$

The nested nature of the grids introduces redundancy if basis functions are included for every point on every level. To avoid this, one introduces alternate grids W_j for $j > 0$, consisting only of the points that first appear at level j:

$$W_j = \left\{ x^{j,k} : x^{j,k} = 2^{-j} k \Delta x, \quad k \text{ odd} \right\}. \tag{E.10}$$

Thus the collection $\{V_0, W_1, W_2, \ldots\}$ represents each grid point exactly once. The corresponding index sets are

$$S_0 = \{0, 1, \ldots, N\}, \qquad S_j = \{1, 3, \ldots, 2^j N - 1\}, \tag{E.11}$$

where S_0 labels the base grid and S_j labels the points on level j.

This nested basis construction allows the field to be represented by a coarse contribution together with corrections from the spaces. A field can be expanded as

$$u(x) = \sum_{k \in S_0} u^{0,k} \, \phi_{0,k}(x) + \sum_{j=1}^{J} \sum_{k \in S_j} d^{j,k} \, \phi_{j,k}(x), \tag{E.12}$$

where $u^{0,k}$ are the base level field values and $d^{j,k}$ are the wavelet coefficients. This is the interpolating wavelet representation of the field. The first term gives the coarse representation on the base grid, while the remaining terms give the corrections required at successively finer levels.

The wavelet coefficient at a point is computed by comparing the actual field value to the value predicted by interpolation from the previous refinement level,

$$d^{j,k} = u^{j,k} - P\left(x^{j,k}, j-1\right), \tag{E.13}$$

where $P(x^{j,k}, j-1)$ denotes the interpolated value obtained from level $j-1$. Thus, the coefficient $d^{j,k}$ measures the local interpolation error associated with omitting the point $x^{j,k}$. This transformation from point values to wavelet coefficients is the forward wavelet transformation. It can be inverted by reconstructing the field values from the base level values and the wavelet coefficients.

There are therefore two equivalent descriptions of the field on the multi level grid. The first is the point representation, given by the field values $\{u^{j,k}\}$. The second is the wavelet representation, given by the base values and wavelet coefficients,

$$\{u^{0,k}, d^{j,k}\}. \tag{E.14}$$

The wavelet representation can be made sparse by discarding points whose wavelet coefficients are below a prescribed tolerance ε,

$$|d^{j,k}| < \varepsilon. \tag{E.15}$$

These discarded points are well approximated by interpolation from coarser levels. The retained points are called essential points, and all base level points are always retained. The field values on the essential points form the sparse point representation. For interpolating wavelets, thresholding the coefficients also provides an a priori control of the representation error, with constants determined by the interpolation basis.

In practice, refinement is controlled by the tolerance ε. Points are retained when,

$$|d^{j,k}| > \varepsilon. \tag{E.16}$$

This criterion provides an efficient adaptive representation: smooth regions of the solution are represented using relatively few points, while regions containing sharp gradients, wave propagation, or strong field structure are refined automatically.

In higher dimensions the construction is extended by taking tensor products of the one dimensional basis functions. In three spatial dimensions,

$$\phi_{j,\mathbf{k}}(x,y,z) = \phi_{j,k_x}(x)\phi_{j,k_y}(y)\phi_{j,k_z}(z), \tag{E.17}$$

where $\mathbf{k} = (k_x, k_y, k_z)$. Equivalently, interpolation can be performed directly in multiple dimensions. Depending on the location of the new point relative to the previous level, the interpolation stencil may involve p, p^2, or p^3 points, where p is the one dimensional interpolation order. The wavelet coefficient is again computed as the difference between the actual field value and the value interpolated from the previous level.

The multidimensional wavelet expansion has the form

$$u(x,y,z) = \sum_{\mathbf{k}} u^{0,\mathbf{k}} \phi_{0,\mathbf{k}}(x,y,z) + \sum_{j=1}^{J} \sum_{\mathbf{k}} d^{j,\mathbf{k}} \phi_{j,\mathbf{k}}(x,y,z). \tag{E.18}$$

As in one dimension, sparse refinement is obtained by retaining only those points whose wavelet coefficients exceed the chosen tolerance. This produces an adaptive multi level grid that concentrates resolution where the evolved fields are not accurately represented by interpolation from coarser levels.

E.2 Sobolev Informed Refinement

The term "Sobolev" is used here because the refinement criterion is informed by the same idea underlying Sobolev norms, which measure both a function and its derivatives. For example, a Sobolev H^s norm takes the form

$$\|u\|_{H^s}^2 = \sum_{|\alpha| \le s} \|D^\alpha u\|_{L^2}^2. \tag{E.19}$$

Our refinement criterion does not compute this norm directly, but instead uses derivative information in a Sobolev-informed manner to identify regions requiring additional resolution.

In addition to the standard wavelet criterion, we also consider a Sobolev inspired modification of the refinement. The purpose of this modification is to make the refinement criterion sensitive not only to the local wavelet truncation estimate of a field, but also to local derivative content. This is useful in regions where the field itself may be relatively smooth in amplitude, while its gradients or curvature contain dynamically relevant structure.

For a given evolved variable u, the standard wavelet refinement indicator may be denoted by

$$I_{\mathrm{W}}(u). \tag{E.20}$$

This quantity represents the usual WAMR estimate based on the magnitude of the local wavelet coefficient. The Sobolev informed indicator augments this quantity by including contributions from first and second spatial derivatives. We use an indicator of the form

$$I_{\mathrm{Sob}}(u) = \sqrt{I_{\mathrm{W}}(u)^2 + (w_1 I_{\nabla u})^2 + (w_2 I_{\nabla^2 u})^2}, \tag{E.21}$$

where $I_{\nabla u}$ is a local first derivative indicator, $I_{\nabla^2 u}$ is a local second derivative indicator, and w_1 and w_2 are user controlled weights. Any required normalization of the derivative indicators is absorbed into these weights. The first derivative term increases refinement in regions with large gradients, while the second derivative term increases refinement in regions with large curvature or rapidly changing gradients.

This refinement measure should be interpreted as a *Sobolev inspired* or *Sobolev weighted* WAMR indicator rather than as an exact discrete Sobolev norm. A true Sobolev norm is a global or integrated quantity involving a function and its derivatives. By contrast, Eq. (E.21) is a local AMR indicator designed to guide mesh refinement. Its role is to bias the adaptive mesh toward regions where derivative information suggests that additional resolution may be needed, while retaining the efficiency of the underlying wavelet based refinement algorithm.

We demonstrate this with code by evolving Maxwell's equations. The following figures compare the baseline wavelet adaptive mesh refinement criterion with the Sobolev informed refinement criterion. These diagnostics compare both the constraint behavior of the two refinement strategies and the computational cost associated with the resulting adaptive meshes.

Figure E.2 shows a representative side-by-side comparison of the adaptive mesh structure produced by the two refinement criteria. For both simulations, the computational domain extends from -20 to 20 in each spatial direction, with a finest-level grid spacing of $h = 0.04$. We evolve the same dipole initial data used for the Maxwell test described in Sec. 5.4.2, so that the two simulations differ only in the refinement criterion used to construct the adaptive mesh.

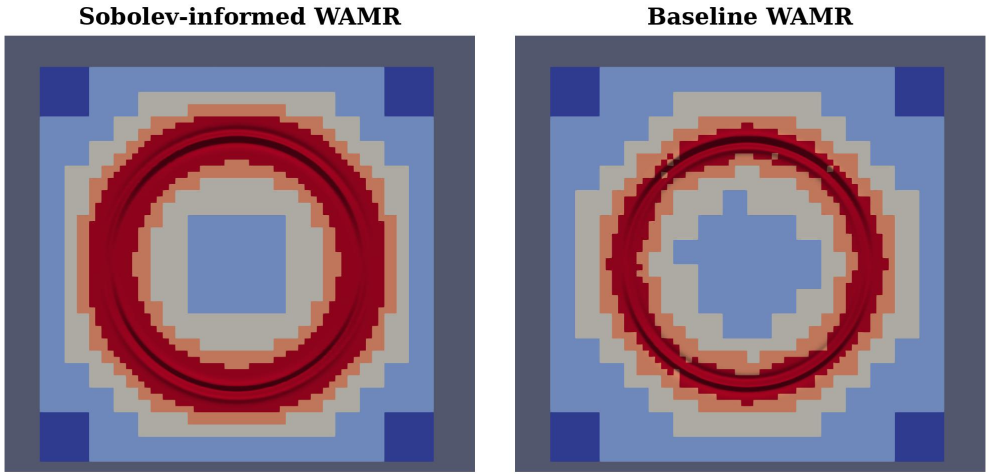
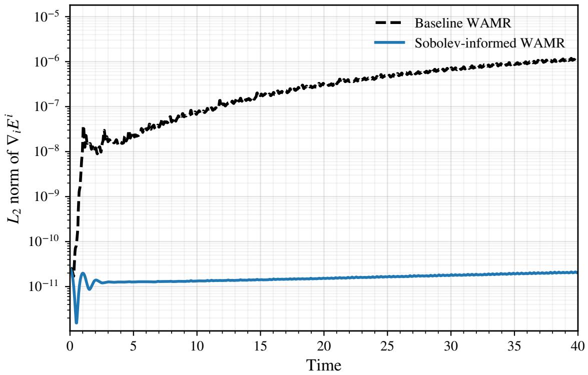
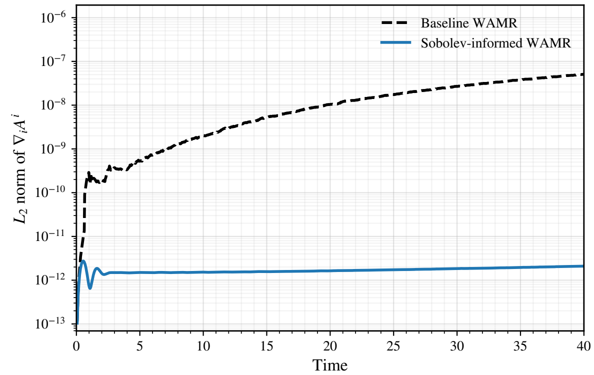


Figure E.2 Representative comparison of the adaptive meshes generated by the baseline wavelet AMR criterion and the Sobolev-informed AMR criterion. Red regions indicate areas of higher refinement, while blue regions indicate areas of lower refinement. The Sobolev-informed criterion includes derivative information in the refinement indicator, which changes where grid points are allocated. The ring corresponds to the outward-propagating dipole pulse. The Sobolev-informed refinement produces a more symmetric mesh and better targets the region of interest, although it is more computationally expensive.

Figure E.3 shows the divergence constraints for the electric and vector-potential fields. These quantities provide a direct measure of the constraint violation during the evolution. The comparison shows how the Sobolev-informed refinement criterion affects the constraint-preserving behavior of the evolution relative to the baseline wavelet criterion. The Sobolev-informed criterion produces substantially lower constraint violations than the baseline case.



(a) L_2 norm of $\nabla_i E^i$.



(b) L_2 norm of $\nabla_i A^i$.

Figure E.3 Constraints for the baseline wavelet AMR and Sobolev-informed AMR refinement criteria. The left panel shows the L_2 norm of the electric-field divergence constraint, while the right panel shows the corresponding divergence constraint for the vector potential. These quantities are used to assess whether the modified refinement indicator improves or degrades the constraint behavior of the evolution.

Figure E.4 compares the number of active mesh points and a measure of the grid efficiency. The number of active mesh points, N_{mesh}, measures the size of the adaptive grid and is therefore a proxy for the computational cost of the evolution. A refinement strategy that achieves comparable or smaller constraint violations using fewer mesh points is more efficient.

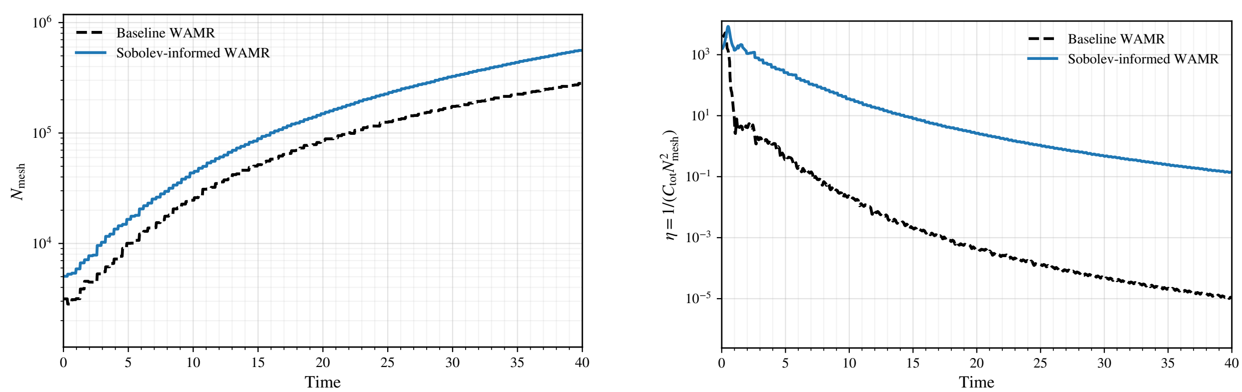
To quantify this tradeoff, we define the grid efficency as:

$$\eta = \frac{1}{C_{\text{tot}} N_{\text{mesh}}^2}, \tag{E.22}$$

where C_{tot} is the total constraint measure and N_{mesh} is the number of active mesh points. In this work, the total constraint measure is constructed from the divergence constraints as

$$C_{\text{tot}} = C_{\nabla E} + C_{\nabla A}, \tag{E.23}$$

where $C_{\nabla E}$ and $C_{\nabla A}$ denote the L_2 norms of the corresponding divergences. With this definition, larger values of η indicate that a simulation achieves smaller constraint violation for a given mesh size. The factor of N_{mesh}^2 penalizes simulations that reduce the constraint violation only by using substantially more grid points.



(a) Number of active mesh points.

(b) Grid efficiency.

Figure E.4 Mesh size and grid efficiency for the baseline wavelet AMR and Sobolev-informed AMR refinement criteria. The left panel shows the number of active mesh points N_{mesh}, which serves as a proxy for computational cost. The right panel shows $\eta = 1/(C_{\text{tot}}N^2_{\text{mesh}})$. Larger values of η indicate smaller constraint violations for a given mesh size and therefore a more efficient use of grid points.

Appendix F

CFD Notes

F.1 CFD Stability Tests

In addition to the eigenvalue stability analysis discussed in the main chapter, we performed numerical stability tests of the compact operators using both Maxwell's equations and the BSSN equations. Representative results from these tests are presented below.

For the Maxwell test, the computational domain extends over $x, y, z \in [-20, 20]$. We use the same initial data and numerical parameters described in the computational fluid dynamics chapter and perform the evolution on the corresponding low resolution uniform grid. The numerical error remains bounded throughout the evolution and decreases cleanly after the initial electromagnetic pulse propagates across the domain. We find no evidence of growing numerical modes for any of the operators shown in Fig. F.1.

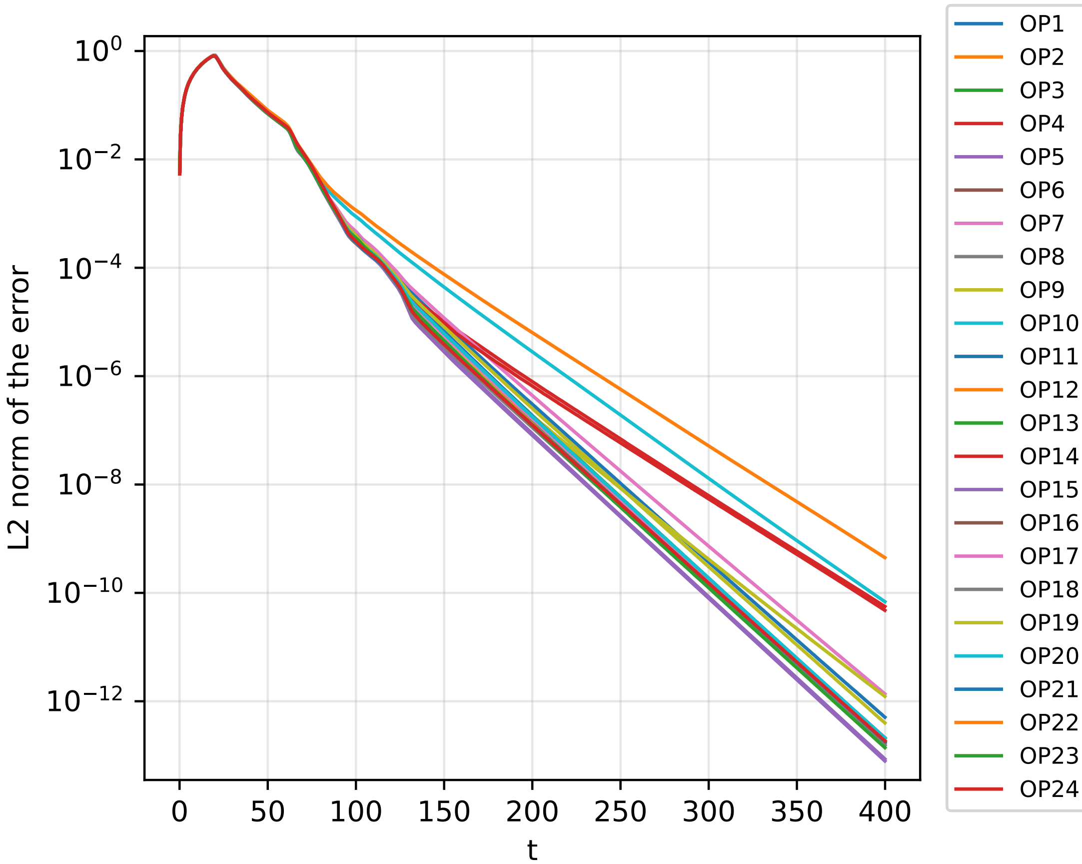


Figure F.1 Evolution of the numerical errors for the finite difference operators applied to the Maxwell test problem. The simulations use a uniform grid on the domain $x, y, z \in [-20, 20]$ with the same initial data and numerical parameters used in the Maxwell tests presented in the main text. The errors remain bounded and decay cleanly as the electromagnetic pulse propagates out of the computational domain, with no evidence of a numerical instability.

We also test the operators by evolving Teukolsky wave initial data with the full BSSN equations. The computational domain extends over $x, y, z \in [-40, 40]$. We use outgoing, even parity initial data in the quadrupolar mode $(\ell, m) = (2, 0)$ with amplitude $\mathcal{A} = 10^{-2}$. The initial perturbation is centered at $r_0 = 20$ and is constructed from a compactly supported C^k radial profile with $k = 6$ and half width $w = 2$.

The simulations use the same remaining numerical parameters as the Teukolsky wave tests described in the main chapter. We evolve the solution for more than ten characteristic crossing times in order to identify any slowly growing unstable modes. As shown in Fig. F.2, the momentum constraint errors remain bounded and decay to small values throughout the evolution. This smooth long term behavior provides further evidence that the compact operators remain stable when applied to the BSSN system.

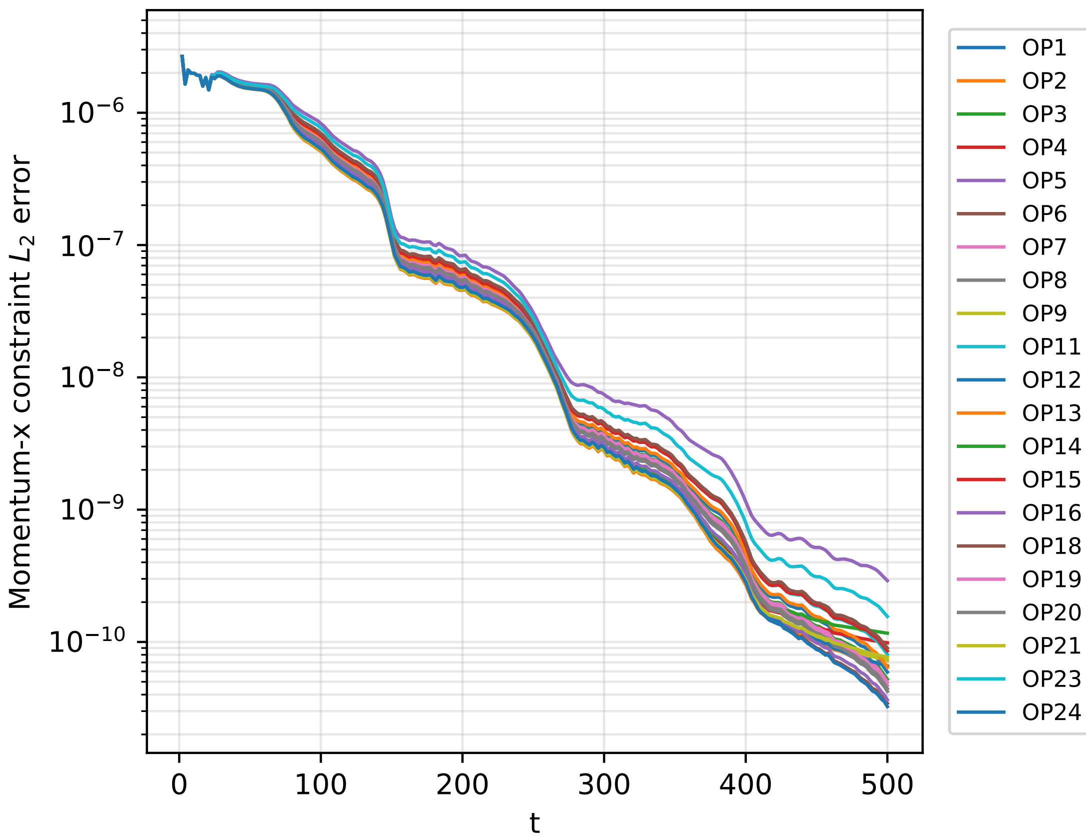


Figure F.2 Evolution of the momentum constraint error during the Teukolsky wave stability test for the finite difference operators considered in this work. The computational domain is $x, y, z \in [-40, 40]$, and the initial data describe an outgoing, even-parity Teukolsky wave in the $(\ell, m) = (2, 0)$ mode with amplitude $\mathcal{A} = 10^{-2}$. The initial perturbation is centered at $r_0 = 20$ and uses a compactly supported C^k radial profile with $k = 6$ and half-width $w = 2$. The errors remain bounded and decay smoothly to small values over more than ten characteristic crossing times, with no evidence of a growing numerical instability.

F.2 Black Hole Merger Tests with CFDs

We also compare the performance of compact and explicit finite difference operators in an inspiraling equal mass binary black hole merger. The binary has mass ratio $q = 1$, an initial separation of $8M$, and quasicircular initial data.

We compare evolutions using fourth-order compact and explicit finite-difference operators with a finest resolution of (h=0.0325) against a more highly resolved sixth-order explicit finite-difference evolution with a finest resolution of (h=0.016).

The resulting gravitational waveforms are shown in Figs. F.3 and F.4. The fourth order explicit evolution is underresolved, which produces a noticeable shift in the merger time. Differences also remain after aligning the waveforms at their peaks.

These results show that the compact operators can outperform explicit operators of the same nominal order in a fully nonlinear numerical relativity simulation. In particular, the compact evolution more closely follows the higher resolution sixth order explicit reference waveform. Further tests are ongoing to determine how robust these improvements remain across different resolutions, binary parameters, and longer inspiral evolutions.

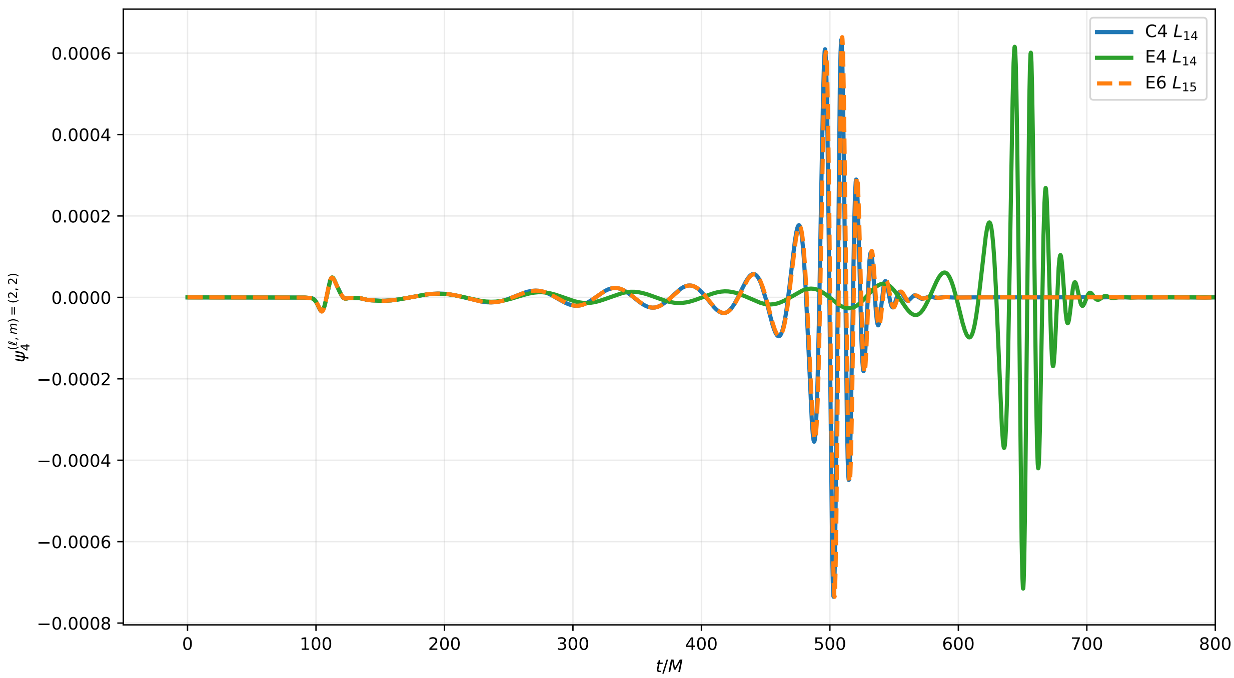


Figure F.3 Comparison of the gravitational waveforms from the equal-mass binary black hole merger evolved with compact and explicit finite difference operators. The binary has mass ratio $q = 1$, initial separation $8M$, and quasicircular initial data. A more highly resolved sixth-order explicit evolution is included as a reference. The underresolved fourth-order explicit evolution shows a noticeable shift in merger time, while the compact evolution remains closer to the reference waveform.

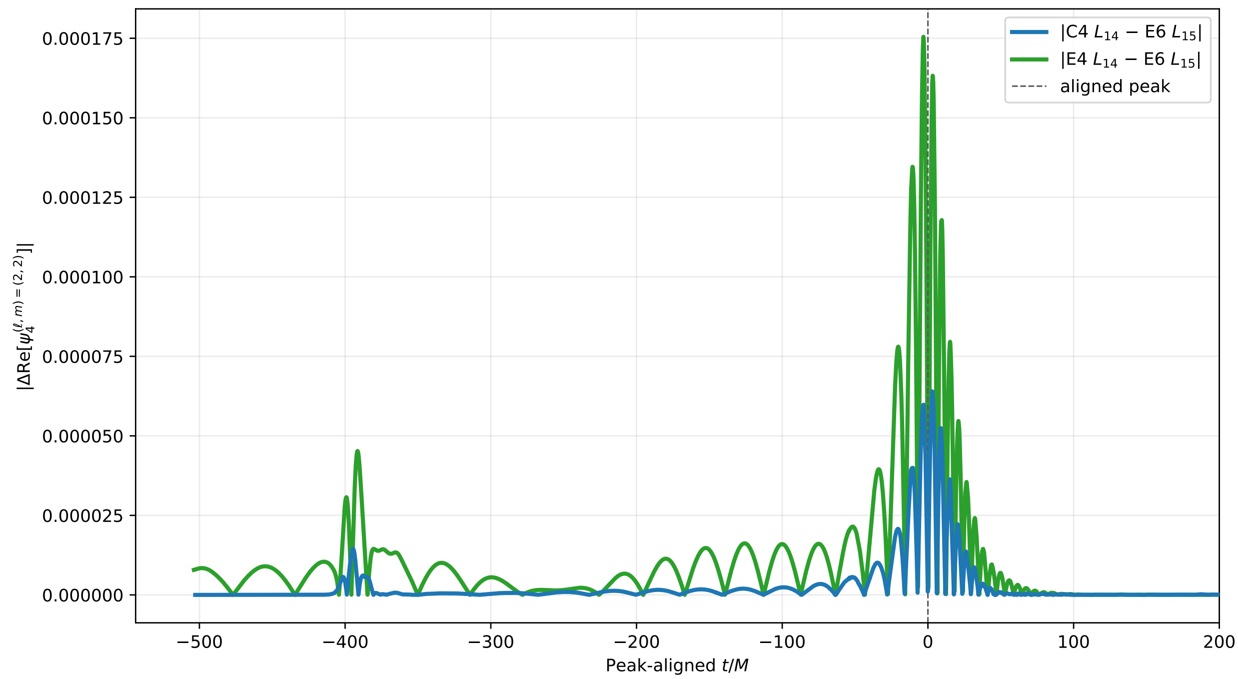


Figure F.4 Differences between the peak-aligned gravitational waveforms from the compact and explicit finite difference evolutions and the more highly resolved sixth-order explicit reference waveform. Peak alignment removes the leading merger-time offset and highlights differences in waveform amplitude and phase. The compact evolution agrees more closely with the reference waveform than the underresolved fourth-order explicit evolution.

Appendix G

Running the EMDA Code on MaryLou

G.1 Building `emda-gr`

To build `emda-gr`, the following external packages are required:

- **C/C++11 or higher** (tested with GNU and Intel compilers)
- **MPI implementation** (tested with `MVAPICH2`, `OpenMPI`, and Intel MPI)
- **BLAS/LAPACK** (tested with `OpenBLAS` and Intel MKL)
- **GSL** (GNU Scientific Library); set `GSL_ROOT_DIR` in `CMake` to enable automatic detection

On MaryLou, the recommended module environment is:

```
andrewc0@login03:$ ml
Currently Loaded Modules:

1. gcc/14.1
2. python/3.12
```

3. cmake/3.28
4. gcc-runtime/13.2.0-g4hc7ev
5. eigen/3.4.0-6kqviac
6. openmpi5-GR
7. gsl-GR
8. mkl-GR
9. rclone/1.65.2-tyg4lvy

To ensure that these dependencies are always available, it is convenient to load them automatically in your shell configuration. Open your `~/.bash_profile`:

```
vim ~/.bash_profile
```

An example `.bash_profile` is shown below. Additional customizations may be added as needed.

```
# --- Modules ---

module purge
module use /grphome/grp_GR/.modulefiles
module load gcc/14.1 python/3.12 cmake/3.28 eigen/3.4.0-6kqviac
module load openmpi5-GR gsl-GR mkl-GR rclone

export HCOLL_DIR=/vapps/rhel9/x86_64/HPC-X/hpcx-v2.19-gcc-inbox-redhat9-cuda12-x86_64/
export LD_LIBRARY_PATH="${HCOLL_DIR}:${LD_LIBRARY_PATH:-}"
```

```
# --- GSL ---

export GSL_DIR=/grphome/grp_GR/apps/gsl-2.8
export GSL_ROOT_DIR=/grphome/grp_GR/apps/gsl-2.8
export LD_LIBRARY_PATH="${GSL_DIR}/lib:${LD_LIBRARY_PATH:-}"

# --- MKL ---

if [ -n "${MKLROOT:-}" ] && [ -d "${MKLROOT}/lib/intel64" ]; then
export LD_LIBRARY_PATH="${MKLROOT}/lib/intel64:${LD_LIBRARY_PATH:-}"
fi

# --- GCC 14 runtime: libstdc++ and libgcc_s ---

# Required to avoid GLIBCXX and CXXABI runtime errors.

if command -v g++ >/dev/null 2>&1; then
GCC_LIBDIR="$(dirname "$(g++ -print-file-name=libstdc++.so.6)")"
GCC_GCCLIBDIR="$(dirname "$(g++ -print-file-name=libgcc_s.so.1)")"

if [ -d "$GCC_LIBDIR" ]; then
export LD_LIBRARY_PATH="$GCC_LIBDIR:${LD_LIBRARY_PATH:-}"
fi
if [ -d "$GCC_GCCLIBDIR" ]; then
export LD_LIBRARY_PATH="$GCC_GCCLIBDIR:${LD_LIBRARY_PATH:-}"
```

```
fi
fi

# --- History settings ---

HISTSIZE=1000000
HISTFILESIZE=2000000
HISTCONTROL=ignoredups:erasedups
HISTIGNORE="ls:bg:fg:history"
shopt -s histappend

# Save history after every command.

PROMPT_COMMAND="history -a; history -c; history -r; $PROMPT_COMMAND"
```

After reloading the shell, verify that the modules have been loaded correctly:

```
ml
```

The output should match the module list shown above.

G.2 Building the Solver

First, clone the EMDA repository. Create a project directory if desired, then run:

```
git clone [git@github.com](mailto:git@github.com):dfvankomen/emda-gr.git
cd emda-gr
```

CMake will automatically fetch `dendrolib` during configuration.

Create and enter a build directory:

```
mkdir build
cd build
```

Configure the project using:

```
ccmake ..
```

CMake will locate the required libraries and open an interactive configuration interface. The main settings that may need to be adjusted are the system of equations and the target CPU architecture.

The available equation systems include EMDA, EMD, and EMGR. For a first test run, choose the EMDA system.

The CPU architecture should match the compute nodes on which the job will run. On MaryLou, the `physics2` and `m12` partitions use 64-core AMD EPYC 7763 processors. These are Zen 3 CPUs, so set

```
CPU_ARCH = znver3
```

After configuration is complete, generate the build files from within `ccmake`. Then compile the code using:

```
make -j
```

G.2.1 Build outputs

Once the build completes, the `emdaSolver` and `tpid` binaries should be located in the solver build directory, for example:

```
emda-gr/build/emdaSolver/
```

At this point, the solver is ready to be used for simulations.

G.3 Runs and Job Scripts

After building the solver, move from the home directory to the scratch directory. The location of the scratch directory depends on the supercomputer. On MaryLou, the scratch area is located under `/nobackup/autodelete/usr/`.

First, change into your scratch area:

```
cd /nobackup/autodelete/usr/YOURUSERNAME/
```

Create a directory for the EMDA runs and enter it:

```
mkdir EMDA_runs
cd EMDA_runs
```

Next, copy the solver binaries and a parameter file into the run directory. In the commands below, replace `~/path/to/` with the location of your local `emda-gr` checkout.

Copy `emdaSolver`:

```
cp ~/path/to/emda-gr/build/emdaSolver/emdaSolver ./
```

Copy `tpid`:

```
cp ~/path/to/emda-gr/build/emdaSolver/tpid ./
```

Copy the parameter file:

```
cp ~/path/to/emda-gr/emda_simplified.param.toml ./
```

Verify that everything copied correctly:

```
ls
```

You should see:

```
emda_simplified.param.toml  emdaSolver  tpid
```

Now create the standard output directories for the run:

```
mkdir cp prof vtu bah
```

These directories store the main simulation outputs:

- `cp`: checkpoint files, which allow the run to be restarted
- `prof`: waveform extraction, ADM quantities, charges, and constraint outputs
- `vtu`: visualization files that can be opened in ParaView
- `bah`: apparent-horizon data

A later section will describe the most important parameter-file settings needed to control these outputs.

G.3.1 Submitting a run with slurm

To run the simulation on the cluster, create a Slurm job script. For example:

```
vim Job.sb
```

Paste the following into the file:

```
#!/bin/bash
#SBATCH --time=72:00:00
#SBATCH --ntasks=256
#SBATCH --nodes=2
```

```
#SBATCH --mem-per-cpu=4096M
#SBATCH -J EMDA_RUN
#SBATCH --mail-user=[youremail@email.com](mailto:youremail@email.com)
#SBATCH --mail-type=END,FAIL
#SBATCH --partition=m12

echo "Starting run"

source ~/.bash_profile

./tpid emda_simplified.param.toml 256 &> TPID.txt
mpirun -np 256 ./emdaSolver emda_simplified.param.toml &>> Inspiral.txt

echo "Run completed"
```

The `tpid` command generates the initial data, while `emdaSolver` evolves that data forward in time. Since these commands are written on separate lines and the `tpid` command is not placed in the background, the script waits for `tpid` to finish before starting `emdaSolver`. This ordering is necessary because `tpid` produces the binary initial-data file, `rit_q2_tpid_sol.bin`, which is then read by `emdaSolver`.

On the `m12` partition, each node has 128 CPU cores. Therefore, a run using 256 MPI tasks typically uses 2 nodes. If the number of MPI tasks is changed, the number of requested nodes should be adjusted accordingly.

Submit the job with:

```
sbatch Job.sb
```

Once submitted, the job will enter the queue and begin running when resources become available. You can check the status of your jobs with:

```
squeue | grep yourusername
```

After the job starts, you can monitor the output by inspecting the text files generated during execution. For example:

```
cat TPID.txt
```

or

```
tail -f Inspiral.txt
```

At this point, the simulation is running.

G.4 Parameter-File Explanation

The parameter file is read by both `tpid` and `emdaSolver`. It allows many run settings to be changed without rebuilding the code, which makes it easy to adjust output frequencies, grid settings, checkpointing behavior, and initial-data parameters.

This section summarizes some of the most important parameters. Parameters not discussed here can usually be left at their default values.

```
# Whether or not the solver should attempt to restore from a checkpoint

EMDA_RESTORE_SOLVER = false
```

This option determines whether `emdaSolver` should attempt to restart from an existing checkpoint. For a fresh run, this should be set to `false`. To continue a previous run from checkpoint data, set this to `true`.

```
EMDA_IO_OUTPUT_FREQ = 400
```

This controls how often VTU output is written. Smaller values produce output more frequently, but they also increase storage usage. This parameter should be chosen based on how much visualization data is needed.

```
EMDA_REMESH_TEST_FREQ = 200
```

This sets how often the solver invokes the adaptive mesh refinement algorithm to determine whether the grid should be refined or coarsened. If this value is too small, remeshing can occur too frequently and may introduce unnecessary grid noise. If it is too large, the grid may not adapt quickly enough to resolve the dynamics. A value around `200` has worked well in typical EMDA runs.

```
EMDA_GW_EXTRACT_FREQ = 25
```

This controls how often gravitational-wave quantities are extracted. The same output step also includes other useful quantities, such as black hole positions and masses. This output is not especially expensive compared with VTU output, so values in the range `25–100` are usually reasonable.

```
EMDA_CHECKPT_FREQ = 100
```

This specifies the number of time steps between checkpoints. Checkpointing is relatively inexpensive compared with full visualization output, so this value can be kept fairly small. More frequent checkpoints make it easier to restart a run after a crash or wall-time limit.

```
# VTU file prefix

EMDA_VTU_FILE_PREFIX = "vtu/emda_gr"

# Checkpoint file prefix

EMDA_CHKPT_FILE_PREFIX = "cp/emda_cp"

# Profiling file prefix

EMDA_PROFILE_FILE_PREFIX = "prof/emda_prof"
```

These options control where different types of output data are written. The `vtu` directory stores visualization output, the `cp` directory stores checkpoint files, and the `prof` directory stores extracted waveforms and other reduced data products.

```
# Which evolution variables to output

EMDA_VTU_OUTPUT_EVOL_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,
25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35]

# Number of evolution variables to output

EMDA_NUM_EVOL_VARS_VTU_OUTPUT = 36
```

These settings determine which evolution variables are written to the VTU files. These variables include quantities such as the metric components, extrinsic-curvature components, gauge variables, and matter fields. If storage usage becomes a concern, some variables may be omitted from the VTU output.

```
# Which constraint variables to output

EMDA_VTU_OUTPUT_CONST_INDICES = [0, 1, 2, 3, 4, 5, 6, 7,
8, 9, 10, 11, 12, 13]

# Number of constraint variables to output

EMDA_NUM_CONST_VARS_VTU_OUTPUT = 14
```

These options are analogous to the evolution-variable output settings, but they apply to constraint quantities. These include quantities such as the Hamiltonian constraint, the momentum constraints, and additional constraints associated with the electromagnetic sector.

```
# Whether to isolate an X slice

EMDA_VTU_X_SLICE = false

# Whether to isolate a Y slice

EMDA_VTU_Y_SLICE = false
```

```
# Whether to isolate a Z slice

EMDA_VTU_Z_SLICE = true

# Any combination may be set to true
```

Writing VTU output for the full 3-D grid can be expensive, so it is often preferable to output a 2-D slice instead. For most visualization purposes, a single slice is sufficient. In typical runs, the Z slice is a good default choice.

Although multiple slice options can be enabled at the same time, doing so may produce output that is less straightforward to interpret. For simple visualization, enable only one slice direction at a time.

```
EMDA_DENDRO_GRAIN_SZ = 50
```

This controls the approximate number of grid blocks assigned to each processor. In practice, this parameter should usually be kept between `25` and `200`. Very small or very large values may reduce parallel efficiency.

```
EMDA_DENDRO_AMR_FAC = 0.2
```

This controls the sensitivity of the adaptive mesh refinement algorithm. Smaller values generally make the grid more sensitive to refinement, while larger values make refinement less aggressive. This parameter should be adjusted with care, since it affects both accuracy and computational cost.

```
EMDA_ELE_ORDER = 6
```

This sets the finite-element order used by the Dendro infrastructure. Loosely speaking, it controls how much information each block must exchange with its neighbors. Higher values increase communication costs and may slow the code. For the current EMDA implementation, this should generally be left at 6.

```
# Minimum depth of the mesh/octree

EMDA_MINDEPTH = 4

# Maximum depth of the mesh/octree

EMDA_MAXDEPTH = 15
```

These parameters control the minimum and maximum refinement levels of the octree mesh. The minimum depth sets the coarsest allowed grid resolution, while the maximum depth sets the finest allowed grid resolution. Increasing `EMDA_MAXDEPTH` improves the maximum available resolution, but it also increases memory usage and computational cost.

The CFL condition is set by the finest level of the grid, so increasing `EMDA_MAXDEPTH` can significantly increase the runtime. Each increase in `EMDA_MAXDEPTH` halves the grid spacing on the finest level. As a rule of thumb, each black hole should be resolved by at least roughly 50 points across the horizon. For an equal-mass binary with total mass $M = 1$ on a ± 400 grid, `EMDA_MAXDEPTH = 15` is usually adequate. For larger mass ratios, the smaller black hole requires additional refinement. For example, a $q = 2$ binary typically requires `EMDA_MAXDEPTH = 16`.

```
########################################################################
```

```
# REFINEMENT STRATEGY

# Which refinement mode to use

# 0 - WAMR

# 1 - EH

# 2 - EH_WAMR

# 3 - BH_LOC

# 4 - BH_WAMR

EMDA_REFINEMENT_MODE = 4
```

This parameter controls the refinement strategy used by the code. The recommended choice is currently `EMDA_REFINEMENT_MODE = 4`, which uses black hole-centered refinement together with wavelet adaptive mesh refinement.

```
EMDA_AMR_R_RATIO = 1.618033988749895
```

This parameter determines how quickly the forced refinement around each black hole falls off with radius. The default value is reasonable and should usually be left unchanged for initial runs.

```
# Wavelet Tolerance Function to use (see grUtils.cpp for adding more)
```

```
# 1,2 - deprecated

# 3 - center highly refined; low refinement elsewhere eases over time

# 4 - start with refinement centered at BHs, then widen to r=0 focused

# 5 - focus refinement in self-similar ways about each BH

# 6 - refine well in r <= t and coarsely in r > t

EMDA_USE_WAVELET_TOL_FUNCTION = 6
```

This parameter controls how the wavelet refinement tolerance varies across the computational domain. Function 6, developed by William Black, has worked well in practice and is the recommended choice for current EMDA runs. More details can be found in Ref. [30].

```
# Minimum wavelet tolerance

EMDA_WAVELET_TOL = 0.00001

# Gravitational wave refinement tolerance

EMDA_GW_REFINE_WTOL = 0.0001

# Maximum wavelet tolerance
```

```
EMDA_WAVELET_TOL_MAX = 0.001

# Wavelet tolerance radii (for shell-based tolerance)

EMDA_WAVELET_TOL_FUNCTION_R0 = 3.0
EMDA_WAVELET_TOL_FUNCTION_R1 = 12.0
```

The minimum wavelet tolerance, `EMDA_WAVELET_TOL`, is the most important of these parameters because it sets the strictest refinement criterion used by the grid. Smaller values force more refinement and generally improve accuracy, but they also increase memory usage and runtime. The remaining parameters depend on the chosen wavelet tolerance function. The default values are reasonable starting points, but they may be adjusted for higher-resolution runs or for runs in which the wave-extraction region requires additional refinement.

```
# Indices of variables used for refinement (see grDef.h for definitions)

EMDA_REFINE_VARIABLE_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,
25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35]

# Number of variables used for refinement

EMDA_NUM_REFINE_VARS = 36
```

These parameters specify which evolution variables are used when deciding whether the grid should be refined. It is not always necessary to use every evolution variable as a refinement variable, but the full list is a safe default for general-purpose EMDA runs.

```
# CFL (Courant--Friedrichs--Lewy) factor

EMDA_CFL_FACTOR = 0.25

# Simulation start time

EMDA_RK_TIME_BEGIN = 0

# Simulation end time

EMDA_RK_TIME_END = 1000
```

These parameters control the CFL factor and the total coordinate time of the simulation. The CFL factor sets the time step relative to the grid spacing on the finest refinement level. The parameters `EMDA_RK_TIME_BEGIN` and `EMDA_RK_TIME_END` determine the start and end times of the evolution. The remaining Runge–Kutta settings can usually be left at their default values.

```
############################################

# EMDA THEORY PARAMETERS

############################################
EMDA_LAMBDA = [1, 1, 1, 1]
EMDA_ALPHA_THEORY = [1.0, 1.0]
EMDA_LF = [1.0, 0.0]
EMDA_P_EXPO = -1.0
```

```
EMDA_ALPHA_POS_NEG_FLOOR = false

EMDA_ETA = [0.4, 0.4]

EMDA_XI = [0, 0, 0]

CHI_FLOOR = 0.00001

EMDA_TRK0 = 0.0

EMDA_ETA_R0 = 1.31

EMDA_ETA_POWER = [2.0, 2.0]

EMDA_ETADAMP = 1.0


ETA_CONST = 2.0

ETA_R0 = 30.0

ETA_DAMPING = 1.0

ETA_DAMPING_EXP = 1.0
```

These parameters define the theory and gauge choices used in the evolution. The main parameters that are typically changed are the two theory couplings α_0 and α_1, stored in `EMDA_ALPHA_THEORY`. The choices

$$(\alpha_0, \alpha_1) = (1, 1)$$

correspond to EMDA,

$$(\alpha_0, \alpha_1) = (1, 0)$$

correspond to EMD, and

$$(\alpha_0, \alpha_1) = (0, 0)$$

correspond to Einstein–Maxwell. The remaining gauge parameters are generally safe to leave at their default values.

```
# Kreiss--Oliger dissipation parameter

KO_DISS_SIGMA = 0.4
```

This parameter controls the strength of Kreiss–Oliger numerical dissipation. Some dissipation is needed for stability, and `KO_DISS_SIGMA = 0.4` is a reasonable starting value.

```
#########################################################################

# Select initial data type

# See the bottom of the file for reference

EMDA_ID_TYPE = 0
```

This parameter selects the type of initial data used by the evolution. The choice `EMDA_ID_TYPE = 0` uses the TPID solver. Other black hole initial-data options are available, but TPID is the standard choice for the binary black hole simulations described here.

```
#########################################################################

# Set up EMDA grid extents

EMDA_GRID_MIN_X = -400.0
```

```
EMDA_GRID_MAX_X = 400.0

EMDA_GRID_MIN_Y = -400.0

EMDA_GRID_MAX_Y = 400.0

EMDA_GRID_MIN_Z = -400.0

EMDA_GRID_MAX_Z = 400.0
```

These parameters define the extent of the computational domain. The values shown here are reasonable defaults for many binary simulations. The outer boundary should be placed far enough from the wave-extraction region that boundary reflections do not contaminate the extracted waveforms during the time interval of interest.

```
#########################################################################


# Black hole parameters


EMDA_BH1_AMR_R = 1.0

EMDA_BH1_CONSTRAINT_R = 2.0

EMDA_BH1_MAX_LEV = 15

EMDA_BH1 = {MASS = 0.5, X = 5.0, Y = 0.0, Z = 0.0,

V_X = -0.00100282, V_Y = 0.095775311, V_Z = 0.0,

SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,

ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}


EMDA_BH2_AMR_R = 1.0

EMDA_BH2_CONSTRAINT_R = 2.0

EMDA_BH2_MAX_LEV = 15

EMDA_BH2 = {MASS = 0.5, X = -5.0, Y = 0.0, Z = 0.0,
```

```
V_X = 0.00100282, V_Y = -0.095775311, V_Z = 0.0,
SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,
ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}
```

These parameters define the black hole properties used by the evolution and by the refinement infrastructure. The parameter EMDA_BH1_AMR_R sets the radius within which refinement is forced around the first black hole, while EMDA_BH1_CONSTRAINT_R sets the radius excluded when computing constraint norms. The corresponding BH2 parameters play the same role for the second black hole. The parameters EMDA_BH1_MAX_LEV and EMDA_BH2_MAX_LEV set the maximum refinement level allowed near each black hole.

The X, Y, and Z entries do not directly determine the black holes' initial positions in the TPID solve. Instead, they specify where the evolution code begins searching for the black hole locations and where it centers the black hole refinement regions. For equal-mass binaries, these values should match the corresponding $\pm b$ positions used in the TPID parameters.

The electric charge should remain below the black hole mass in order to avoid exceeding the extremal limit. Magnetic charge is still under development and should not be used for standard production runs. The spin parameter corresponds to the dimensionless spin χ, with angular momentum $S = \chi m^2$.

TPID reads the charge and spin from these black hole parameter blocks, but it does not use the listed mass or position values in the same way as the evolution code. The TPID mass and separation are set separately by the TPID parameters described below.

Finally, note that V_X, V_Y, and V_Z represent the black hole momenta, not their coordinate velocities. This distinction is important when setting up binary initial data.

```
#########################################################################


# GRAVITATIONAL WAVE EXTRACTION PARAMETERS
```

```
EMDA_GW_NUM_RADAII = 6
EMDA_GW_RADAII = [50.0, 60.0, 70.0, 80.0, 90.0, 100.0]

EMDA_GW_NUM_LMODES = 7
EMDA_GW_L_MODES = [2, 3, 4, 5, 6, 7, 8]

# Radius at which conserved quantities are extracted

EMDA_GW_EXTRACTION_RADIUS = 300.0

# Modes for EM, dilaton, and axion radiation

EMDA_RAD_NUM_LMODES = 6
EMDA_RAD_L_MODES = [1, 2, 3, 4, 5, 6]
########################################################################
```

These parameters control the extraction of gravitational, electromagnetic, dilaton, and axion radiation. Radiation is extracted on spherical surfaces at the radii listed in `EMDA_GW_RADAII`. The gravitational-wave modes are set by `EMDA_GW_L_MODES`, while the modes for the electromagnetic, dilaton, and axion fields are set by `EMDA_RAD_L_MODES`.

The mode numbers correspond to the spherical-harmonic multipoles being extracted. Although modes can be extracted up to roughly $\ell \approx 10$, the dominant contributions are usually contained in the lower modes, especially $\ell \leq 4$.

```
# Time steps between apparent-horizon (AEH) solves
AEH_SOLVER_FREQ = 50
```

This parameter controls how often the apparent-horizon solver is called. The remaining AEH-related settings should generally be left at their default values unless the apparent-horizon finder fails or additional horizon information is needed.

```
TPID_PAR_B = 5.0
TPID_TARGET_M_PLUS  = 0.48594
TPID_TARGET_M_MINUS = 0.48594
```

The parameter `TPID_PAR_B` sets the coordinate half-separation used by TPID. For example, `TPID_PAR_B = 5.0` corresponds to punctures located near $x = \pm 5$. The target masses are the values used internally by TPID when constructing the initial data.

```
TPID_CENTER_OFFSET = {X = 0.0, Y = 0.0, Z = 0.0}
```

For equal-mass binaries, this should usually be left at zero. For unequal-mass binaries, the system may need to be recentered so that the center of mass is located at the origin.

```
TPID_NPOINTS_A   = 40
TPID_NPOINTS_B   = 40
TPID_NPOINTS_PHI = 20
```

These parameters control the resolution of the TPID grid. The values shown here are reasonable defaults for a q=1 black hole, but higher values may be needed for more accurate initial data and higher mass ratios.

```
TPID_GIVE_BARE_MASS = 1
```

This parameter determines whether the puncture masses are fixed or solved for. When `TPID_GIVE_BARE_MASS = 0`, TPID solves for the bare masses. When `TPID_GIVE_BARE_MASS = 1`, TPID uses the specified target masses.

A useful procedure is to first estimate the desired masses, run TPID with mass solving enabled, then update the target masses and rerun with `TPID_GIVE_BARE_MASS = 1` to finalize the initial data.

```
TPID_REPLACE_LAPSE_WITH_SQRT_CHI = true
```

This option should be enabled when using the slow-start lapse.

```
# Include dilatonic contributions in rho during TPID solve
TPID_USE_DILATON = true

# Interpolated initial data should not be used with TPID_USE_DILATON
# You should also be using the EMD equations in this case
TPID_USE_INTERPOLATED_SOLUTION = false
```

These parameters control how scalar-field contributions are included in the initial-data solve. If both options are set to `false`, TPID generates Einstein–Maxwell initial data. Setting `TPID_USE_INTERPOLATED_SO` `= true` produces EMD-type initial data. Setting `TPID_USE_DILATON = true` includes dilatonic contributions directly in the TPID solve.

The initial-data choice should be consistent with both the theory parameters in `EMDA_ALPHA_THEORY` and the system of equations selected when the code was configured with CMake.

Appendix H

ParaView

This chapter provides a brief introduction to several basic ParaView workflows used to inspect Dendro simulation output. We begin by loading simulation files, then describe how to create slices, visualize fields, use the Calculator filter, adjust color maps, export data, and locate regions of interest.

H.1 Opening Simulation Data

To load data into ParaView, click the **Open File** button in the top-left toolbar. This button is highlighted in Fig. **??**.

After clicking the **Open File** button, navigate to the directory where the simulation output is saved. Dendro commonly writes visualization data in `.vtu` format, which stands for *VTK Unstructured Grid*. For parallel runs, the output is often accompanied by a `.pvtu` file. The `.pvtu` file acts as a wrapper that tells ParaView how the individual `.vtu` pieces fit together.

In most cases, one should open the `.pvtu` file rather than opening the individual `.vtu` files by hand. This allows ParaView to load the full parallel dataset as a single object.

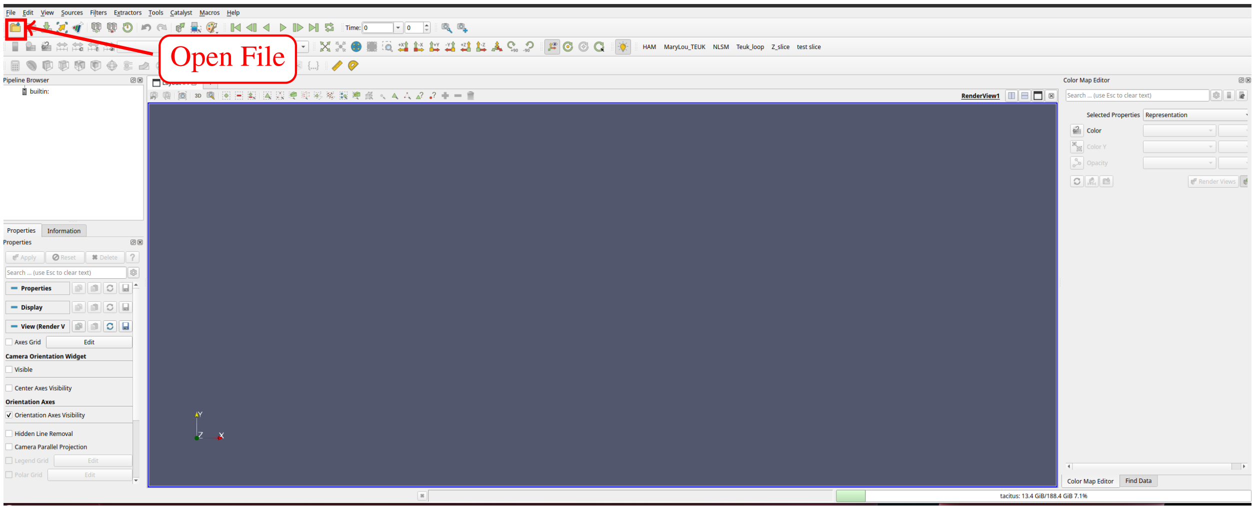


Figure H.1 The ParaView interface with the **Open File** icon highlighted.

H.2 Creating a Slice

Once the data has been loaded, a common operation is to create a two-dimensional slice through the three-dimensional dataset. In this example, the **Slice** tool is used to create a slice through the data, as shown in Fig. **??**.

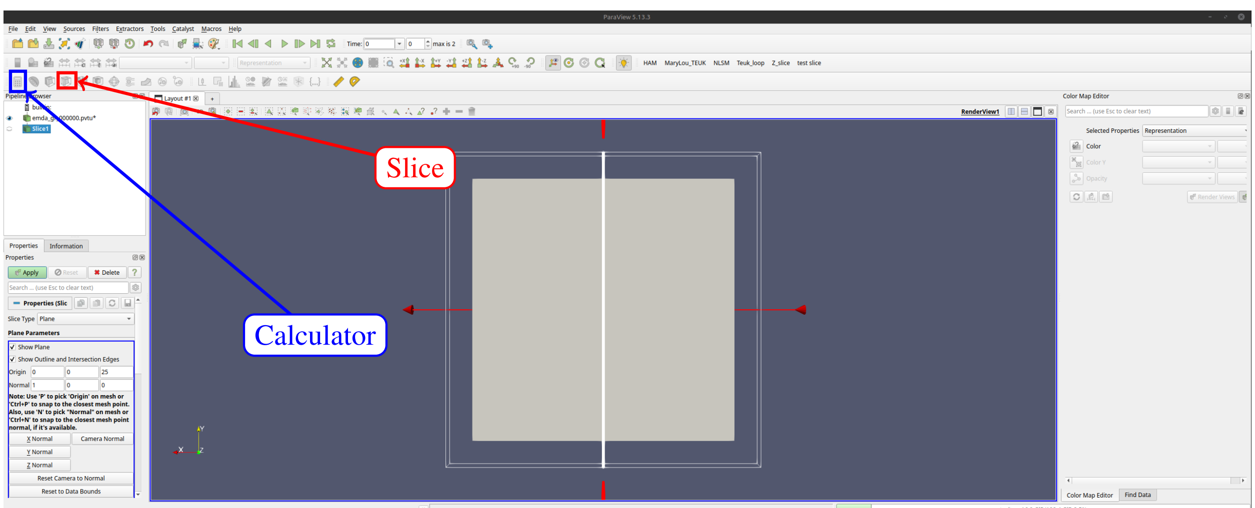


Figure H.2 The ParaView interface with the **Slice** and **Calculator** tools highlighted.

After selecting the **Slice** tool, ParaView displays the slice settings in the Properties panel. These settings determine the location and orientation of the slicing plane.

H.3 Adjusting the Slice and Viewing Data

ParaView allows the user to choose how the data should be sliced. Often, we want to create z-slices, since this is the default orientation used for many Dendro visualizations. To do this, set the slice normal to $(0,0,1)$. We also set the origin to $(0,0,10^{-5})$. The small nonzero z-value is used because setting the slice exactly at $z = 0$ can sometimes cause ParaView to miss cells lying on the symmetry plane.

The **Show Plane** option should be turned off so that only the data slice is displayed. After changing these settings, click **Apply**.

Once the slice is displayed, the visualized field can be changed using the field-selection menu. The available time steps can be explored using the time controls, and the representation menu can be used to switch between viewing modes such as **Surface** and **Wireframe**. These controls are highlighted in Fig. **??**.

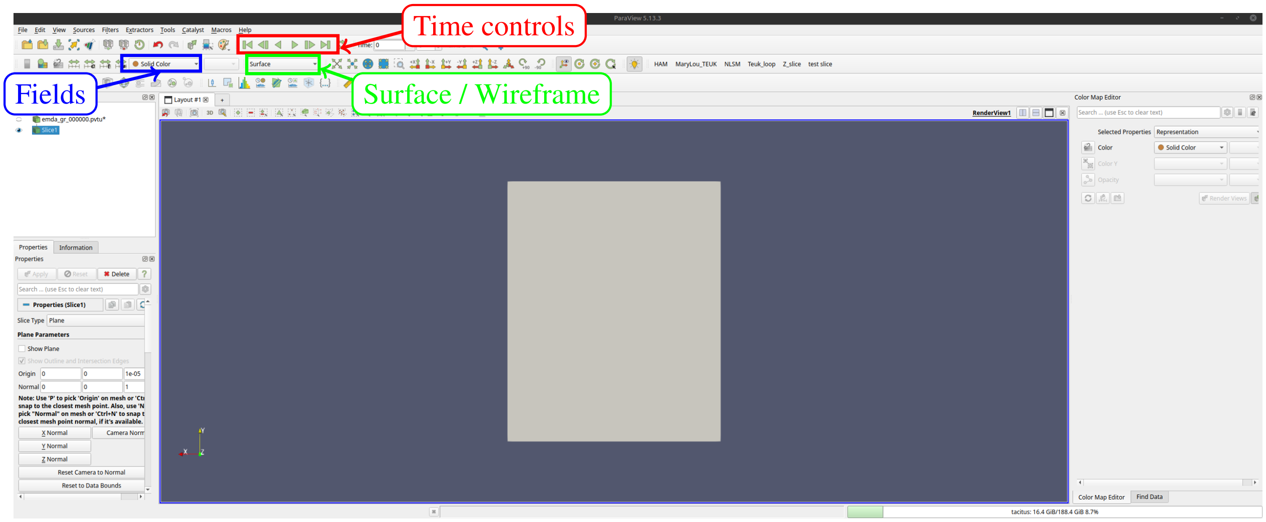


Figure H.3 ParaView controls for stepping through time, selecting the displayed field or color option, and switching between representations such as **Surface** and **Wireframe**.

H.4 Using the Calculator

The **Calculator** filter can be used to construct derived quantities from the fields stored in the simulation output. After selecting the loaded dataset or slice, click the **Calculator** button. In the Calculator panel, choose the appropriate scalar or vector fields from the available data arrays. Mathematical operations can then be applied to these fields to define a new quantity for visualization or export.

For example, the Calculator can be used to rescale a field, combine multiple fields, or construct diagnostic quantities that are not stored directly in the output files. After entering the desired expression, click **Apply** to create the new field.

H.5 Warping by Scalar

The **Warp By Scalar** filter can be used to displace a surface according to the value of a scalar field. This is useful for visualizing the relative amplitude of a field on a slice.

To use the filter, click **Filters** and search for **Warp By Scalar**. After selecting the filter, its settings appear in the Properties panel, in the same general location as the slice settings.

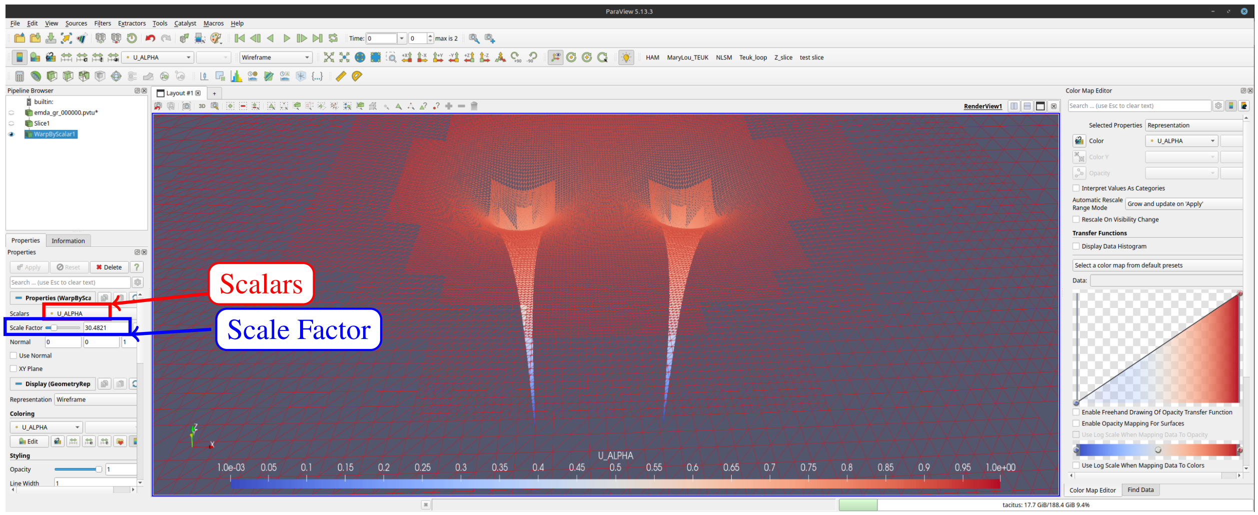


Figure H.4 The **Warp By Scalar** settings in the Properties panel. The **Scalars** menu selects the field used for the warping, and the **Scale Factor** controls the strength of the warp.

In the **Scalars** menu, select the field that should determine the warp. The **Scale Factor** controls the strength of the displacement. Large scale factors can make small-amplitude fields easier to see, but they can also distort the visualization if chosen too large.

H.6 Using the Color Map Editor

The **Color Map Editor** controls how ParaView maps data values to colors. It can be used to change the color scale, adjust the visible data range, modify opacity, and choose a different color preset.

By default, the color mapping may not automatically update when stepping through time. The easiest way to ensure that the colors match the data range at each time step is to rescale the color range when the time step changes. This prevents the visualization from using an outdated color range from a previous time step.

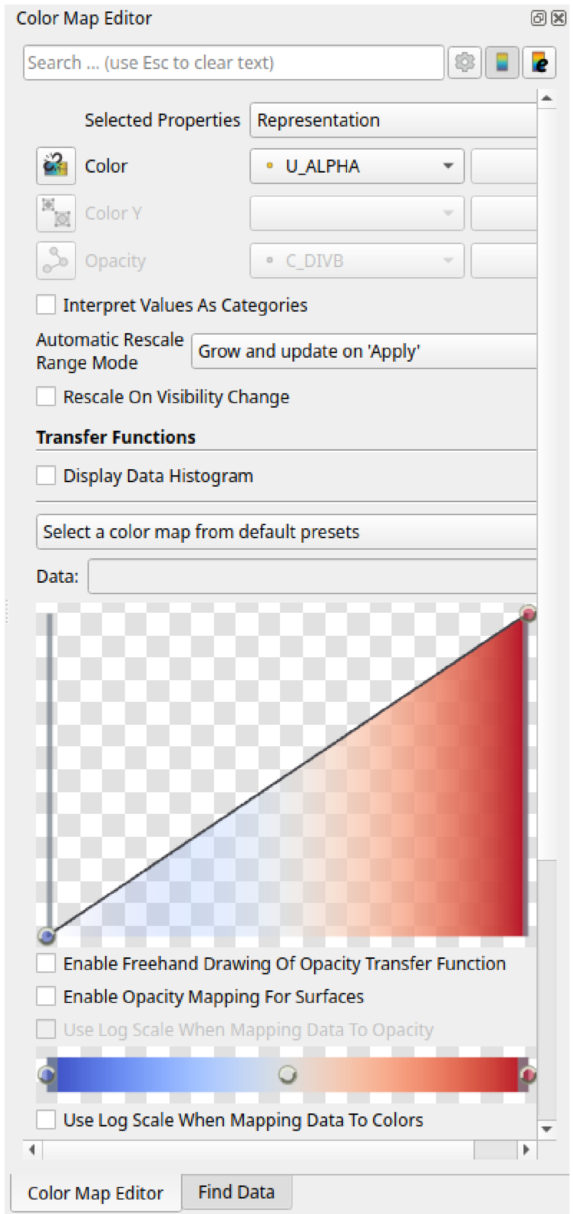


Figure H.5 The **Color Map Editor** panel in ParaView. This panel controls the color scale, opacity, and transfer function used to visualize the selected field.

H.7 Macros, Animations, CSV Output, and Finding Data

Often, we want to repeat the same visualization or post-processing steps for many datasets. ParaView can automate these repetitive tasks using macros. To create a macro, click **Tools** and select **Start Trace**. Then perform the sequence of actions that should be automated. When finished, click **Stop Trace**. ParaView will then allow the recorded trace to be saved as a Python script or macro, which can later be reused.

ParaView can also save the currently loaded data as a CSV file. To do this, click **File** and select **Save Data**. This saves the data from the currently active object, such as the current slice. The resulting CSV file can then be loaded into Python, MATLAB, Mathematica, or another analysis tool.

Animations can be saved by clicking **File** and selecting **Save Animation**. This is useful for producing movies that show the time evolution of a field.

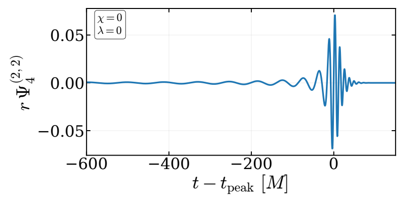
Finally, ParaView can be used to find data that satisfies certain conditions. To do this, click **View** and select **Find Data**. In the **Find Data** panel, choose the dataset to search. The element type can usually be left as **Point**. Then choose the field to search, set the desired condition and value, and click **Find Data**. The matching region will be highlighted in the render view.

Appendix I

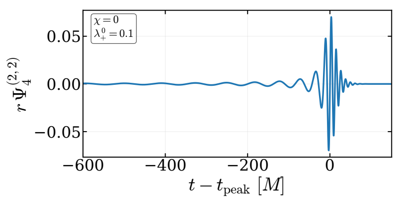
Waveform Catalog

I.1 Gravitational Waveform Catalog

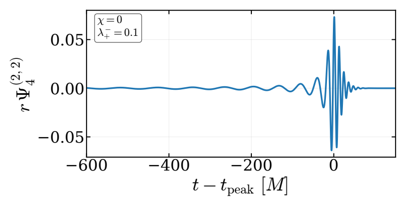
The following figures show the extracted $r\Psi_4^{(2,2)}$ waveforms at $r \simeq 100M$. For each mass ratio and spin, the neutral case is shown above the charged configurations. The charged cases are ordered by charge magnitude and charge configuration.



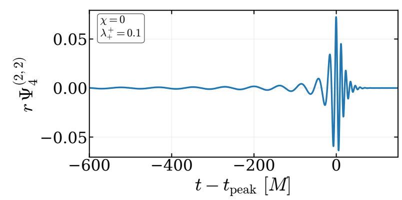
(a) Neutral



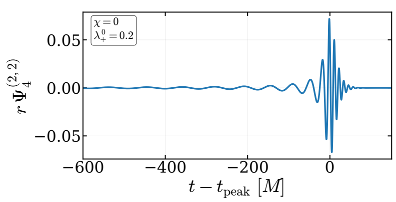
(b) ($\lambda^+_0 = 0.10$)



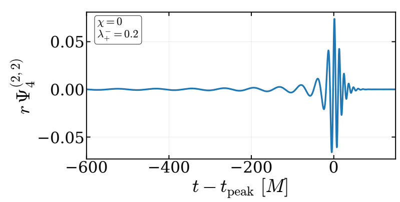
(c) ($\lambda^+ * - = 0.10$)



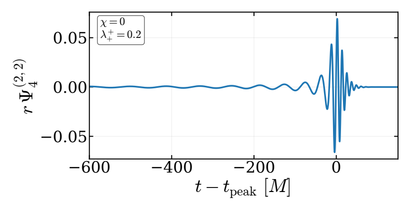
(d) ($\lambda^+_+ = 0.10$)



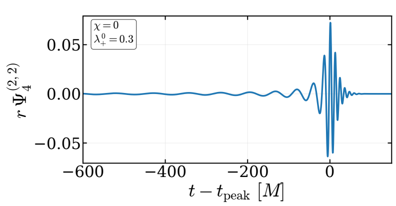
(e) ($\lambda^+_0 = 0.20$)



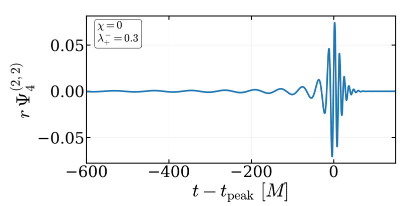
(f) ($\lambda^+ * - = 0.20$)



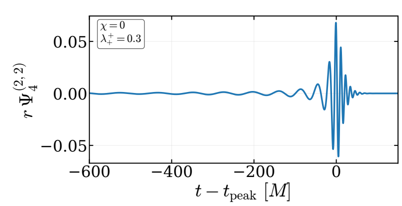
(g) ($\lambda^+_+ = 0.20$)



(h) ($\lambda^+_0 = 0.30$)

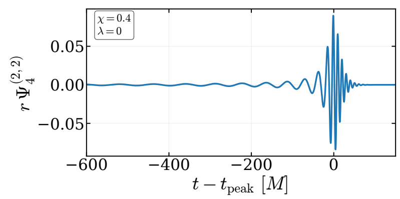


(i) ($\lambda^+ * - = 0.30$)

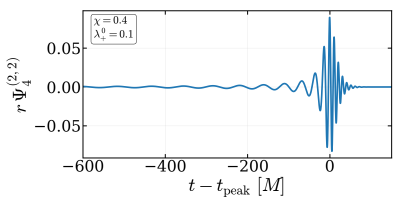


(j) ($\lambda^+_+ = 0.30$)

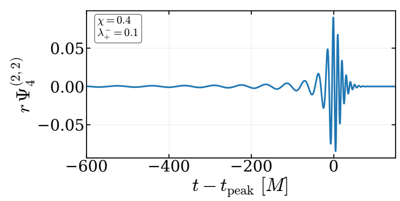
Figure I.1 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 1$ and $\chi = 0$. The neutral case is shown at the top, followed by charged configurations ordered by increasing charge magnitude and charge configuration.



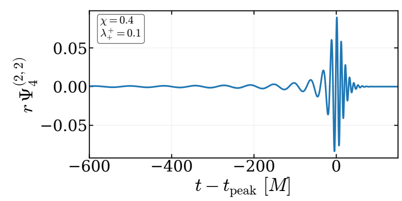
(a) Neutral



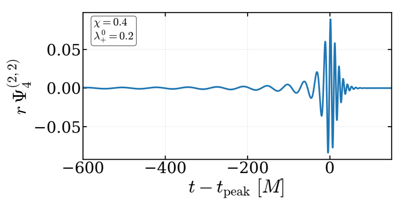
(b) ($\lambda_0^+ = 0.10$)



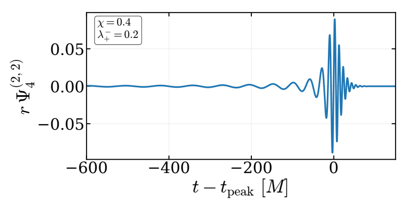
(c) ($\lambda^+ * - = 0.10$)



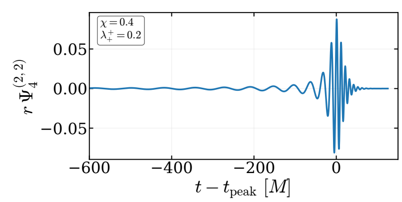
(d) ($\lambda_+^+ = 0.10$)



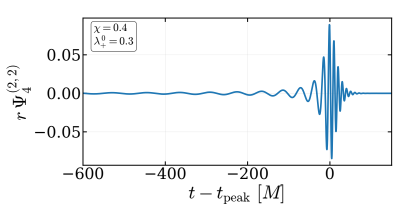
(e) ($\lambda_0^+ = 0.20$)



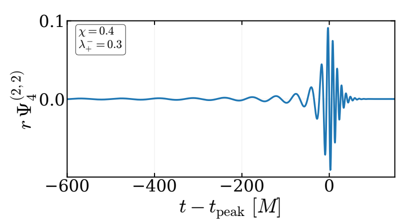
(f) ($\lambda^+ * - = 0.20$)



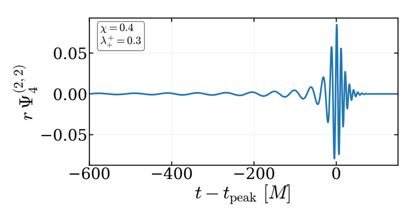
(g) ($\lambda_+^+ = 0.20$)



(h) ($\lambda_0^+ = 0.30$)

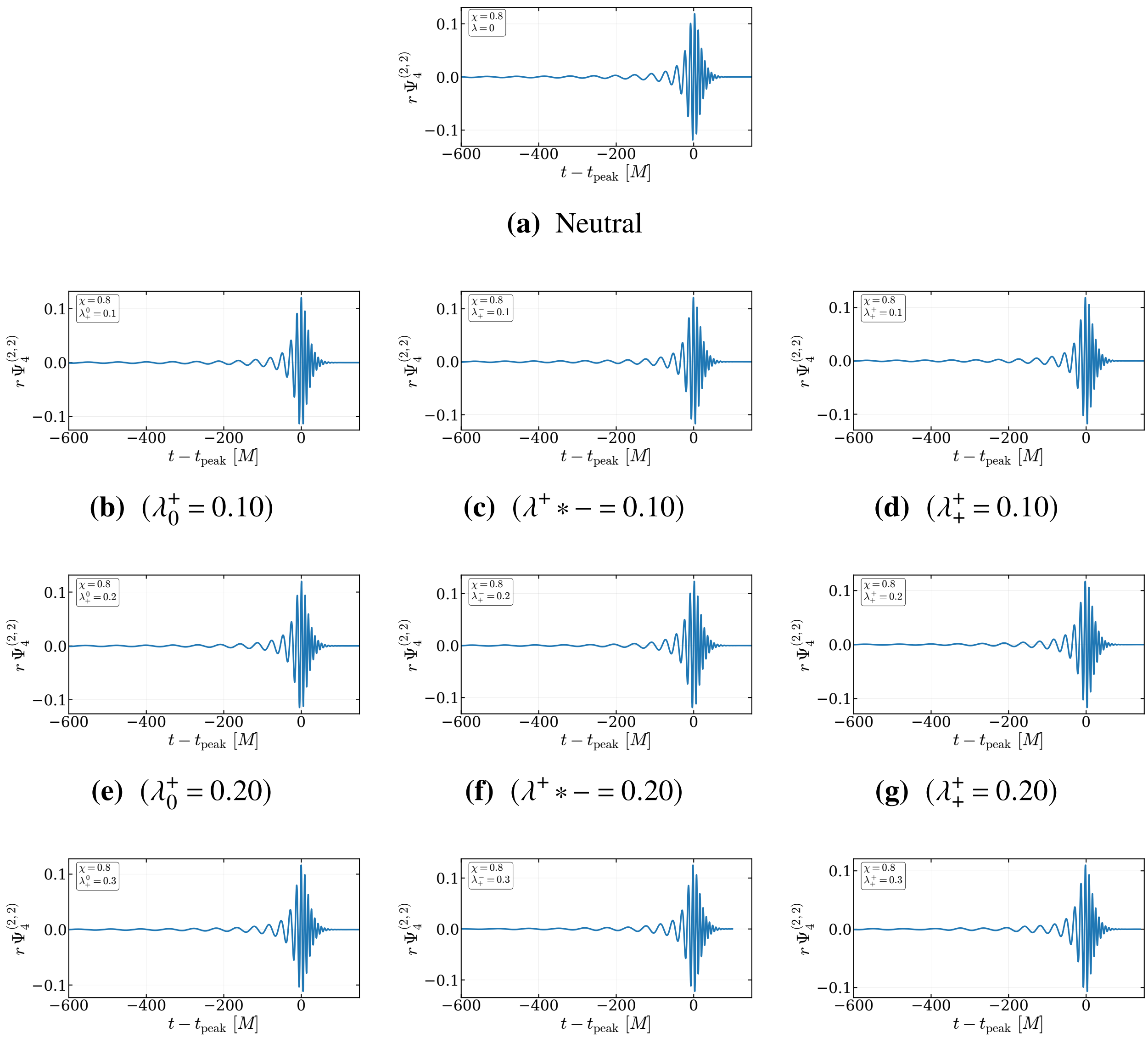


(i) ($\lambda^+ * - = 0.30$)



(j) ($\lambda_+^+ = 0.30$)

Figure I.2 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 1$ and $\chi = 0.4$. The neutral case is shown at the top, followed by charged configurations ordered by increasing charge magnitude and charge configuration.



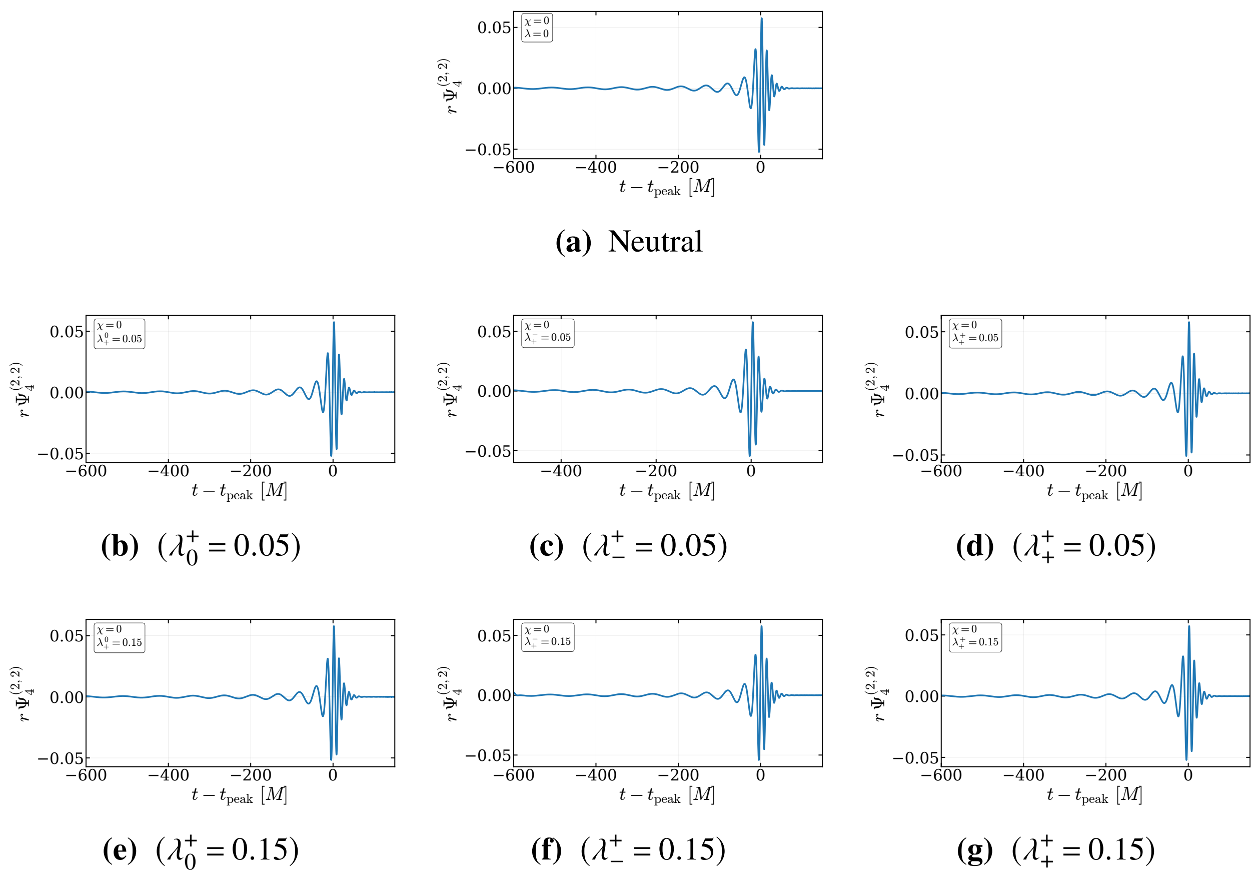
(a) Neutral

(b) $(\lambda_0^+ = 0.10)$ **(c)** $(\lambda^+ * - = 0.10)$ **(d)** $(\lambda_+^+ = 0.10)$

(e) $(\lambda_0^+ = 0.20)$ **(f)** $(\lambda^+ * - = 0.20)$ **(g)** $(\lambda_+^+ = 0.20)$

(h) $(\lambda_0^+ = 0.30)$ **(i)** $(\lambda^+ * - = 0.30)$ **(j)** $(\lambda_+^+ = 0.30)$

Figure I.3 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 1$ and $\chi = 0.8$. The neutral case is shown at the top, followed by charged configurations ordered by increasing charge magnitude and charge configuration.

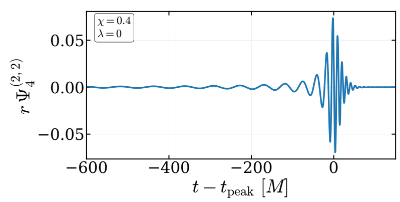


(a) Neutral

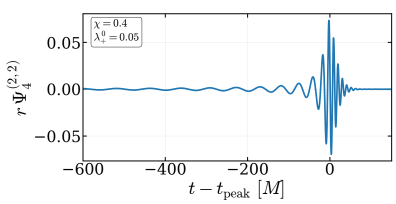
(b) $(\lambda_0^+ = 0.05)$ **(c)** $(\lambda_-^+ = 0.05)$ **(d)** $(\lambda_+^+ = 0.05)$

(e) $(\lambda_0^+ = 0.15)$ **(f)** $(\lambda_-^+ = 0.15)$ **(g)** $(\lambda_+^+ = 0.15)$

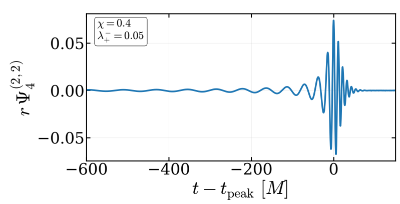
Figure I.4 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 2$ and $\chi = 0$. The neutral case is shown at the top, followed by charged configurations ordered by increasing charge magnitude and charge configuration.



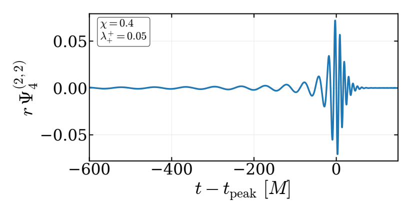
(a) Neutral



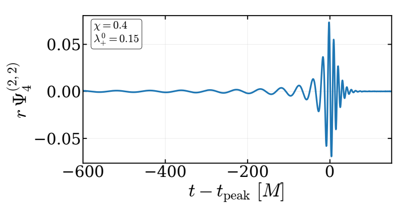
(b) $(\lambda_0^+ = 0.05)$



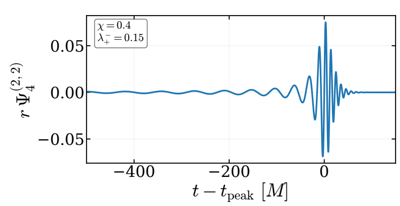
(c) $(\lambda^+ * - = 0.05)$



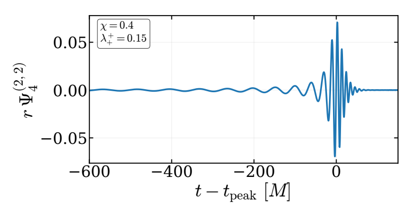
(d) $(\lambda_+^+ = 0.05)$



(e) $(\lambda_0^+ = 0.15)$

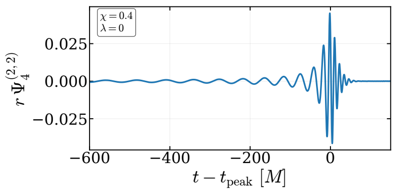


(f) $(\lambda^+ * - = 0.15)$

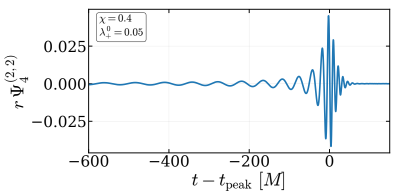


(g) $(\lambda_+^+ = 0.15)$

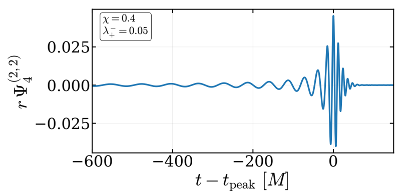
Figure I.5 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 2$ and $\chi = 0.4$. The neutral case is shown at the top, followed by charged configurations ordered by increasing charge magnitude and charge configuration.



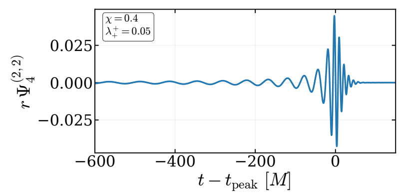
(a) Neutral



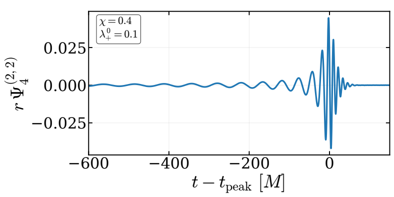
(b) ($\lambda_0^+ = 0.05$)



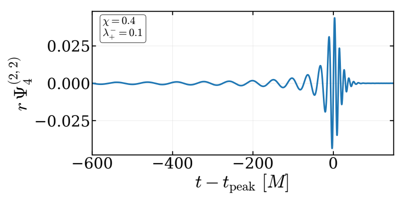
(c) ($\lambda_-^+ = 0.05$)



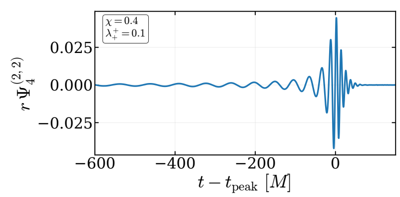
(d) ($\lambda_+^+ = 0.05$)



(e) ($\lambda_0^+ = 0.10$)

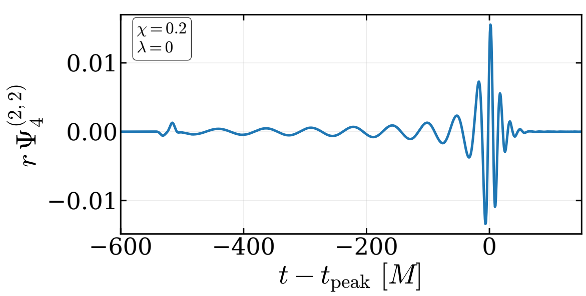


(f) ($\lambda_-^+ = 0.10$)

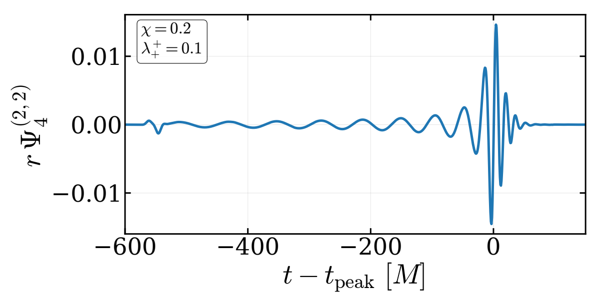


(g) ($\lambda_+^+ = 0.10$)

Figure I.6 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 4$ and $\chi = 0.4$. The neutral case is shown at the top, followed by charged configurations with $|\lambda_i| = 0.05$ and $|\lambda_i| = 0.10$.



(a) Neutral

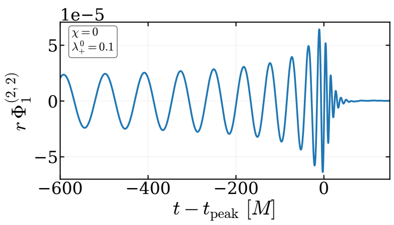


(b) $\lambda_+^+ = 0.10$

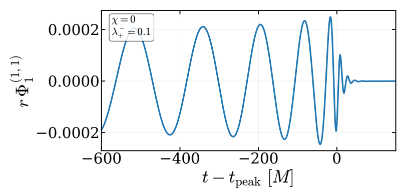
Figure I.7 Extracted $r\Psi_4^{(2,2)}$ waveforms for $q = 8$ and $\chi = 0.2$.

I.2 Φ_1 Waveform Catalog

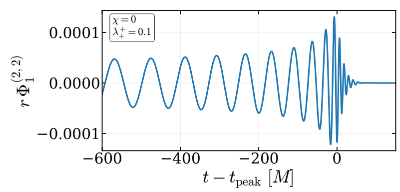
The following figures show the extracted Φ_1 waveforms at $r \simeq 100M$. Since there is no radiation in the neutral case, only charged configurations are shown. The charged cases are ordered by charge magnitude and charge configuration. For the $+-$ cases, we show the $(\ell, m) = (1, 1)$ mode, while for the $+0$ and $++$ cases we show the $(\ell, m) = (2, 2)$ mode.



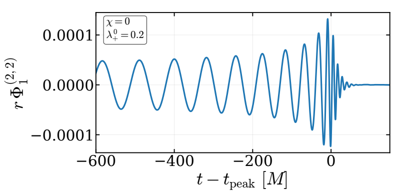
(a) $\lambda_0^+ = 0.10$



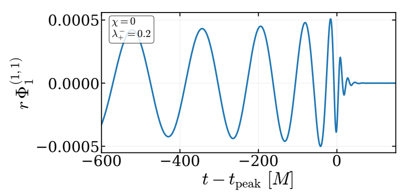
(b) $\lambda_-^+ = 0.10$



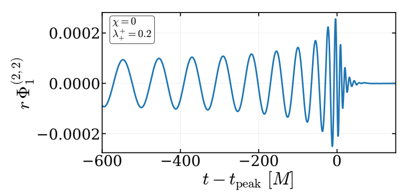
(c) $\lambda_+^+ = 0.10$



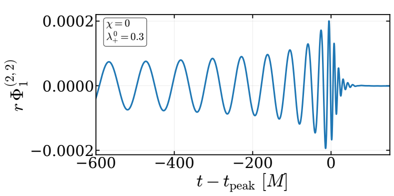
(d) $\lambda_0^+ = 0.20$



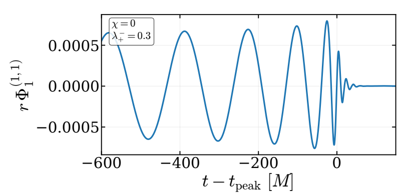
(e) $\lambda_-^+ = 0.20$



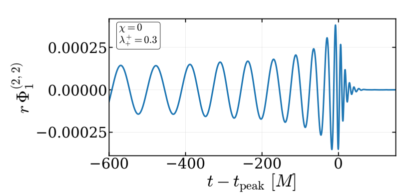
(f) $\lambda_+^+ = 0.20$



(g) $\lambda_0^+ = 0.30$

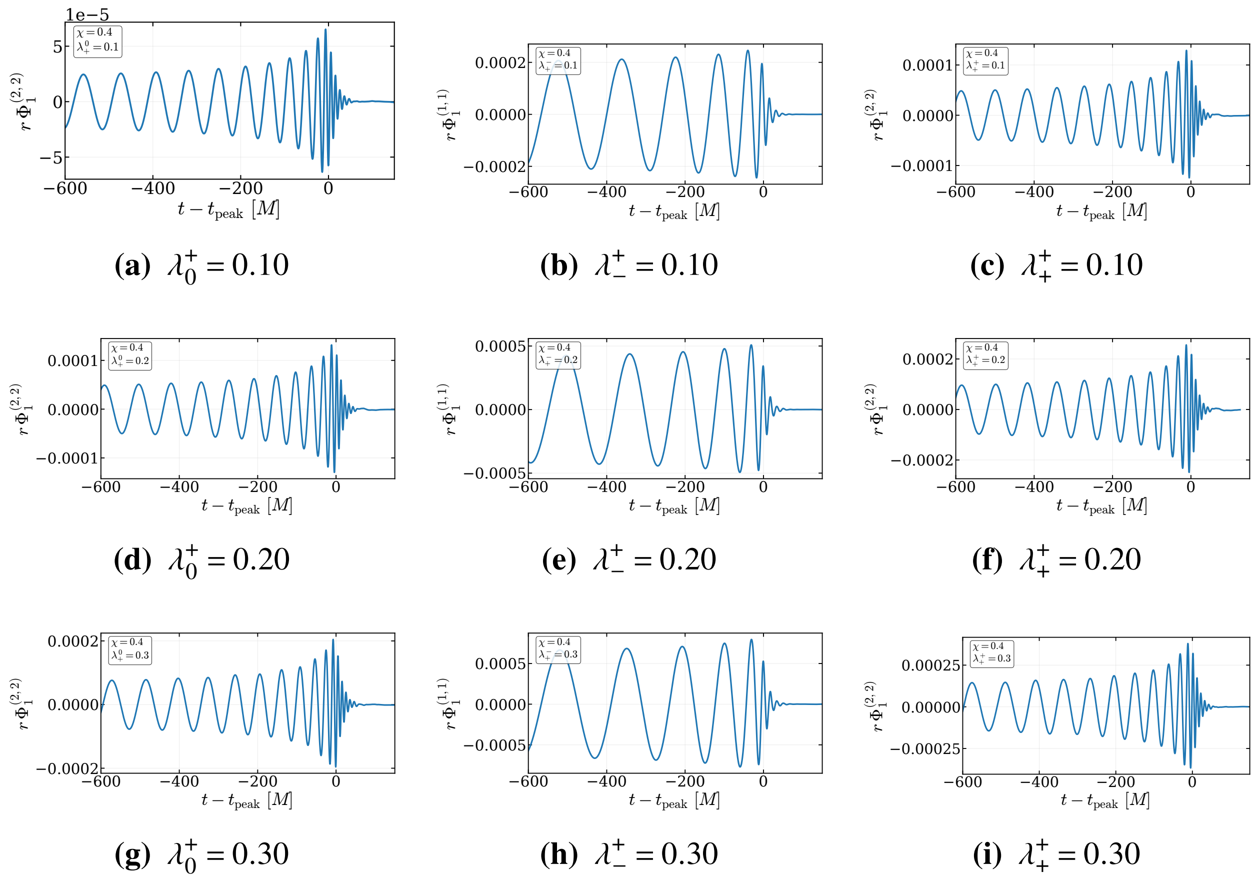


(h) $\lambda_-^+ = 0.30$



(i) $\lambda_+^+ = 0.30$

Figure I.8 Extracted Φ_1 waveforms for $q = 1$ and $\chi = 0$.

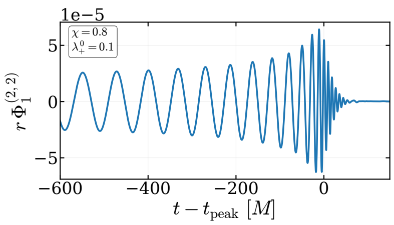


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

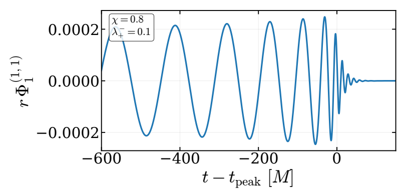
(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

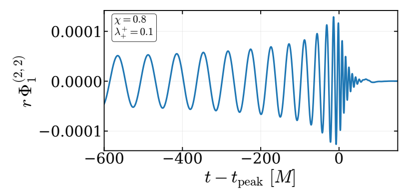
Figure I.9 Extracted Φ_1 waveforms for $q = 1$ and $\chi = 0.4$.



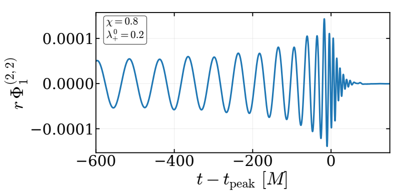
(a) $\lambda_0^+ = 0.10$



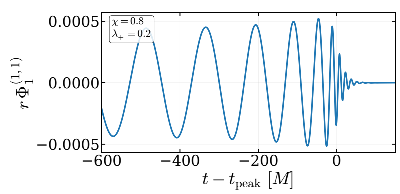
(b) $\lambda_-^+ = 0.10$



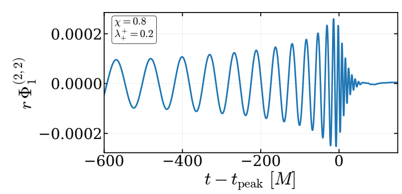
(c) $\lambda_+^+ = 0.10$



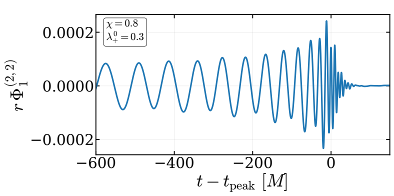
(d) $\lambda_0^+ = 0.20$



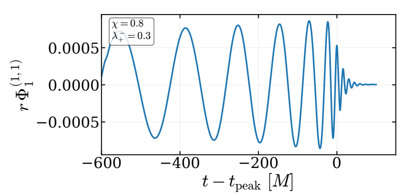
(e) $\lambda_-^+ = 0.20$



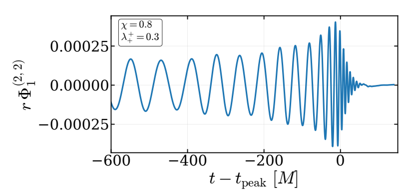
(f) $\lambda_+^+ = 0.20$



(g) $\lambda_0^+ = 0.30$

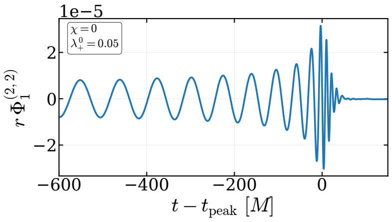


(h) $\lambda_-^+ = 0.30$

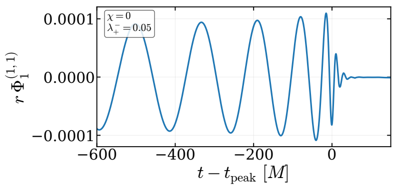


(i) $\lambda_+^+ = 0.30$

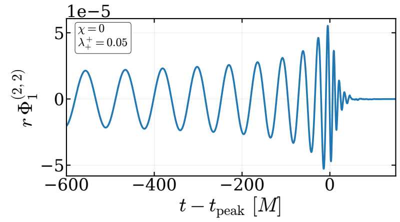
Figure I.10 Extracted Φ_1 waveforms for $q = 1$ and $\chi = 0.8$.



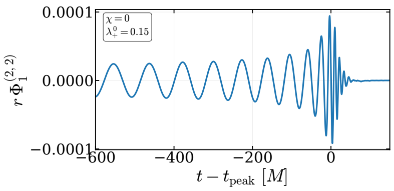
(a) $\lambda_0^+ = 0.05$



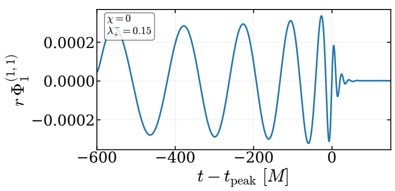
(b) $\lambda_-^+ = 0.05$



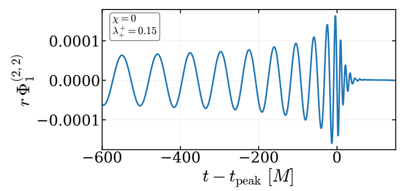
(c) $\lambda_+^+ = 0.05$



(d) $\lambda_0^+ = 0.15$

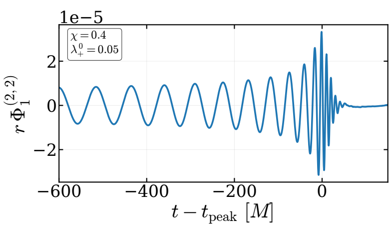


(e) $\lambda_-^+ = 0.15$

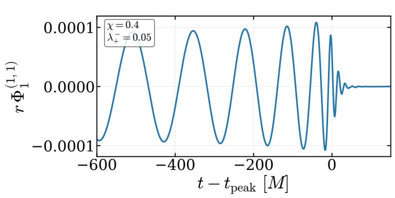


(f) $\lambda_+^+ = 0.15$

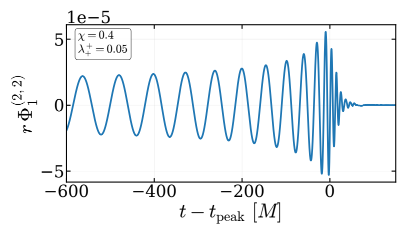
Figure I.11 Extracted Φ_1 waveforms for $q = 2$ and $\chi = 0$.



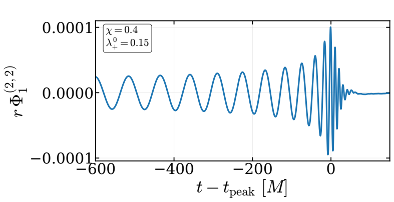
(a) $\lambda_0^+ = 0.05$



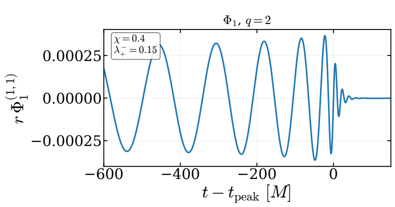
(b) $\lambda_-^+ = 0.05$



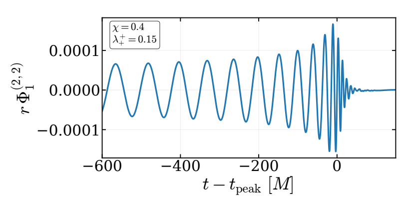
(c) $\lambda_+^+ = 0.05$



(d) $\lambda_0^+ = 0.15$

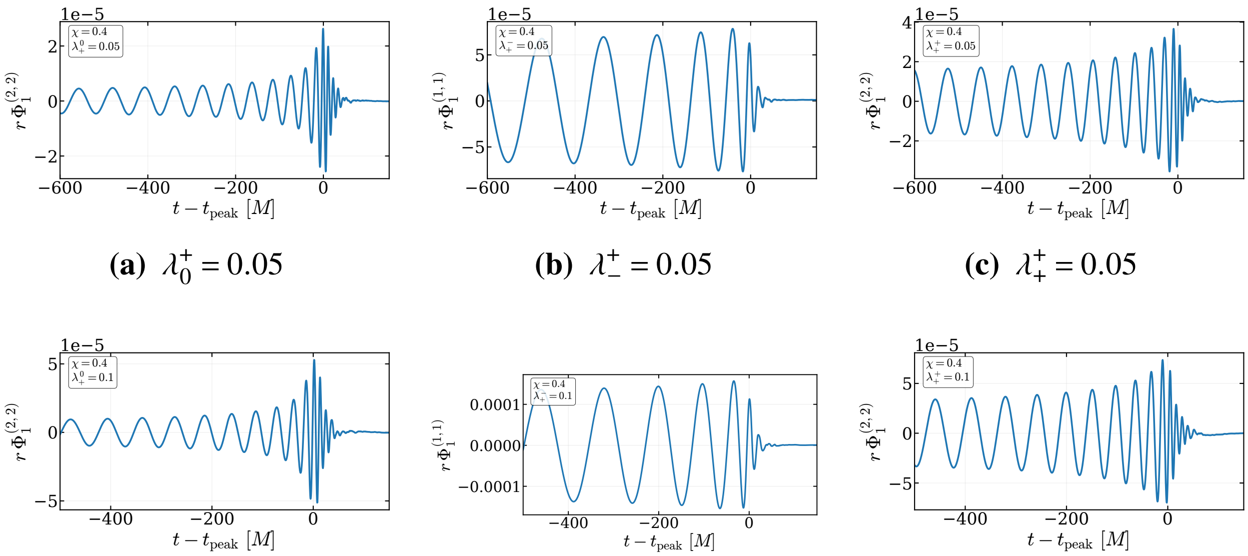


(e) $\lambda_-^+ = 0.15$



(f) $\lambda_+^+ = 0.15$

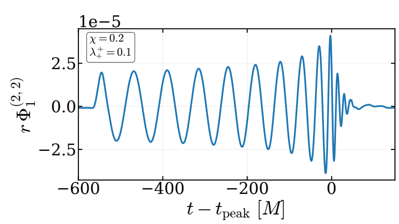
Figure I.12 Extracted Φ_1 waveforms for $q = 2$ and $\chi = 0.4$.



(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

(d) $\lambda_0^+ = 0.10$ **(e)** $\lambda_-^+ = 0.10$ **(f)** $\lambda_+^+ = 0.10$

Figure I.13 Extracted Φ_1 waveforms for $q = 4$ and $\chi = 0.4$.

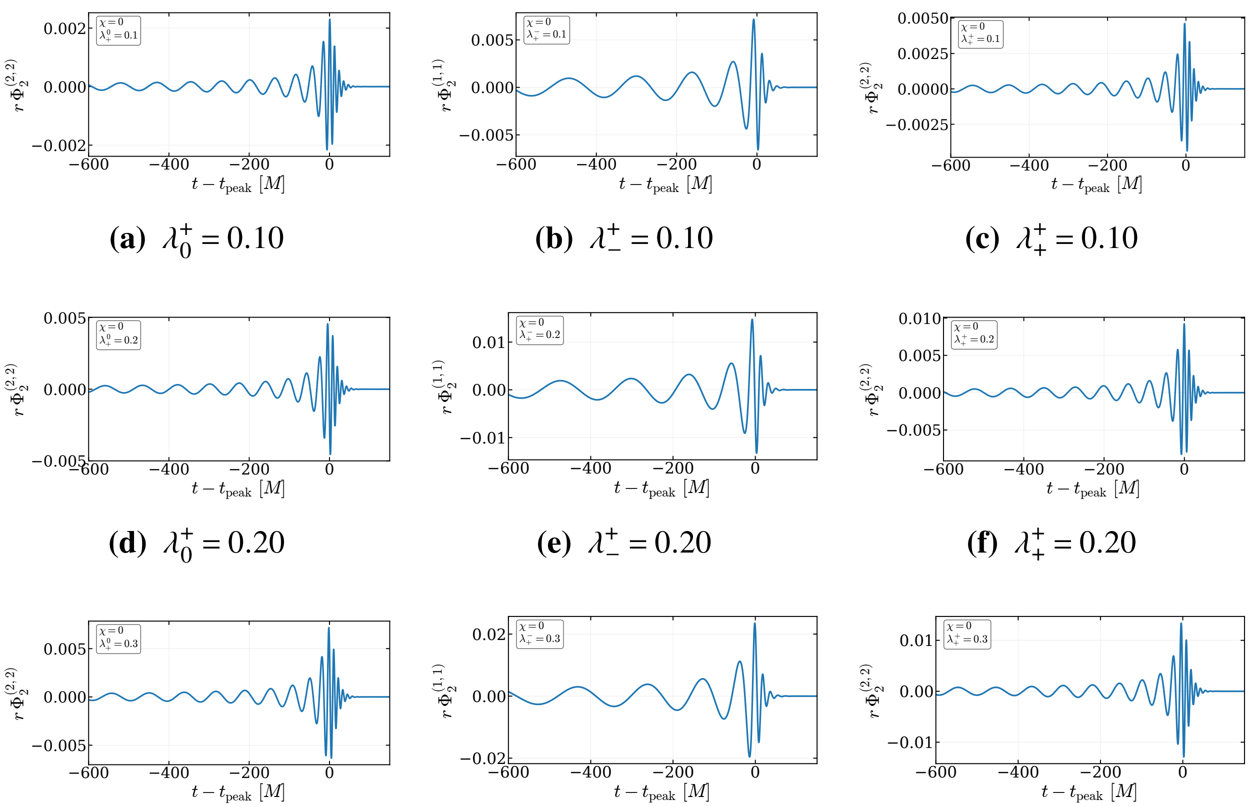


(a) $\lambda_+^+ = 0.10$

Figure I.14 Extracted Φ_1 waveforms for $q = 8$ and $\chi = 0.2$.

I.3 Φ_2 Waveform Catalog

The following figures show the extracted Φ_2 waveforms at $r \simeq 100M$. Since there is no radiation in the neutral case, only charged configurations are shown. The charged cases are ordered by charge magnitude and charge configuration. For the $+-$ cases, we show the $(\ell, m) = (1, 1)$ mode, while for the $+0$ and $++$ cases we show the $(\ell, m) = (2, 2)$ mode.

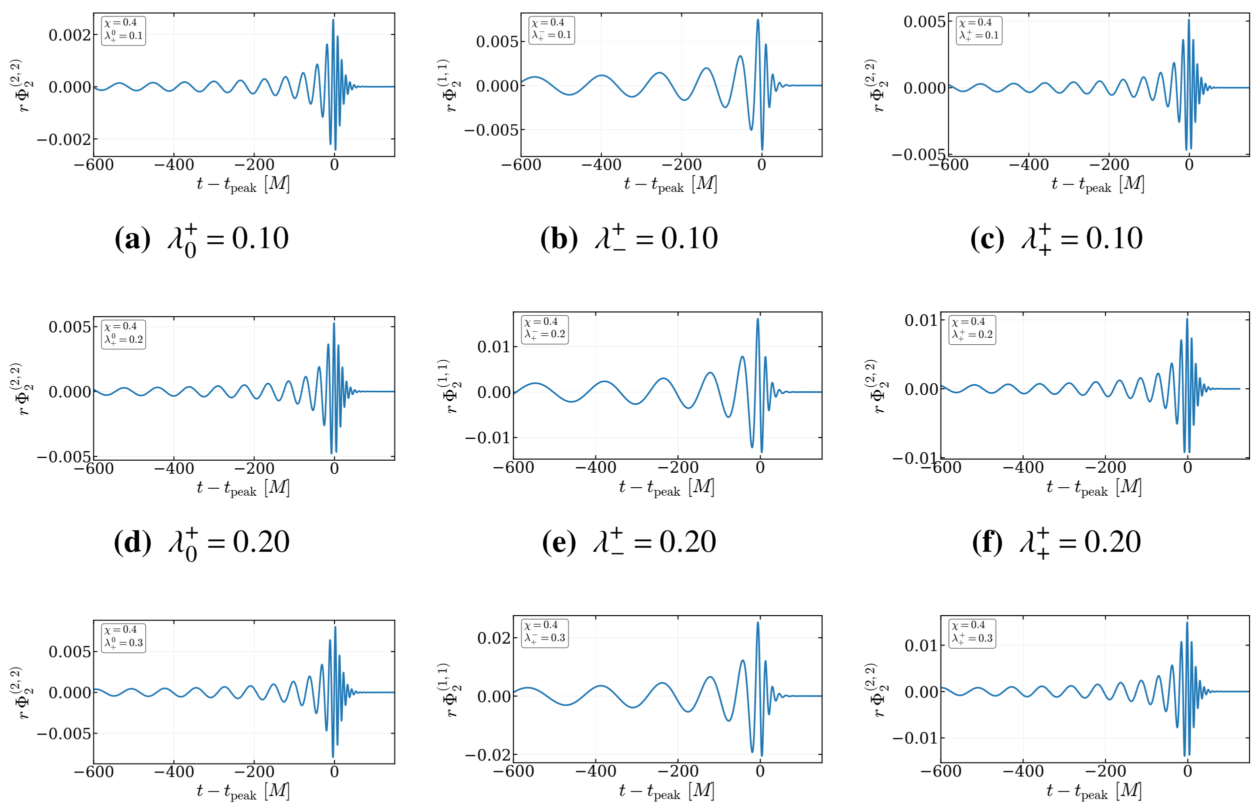


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

Figure I.15 Extracted Φ_2 waveforms for $q = 1$ and $\chi = 0$.



(a) $\lambda_0^+ = 0.10$

(b) $\lambda_-^+ = 0.10$

(c) $\lambda_+^+ = 0.10$

(d) $\lambda_0^+ = 0.20$

(e) $\lambda_-^+ = 0.20$

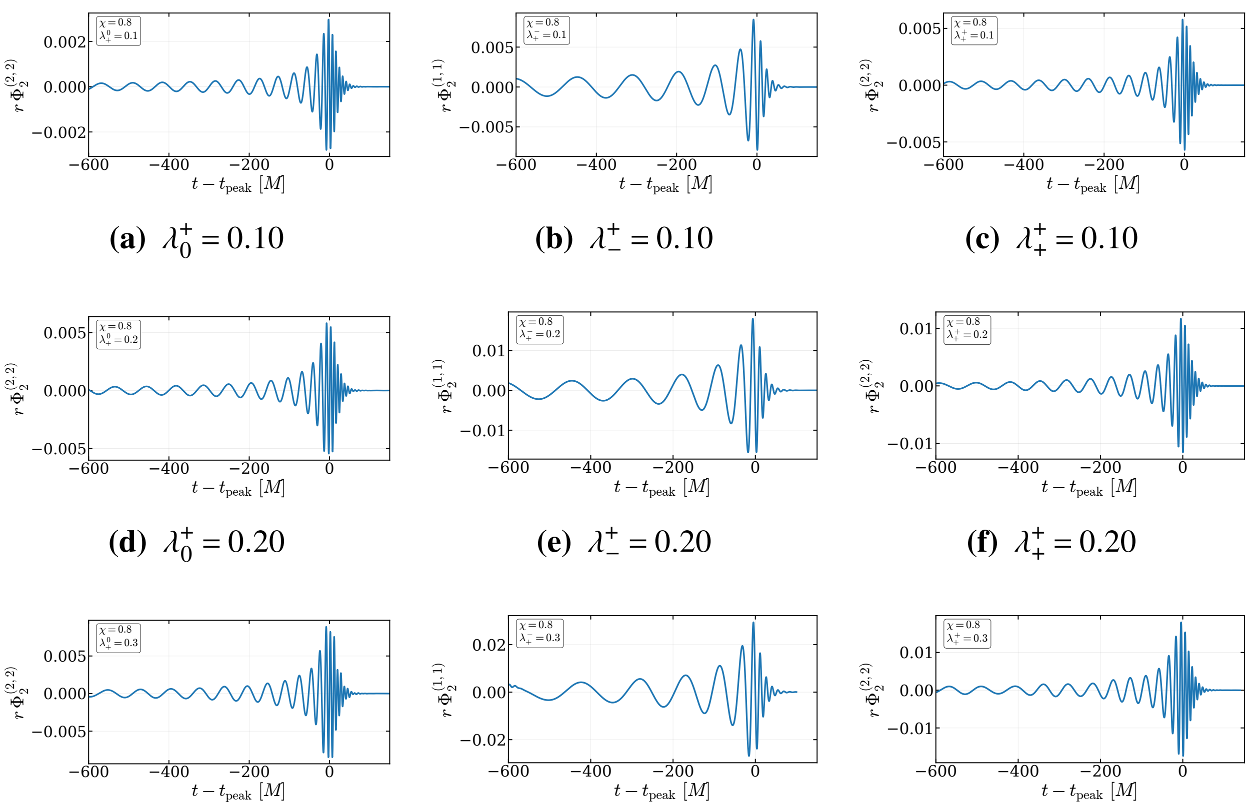
(f) $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$

(h) $\lambda_-^+ = 0.30$

(i) $\lambda_+^+ = 0.30$

Figure I.16 Extracted Φ_2 waveforms for $q = 1$ and $\chi = 0.4$.

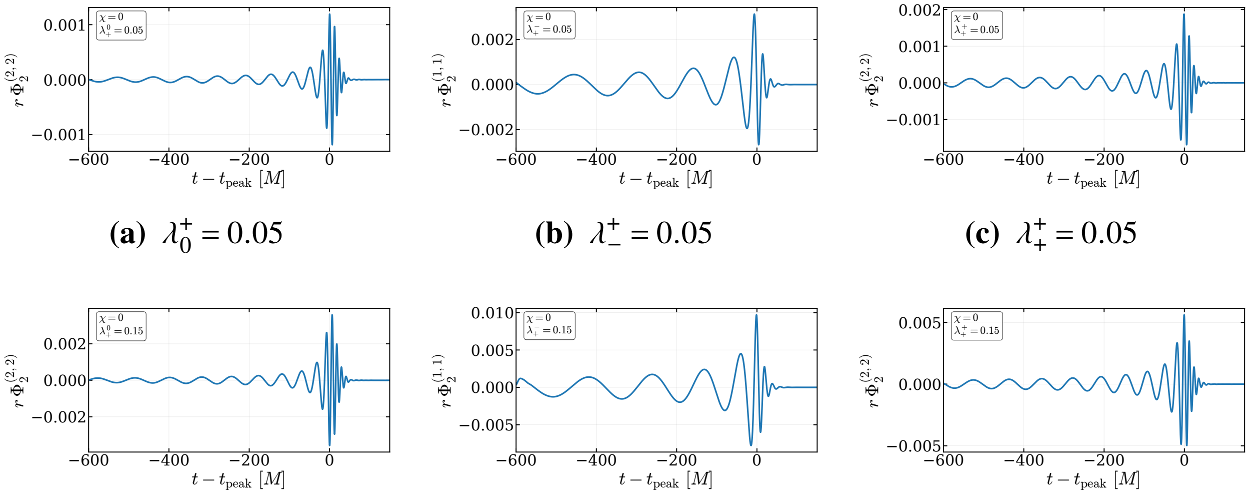


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

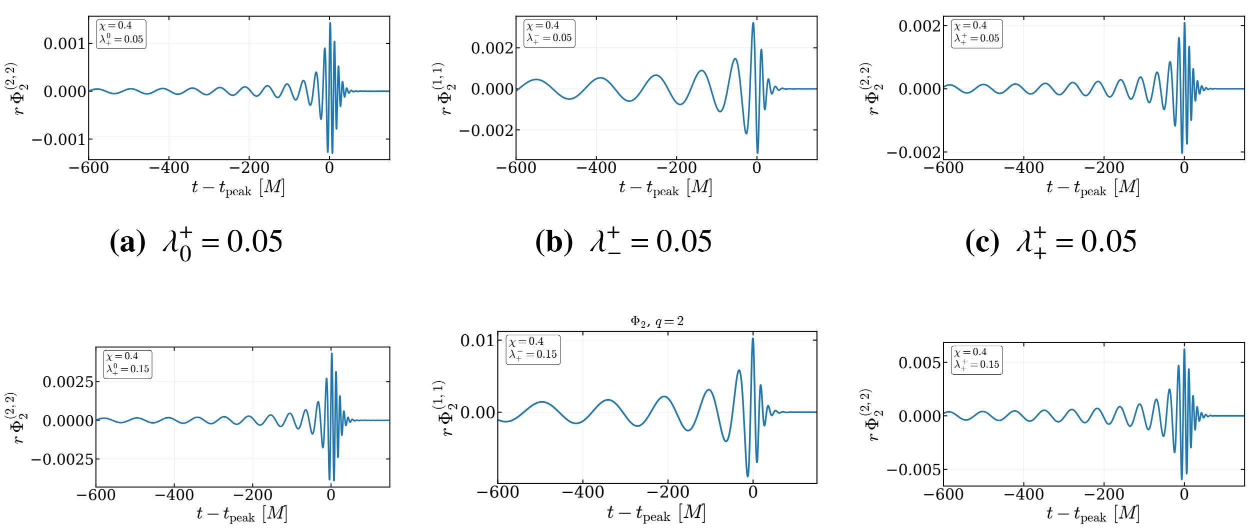
Figure I.17 Extracted Φ_2 waveforms for $q = 1$ and $\chi = 0.8$.



(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

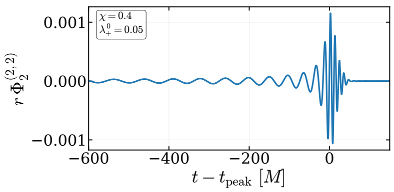
Figure I.18 Extracted Φ_2 waveforms for $q = 2$ and $\chi = 0$.



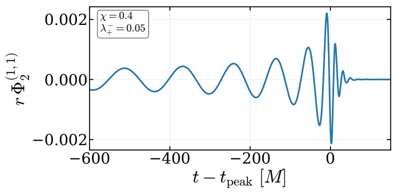
(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

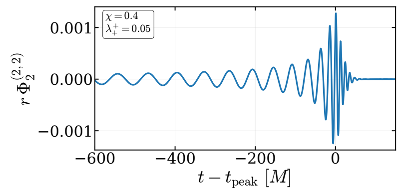
Figure I.19 Extracted Φ_2 waveforms for $q = 2$ and $\chi = 0.4$.



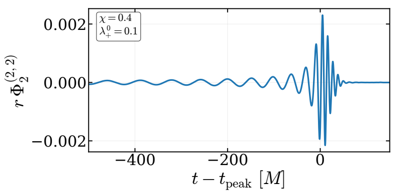
(a) $\lambda_0^+ = 0.05$



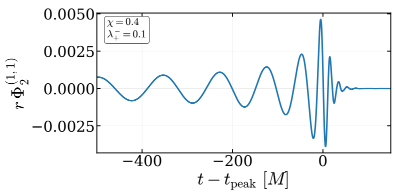
(b) $\lambda_-^+ = 0.05$



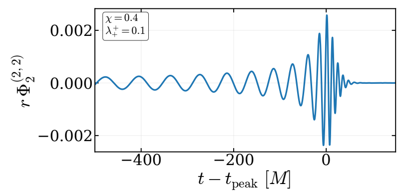
(c) $\lambda_+^+ = 0.05$



(d) $\lambda_0^+ = 0.10$

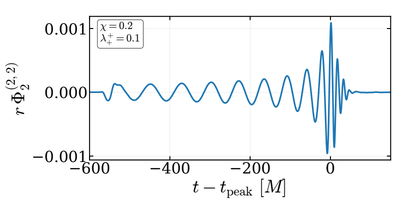


(e) $\lambda_-^+ = 0.10$



(f) $\lambda_+^+ = 0.10$

Figure I.20 Extracted Φ_2 waveforms for $q = 4$ and $\chi = 0.4$.

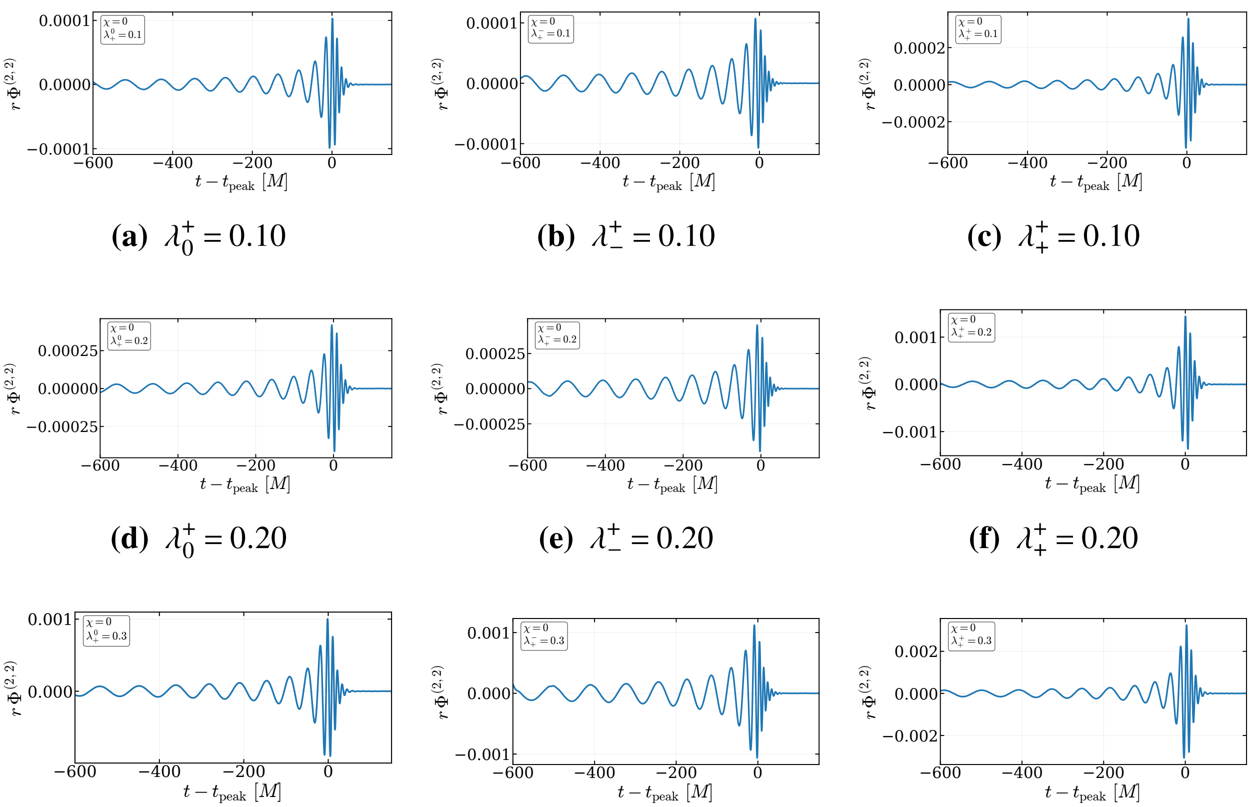


(a) $\lambda_+^+ = 0.10$

Figure I.21 Extracted Φ_2 waveforms for $q = 8$ and $\chi = 0.2$.

I.4 Dilaton ϕ Waveform Catalog

The following figures show the extracted dilaton waveforms at $r \simeq 100M$. Only charged configurations are shown. The charged cases are ordered by charge magnitude and charge configuration. We show the $(\ell, m) = (2, 2)$ dilaton mode.

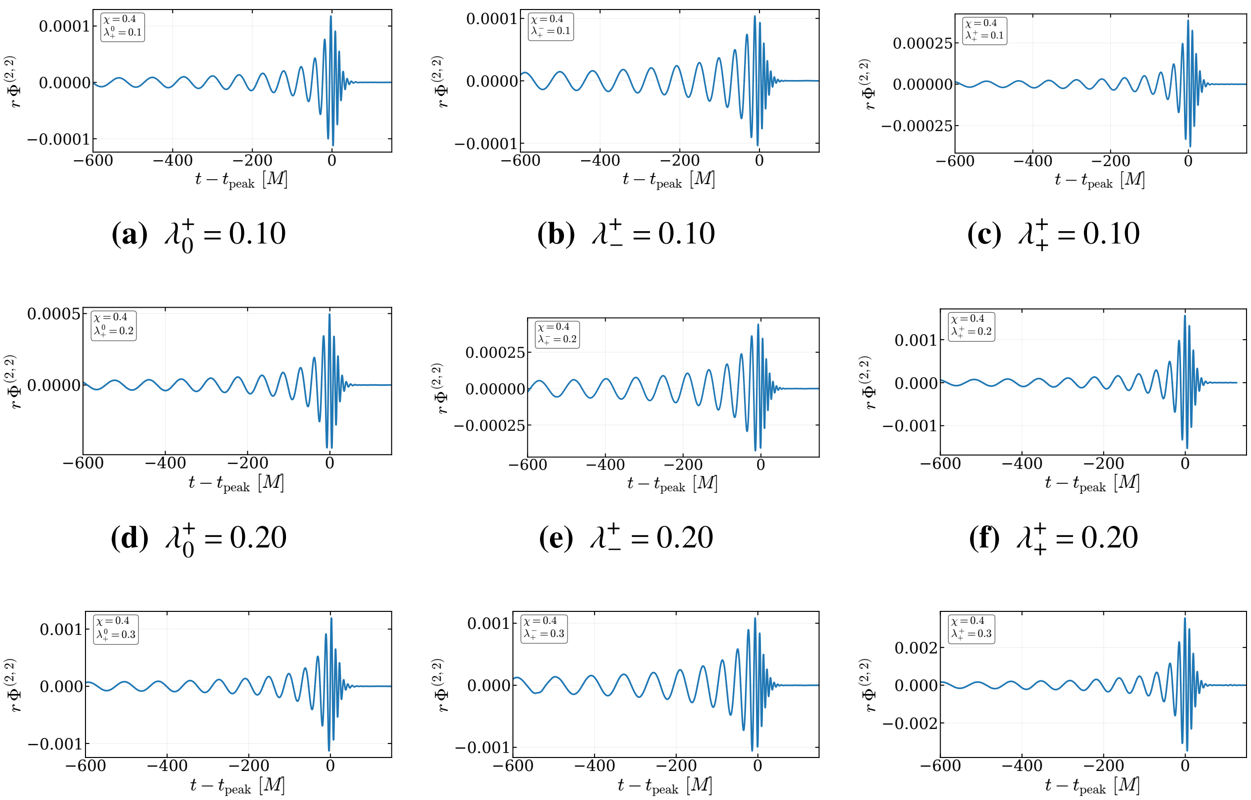


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

Figure I.22 Extracted dilaton waveforms for $q = 1$ and $\chi = 0$.

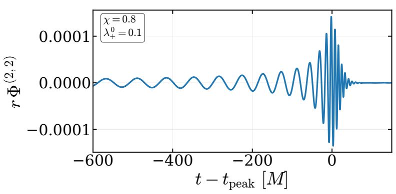


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

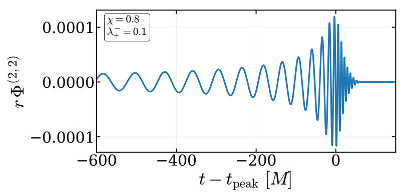
(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

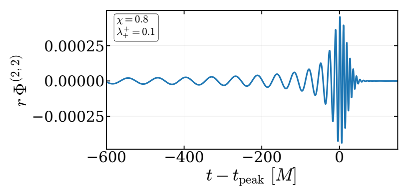
Figure I.23 Extracted dilaton waveforms for $q = 1$ and $\chi = 0.4$.



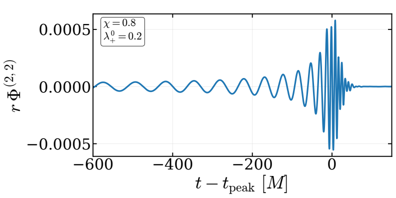
(a) $\lambda_0^+ = 0.10$



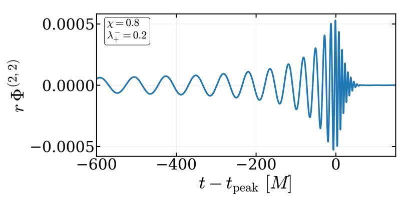
(b) $\lambda_-^+ = 0.10$



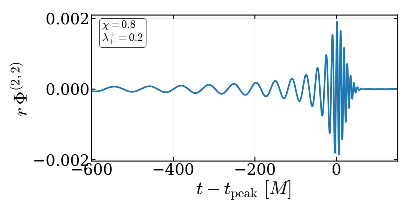
(c) $\lambda_+^+ = 0.10$



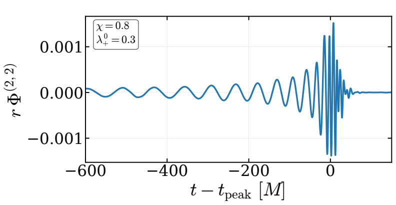
(d) $\lambda_0^+ = 0.20$



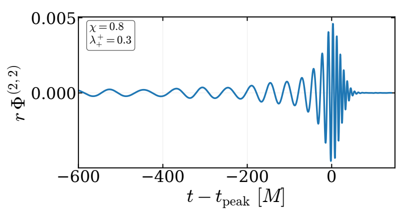
(e) $\lambda_-^+ = 0.20$



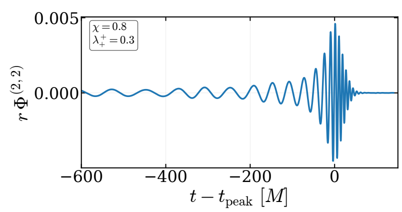
(f) $\lambda_+^+ = 0.20$



(g) $\lambda_0^+ = 0.30$

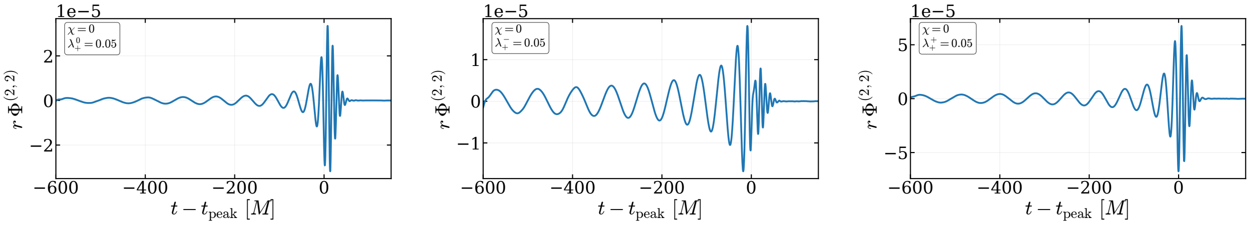


(h) $\lambda_-^+ = 0.30$

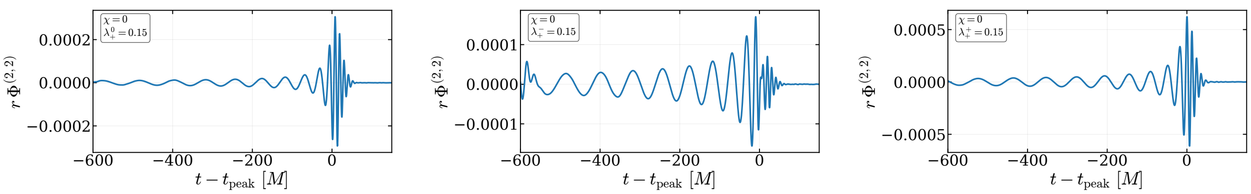


(i) $\lambda_+^+ = 0.30$

Figure I.24 Extracted dilaton waveforms for $q = 1$ and $\chi = 0.8$.

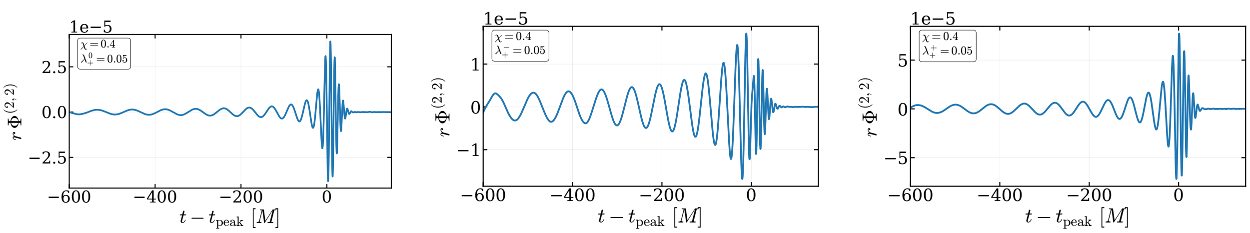


(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

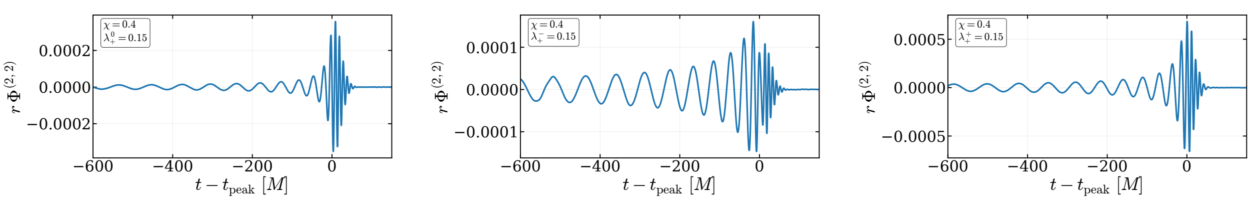


(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

Figure I.25 Extracted dilaton waveforms for $q = 2$ and $\chi = 0$.

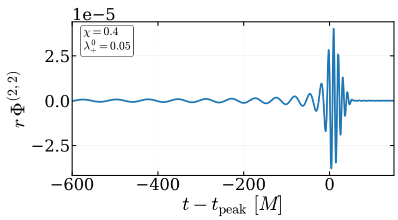


(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

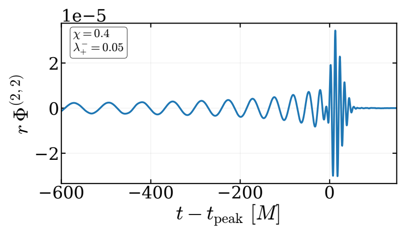


(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

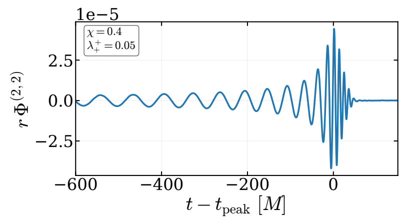
Figure I.26 Extracted dilaton waveforms for $q = 2$ and $\chi = 0.4$.



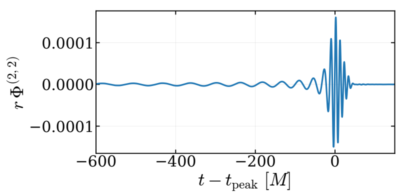
(a) $\lambda_0^+ = 0.05$



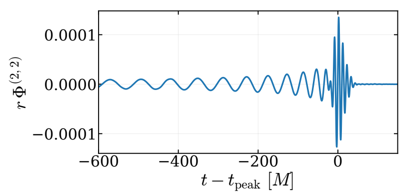
(b) $\lambda_-^+ = 0.05$



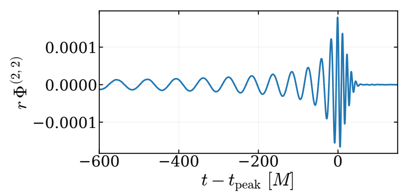
(c) $\lambda_+^+ = 0.05$



(d) $\lambda_0^+ = 0.10$

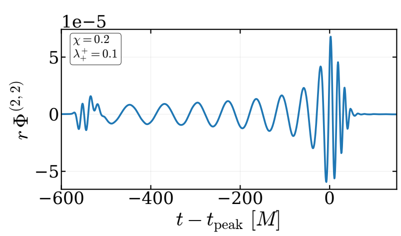


(e) $\lambda_-^+ = 0.10$



(f) $\lambda_+^+ = 0.10$

Figure I.27 Extracted dilaton waveforms for $q = 4$ and $\chi = 0.4$.

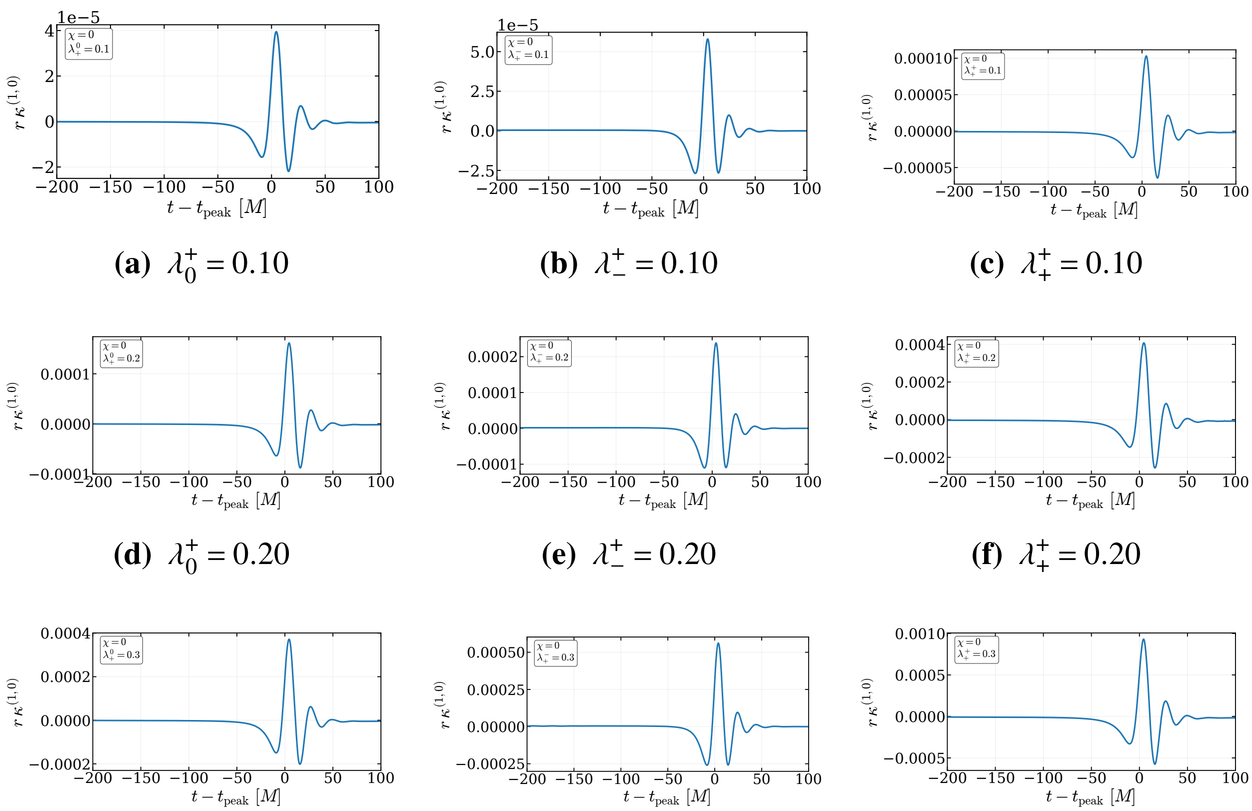


(a) $\lambda_+^+ = 0.10$

Figure I.28 Extracted dilaton waveform for $q = 8$ and $\chi = 0.2$.

I.5 Axion κ Waveform Catalog

The following figures show the extracted axion waveforms at $r \simeq 100M$. Only charged configurations are shown. The charged cases are ordered by charge magnitude and charge configuration. We show the $(\ell, m) = (1, 0)$ axion mode.

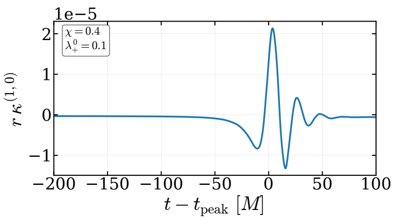


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

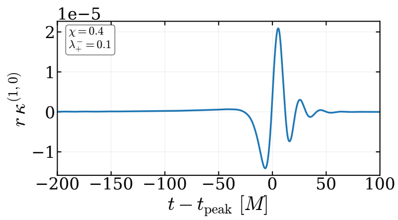
(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

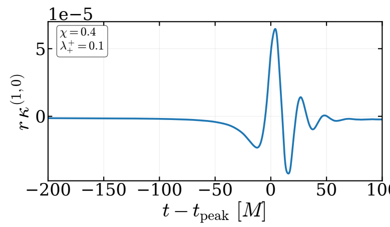
Figure I.29 Extracted axion waveforms for $q = 1$ and $\chi = 0$.



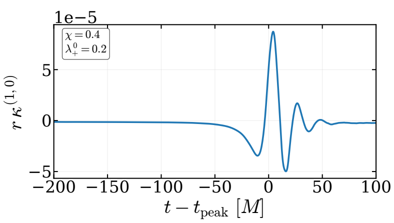
(a) $\lambda_0^+ = 0.10$



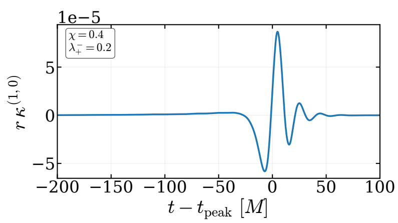
(b) $\lambda_-^+ = 0.10$



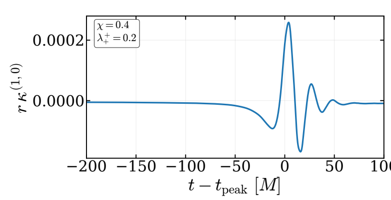
(c) $\lambda_+^+ = 0.10$



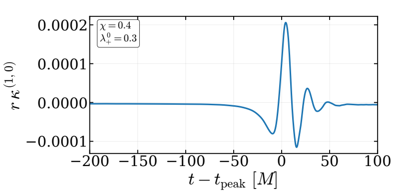
(d) $\lambda_0^+ = 0.20$



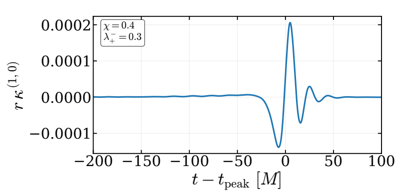
(e) $\lambda_-^+ = 0.20$



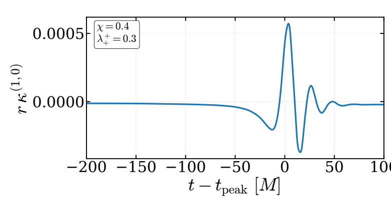
(f) $\lambda_+^+ = 0.20$



(g) $\lambda_0^+ = 0.30$

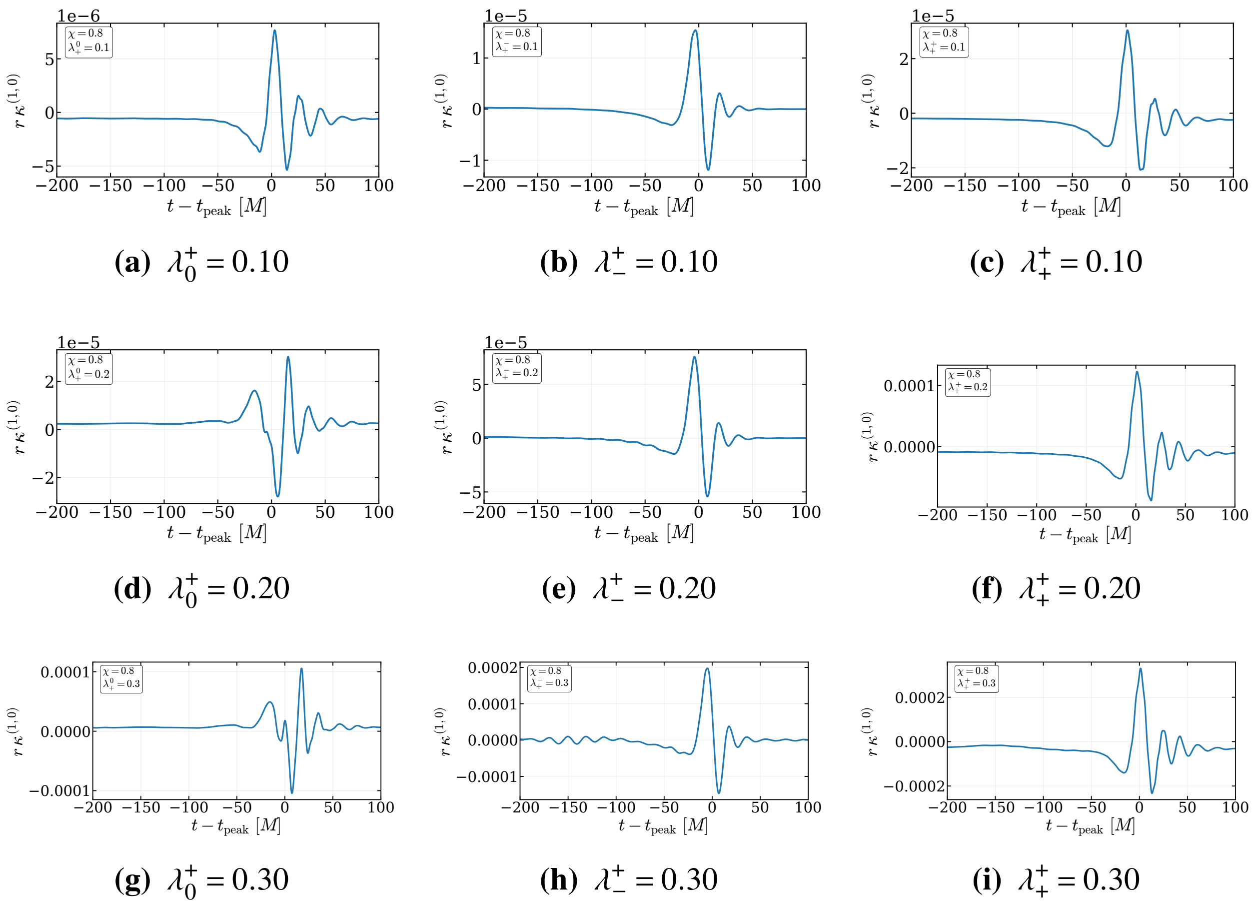


(h) $\lambda_-^+ = 0.30$



(i) $\lambda_+^+ = 0.30$

Figure I.30 Extracted axion waveforms for $q = 1$ and $\chi = 0.4$.

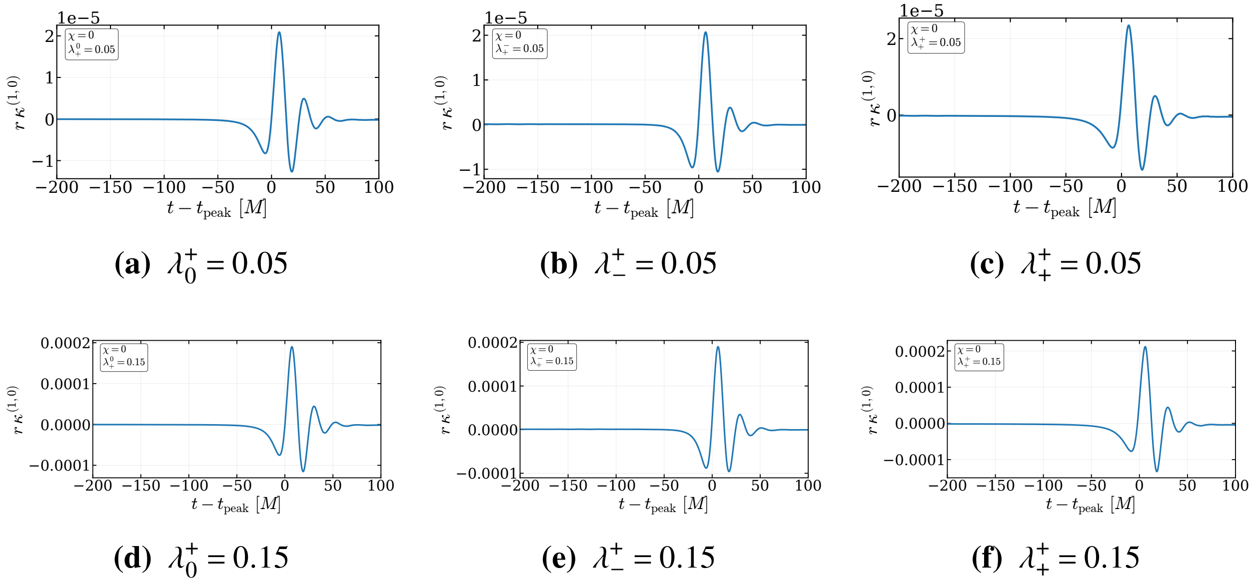


(a) $\lambda_0^+ = 0.10$ **(b)** $\lambda_-^+ = 0.10$ **(c)** $\lambda_+^+ = 0.10$

(d) $\lambda_0^+ = 0.20$ **(e)** $\lambda_-^+ = 0.20$ **(f)** $\lambda_+^+ = 0.20$

(g) $\lambda_0^+ = 0.30$ **(h)** $\lambda_-^+ = 0.30$ **(i)** $\lambda_+^+ = 0.30$

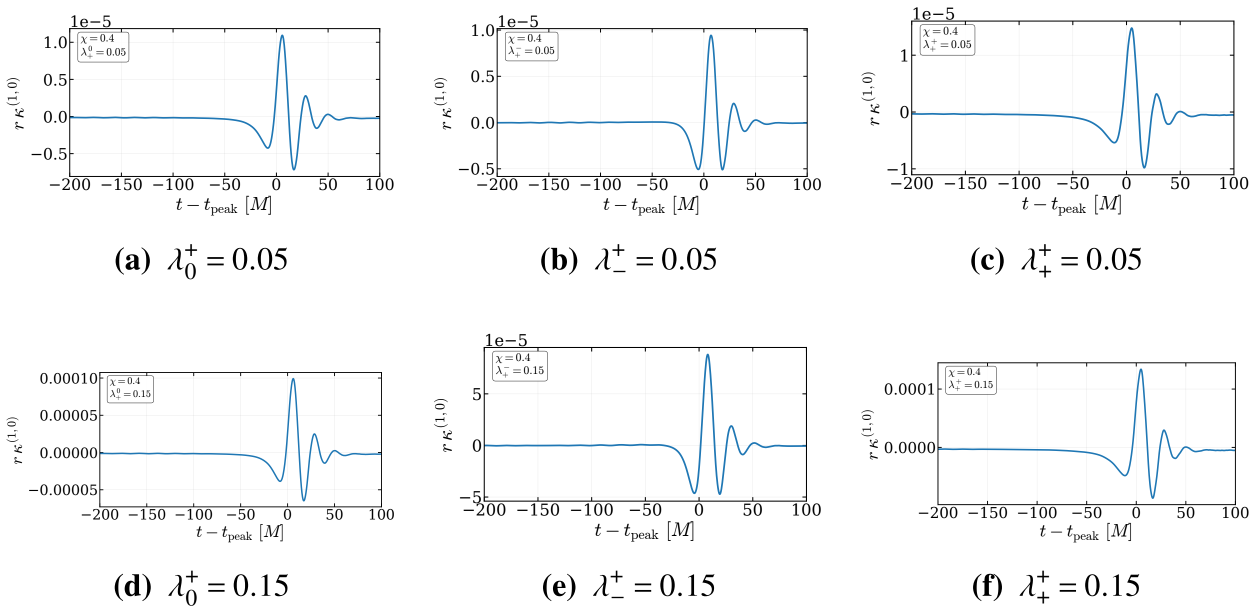
Figure I.31 Extracted axion waveforms for $q = 1$ and $\chi = 0.8$.



(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

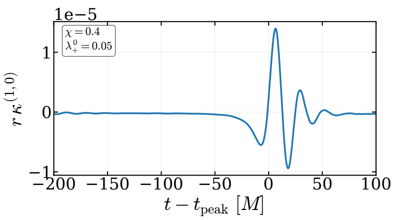
Figure I.32 Extracted axion waveforms for $q = 2$ and $\chi = 0$.



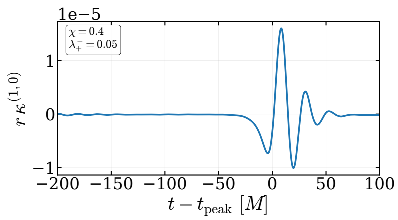
(a) $\lambda_0^+ = 0.05$ **(b)** $\lambda_-^+ = 0.05$ **(c)** $\lambda_+^+ = 0.05$

(d) $\lambda_0^+ = 0.15$ **(e)** $\lambda_-^+ = 0.15$ **(f)** $\lambda_+^+ = 0.15$

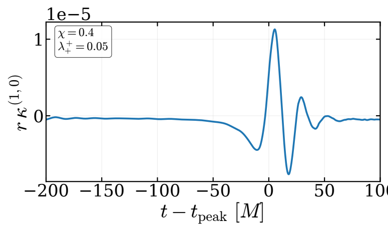
Figure I.33 Extracted axion waveforms for $q = 2$ and $\chi = 0.4$.



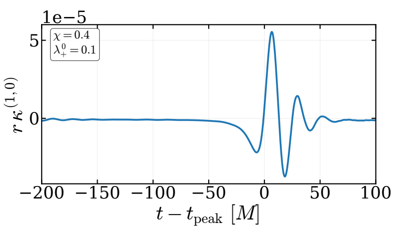
(a) $\lambda_0^+ = 0.05$



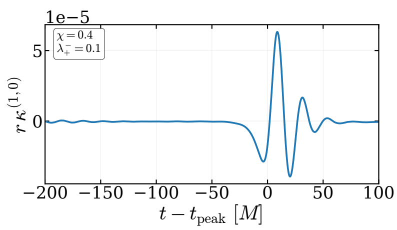
(b) $\lambda_-^+ = 0.05$



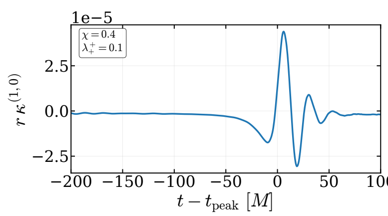
(c) $\lambda_+^+ = 0.05$



(d) $\lambda_0^+ = 0.10$

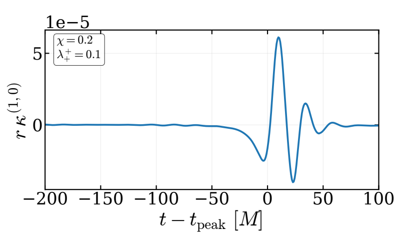


(e) $\lambda_-^+ = 0.10$



(f) $\lambda_+^+ = 0.10$

Figure I.34 Extracted axion waveforms for $q = 4$ and $\chi = 0.4$.



(a) $\lambda_+^+ = 0.10$

Figure I.35 Extracted axion waveform for $q = 8$ and $\chi = 0.2$.

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